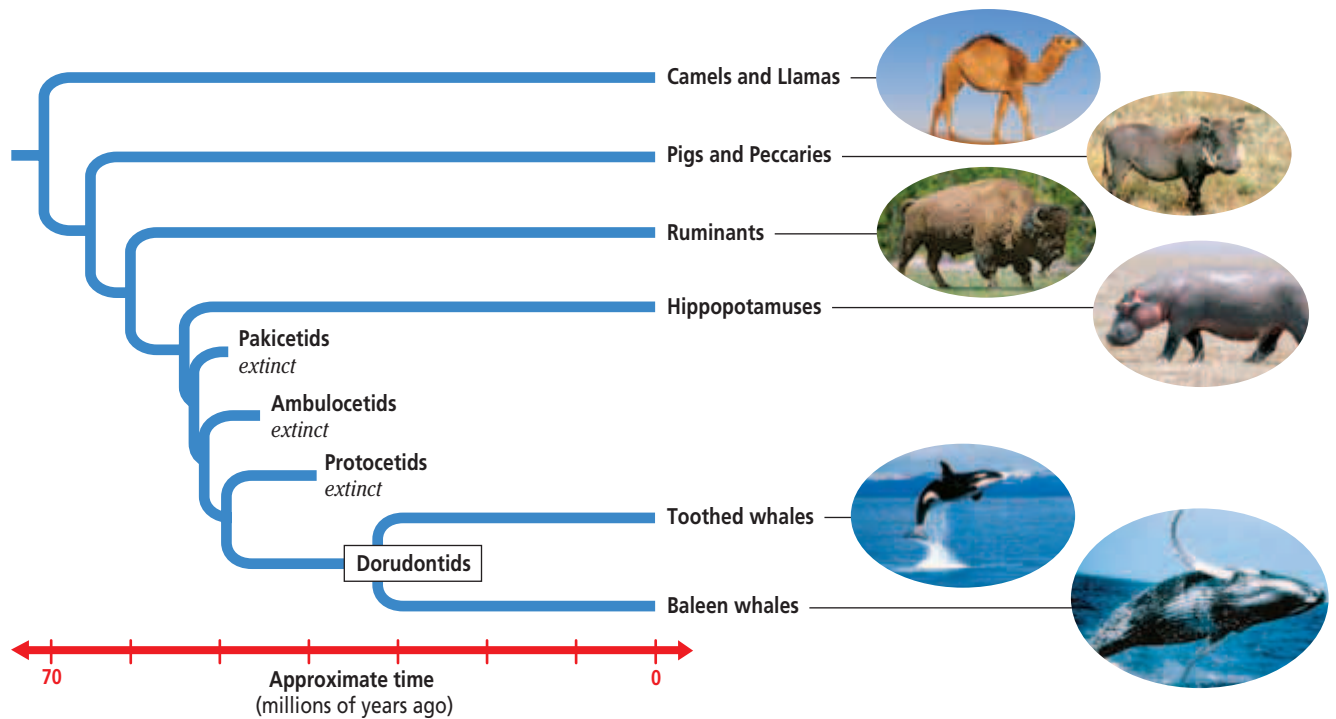




For example, scientists try to model **phylogeny**, the relationships by ancestry among groups of organisms. To do so, they must analyze many kinds of evidence. Sometimes, different evidence supports different models. Figure 15-10 shows a phylogenetic diagram, or “tree,” which models a hypothesized phylogeny. This particular phylogeny was based on an analysis of biological molecules in animals that seem closely related to whales. The “trunk” represents a past species that could have been the ancestor of all these animals. Each “branch” over time represents a separate population or lineage. More closely related groups appear closer to each other on a branch. For example, whales are more closely related to hippopotamuses than to camels. Hypotheses such as these are supported when independent anatomical and biochemical data match the same model.

FIGURE 15-10

This phylogenetic diagram, or “tree,” shows one hypothesis of the ancestral relationships among whales and the group of hoofed mammals known as *artiodactyls*. Scientists construct models such as these by comparing the anatomical or biochemical traits of many organisms. For updates on phylogenetic information, visit go.hrw.com and enter the keyword HM6 Phylo.

SECTION 2 REVIEW

1. Relate several inferences about the history of life that are supported by geologic evidence.
2. What evidence supports the hypothesis that whales evolved from land-dwelling mammals?
3. Compare the concepts of homologous structures, analogous structures, and vestigial structures.
4. Explain the evidence of evolution presented by the mammals of Australia.
5. Compare the use of biological molecules to other types of analysis of phylogeny.

CRITICAL THINKING

6. **Analyzing Theories** Does natural selection act on vestigial structures? Support your answer.
7. **Analyzing Models** If the DNA of a whale, a hippopotamus, and a camel were compared, what finding would support the model in Figure 15-10 above?
8. **Inferring Relationships** Fly embryos and frog embryos differ from each other more than frog embryos and ape embryos do. What does this imply about how these groups may be related?

SECTION 3

OBJECTIVES

- **Describe** how convergent evolution can result among different species.
- **Explain** how divergent evolution can lead to species diversity.
- **Compare** artificial selection and natural selection.
- **Explain** how organisms can undergo coevolution.

VOCABULARY

convergent evolution
divergent evolution
adaptive radiation
artificial selection
coevolution

EVOLUTION IN ACTION

Evolution is a continuous process. Evolution is going on today in populations of living species and can be observed, recorded, and tested. Patterns of evolution repeat in different times and places. Interactions between species, including humans, affect their ongoing evolution.

CASE STUDY: CARIBBEAN ANOLE LIZARDS

Often, when scientists compare groups of species, the scientists find patterns that are best explained as evolution in progress. An example is the comparison of anole lizard species (genus *Anolis*) on the Caribbean islands of Cuba, Hispaniola, Jamaica, and Puerto Rico. Among these lizards, each species' body type correlates with the habitat in which the species lives, as shown in Figure 15-11. For example, anole species that live mainly on tree trunks have stocky bodies and long legs. In contrast, those that reside on slender twigs have thin bodies, short legs and tails, and large toe pads. Grass-dwelling anoles tend to be slender and have very long tails. In all, there are at least six anole body types that are each adapted to their environment in a unique way. Also, distinct species of anoles with the same body types occur on different islands. For example, a distinct species of twig-dwelling anole is found on each island.

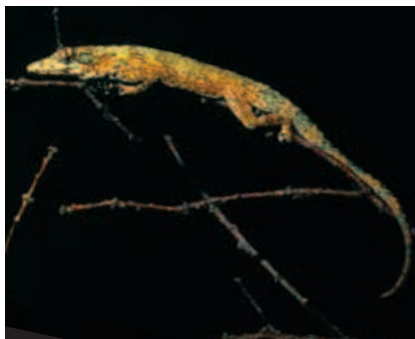
Many different hypotheses could explain these observations. Two possibilities are that (1) an ancestral anole species specialized for living on twigs originally lived on one island and later migrated to other islands or that (2) each twig-dwelling species evolved independently on each island from distinct ancestor anole species.

FIGURE 15-11

Each of these lizards is a member of the genus *Anolis* and lives on the island of Hispaniola in the Caribbean. One species (a) dwells mainly on tree trunks and on the ground and has much longer legs than a species (b) that mostly inhabits tree branches. Another species (c) stays mainly in the grass and has a long tail.



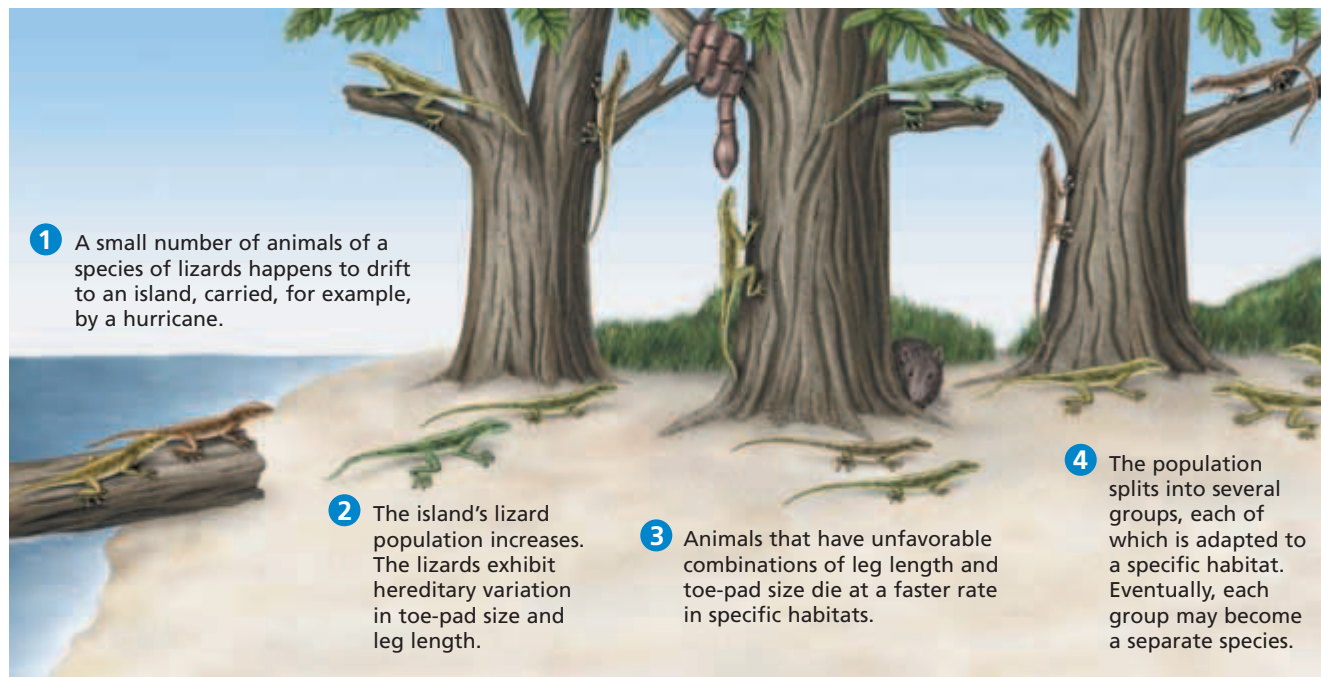
(a) *Anolis cybotes*



(b) *Anolis insolitus*



(c) *Anolis pulchellus*



Biologists tested these hypotheses by comparing DNA from various species to look for closely related species. The DNA evidence supported hypothesis 2—twig-dwelling species evolved independently on each island. In other words, each twig-dwelling species came from different ancestors but evolved similar adaptations to similar habitats. The process by which different species evolve similar traits is called **convergent evolution**. Many other examples of convergent evolution can be found in nature.

FIGURE 15-12

This diagram shows a possible scenario to explain, through natural selection, the evolution of a variety of anole lizard species in the Caribbean islands by descent from common ancestors.

Divergence and Radiation

A model of Caribbean anole evolution must also explain how the lizards became adapted to their particular habitats. Studies showed that long-legged trunk-dwelling species could run faster on flat surfaces than short-legged twig-dwelling species, but the twig-dwelling species could cling to twigs better and did not fall as often. However, both kinds of lizards on each island were closely related.

The best explanation for this pattern of phylogeny is that divergent evolution occurred on each island. **Divergent evolution** is a process in which the descendants of a single ancestor diversify into species that each fit different parts of the environment. Lizards with genes for large toe pads and short legs ran so slowly on the trunk and ground that predators often caught them, and lizards with long legs and small toe pads often slipped if they climbed thin branches.

Sometimes, a new population in a new environment, such as an island, will undergo divergent evolution until the population fills many parts of the environment. This pattern of divergence is called **adaptive radiation**. Figure 15-12 illustrates a possible scenario for the evolution of Caribbean anole lizards. Fossil evidence suggests many cases of adaptive radiation on the geologic time scale.

Quick Lab

Observing Adaptations Around You

Materials paper and pencil

Procedure Observe organisms around your school grounds or around your home. Describe any traits that seem to be adaptations to a particular environment or way of life. Also, look for and describe variations within groups of organisms that you see. Explain your reasoning for each inference you make about adaptations.

Analysis Which variations in the traits that you observed might increase or decrease the fitness of the organisms? Explain your reasoning.



FIGURE 15-13

Recent DNA evidence shows that despite the enormous variation among domestic dogs, all varieties descended from Asian wolves. By artificially selecting the dogs that will be the parents of the next generation, people have increased the rate of divergent evolution among domestic dogs.

ARTIFICIAL SELECTION

Darwin started his famous book with a chapter on **artificial selection**. This process occurs when a human breeder chooses individuals that will parent the next generation. For example, humans may choose to breed oat plants that yield more grain per stalk or greyhounds that run faster. Because of the immense differences among varieties of dogs, as shown in Figure 15-13, Darwin doubted that all domestic dog breeds arose from the same wild species. But in the 2000s, geneticists analyzed DNA from 654 dog breeds, including ancient dog remains. Their findings indicated that all breeds of dogs share DNA similarities with wolves in East Asia. These findings support the hypothesis that humans first selected domestic dogs from a wolf population about 15,000 years ago.

COEVOLUTION

It is important to keep in mind that evolution is ongoing and that in a given environment, many species may be evolving at once. Each species is part of the forces of natural selection that act upon the other species. When two or more species have evolved adaptations to each other's influence, the situation is called **coevolution**.

Through coevolution, some species have evolved strategies to avoid being eaten, while the animals that eat them have evolved strategies to keep eating them. Many flowering plants have evolved such that specific insects carry pollen to other plants. Some microbes have evolved to live within certain animals, while these animals have adapted to either benefit from or avoid the microbes.

Humans are also involved in many cases of coevolution. For example, humans have developed and used antibiotics, such as penicillin, to kill disease-causing bacteria. But as antibiotic use has increased, many populations of bacteria have evolved adaptations to resist the effects of some antibiotics. This kind of adaptation is called *resistance*. Similarly, the evolution of resistance to pesticides is observed among populations of insects in agricultural settings.

SECTION 3 REVIEW

1. Explain how the anole lizard species on Caribbean islands demonstrate both convergent and divergent evolution.
2. What are the key differences and similarities between natural selection and artificial selection?
3. Give examples of species that are likely to be coevolving. Describe how each species influences the evolution of the other species.

CRITICAL THINKING

4. **Inferring Meaning** What is the meaning of *radiation* as used in the term *adaptive radiation*?
5. **Constructing Models** Draw a phylogenetic tree to match each of the two proposed hypotheses for the evolution of the anole lizards.
6. **Analyzing Patterns** Propose a reason why some Caribbean islands lack lizard species.

CHAPTER HIGHLIGHTS

SECTION 1

History of Evolutionary Thought

- Evolution is the process of change in the inherited characteristics within populations over generations such that new types of organisms develop from preexisting types.
- Scientific understanding of evolution began to develop in the 17th and 18th centuries as geologists and naturalists compared geologic processes and living and fossil organisms around the world.
- Among geologists, Cuvier promoted the idea of catastrophism, and Lyell promoted uniformitarianism. Among naturalists, Lamarck proposed the inheritance of acquired characteristics as a mechanism for evolution.
- After making many observations and considering ideas of other scientists, both Darwin and Wallace proposed the theory of natural selection to explain evolution.
- Darwin wrote *On the Origin of Species*, in which he argued that descent with modification occurs, that all species descended from common ancestors, and that natural selection is the mechanism for evolution.
- Organisms in a population adapt to their environment as the proportion of individuals with genes for favorable traits increases. Those individuals that pass on more genes are considered to have greater fitness.

Vocabulary

evolution (p. 297)
strata (p. 298)

natural selection (p. 300)

adaptation (p. 300)

fitness (p. 301)

SECTION 2

Evidence of Evolution

- Evidence of evolution can be found by comparing several kinds of data, including the fossil record, biogeography, anatomy and development, and biological molecules. Evolutionary theories are supported when several kinds of evidence support similar conclusions.
- Geologic evidence supports theories about the age and development of Earth. The fossil record shows that the types and distribution of organisms on Earth have changed over time. Fossils of transitional species show evidence of descent with modification.
- Biogeography, the distribution of organisms, shows evidence of descent with modification.
- In organisms, analogous structures are similar in function but have different evolutionary origins. Homologous structures have a common evolutionary origin. A species with a vestigial structure probably shares ancestry with a species that has a functional form of the structure. Related species show similarities in embryological development.
- Similarity in the subunit sequences of biological molecules such as RNA, DNA, and proteins indicates a common evolutionary history.
- Modern scientists integrate Darwin's theory with other advances in biological knowledge. Theories and hypotheses about evolution continue to be proposed and investigated.

Vocabulary

fossil (p. 302)
superposition (p. 302)
relative age (p. 302)

absolute age (p. 303)
biogeography (p. 305)

homologous structure (p. 305)
analogous structure (p. 305)

vestigial structure (p. 306)
phylogeny (p. 307)

SECTION 3

Evolution in Action

- Ongoing examples of evolution among living organisms can be observed, recorded, and tested.
- In divergent evolution, related populations become less similar as they respond to different environments. Adaptive radiation is the divergent evolution of a single group of organisms in a new environment.
- In convergent evolution, organisms that are not closely related resemble each other because they have responded to similar environments.
- The great variety of dog breeds is an example of artificial selection.
- The increasing occurrence of antibiotic resistance among bacteria is an example of coevolution in progress.

Vocabulary

convergent evolution (p. 309)
divergent evolution (p. 309)

adaptive radiation (p. 309)

artificial selection (p. 310)

coevolution (p. 310)

CHAPTER REVIEW

USING VOCABULARY

- Use the following terms in the same sentence: *evolution*, *natural selection*, *adaptation*, and *fitness*.
- For each pair of terms, explain how the meanings of the terms differ.
 - acquired trait* and *inherited trait*
 - homologous structure* and *analogous structure*
 - relative age* and *absolute age*
 - divergent evolution* and *convergent evolution*
 - artificial selection* and *natural selection*
- Word Roots and Origins** The word *radiation* is derived from the Latin *radius*, which means “rod” or “ray.” Using this information, explain the meaning of *adaptive radiation*.

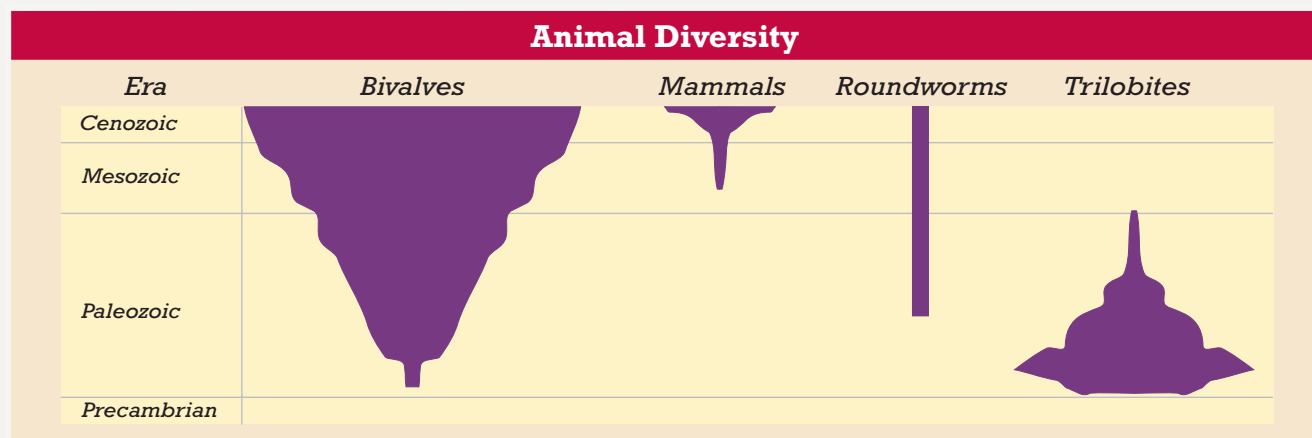
UNDERSTANDING KEY CONCEPTS

- Define** the biological process of evolution.
- Contrast** Cuvier’s catastrophism with Lyell’s uniformitarianism.
- Describe** how the finch species of the Galápagos Islands illustrate descent with modification.
- List** the steps of reasoning that Darwin gave to explain the process of natural selection.
- Identify** several factors that could limit the growth of populations.
- Relate** the roles of adaptation and fitness in the theory of natural selection.
- State** several inferences about evolution that are supported by fossil evidence.
- Describe** evidence from biogeography that species evolve adaptations to their environments.
- Identify** an example of a vestigial structure.

- Explain** how biological molecules indicate relatedness between species.
- Summarize** the examples of convergent and divergent evolution seen in Caribbean lizards.
- Describe** an example of coevolution.
- Explain** why antibiotics are not consistently effective against infections of bacteria.
- 
CONCEPT MAPPING Use the following terms to create a concept map: *struggle to survive*, *theory*, *inheritable variation*, *Darwin*, *overpopulation*, *natural selection*, and *successful reproduction*.

CRITICAL THINKING

- Analyzing Concepts** Could a characteristic that is not controlled by heredity be subject to natural selection? Explain your answer.
- Making Predictions** Suppose that an island in the Pacific Ocean was just formed by a volcano. Describe a possible scenario for the kinds of species that could be found on this island over the next million years.
- Interpreting Graphics** The graph below shows the diversity among different groups of animals over time. Of the four eras listed, the Cenozoic era is the most recent. The changing width of the bar for each group reflects the changing number of known subgroups over geologic time. Use the graph to answer the following questions:
 - Which group most recently evolved?
 - Which group is or was the most diverse? the least diverse?
 - Which group diversified rapidly soon after it first appeared?
 - Which group(s) became extinct?





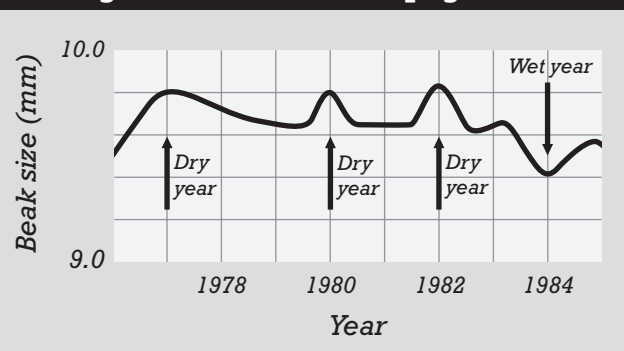
Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What is the term for the idea that geologic processes occurring now on Earth are much the same as those that occurred long ago?
 - catastrophism
 - uniformitarianism
 - adaptive radiation
 - convergent evolution
- What is the term for the biological process by which the kinds of organisms on Earth change over time?
 - evolution
 - superposition
 - biogeography
 - uniformitarianism
- When the internal structures of two species are very similar, what can be inferred about both species?
 - They share similar environments.
 - They evolved in similar environments.
 - They have similar external structures.
 - They evolved from a common ancestor.

INTERPRETING GRAPHICS: The graph below shows the variation in average beak size in a group of finches in the Galápagos Islands over time. These finches eat mostly seeds. Use the graph to answer the question that follows.

Average Beak Size in Galápagos Finches

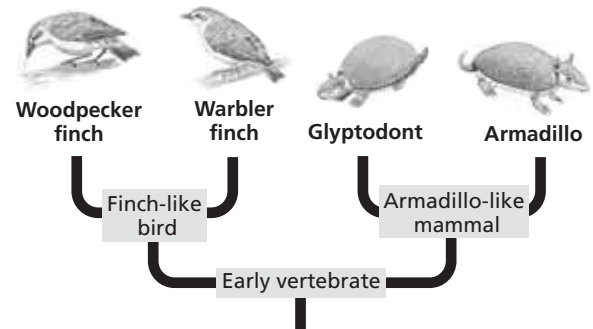


- Beak size in these finches is correlated to the size of seeds they can eat. What can be inferred from the graph?
 - In wet years, the finches that survive are mostly those that can eat larger seeds.
 - In dry years, the finches that survive are mostly those that can eat larger seeds.
 - In all years, the finches that survive are mostly those that can eat larger seeds.
 - In all years, the finches that survive are mostly those that can eat smaller seeds.

DIRECTIONS: Complete the following analogy.

- vestigial : functional :: vacated :
 - used
 - visceral
 - broken
 - occupied

INTERPRETING GRAPHICS: The diagram below shows possible evolutionary relationships between some organisms. Use the diagram to answer the question that follows.



- What does the diagram imply about warbler finches and armadillos?
 - They are unrelated.
 - They are equally related to glyptodonts.
 - They share a common ancestor.
 - They did not evolve from older forms of life.

SHORT RESPONSE

The human body has a tailbone but no tail. It also has an organ called the appendix, which is attached to the intestines but does not serve a function in digestion.

How would an evolutionary biologist explain the presence of these structures in the human body?

EXTENDED RESPONSE

An example of an acquired characteristic is large leg muscles built up in an individual by frequent running. An example of an inherited characteristic is the maximum height to which an individual can grow.

Part A Explain the difference between inherited and acquired characteristics.

Part B Contrast two historical theories that explained evolution, based on either acquired characteristics or inherited characteristics.

Test TIP

When answering questions based on graphs, look for answers that are supported by evidence from the graphs.

Modeling Selection

OBJECTIVES

- Simulate the generation of variation.
- Model the selection of favorable traits in new generations.

PROCESS SKILLS

- observing
- testing
- measuring

MATERIALS

- construction paper
- cellophane tape
- soda straws
- penny, or other coin
- six-sided die
- scissors
- meterstick or tape measure
- metric ruler

Background

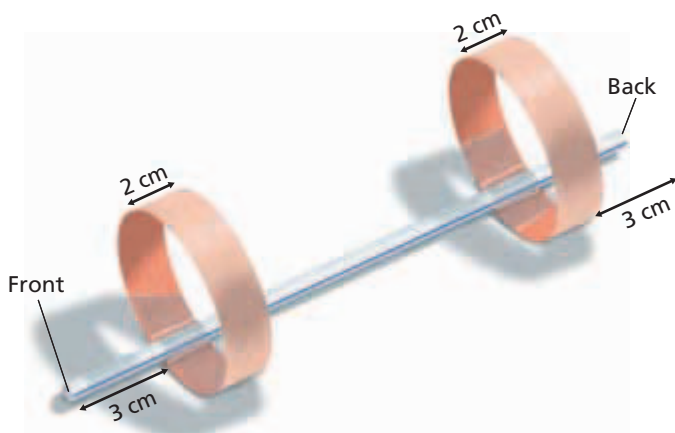
1. The fictitious Egyptian origami bird (*Avis papyrus*) lives in arid regions of North Africa. Only the birds that can fly the long distances between oases live long enough to breed successfully.
2. Successful evolution requires the generation of inheritable variations by mutation and then selection by the environment of the most-fit individuals.

PART A Parent Generation

1. Make a "parent bird." First, cut two strips of paper, 2 cm $\times$ 20 cm each. Make a loop with one strip of paper, letting the paper overlap by 1 cm, and tape the loop closed. Repeat for the other strip. Tape one loop 3 cm from each end of the straw, as shown in the figure. Mark the front end of the bird with a felt-tip marker. This "bird" will represent the parent generation.
2. In your lab report, prepare a data table like the one shown below.

DATA TABLE

Bird	Coin flip (H or T)	Die throw (1–6)	Anterior wing (cm)			Posterior wing (cm)			Average distance flown (m)
			Width	Circum.	Distance from front	Width	Circum.	Distance from back	
Parent	NA	NA	2	19	3	2	19	3	
Generation 1									
Chick 1									
Chick 2									
Chick 3									
Generation 2									
Chick 1									
Chick 2									
Chick 3									



3. Test how far your parent bird can fly by releasing it with a gentle overhand pitch. Test the bird twice. Record the bird's average flight distance in your data table.

PART B First (F_1) Generation

4. Model the breeding of offspring. Each origami bird lays a clutch of three eggs. Assume that the first chick is a clone of the parent. Fill in the table with data for this chick.
5. Assume that the next two chicks have mutations. Follow steps A–C below to determine the effects of each mutation.

Step A Flip a coin to determine which part of the bird is affected by the mutation.
 Heads = anterior (front)
 Tails = posterior (back)

Step B Throw a die to determine how the mutation affects the wing.



(1) = The wing position moves 1 cm toward the end of the straw.



(2) = The wing position moves 1 cm toward the middle of the straw.



(3) = The circumference of the wing increases 2 cm.



(4) = The circumference of the wing decreases by 2 cm.



(5) = The width of the wing increases by 1 cm.



(6) = The width of the wing decreases by 1 cm.

Step C A mutation is lethal if it results in a wing falling off or a wing with a circumference smaller than that of the straw. If you get a lethal mutation, disregard it, and breed another chick. Record the mutations and the dimensions of each offspring in your data table. The circumference of the wings can be calculated by measuring the length of the strips of paper used to form the wings and subtracting 1 cm for the overlap.

6. Test the birds. Release each bird with a gentle overhand pitch. It is important to release the birds as uniformly as possible. Test each bird at least twice.
7. The most successful bird is the one that flies the farthest. Record the flight distance of each offspring bird in your data table.

PART C Subsequent Generations

8. Assume that the most successful bird in the F_1 generation is the sole parent of the next (F_2) generation. Continue to breed, test, and record data for 8 more generations.
9.   Clean up your materials before leaving the lab.

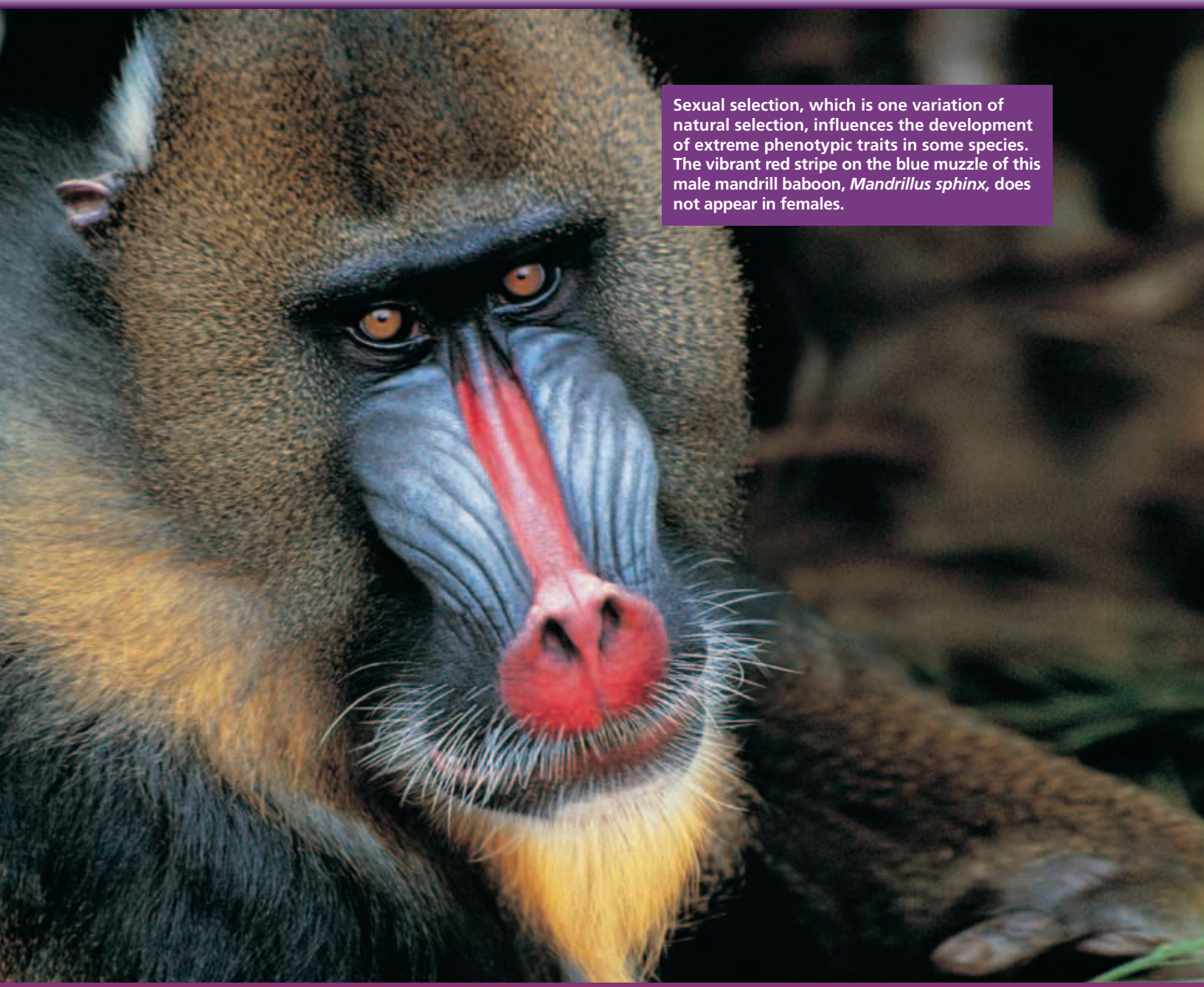
Analysis and Conclusions

1. Did your selection process result in birds that fly better?
2. Describe two aspects of this investigation that model evolution of biological organisms.
3. Your most successful bird may have a different lineage from the most successful bird of your neighboring groups. Compare your winning bird with those of your neighbors. How does it differ?
4. What might happen to your last bird if the environmental conditions change?
5. How might this lab help explain the observations Darwin made about finches on the Galápagos Islands?

Further Inquiry

A flock of origami birds is blown off the mainland and onto a very small island. These birds face little danger on the ground, but they experience significant risk when flying because they can be blown off the island. Birds that cannot fly at all are most likely to survive and reproduce. Continue the experiment for several generations, selecting birds that can't fly.

POPULATION GENETICS AND SPECIATION



Sexual selection, which is one variation of natural selection, influences the development of extreme phenotypic traits in some species. The vibrant red stripe on the blue muzzle of this male mandrill baboon, *Mandrillus sphinx*, does not appear in females.

SECTION 1 *Genetic Equilibrium*

SECTION 2 *Disruption of Genetic Equilibrium*

SECTION 3 *Formation of Species*

GENETIC EQUILIBRIUM

By the time of Darwin's death, in 1882, the idea of evolution by natural selection had gained wide acceptance among scientists. Within the next century, an increasing scientific understanding of genetics became strongly linked with theories of evolution and natural selection.

VARIATION OF TRAITS WITHIN A POPULATION

Population genetics is the study of evolution from a genetic point of view. Evolution at the genetic level is sometimes called **microevolution**, defined as a change in the collective genetic material of a population. Recall that the genetic material of organisms consists of many alleles—or variations—of many genes that code for various traits. Recall that a population consists of a group of individuals of the same species that routinely interbreed. Populations are important to the study of evolution because a population is the smallest unit in which evolution occurs.

Within a population, individuals may vary in observable traits. For example, fish of a single species in a pond may vary in size. Biologists often study variation in a trait by measuring that trait in a large sample. Figure 16-1 shows a graph of the frequency of lengths in a population of mature fish. Because the shape of the curve looks like a bell, it is called a **bell curve**. The bell curve shows that whereas a few fish in this population are very short and a few are very long, most are of average length. In nature, many quantitative traits in a population—such as height and weight—tend to show variation that follows a bell curve pattern.

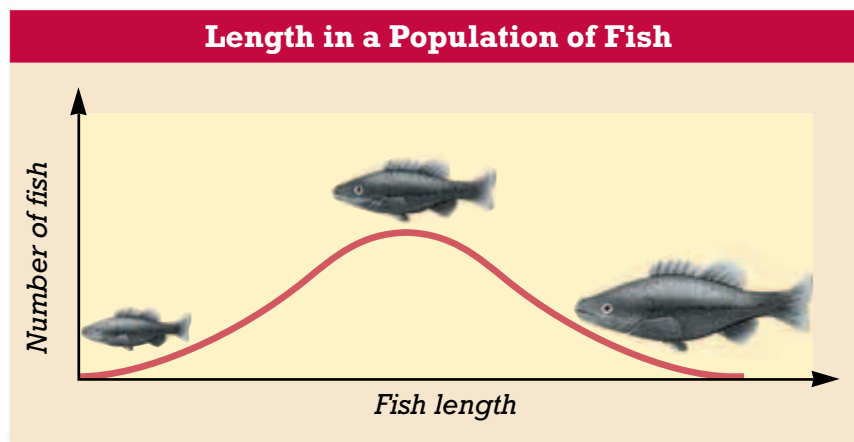


FIGURE 16-1

A bell curve illustrates that most members of a population have similar values for a given, measurable trait. Only a few individuals display extreme variations of the trait.

OBJECTIVES

- **Identify** traits that vary in populations and that may be studied.
- **Explain** the importance of the bell curve to population genetics.
- **Compare** three causes of genetic variation in a population.
- **Calculate** allele frequency and phenotype frequency.
- **Explain** Hardy-Weinberg genetic equilibrium.

VOCABULARY

population genetics
microevolution
bell curve
gene pool
allele frequency
phenotype frequency
Hardy-Weinberg genetic equilibrium



FIGURE 16-2

Many varied but similar phenotypes occur within families because members of a family share some alleles but not others.

Causes of Variation

What causes variation in traits? Some variations are influenced by environmental factors, such as the amount or quality of food available to an organism. Variation is also influenced by heredity. Some variations occur as a range of phenotypic possibilities (such as a range of body sizes), whereas others occur as a set of specific phenotypes (such as two possible flower colors).

To consider variability, think about phenotypes within a single human family. Two parents, each with a distinct genotype, may produce several children. In the picture of the family in Figure 16-2, the two young-adult brothers are not identical to each other, even though their genotypes are combinations of the genotypes of the same two parents. Both young men resemble their father, though in different traits. The baby resembles his young father, his grandfather, and his uncle. Thus, these males representing three generations look similar but not identical.

What causes genes to vary? Variations in genotype arise in three main ways. (1) *Mutation* is a random change in a gene that is passed on to offspring. (2) *Recombination* is the reshuffling of genes in a diploid individual. Recall that recombination occurs during meiosis by independent assortment and crossing-over of genes on chromosomes. (3) The *random pairing of gametes* occurs because each organism produces large numbers of gametes. So, the union of a particular pair of gametes is partly a matter of chance.

Scientists are still exploring other causes of variation in traits. For example, the expression of some genes depends upon the presence or absence of other genes or factors in the environment. The net result of having many alleles of many genes is the variety of unique genotypes and phenotypes that we see in populations.

THE GENE POOL

Population geneticists use the term **gene pool** to describe the total genetic information available in a population. It is easy to imagine genes for the next generation as existing in an imaginary pool. If you could inventory this pool and know all of the alleles that are present, then you could apply a simple set of rules based on probability theory to predict expected genotypes and their frequencies for the next generation.

Suppose, for example, that there are two alleles of a hypothetical gene, *A* and *a*, in a set of 10 gametes. If half the gametes in the set (5 gametes) carry the allele *A*, we would say that the allele frequency of the *A* allele is 0.5, or 50 percent. **Allele frequency** is determined by dividing the number of a certain allele (five instances of the *A* allele) by the total number of alleles of all types in the population (10 gametes, each with either an *A* or an *a* allele). Remember that a gamete is haploid and therefore carries only one allele for each gene.

Predicting Phenotype

The population of four o'clock flowers, shown in Figure 16-3, illustrates how phenotype can change from generation to generation. Homozygous RR flowers are red. Homozygous rr flowers are white. Heterozygous Rr flowers are pink rather than red, as you might expect. These flowers show incomplete dominance for color, meaning heterozygotes show a trait that falls between the dominant trait and the recessive trait. Thus, homozygotes and heterozygotes can be easily identified by observing the phenotype.

Compare the parent generation with the offspring generation of the four o'clock flowers shown in Figure 16-3. There are equal numbers of plants with the RR genotype and the Rr genotype in the first generation. You can compute the phenotype frequencies from the figure. A **phenotype frequency** is equal to the number of individuals with a particular phenotype divided by the total number of individuals in the population. Phenotype frequencies in the first generation are 0.5 pink (4 pink plants out of a total of 8 plants), 0.5 red (4 red plants out of a total of 8 plants), and 0.0 white. Recall that allele frequencies are computed using the same principle: the allele frequencies in the first-generation plants are 0.75 R (12 R alleles out of a total of 16 alleles) and 0.25 r (4 r alleles out of a total of 16 alleles).

We now can predict the genotypes and phenotypes of the second generation. If a male gamete encounters a female gamete, they will produce a new four o'clock plant whose genotype is the combination of both parental gametes. Thus, an R male gamete combined with an R female gamete will produce a plant with the RR genotype, which has red flowers. According to the laws of probability, the chance of an R gamete (a single allele) meeting with another R gamete is the arithmetic product of their allele frequencies in the gene pool:

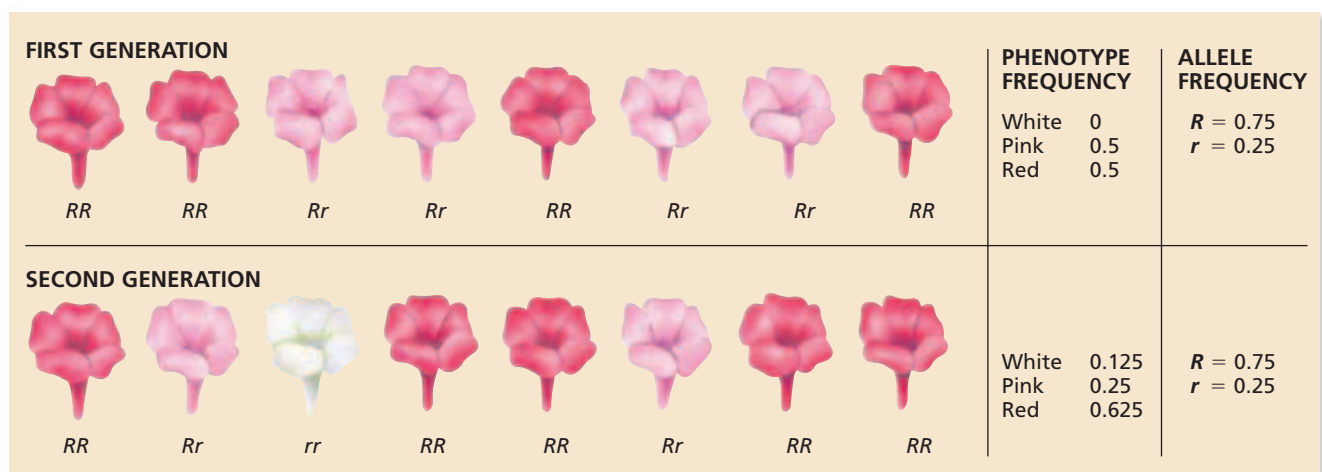
$$\begin{aligned}\text{frequency of } R \times \text{frequency of } R &= \text{frequency of } RR \text{ pair} \\ 0.75 \times 0.75 &= 0.5625\end{aligned}$$

The expected frequency of the rr genotype is then

$$\begin{aligned}\text{frequency of } r \times \text{frequency of } r &= \text{frequency of } rr \text{ pair} \\ 0.25 \times 0.25 &= 0.0625\end{aligned}$$

FIGURE 16-3

Although the four o'clock flowers differ phenotypically from generation to generation, the allele frequencies tend to remain the same.



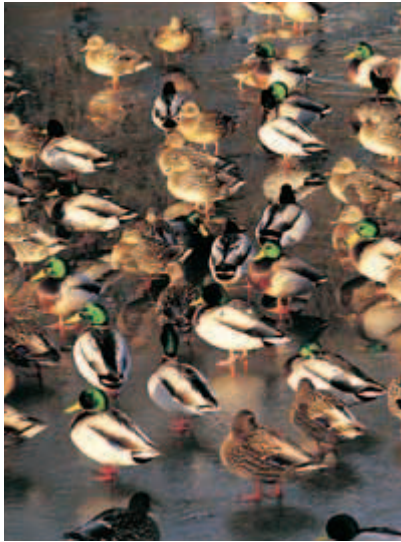


FIGURE 16-4

This flock of mallards, *Anas platyrhynchos*, likely violates some or all of the conditions necessary for Hardy-Weinberg genetic equilibrium.

The frequencies of all genotypes expected in the second generation must add up to 1.0, just as fractions of a whole must add up to 1. Having established the probabilities of getting an RR and an rr plant, we can compute the expected frequency of the Rr plants. All those plants that are neither RR nor rr will be Rr , so

$$1.0 - \text{frequency of } RR - \text{frequency of } rr = \text{frequency of } Rr$$

$$1.0 - 0.5625 - 0.0625 = 0.375$$

HARDY-WEINBERG GENETIC EQUILIBRIUM

It is clear from the example of the four o'clock flowers that phenotype frequencies can change dramatically from generation to generation. But what happens to allele frequencies over generations? A German physician, Wilhelm Weinberg (1862–1937), and a British mathematician, Godfrey Hardy (1877–1947), independently showed that genotype frequencies in a population tend to remain the same from generation to generation unless acted on by outside influences. This principle is referred to as **Hardy-Weinberg genetic equilibrium**, and it is based on a set of assumptions about an ideal hypothetical population that is not evolving:

1. No net mutations occur; that is, the alleles remain the same.
2. Individuals neither enter nor leave the population.
3. The population is large (ideally, infinitely large).
4. Individuals mate randomly.
5. Selection does not occur.

Bear in mind that true genetic equilibrium is a theoretical state. Real populations, such as the flock of mallards in Figure 16-4, may not meet all of the conditions necessary for genetic equilibrium. By providing a model of how genetic equilibrium is maintained, the Hardy-Weinberg principle allows us to consider what forces disrupt genetic equilibrium.

Word Roots and Origins

equilibrium

from the Latin *aequilibrium*, meaning "equal balance"

SECTION 1 REVIEW

1. How does the distribution of traits in a population look when displayed as a graph?
2. Describe three causes of genetic variation in a population.
3. What is meant by the term *human gene pool*?
4. How is phenotype frequency computed?
5. What are the conditions that a population must meet in order to have genetic equilibrium?

CRITICAL THINKING

6. **Evaluating Methods** By observation only, is it easier to deduce the genotype of organisms for an allele that has complete dominance or incomplete dominance?
7. **Making Calculations** Half of a population of four o'clocks has red flowers, and half has white flowers. What is the frequency of the r allele?
8. **Relating Concepts** How does the pairing of gametes produce genotypic variation?

DISRUPTION OF GENETIC EQUILIBRIUM

Evolution is the change in a population's genetic material over generations, that is, a change of the population's allele frequencies or genotype frequencies. Any exception to the five conditions necessary for Hardy-Weinberg equilibrium can result in evolution.

MUTATION

The first requirement for genetic equilibrium is that allele frequencies not change overall because of mutations. Spontaneous mutations occur constantly, at very low rates under normal conditions. But if an organism is exposed to mutagens—mutation-causing agents such as radiation and certain chemicals—mutation rates can increase significantly. Mutations can affect genetic equilibrium by producing totally new alleles for a trait. Many mutations are harmful, although some have no effect. Because natural selection operates only on genes that are expressed, it is very slow to eliminate harmful recessive mutations. In the long run, however, beneficial mutations are a vital part of evolution.

GENE FLOW

The second requirement for genetic equilibrium is that the size of the population remains constant. If individuals move, genes move with them. **Immigration** is the movement of individuals into a population, and **emigration** is the movement of individuals out of a population.

The behavioral ecology of some animal species encourages immigration and emigration. Common baboons live on the savannas of eastern Africa in social and breeding groups called *troops*. A troop is dominated by a few adult males, and it may have from 10 to 200 members. Females tend to remain with the troop they are born into; however, younger or less dominant males leave their birth troop, eventually joining another troop. This constant movement of male animals ensures gene flow. **Gene flow** is the process of genes moving from one population to another. Gene flow can occur through various mechanisms, such as the migration of individuals or the dispersal of seeds or spores.

OBJECTIVES

- **List** five conditions under which evolution may take place.
- **Explain** how migration can affect the genetics of populations.
- **Explain** how genetic drift can affect populations of different sizes.
- **Contrast** the effects of stabilizing selection, directional selection, and disruptive selection on populations over time.
- **Identify** examples of nonrandom mating.

VOCABULARY

immigration
emigration
gene flow
genetic drift
sexual selection
stabilizing selection
disruptive selection
directional selection

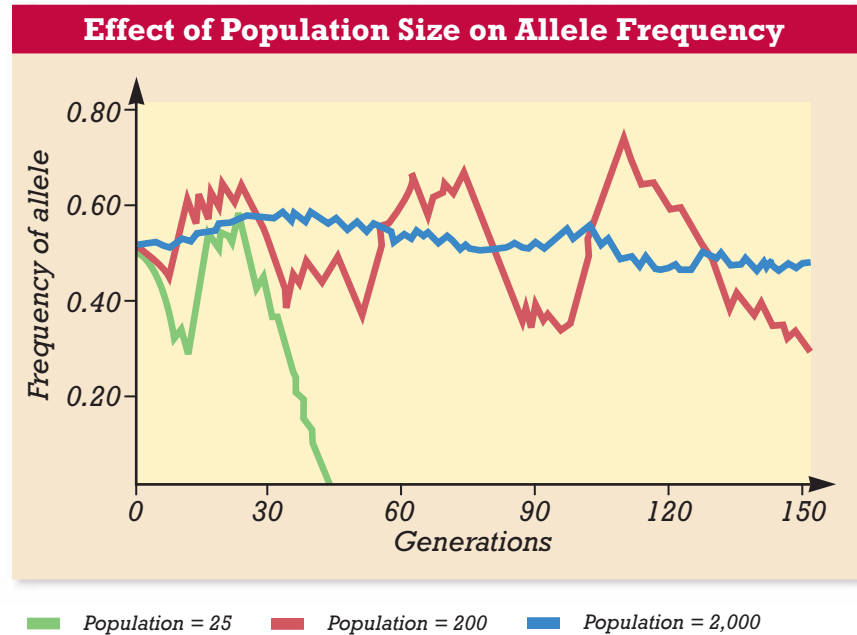
Word Roots and Origins

immigration

from the Latin *immigrare*,
meaning "to go into"

FIGURE 16-5

Genetic drift is significant only in small and medium-sized populations. In a small population, a particular allele may disappear completely over a few generations. In a larger population, a particular allele may vary widely in frequency due to chance but still be present in enough individuals to be maintained in the population. In a much larger population, the frequency of a particular allele may vary slightly by chance but remain relatively stable over generations.



GENETIC DRIFT

FIGURE 16-6

Populations of the once nearly extinct northern elephant seal, *Mirounga angustirostris*, have lost genetic variability—individuals are homozygous for all of their genes that have been tested. This result of genetic drift could make the species vulnerable to extinction.



The third requirement of genetic equilibrium is the presence of a large population. The Hardy-Weinberg principle is based on the laws of probability, which are less applicable to smaller populations. **Genetic drift** is the phenomenon by which allele frequencies in a population change as a result of random events, or chance. In small populations, the failure of even a single organism to reproduce can significantly disrupt the allele frequency of the population, as can greater-than-normal reproduction by an individual, resulting in genetic drift. Because it can result in significant changes within a population, genetic drift is thought to be another possible mechanism for the evolution of new species.

Figure 16-5 shows a graph of genetic drift in populations of three differing sizes. Small populations can undergo abrupt changes in allele frequencies, exhibiting a large degree of genetic drift, whereas large populations retain fairly stable allele frequencies, maintaining a small degree of genetic drift. In the smallest population shown in the graph, the frequency of the example allele reaches zero at about the 45th generation. If we assume that the population started with two alleles for a trait, then only one allele is left, and every individual is homozygous for that trait. Once this change happens, the population is in danger of becoming extinct because there is no variation for natural selection to act on. For example, a natural disaster or a new disease could wipe out the entire population. For this reason, endangered species, such as the northern elephant seal, as shown in Figure 16-6, remain in peril of extinction even as their numbers increase.

NONRANDOM MATING

The fourth requirement of genetic equilibrium is random matings, without regard to genetic makeup. However, many species do not mate randomly. Mate selection is often influenced by geographic proximity, which can result in mates with some degree of kinship. Matings of related individuals can amplify certain traits and can result in offspring with disorders caused by recessive genes, which, although rare, may be present in the genomes of related individuals.

In another example of nonrandom mating, individuals may select a mate that has traits similar to their own traits. This mate would probably have some similar genes. The selection of a mate based on similarity of traits is called *assortative mating*. Nonrandom mating affects which alleles will be combined within individuals, but it does not affect overall allele frequencies within a population.

Sexual Selection

In many species of birds, the males are brightly colored and often heavily plumed, such as the peacock shown in Figure 16-7. These elaborately decorated males are easy for predators to see. Why would natural selection work in favor of an organism being conspicuous to a predator? Females tend to choose the males they mate with based on certain traits. This tendency is referred to as **sexual selection**. In order to leave offspring, a male must be selected by the female. The peacock's gaudy plumage increases his chances of being selected. Extreme traits, such as heavy, brightly colored plumage, may give the female an indication of the quality of the male's genes or his fitness in his environment. Remember that natural selection acts upon differences in survival *and* reproduction. Natural selection favors an increase in the genes of successful *reproducers*, rather than merely those of successful *survivors*.



FIGURE 16-7

Males sometimes display extreme traits, such as the large tail of this peacock, *Pavo cristatus*. This trait is favorable if it attracts females and increases the reproductive fitness of the male.



Quick Lab

Evaluating Selection

Materials unlined paper, colored pencils, 25 colored candies

Procedure

1. Fold a sheet of unlined paper in half, top over bottom. Using colored pencils, decorate half the paper with different colored circles. Make each colored circle about the size of a quarter.
2. Scatter your "population" of candies over the undecorated half of the sheet of paper. Count and record how many candies match the background color.
3. Now, scatter the candies over the decorated half of the sheet of paper. Count and record how many candies match the background color.
4. Candies that match the background color are camouflaged. Calculate the ratio of camouflaged candies to uncamouflaged candies in steps 2 and 3.
5. Repeat steps 2–4 two times, and average your results.
6. Exchange paper with another group, and repeat steps 2–5.

Analysis Was your population more successfully camouflaged on the white background or on the colored background? How did color diversity affect your population's success on the colored background? Based on your results, predict which type of selection might increase your population's fitness in a multi-colored environment.

NATURAL SELECTION

The fifth requirement of genetic equilibrium is the absence of natural selection. Natural selection is an ongoing process in nature, so it often disrupts genetic equilibrium. As you have learned, natural selection means that some members of a population are more likely than other members to survive and reproduce and thus contribute their genes to the next generation.

Recall that natural selection operates on variations of traits within a population, such as body size or color. When natural selection is at work over time, the distribution of traits in a population may change. In a graph, this kind of change would appear as a shift away from the normal bell curve. Scientists observe three general patterns of natural selection: stabilizing selection, disruptive selection, and directional selection.

Stabilizing Selection

In **stabilizing selection**, individuals with the average form of a trait have the highest fitness. The average represents the optimum for most traits; extreme forms of most traits confer lower fitness on the individuals that have them. Consider a hypothetical species of lizard in which larger-than-average individuals might be more easily spotted, captured, and eaten by predators. On the other hand, lizards that are smaller than average might not be able to run fast enough to escape.

Figure 16-8a shows the effect of stabilizing selection on body size in these lizards. The red curve shows the initial variation in lizard size as a standard bell curve. The blue curve represents the variation in body size several generations after a new predator was introduced. This predator easily captured the large, visible lizards and the small, slower lizards. Thus, selection against these extreme body types reduced the size range of the lizards. Stabilizing selection is the most common kind of selection. It operates on most traits and results in very similar morphology between most members of a species.

Disruptive Selection

In **disruptive selection**, individuals with either extreme variation of a trait have greater fitness than individuals with the average form of the trait. Figure 16-8b shows the effect of disruptive selection on shell color in limpets, which are marine animals. The shell color of limpets varies from pure white to dark tan. White-shelled limpets that are on rocks covered with goose barnacles, which are also white, are at an advantage. Birds that prey on limpets have a hard time distinguishing the white-shelled limpets from the goose barnacles. On bare, dark-colored rocks, dark-shelled limpets are at an advantage. Again, the limpet-eating birds have a hard time locating the dark shells against the dark background. However, the birds easily spot limpets with shells of intermediate color, which are visible against both the white and dark backgrounds.

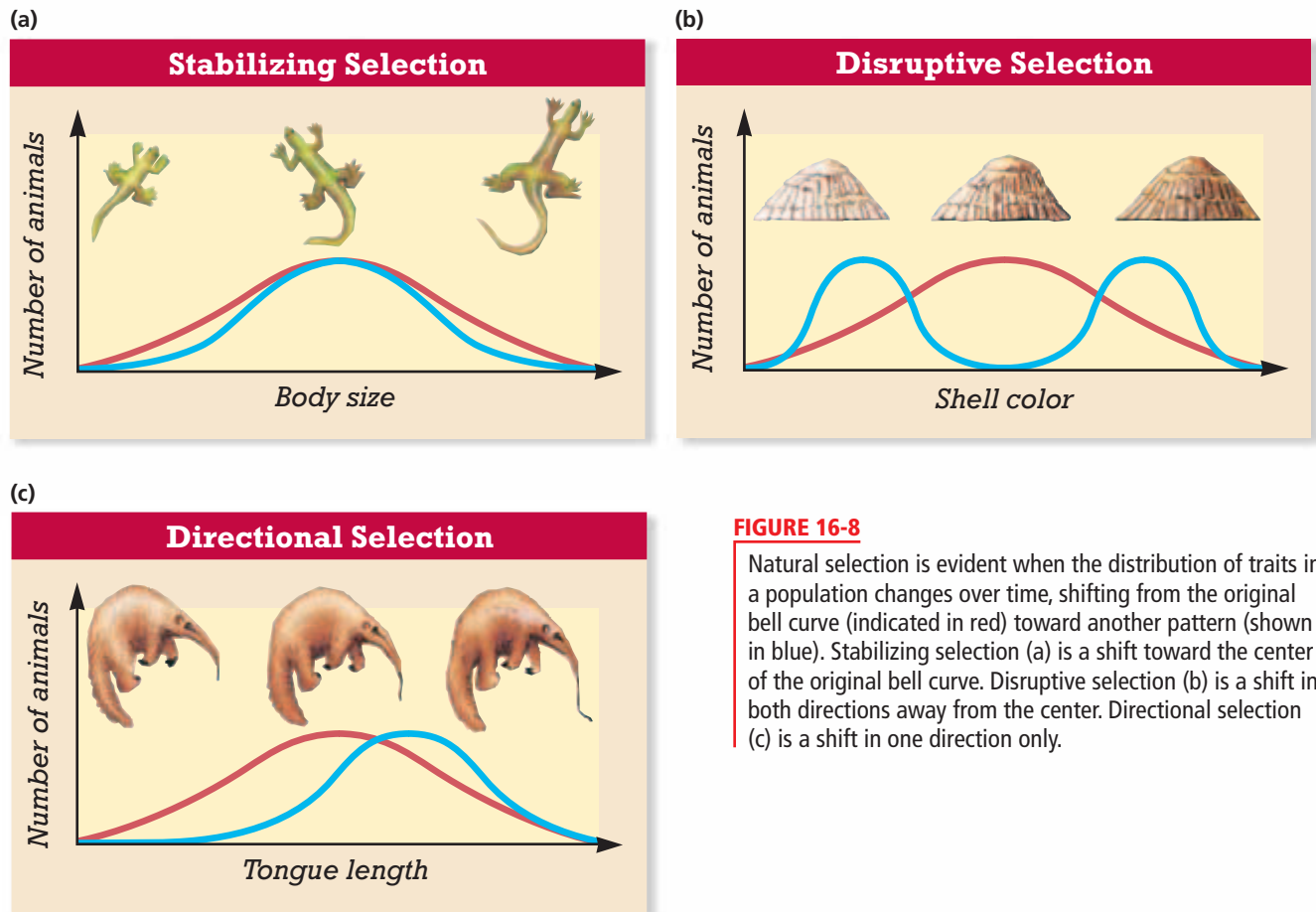


FIGURE 16-8

Natural selection is evident when the distribution of traits in a population changes over time, shifting from the original bell curve (indicated in red) toward another pattern (shown in blue). Stabilizing selection (a) is a shift toward the center of the original bell curve. Disruptive selection (b) is a shift in both directions away from the center. Directional selection (c) is a shift in one direction only.

Directional Selection

In **directional selection**, individuals that display a more extreme form of a trait have greater fitness than individuals with an average form of the trait. Figure 16-8c shows the effects of directional selection on tongue length in anteaters. Anteaters feed by breaking open termite nests, pushing their sticky tongue into the nest, and lapping up termites. Suppose that the termites in an area began to build deeper nests. Anteaters with long tongues could more effectively prey on these termites than could anteaters with short or average tongues. Thus, directional selection would act to direct the trait of tongue length away from the average and toward one extreme.

SECTION 2 REVIEW

1. List five conditions that can disrupt genetic equilibrium and cause evolution to occur.
2. Explain the role of mutations in evolution.
3. Contrast gene flow with genetic drift.
4. Explain why genetic drift is more significant in smaller populations.
5. Contrast stabilizing selection, disruptive selection, and directional selection.

CRITICAL THINKING

6. **Making Inferences** Why might a harmful allele persist in a population for many generations?
7. **Relating Concepts** Give an example of a species that exhibits the effects of sexual selection.
8. **Applying Concepts** For each of the three patterns of natural selection, give an example of a species that exhibits the effects of that selection.

SECTION 3

OBJECTIVES

- **Relate** the biological species concept to the modern definition of species.
- **Explain** how the isolation of populations can lead to speciation.
- **Compare** two kinds of isolation and the pattern of speciation associated with each.
- **Contrast** the model of punctuated equilibrium with the model of gradual change.

VOCABULARY

speciation
morphology
biological species concept
geographic isolation
allopatric speciation
reproductive isolation
prezygotic isolation
postzygotic isolation
sympatric speciation
gradualism
punctuated equilibrium

FIGURE 16-9

The facial features of red-tailed monkeys, *Cercopithecus ascanus*, can differ from individual to individual.



FORMATION OF SPECIES

How many species of organisms exist on Earth today?

Undiscovered species may be so numerous that we have no accurate answer. For example, even small areas of tropical rain forests can contain thousands of species of plants, animals, and microorganisms. New species are discovered and others become extinct at an increasing rate. In this section, you will learn how one species can become two through a process called speciation.

THE CONCEPT OF SPECIES

You have learned that existing species are essentially changed versions of older species. The process of species formation, **speciation** (SPEE-shee-AY-shun), results in closely related species. Some are very similar to their shared ancestral species, whereas other descendant species become quite different over time.

Morphological Concept of Species

For many years, scientists used the internal and external structure and appearance of an organism—its **morphology** (mawr-FAHL-uh-jee)—as the chief criterion for classifying it as a species. Using the morphological concept of species, scientists defined species primarily according to structure and appearance. Because morphological characteristics are easy to observe, making species designations based on morphology proved convenient.

The morphological concept of species has limitations, however. There can be phenotypic differences among individuals in a single population. Notice, for example, the variation between the two red-tailed monkeys shown in Figure 16-9. To further complicate the matter, some organisms that appear different enough to belong to different species interbreed in the wild and produce fertile offspring. In response to the capacity of dissimilar organisms to reproduce, the biological species concept arose.

The Biological Species Concept

According to the **biological species concept**, as proposed by German-born, American biologist Ernst Mayr (1904–2005), a species is a population of organisms that can successfully interbreed but cannot breed with other groups. Although this definition is useful for living animals, the biological species concept does not provide a satisfactory definition for species of extinct organisms, whose reproductive compatibility cannot be tested. Nor is it useful for organisms that do not reproduce sexually. Thus, our modern definition of species includes components of both the morphological and biological species concepts. A species is a single kind of organism. Members of a species are morphologically similar and can interbreed to produce fully fertile offspring. The many species alive today diverged from a smaller number of earlier species.

ISOLATION AND SPECIATION

How do species give rise to other, different species? Speciation begins with isolation. In isolation, two parts of a formerly interbreeding population stop interbreeding. Two important types of isolation frequently drive speciation.

Geographic Isolation

Geographic isolation is the physical separation of members of a population. Populations may be physically separated when their original habitat becomes divided. A deep canyon could develop, a river could change course, or a drying climate in a valley could force surviving fragments of an original population into separate mountain ranges. Once the subpopulations become isolated, gene flow between them stops. Natural selection and genetic drift cause the two subpopulations to diverge, eventually making them incompatible for mating.

In pupfish, small freshwater fish shown in Figure 16-10, speciation following geographic isolation apparently took place in parts of the western United States, including the desert of Death Valley. Death Valley has a number of isolated ponds formed by springs. Each pond contains a species of fish that lives only in that one pond, but the fish species of various ponds in the area are quite similar.

How did these different populations of fish become isolated in Death Valley? Geologic evidence indicates that most of Death Valley was covered by a lake during the last ice age. When the ice age ended, the region became dry, and only small, spring-fed ponds remained. Members of a fish species that previously formed a single population in the lake may have become isolated in different ponds. The environments of the isolated ponds differ enough that the separate populations of fish diverged. Eventually, the fishes in the different ponds diverged enough to be considered separate species.

FIGURE 16-10

These two types of pupfish live in isolated water sources in the western United States. Both types appear to have evolved from a common ancestor after undergoing geographic isolation.



(a) desert pupfish, *Cyprinodon macularius*



(b) Amargosa pupfish, *Cyprinodon nevadensis*



(a) white-tailed antelope squirrel,
Ammospermophilus leucurus



(b) Harris's antelope squirrel,
Ammospermophilus harrisi

FIGURE 16-11

These two closely related squirrels are probably the result of allopatric speciation. The white-tailed antelope squirrel (a) is found on the north rim of the Grand Canyon, and Harris's antelope squirrel (b) is found on the south rim.

Geographic barriers can be formed by canyons, mountain ranges, bodies of water, deserts, or other geographic features that organisms cannot cross. In addition, parts of a population may be accidentally transported to new islands or slowly drift apart on separate continents. On the geologic time scale, the processes of geology frequently rearrange populations.

Whether or not a geographic barrier will isolate a particular group of organisms depends on the organisms' ability to move around. Birds, for example, can easily fly back and forth across a deep canyon. However, a canyon might be a major barrier to a small, crawling mammal. An example of such a barrier is the Grand Canyon in Arizona. The ever-deepening canyon separates the habitats of two closely related populations of squirrels, shown in Figure 16-11. These two populations are different enough to be considered separate species, but similar enough that scientists debate whether they might simply be subspecies. Because their ranges do not overlap, the two populations do not interbreed.

Allopatric Speciation

Allopatric speciation happens when species arise as a result of geographic isolation. *Allopatric* means "different homelands." Populations separated by a geographic barrier no longer experience gene flow between them. So, the gene pools of each separate population may begin to differ due to genetic drift, mutations, and natural selection.

Allopatric speciation is more likely to occur in small populations because a smaller gene pool will be changed more significantly by genetic drift and natural selection. The key question in this type of speciation is whether or not the separated populations become different enough to be reproductively isolated from one another. In other words, if the geographic barrier is removed, could the two groups interbreed and produce fertile offspring?

Reproductive Isolation

Sometimes, groups of organisms within a population become genetically isolated without being geographically isolated. **Reproductive isolation** results from barriers to successful breeding between population groups in the same area. Reproductive isolation and the species formation that follows it may sometimes arise through disruptive selection. Remember that in disruptive selection, the two extremes of a trait in a given population are selected for and the organisms begin to diverge. Once successful mating is prevented between members of the two subpopulations, the effect is the same as what would have occurred if the two subpopulations had been geographically isolated. There are two general types of reproductive isolation: prezygotic (pree-zie-GAHT-ik) isolation and postzygotic isolation. **Prezygotic isolation**, or *premating isolation*, occurs before fertilization, and **postzygotic isolation**, or *postmating isolation*, occurs after fertilization.

Word Roots and Origins

prezygotic

from the Latin *prae*, meaning "before," and the Greek *zygotos*, meaning "yoked"

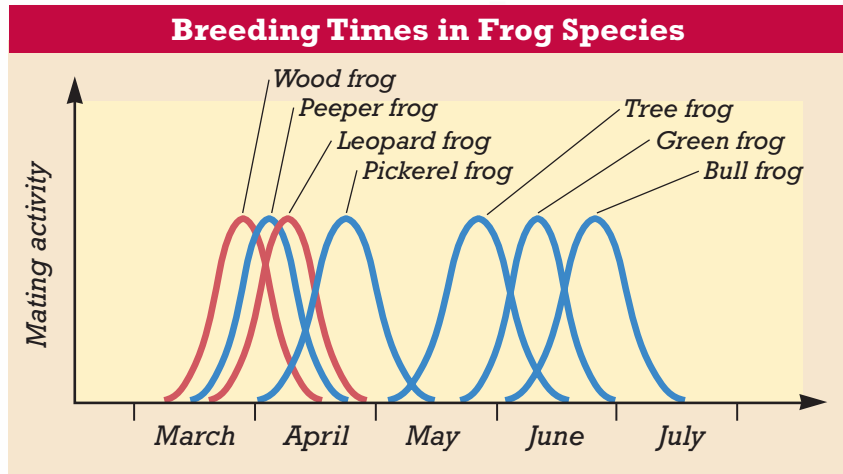


FIGURE 16-12

As the graph shows, frogs that share habitats may be reproductively isolated by differences in timing of mating activity.

If two potentially interbreeding species mate and fertilization occurs, success is measured by the production of healthy, fully fertile offspring. But this may be prevented by one of several types of postzygotic isolation. The offspring of interbreeding species may not develop completely and may die early, or, if healthy, they may not be fertile. From an evolutionary standpoint, if death or sterility of offspring occurs, the parent organisms have wasted their gametes producing offspring that cannot, in turn, reproduce.

In contrast, prezygotic isolating mechanisms can reduce the chance of hybrid formation. For example, a mating call that is not recognized as such by a potential mate can contribute to isolation. Differences in mating times are another type of prezygotic isolation. Both mechanisms are in effect for the frogs shown in Figure 16-12. The time of peak mating activity differs for each frog, reducing the chance of interbreeding. As a result, the wood frog and the leopard frog, shown in Figure 16-13, are reproductively isolated. Though these two frogs interbreed in captivity, they do not interbreed where their ranges overlap in the wild. The wood frog usually breeds in late March, and the leopard frog usually breeds in mid-April.

Sympatric Speciation

Sympatric speciation occurs when two subpopulations become reproductively isolated within the same geographic area. Charles Darwin proposed this model of speciation in the 1850s. He hypothesized that competing individuals within a population could gain an adaptive advantage by using slightly different niches. This specialization could lead each group to become reproductively isolated from the other.

For example, a population of insects might live on a single type of plant. If some of the individuals from this population began to live on another type of plant, they might no longer interbreed with the original population. The two groups of insects would then be able to evolve independently and could eventually become two different species.

FIGURE 16-13

Differences in peak mating times and in mating calls appear to have led to reproductive isolation of the wood frog (a) from its close relative, the leopard frog (b).



(a) wood frog, *Rana sylvatica*



(b) leopard frog, *Rana pipiens*

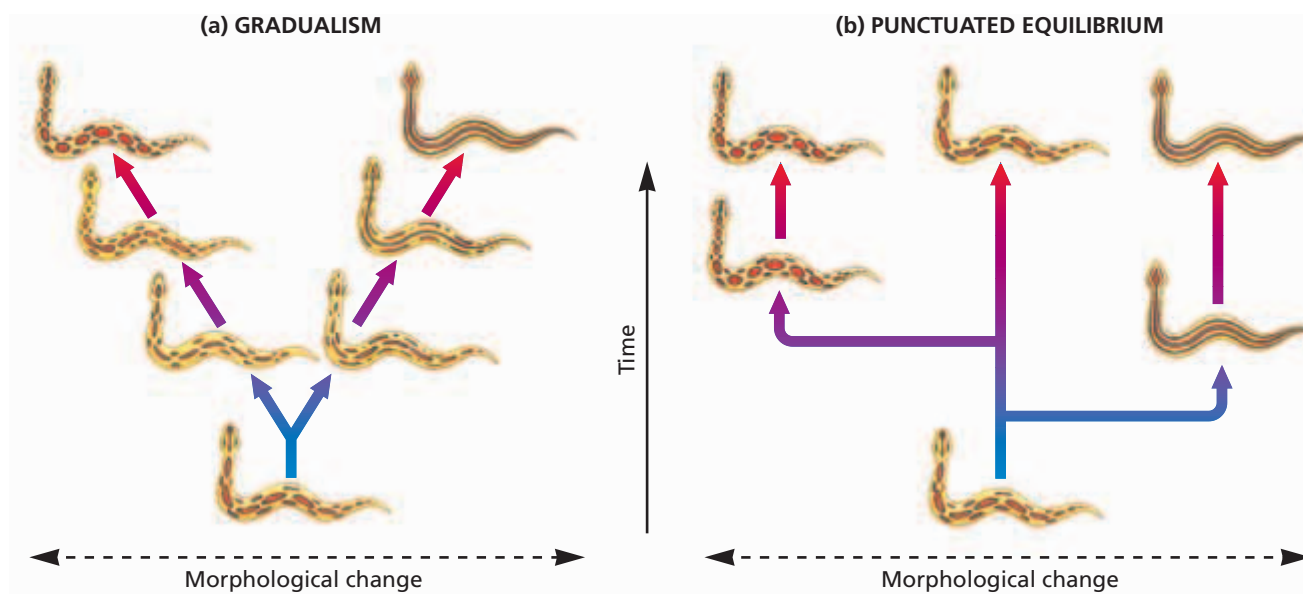


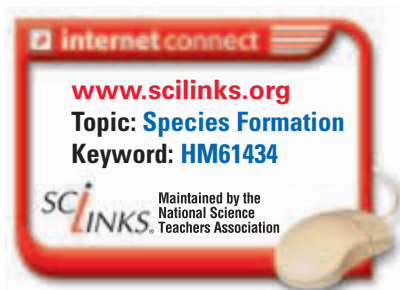
FIGURE 16-14

In the model of speciation shown on the left, species evolve gradually, at a stable rate. In the model of speciation shown on the right, species arise abruptly and differ noticeably from the root species. These species then change little over time.

RATES OF SPECIATION

Speciation sometimes requires millions of years. But apparently some species can form more rapidly. For example, Polynesians introduced banana trees to the Hawaiian Islands about a thousand years ago. Today, there are several species of moths that are unique to the Hawaiian Islands and that feed only on bananas. These species likely descended from ancestral moths during the past thousand years, since bananas were introduced to Hawaii.

The idea that speciation occurs at a regular, gradual rate is called **gradualism**. However, some scientists think that speciation happens in “bursts” relative to the geologic time scale. The fossil record holds evidence that many species existed without change for long periods of time, whereas in some cases a great diversity of new forms seems to have evolved rapidly. That is, change occurred in a few thousand, rather than a few million, years. Scientists call this pattern of species formation **punctuated equilibrium**. The term *punctuated* refers to sudden, rapid change, and *equilibrium* refers to periods of little change. Figure 16-14 illustrates these two contrasting models as they might apply to the evolution of snakes.



SECTION 3 REVIEW

1. What role did Ernst Mayr play in the development of the modern biological species concept?
2. Explain how geographic isolation can lead to allopatric speciation.
3. Explain how reproductive isolation can lead to sympatric speciation.
4. Contrast the model of punctuated equilibrium with the model of gradualism.

CRITICAL THINKING

5. **Critiquing Explanations** What are two shortcomings of the biological species concept?
6. **Analyzing Concepts** Describe one possible scenario of postzygotic reproductive isolation in an animal species.
7. **Drawing Conclusions** How might the generation time of a population affect future speciation?

CHAPTER HIGHLIGHTS

SECTION 1

Genetic Equilibrium

- Population biologists study many different traits in populations, such as size and color.
- Traits vary and can be mapped along a bell curve, which shows that most individuals have average traits, whereas a few individuals have extreme traits.
- Variations in genotype arise by mutation, recombination, and the random fusion of gametes.
- The total genetic information available in a population is called the gene pool.
- Allele frequencies in the gene pool do not change unless acted upon by certain forces.
- The principle of Hardy-Weinberg genetic equilibrium is a theoretical model of a population in which no evolution occurs and the gene pool of the population is stable.

Vocabulary

microevolution (p. 317)

population genetics (p. 317)

bell curve (p. 317)

gene pool (p. 318)

allele frequency (p. 318)

phenotype frequency (p. 319)

Hardy-Weinberg genetic

equilibrium (p. 320)

SECTION 2

Disruption of Genetic Equilibrium

- Evolution can take place when a population is not in a state of genetic equilibrium. Thus, evolution may take place when populations are subject to genetic mutations, gene flow, genetic drift, nonrandom mating, or natural selection.
- Emigration and immigration cause gene flow between populations and can thus affect gene frequencies.
- Genetic drift is a change in allele frequencies due to random events. Genetic drift operates most strongly in small populations.
- Mating is nonrandom whenever individuals may choose partners. Sexual selection occurs when certain traits increase an individual's success at mating. Sexual selection explains the development of traits that improve reproductive success but that may harm the individual.
- Natural selection can influence evolution in one of three general patterns. Stabilizing selection favors the formation of average traits. Disruptive selection favors extreme traits rather than average traits. Directional selection favors the formation of more-extreme traits.

Vocabulary

immigration (p. 321)

emigration (p. 321)

gene flow (p. 321)

genetic drift (p. 322)

sexual selection (p. 323)

stabilizing selection (p. 324)

disruptive selection (p. 324)

directional selection (p. 325)

SECTION 3

Formation of Species

- According to the biological species concept, a species is a population of organisms that can successfully interbreed but cannot breed with other groups.
- Geographic isolation results from the separation of population subgroups by geographic barriers. Geographic isolation may lead to allopatric speciation.
- Reproductive isolation results from the separation of population subgroups by barriers to successful breeding. Reproductive isolation may lead to sympatric speciation.
- In the gradual model of speciation, species undergo small changes at a constant rate. In the punctuated equilibrium model, new species arise abruptly, differ greatly from their ancestors, and then change little over long periods.

Vocabulary

speciation (p. 326)

morphology (p. 326)

biological species concept
(p. 327)

geographic isolation (p. 327)

allopatric speciation (p. 328)

reproductive isolation
(p. 328)

prezygotic isolation (p. 328)

postzygotic isolation (p. 328)

sympatric speciation (p. 329)

gradualism (p. 330)

punctuated equilibrium
(p. 330)

CHAPTER REVIEW

USING VOCABULARY

1. Use each of the following terms in a separate sentence: *bell curve* and *gene flow*.
2. For each pair of terms, explain how the meanings of the terms differ.
 - a. *allele frequency* and *phenotype frequency*
 - b. *stabilizing selection* and *disruptive selection*
 - c. *immigration* and *emigration*
 - d. *geographic isolation* and *reproductive isolation*
 - e. *allopatric speciation* and *sympatric speciation*
 - f. *punctuated equilibrium* and *gradualism*
3. Use the following terms in the same sentence: *nonrandom mating*, *assortative mating*, and *sexual selection*.
4. Use the following terms in the same sentence: *genetic equilibrium*, *gene pool*, and *speciation*.
5. **Word Roots and Origins** The word *disrupt* is derived from the Latin *disruptus*, which means “to break apart.” Using this information, explain the term *disruptive selection*.

UNDERSTANDING KEY CONCEPTS

6. **Compare** the three main causes of variation in the genotypes of organisms.
7. **Identify** the five conditions that are necessary for Hardy-Weinberg genetic equilibrium.
8. **Identify** the five conditions that may cause evolution to occur in a population.
9. **Describe** how immigration and emigration can alter allele frequencies in a population.
10. **List** examples of how mating could be nonrandom in a population.
11. **Contrast** natural selection with sexual selection.
12. **Identify** which type of selection is happening when a population’s bell curve narrows over time.
13. **Explain** why prezygotic isolating mechanisms have an advantage over postzygotic isolating mechanisms.
14. **Relate** the size of a population to the influence of genetic drift on the population’s gene pool.
15. **Compare** the effects of stabilizing selection, disruptive selection, and directional selection.
16. **Model** a situation that might cause the geographic isolation of a subgroup of a population of fish living in a large river.
17. **Explain** why the biological species concept cannot be used to identify fossil organisms.

18. **Identify** the type of isolating mechanism in the following scenario: Where populations of two related species of frogs overlap geographically, their mating calls differ more than they do where the species don’t overlap.
19. **Summarize** the hypothesis of punctuated equilibrium as it relates to the rate of speciation.
20. **CONCEPT MAPPING** Use the following  terms to create a concept map of how new species can form: *natural selection*, *allele frequency*, *geographic isolation*, *reproductive isolation*, and *speciation*.

CRITICAL THINKING

21. **Relating Concepts** Explain the relationship between evolution and natural selection.
22. **Forming Hypotheses** Propose a hypothesis about how pollutants in the environment could influence the evolution of its inhabitants.
23. **Drawing Conclusions** Freeways may provide an effective geographic isolating mechanism for some slow-moving animals. Are such artificial barriers likely to result in complete speciation?
24. **Applying Information** The common biological definition of *species* states that a species is a group of organisms that can interbreed and produce fertile offspring in nature. A mule is the offspring of a horse and a donkey. Mules are always sterile. By the definition above, do a horse and a donkey belong to the same species? Explain your answer.
25. **Interpreting Graphics** From the illustration of four o’clock flowers shown below, calculate the frequency of the *R* and *r* alleles, and state the phenotype frequency.



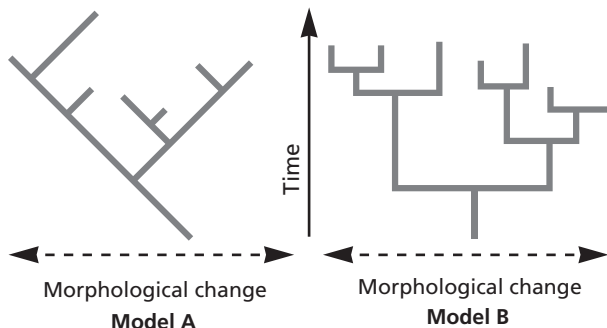


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What is the term for the total genetic information in a population?
 - gene pool
 - allele frequency
 - distribution of traits
 - phenotype frequency
- Saint Bernards and Chihuahuas (two breeds of domestic dogs) cannot normally mate because they differ so much in size. Thus, they are reproductively isolated to some extent. What type of isolating mechanism is operating in this case?
 - artificial
 - prezygotic
 - postzygotic
 - geographic
- How do mutations affect genetic equilibrium?
 - Mutations cause emigration.
 - Mutations cause immigration.
 - Mutations introduce new alleles.
 - Mutations maintain genotype frequency.

INTERPRETING GRAPHICS: The illustration below shows two contrasting models for rates of speciation. Use the illustration to answer the questions that follow.

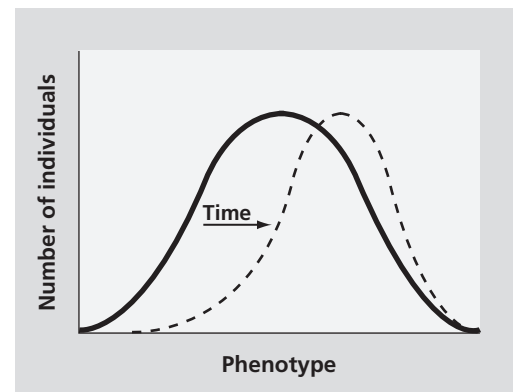


- Which model of speciation rates is illustrated by model A in the graph?
 - gradualism
 - sexual selection
 - disruptive selection
 - punctuated equilibrium
- Which model of speciation rates is illustrated by model B in the graph?
 - gradualism
 - sexual selection
 - disruptive selection
 - punctuated equilibrium

DIRECTIONS: Complete the following analogy:

6. genotype : allele :: phenotype :
- trait
 - mutation
 - gene pool
 - population

INTERPRETING GRAPHICS: The illustration below shows the occurrence of variations in a particular characteristic within a population. The dark line represents an earlier point in time than the dashed line. Use the illustration to answer the question that follows.



- Which type of selection is modeled in the illustration above?
 - sexual selection
 - disruptive selection
 - stabilizing selection
 - directional selection

SHORT RESPONSE

Explain the difference between reproductive isolation and geographic isolation.

EXTENDED RESPONSE

The phrase *Hardy-Weinberg genetic equilibrium* refers to the frequency of genotypes in populations from generation to generation.

Part A Briefly describe what this model predicts about genotype frequencies.

Part B What are the set of assumptions that must be met for the Hardy-Weinberg genetic equilibrium to be valid?

Test TIP

For multiple-choice questions, try to eliminate any answer choices that are obviously incorrect, and then consider the remaining answer choices.

Predicting Allele Frequency

OBJECTIVES

- Demonstrate the effect of natural selection on genotype frequencies.

PROCESS SKILLS

- modeling
- predicting
- calculating
- analyzing

MATERIALS

- 300 black beads
- 300 white beads
- 3 containers
- labeling tape
- marking pen

Background

1. What is natural selection?
2. What is the result of natural selection?

PART A Random Mating

1. Obtain three containers, and label them "Parental," "Offspring," and "Dead."
2. Place 200 black beads and 200 white beads in the "Parental" container. Assume that each black bead

represents a dominant allele for black coat (B) and that each white bead represents a recessive allele for white coat (b) in a hypothetical animal. Assume that the container holds gametes from a population of 200 of these hypothetical animals: 50 BB , 100 Bb , and 50 bb .

3. In your lab report, make a data table like Table A.
4. Without looking, remove two beads from the "Parental" container. What does this simulate?
5. Record the genotype and phenotype of the resulting offspring in your data table. Then, put the alleles into the "Offspring" container. Replace the beads that you removed from the "Parental" container with new beads of the same color.
6. Repeat steps 4 and 5 forty-nine times. Record the genotype and phenotype of each offspring in your data table.
7. Calculate the frequencies of alleles in the offspring. First, make a table in your lab report like Table B. Then, count and record the number of black beads in the "Offspring" container. This number divided by the total number of beads (100) and multiplied by 100% is the frequency of B alleles. Then, count and record the number of white beads in the "Offspring" container. Determine the frequency of b alleles as you did with the B alleles.

TABLE A MATING

Trial	Random mating		Nonrandom mating	
	Offspring genotype	Offspring phenotype	Offspring genotype	Offspring phenotype
1				
2				
3				
4				
5				

TABLE B ALLELE FREQUENCIES

Generation	Number of <i>B</i> alleles	<i>B</i> alleles/ total alleles	<i>B</i> allele frequency	Number of <i>b</i> alleles	<i>b</i> alleles/ total alleles	<i>b</i> allele frequency
Parental	200	200/400	50%	200	200/400	50%
Offspring						

TABLE C PHENOTYPE FREQUENCIES

Generation	Number of black animals	Black animals/ total animals	Frequency of black animals	Number of white animals	White animals/ total animals	Frequency of white animals
Parental	100	150/200	75%	50	50/200	25%
Offspring						

- Calculate the frequencies of phenotypes in the offspring. First, make a table in your lab report like Table C. Then, count and record the number of offspring with black coat color. Divide this number by the total number of offspring (50), and multiply by 100% to determine the frequency of black phenotype. Repeat this calculation to determine the frequency of white coat color among the offspring.

PART B Nonrandom Mating

- Return the beads in the "Offspring" container to the container labeled "Parental."
- Assume that animals with white-coat phenotype are incapable of reproducing. What is the genotype of animals with a white coat? To simulate this situation, remove 100 white beads from the container labeled "Parental," and set them aside.
- Start by removing two beads from the container labeled "Parental," and record the results in your Table A. If the offspring has a white-coat phenotype, put its alleles in the container labeled "Dead." If the offspring has a black-coat phenotype, put its alleles in the container labeled "Offspring."
- If animals with a white-coat phenotype cannot reproduce, predict what would happen to allele frequency if step 11 were repeated until the parental gene pool was empty. Write your prediction in your lab report.
- Repeat step 11 until the parental gene pool is empty. Record the results of each pairing in your lab report in Table A. Compare your results with your prediction.

- Transfer the beads from the "Offspring" container to the "Parental" container. Leave the beads that you have placed in the "Dead" container in that container. Do not return those beads to the parental container.
- Repeat step 11 again until the parental pool is empty. Record your results in your data table.
- Repeat steps 14 and 15 two more times.
- Calculate the frequencies of the final genotypes produced, as you did in Part A. Compare the results with your prediction from step 12.
-  Clean up your materials before leaving the lab.

Analysis and Conclusions

- Compare the frequency of recessive alleles produced in Part A with that produced in Part B. Did you correctly predict the frequencies?
- Did the frequency of the *b* allele change uniformly through all generations? If not, what happened?
- Why did you remove 100 white beads from the "Parental" container in step 10?
- How did this change the phenotype frequency of white animals in the parental generation from the original ratio of 50/200?

Further Inquiry

If you continued Part B, would you eventually eliminate the *b* allele? Form a hypothesis and test it.

CLASSIFICATION OF ORGANISMS

The ground pangolin, *Manis temminckii*, inhabits the forests, grasslands, and savannas of Africa. When threatened, it rolls into a tight ball, which exposes its razor-sharp scales. Its common name is *scaly anteater* because of its feeding habits, but it is not classified with other kinds of anteaters.



SECTION 1 *Biodiversity*

SECTION 2 *Systematics*

SECTION 3 *Modern Classification*

BIODIVERSITY

Biologists have named and classified almost 2 million species. However, they estimate that the total number of species on Earth is much greater. Over time, scientists have created various systems of classification to organize their knowledge of the tremendous number of species. Each system places species into categories based on particular characteristics.

CLASSIFYING ORGANISMS

One important branch of biology investigates **biodiversity**, the variety of organisms considered at all levels from populations to ecosystems. In general, the scientific catalogue of all biodiversity on Earth has increased over time. The number of identified species has grown from hundreds to millions. And scientists are certain that many more remain unidentified.

Since the 1980s, ecologist Terry Erwin and others have been working to catalog insect species in plots of tropical rain forest. To do this, researchers may fog the treetops with insecticide and then catch the falling insects in a net, as shown in Figure 17-1. Then, the scientists will count and classify the insects. Using such methods, Erwin collected over 1,000 species of beetles from just 19 trees of the same species in a limited area of rain forest. Furthermore, Erwin used his data to estimate the total possible number of insect species worldwide. He proposed that there could be more than 30 million species of insects on Earth!

Every year, biologists discover thousands of new species and seek to classify them in a meaningful way. Over the centuries, classification systems for organisms have been proposed and modified to include an increasing understanding of how organisms are structured and how they evolved. Biologists are challenged to organize their knowledge of such diversity in a way that makes sense.

For example, consider the pangolin, a species that is covered with scales and catches ants on its sticky tongue. Should pangolins be grouped in a category with other scaly animals, such as lizards and crocodiles? Biologists have decided against such a classification, because pangolins share many more similarities with mammals such as dogs and cats than with lizards or crocodiles. Should pangolins be grouped in a category with other mammals that use sticky tongues to eat ants? Again, biologists have decided against this grouping, because pangolins share more similarities with dogs and cats than with the other ant-eating mammal species.

OBJECTIVES

- **Relate** biodiversity to biological classification.
- **Explain** why naturalists replaced Aristotle's classification system.
- **Identify** the main criterion that Linnaeus used to classify organisms.
- **List** the common levels of modern classification from general to specific.

VOCABULARY

biodiversity
taxonomy
taxon
kingdom
domain
phylum
division
class
order
family
genus
species
binomial nomenclature
subspecies

FIGURE 17-1

These researchers are fogging a tree with insecticide to collect the insects that live on its leaves and branches.



Word Roots and Origins

taxonomy

from the Greek words *taxis*, meaning “to put in order,” and *nomia*, meaning “law”

TAXONOMY

The science of describing, naming, and classifying organisms is called **taxonomy** (taks-AHN-uh-mee). Any particular group within a taxonomic system is called a **taxon** (plural, *taxa*). Over time, scientists have created taxonomic systems that have different numbers and levels of taxa as well as different names for each taxon.

About 2,400 years ago, the Greek philosopher Aristotle classified organisms into only two taxa—either plants or animals. He grouped animals according to whether they lived on land, in water, or in air and grouped plants based on differences in their stems. As naturalists described more organisms, they realized that Aristotle’s classification system was inadequate.

Early naturalists also found that common names, such as *robin* or *fir tree*, were not useful to identify organisms because common names vary from place to place and many don’t accurately define a species. For example, the common name *jellyfish* is misleading. Although it has a jellylike body, a jellyfish is not a fish. Some naturalists named species by using long descriptions in Latin, but these names were difficult to remember and did not describe relationships among organisms.

The Linnaean System

Swedish naturalist Carolus Linnaeus (1707–1778) devised a system of grouping organisms into hierarchical categories according to their form and structure. Each category represents a level of grouping from larger, more general categories to smaller, more specific categories. Linnaeus’s original system had seven levels. Figure 17-2 and Table 17-1 show a modern classification of different organisms in a hierarchical system similar to that used by Linnaeus.

FIGURE 17-2

Under the modern Linnaean system, the classification of an organism places the organism within a nested hierarchy of taxa. The hierarchy ranges from the most general category (domain) to the most specific (species).

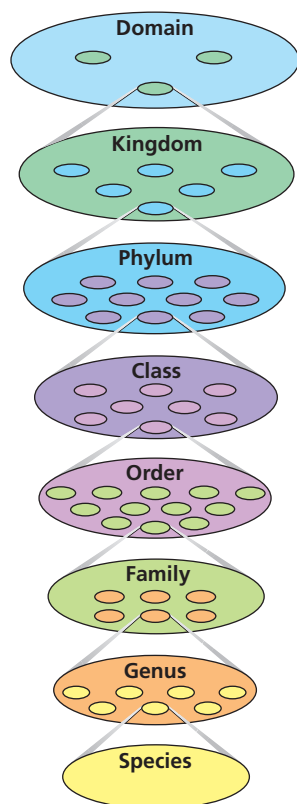


TABLE 17-1 Classification Hierarchy of Organisms

	<u>Pangolin</u>	<u>Dandelion</u>
Domain	Eukarya	Eukarya
Kingdom	Animalia	Plantae
Phylum/division	Chordata	Magnoliophyta
Class	Mammalia	Magnoliopsida
Order	Pholidota	Asterales
Family	Manidae	Asteraceae
Genus	Manis	Taraxacum
Species	<i>Manis temminckii</i>	<i>Taraxacum officinale</i>
		

LEVELS OF CLASSIFICATION

The modern version of Linnaeus's nested hierarchy of organization is shown in Figure 17-2. His largest category was the **kingdom**, which was made up of only two kingdoms—animals and plants. Modern biologists adopted this system but added several other kingdoms as well as **domains**, which are categories above the kingdom level. Subsets below the kingdom level are called **phyla** (FIE-luh) (singular, *phylum*). Each phylum consists of **classes**, and each class contains **orders**. Still smaller groupings are the **family** and the **genus** (JEE-nuhs). The smallest grouping, which contains only a single kind of organism, is the **species** (SPEE-sheez).

Binomial Nomenclature

Linnaeus gave an organism a species name, or scientific name, with two parts: the genus name followed by the *species identifier*. This system of two-part names is known as **binomial nomenclature** (bie-NOH-mee-uhl NOH-muhn-KLAY-chuhr). The species name for humans is *Homo sapiens*. The species name is written in italics with the genus name capitalized. Species names generally come from Latin roots and are intended to be the same for all countries and in every language.

The name of a species is often quite descriptive. One kind of amoeba, which changes its shape seemingly randomly as it crawls in a pond, is *Chaos chaos*. The ground pangolin's species identifier is *temminckii*, which commemorates C. J. Temminck, a 19th-century Dutch naturalist who first collected a specimen of this species.

Biologists refer to variations of a species that live in different geographic areas as **subspecies**. A subspecies name follows the species identifier. For example, *Terrapene carolina triunui* is a subspecies of the common eastern box turtle, *Terrapene carolina*.



Quick Lab

Practicing Classification

Materials paper, pencil

Procedure Using Table 17-1 as a model, classify a fruit or vegetable that you would find in a grocery store. Use all eight levels of classification.

Analysis At which level did you assign the least specific name? At which level did you assign the most specific name? Would Aristotle have classified your item differently? Explain your answer.

SECTION 1 REVIEW

1. Explain how Earth's biodiversity relates to classification.
2. In what ways was Aristotle's classification system inadequate?
3. What criteria did Linnaeus use to classify organisms?
4. What are the eight levels of the modern classification system?
5. What are two reasons that species names are more precise than common names?
6. How is a subspecies given a scientific name?

CRITICAL THINKING

7. **Analyzing Models** Explain why many of Linnaeus's categories are still used.
8. **Calculating Information** Suppose that there are 67 butterfly species for every 22,000 insect species and there are 15,000 to 22,000 species of butterflies worldwide. How many insect species are there?
9. **Applying information** Organize ten produce items from the grocery store into a hierarchy, and make up a species name for each one based on Linnaean principles.

MILESTONES

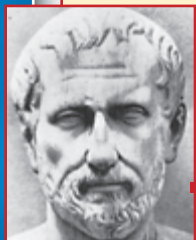
IN

Classification of Organisms

Timeline

Prehistory

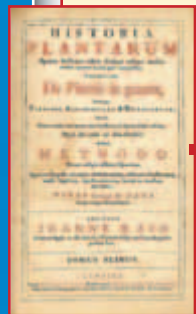
Humans identify plants and animals.



300 B.C.

Theophrastus classifies 500 plants.

1555 Gessner publishes *Historia Animalium*.



1650 Ray publishes *Historia Plantarum*.

1735 Linnaeus proposes taxonomic system.

1866 Haeckel draws a "Tree of Life."



1959–1969 Whittaker proposes four-kingdom and then five-kingdom system.

1966 Hennig proposes cladistics.

1977–1990

Woese names sixth kingdom and three domains.



For uncounted millennia, humans have been observing Earth's organisms and trying to make sense of their diversity. Over time, a naming system that lumped all life-forms into "plants" or "animals" gave way to more sophisticated and accurate approaches to classification based on shared traits and ancestry. The modern approach can logically organize millions of species, including those yet to be discovered.

Prehistoric humans survived by hunting animals and gathering plants. Their lives depended on recognizing the edible and the harmful. Indeed, native cultures have passed along a deep store of knowledge about thousands of plant and animal species organized by utility, such as edibility, toxicity, and medicinal value.

The Greek philosopher Aristotle observed and recorded nature in a scholarly way. His student Theophrastus, who lived from 370 BCE to 285 BCE, recorded 500 plant types, classified into herbs, shrubs, "pre-shrubs," and trees.

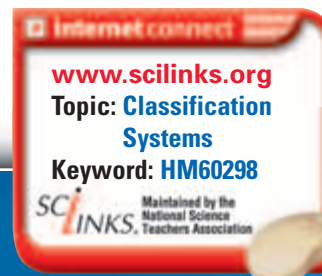
Advances in transportation, navigation, and exploration in the Middle Ages greatly expanded people's lists of observed organisms. In 1555, Swiss naturalist Conrad Gessner published *Historia Animalium*, categorizing thousands of animals into quadrupeds, birds, fish, and snakes. One hundred years later in his *Historia Plantarum*, English naturalist John Ray organized thousands of plants based on visual similarities and differences. In 1735, Swedish naturalist Carolus Linnaeus produced *Systema Naturae*, which categorized thousands of organisms into a hierarchy starting with genus and species and building to higher taxa including two kingdoms.

Several refinements have come to these higher taxa. German biologist Ernst Haeckel organized a "Tree of Life" in 1866. In 1894,

he redrew its branches into three kingdoms. In 1959, American ecologist R. H. Whittaker established a five-kingdom taxonomy. In 1966, German biologist Willi Hennig invented cladistics, which classifies organisms based on their shared, derived traits in order to reflect their evolutionary history. Molecular data obtained by using ribosomal RNA sequences as an evolutionary measure allowed American molecular biologist Carl Woese to propose in 1977 a six-kingdom system that divided the then-existing kingdom Monera into two new kingdoms: Archaeobacteria and Eubacteria. In 1990, Woese introduced the three-domain system that is used today.

Review

1. Describe how prehistoric humans "classified" species.
2. Why has the need for a system of classification grown over the centuries?
3. How do modern classification systems differ from the system of classification used by John Ray?
4. Describe a scientific advancement that enabled Woese to introduce the six-kingdom system.



SYSTEMATICS

*More than 200 years ago, Linnaeus grouped organisms according to similarities that he could readily see. Modern biologists consider not only visible similarities but also similarities in embryos, chromosomes, proteins, and DNA. In **systematics**, the goal is to classify organisms in terms of their natural relationships. This section will discuss some modern methods of systematics.*

PHYLOGENETICS

Biologists today may choose among and sometimes combine several systems of classification. Increasingly, systematic taxonomists agree that an organism's classification should reflect phylogeny. Recall that *phylogeny* is the evolutionary history of a species or a taxon. Thus, modern taxonomists are often involved in **phylogenetics**, the analysis of the evolutionary or ancestral relationships among taxa.

Systematists usually use several types of evidence to hypothesize about phylogenetics. Often, they compare the visible similarities among currently living species or fossils from extinct organisms. Biologists may also compare patterns of embryonic development and the ways in which the embryos of different species express similar genes. Furthermore, they may compare similar chromosomes and macromolecules, such as DNA or RNA, from different species to deduce phylogenetic relationships.

Systematists often represent their hypotheses in the form of a **phylogenetic diagram**, also called a *phylogenetic tree*. This type of diagram looks like a family tree and has a branching pattern that indicates how closely related a subset of taxa are thought to be. As with all hypotheses, phylogenetic diagrams may change whenever new discoveries and investigations cause scientists to revise their hypotheses.

Evidence of Shared Ancestry

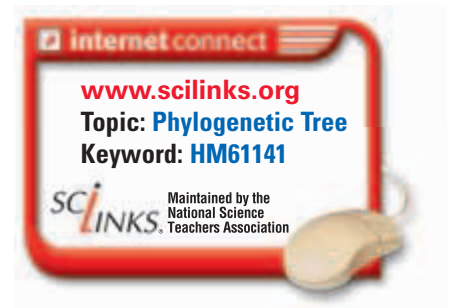
Biologists use fossils as important clues for the timing of evolutionary changes and divergence. The use of fossils in the study of evolutionary relationships is critical, but the fossil record may lack evidence about many kinds of organisms, especially small and soft-bodied organisms, such as worms, fungi, or bacteria. The fossil record can help provide the framework of a phylogenetic diagram, but a systematist would test inferred relationships with additional evidence.

OBJECTIVES

- **Identify** the kinds of evidence that modern biologists use in classifying organisms.
- **Explain** what information a phylogenetic diagram displays.
- **State** the criteria used in cladistic analysis.
- **Describe** how a cladogram is made.
- **Discuss** how proteins and chromosomes are used to classify organisms.
- **Explain** cladistic taxonomy, and identify one conclusion that is in conflict with classical taxonomy.

VOCABULARY

systematics
phylogenetics
phylogenetic diagram
cladistics
shared character
derived character
clade
cladogram



Word Roots and Origins

clade

from the Greek *klados*,
meaning "a branch" or "a shoot"

Systematists also compare *homologous* features, items that share a common ancestry. For example, the jaws of pangolins and of dogs are homologous. It is essential to separate homologous features from *analogous* features (features that are similar because they have a similar function rather than a similar lineage). Scales are an example of analogous features. Both pangolins and snakes have scales, but the fossil record shows that scales evolved independently in the two taxa. The greater the number of homologous features shared by two organisms, the more closely related the organisms are.

Embryological evidence also helps establish phylogenetic relationships. For example, a fluid-filled sac called an *amnion* surrounds the embryos of reptiles, birds, and mammals. The embryos of other vertebrates lack an amnion. This shared and homologous embryological feature is used to define *amniotes*. All amniotes are combined into one taxon that includes all reptiles, birds, and mammals but excludes other vertebrates.

CLADISTICS

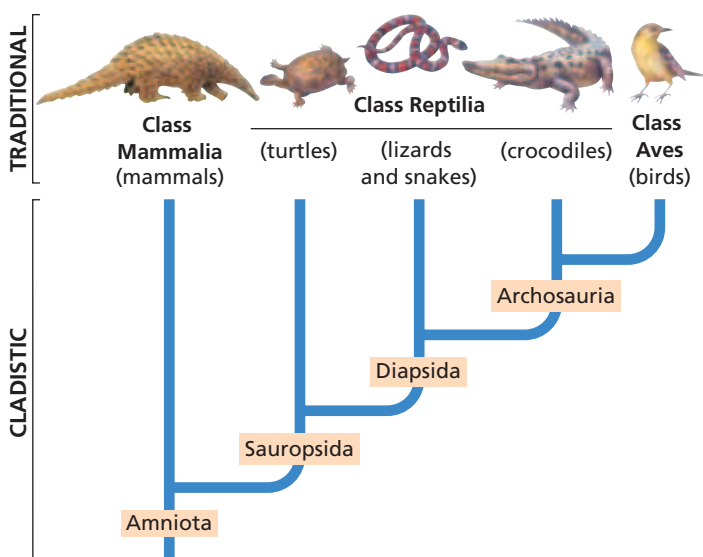
In 1966, German biologist Willi Hennig developed **cladistics** (kluh-DIS-tiks), a system of phylogenetic analysis that uses shared and derived characters as the only criteria for grouping taxa. A **shared character** is a feature that all members of a group have in common, such as hair in mammals or feathers in birds. A **derived character** is a feature that evolved only within the group under consideration. For example, feathers are thought to be a derived character for birds. Among living and fossil animals, the only animals that have feathers are birds and a few extinct reptiles that were very similar to birds in other ways. Therefore, it is reasonable to hypothesize that feathers evolved only within the bird lineage and were not inherited from the ancestors that birds share with reptiles.

Cladists assume that organisms that share one or more derived characters probably inherited those characters from a common ancestor. Cladists use the term **clade** for the group of organisms that includes an ancestor plus all of its descendants. Notice that clades do not have category names such as "class" or "phylum." Cladists create phylogenetic diagrams called **cladograms** (KLAD-uh-GRAMZ), such as the one in Figure 17-3.

Because cladists use strict criteria, their taxonomies may differ from those of traditional systematists. For example, traditional systematists grouped crocodiles with turtles, lizards, and snakes in the class Reptilia, but placed birds in their own class, Aves.

FIGURE 17-3

Traditional systematists placed crocodiles in the class Reptilia, but placed birds in the class Aves. In contrast, cladistic taxonomists have grouped crocodiles and birds together in a clade named *Archosauria*. Notice that clades do not have category names such as "class" or "phylum."



In contrast, cladistic taxonomists have grouped crocodiles and birds together in a clade named *Archosauria*. This grouping reflects the hypothesis that crocodiles and birds share a more recent ancestor than either group shares with any other animals. Archosauria is then grouped with successively less-related clades of animals.

Constructing a Cladogram

Because cladistic analysis is comparative, the analysis deliberately includes an organism that is only distantly related to the other organisms. This organism is called an *out-group*. The out-group is a starting point for comparisons with the other organisms being evaluated.

To make a cladogram like Figure 17-4, start by making a data table like Table 17-2. Place organisms in a column at the left and their characteristics in a row at the top. Choose an out-group organism. In our example, moss has fewer traits in common with and seems most distantly related to the other groups in the table. So, moss is the choice for an out-group. Score the characters that are lacking in the out-group as a 0, and score the presence of a derived character as a 1. For example, mosses lack seeds, so they receive a 0 for the “seeds” character. Only pines and flowering plants have seeds, so they receive a 1 for that character.

The table now reveals the derived character that is shared by most of the taxa. In this example, that character is vascular tissue—vessels that transport water and sugars. This character is shown at the base of the first branch of the cladogram. The second most common character is seeds. Because ferns lack seeds, they are placed on the second branch of the cladogram. The least common character is flowers. Because pines lack flowers, they are placed on the third branch. Finally, flowering plants are placed on the last branch.

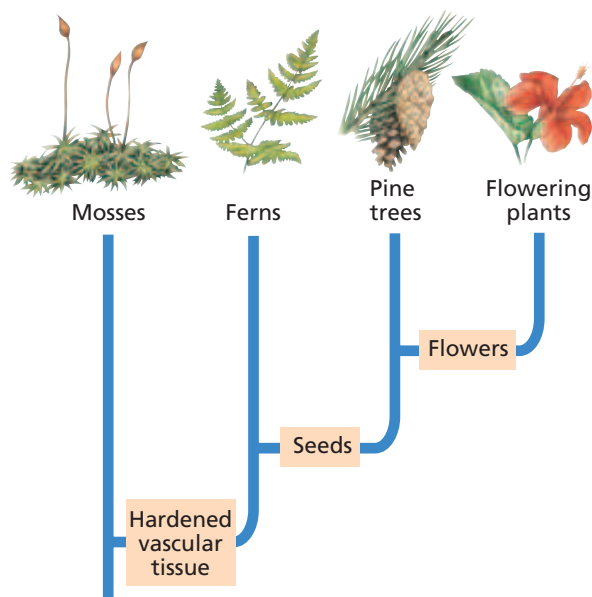
The resulting “tree” is a cladistic hypothesis for the evolutionary relationships among these plants. In addition to considering obvious physical characters, such as seeds and flowers, cladists may consider molecular characters, such as an individual nucleotide in a gene sequence or an amino acid sequence within a protein.

TABLE 17-2 Data Table for Cladogram

Group of organisms	Characters		
	Vascular tissue	Seeds	Flowers
Mosses (out-group)	0	0	0
Ferns	1	0	0
Pine trees and other conifers	1	1	0
Flowering plants	1	1	1

FIGURE 17-4

This cladogram groups several major kinds of plants according to their shared, derived characters. The most common character (vascular tissue) is shared by all groups. The least common character (flowers) separates flowering plants from all the other plants.



Amino Acid Sequence of a Petal-Forming Gene in Different Plants

Aster	M	G	R	G	K	I	E	I	K	I	E	N	N	T	N	R	Q	V	T	Y	S	K	R	R	N	G	I	F	K	K	A	H	E	L	T	V	L	C	D	A	K	V	S	L	I	M	F	S	N	T	G	K	F	H	E	Y	
Tomato	-	-	-	G	K	I	E	I	K	I	E	N	S	T	N	R	Q	V	T	Y	S	K	R	R	N	G	I	F	K	K	R	K	E	L	T	V	L	C	D	A	K	I	S	L	I	M	L	S	S	T	R	K	Y	H	E	Y	
Snapdragon	M	A	R	G	K	I	Q	I	K	R	I	E	N	Q	T	N	R	Q	V	T	Y	S	K	R	R	N	G	L	F	K	K	A	H	E	L	S	V	L	C	D	A	K	V	S	I	I	M	I	S	S	T	Q	K	L	H	E	Y
Rice	M	G	R	G	K	I	E	I	K	R	I	E	N	A	T	N	R	Q	V	T	Y	S	K	R	R	T	G	I	M	K	K	A	R	E	L	T	V	L	C	D	A	Q	V	A	I	I	M	F	S	S	T	G	K	Y	H	E	F

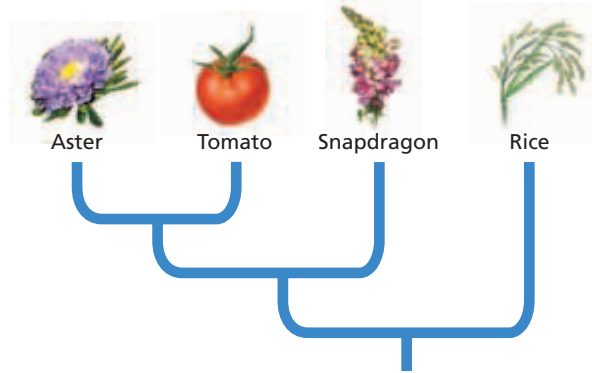


FIGURE 17-5

This cladogram is based on similar amino acid sequences in a specific protein produced by these plants. The initials M, G, and so on indicate different amino acids. The yellow squares indicate differences within the otherwise-identical sequences.

Molecular Cladistics

A biologist can count the shared, derived amino acids at each position in a protein and, from the analysis, construct a tree that hypothesizes relationships between various species. On a molecular cladogram, branch lengths are proportional to the number of amino acid changes. Such molecular data are independent of physical similarities or differences. The analysis shown in Figure 17-5 is of the amino acid sequence of a protein involved in flower development.

Biologists have used evolutionary changes in the sequence of macromolecules, such as DNA, RNA, and proteins, as a form of *molecular clock*, a tool for estimating the sequence of past evolutionary events. The molecular clock hypothesis suggests that the greater the differences between a pair of sequences, the longer ago those two sequences diverged from a common ancestor. A researcher who matches a molecular clock carefully with the fossil record can use it to hypothesize when various characteristics arose and when organisms diverged from ancestral groups.

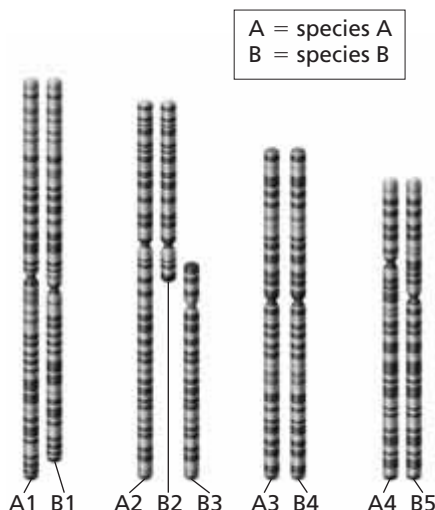
Chromosomes

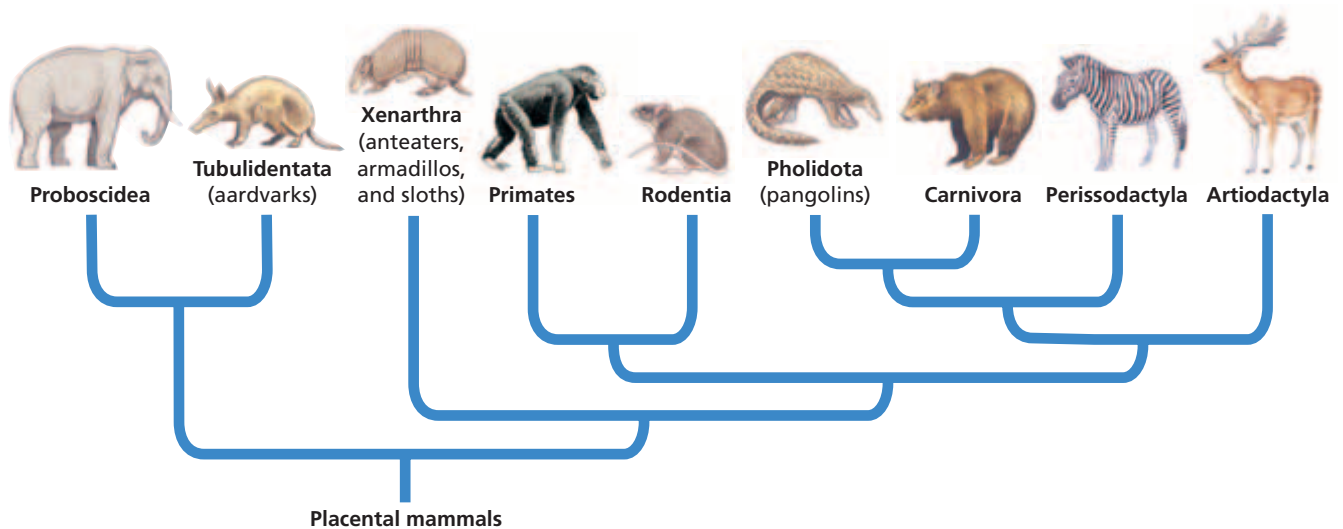
Analyzing karyotypes can provide still more information on evolutionary relationships. As Figure 17-6 shows, chromosomes can be stained to reveal a pattern of bands. If two species have the same banding pattern in regions of similar chromosomes, the regions are likely to have been inherited from a single chromosome in the last common ancestor of the two species. Karyotypic data are totally independent of both physical similarities and molecular data.

For example, the chromosomes of two species are shown in Figure 17-6. Several of the chromosomes have similar banding patterns, suggesting that the chromosomes are homologous. In addition, two of species B's chromosomes are similar to parts of one of species A's chromosomes. In such cases, biologists may still hypothesize that all of the chromosomes were inherited from the same ancestor. It is possible that in one of the descendants, one chromosome became two or two chromosomes became one.

FIGURE 17-6

In this example, the karyotype of species A is very similar to that of species B. The diagram compares 4 of species A's chromosomes with 5 of species B's chromosomes. The banding pattern similarities suggest that chromosomes B2 and B3, combined, are homologous to chromosome A2.





PUTTING IT ALL TOGETHER

To classify an organism and represent its systematics in an evolutionary context, biologists use many types of information to build and revise phylogenetic models. Systematists will use data about physical features, embryos, genes in the nucleus, mitochondrial DNA, and ribosomal RNA.

Consider the classification of pangolins that is shown in Figure 17-7. African and Asian pangolins share several adaptations with other mammals that eat ants, including African aardvarks and South American anteaters. These adaptations include narrow snouts, strong digging claws, and long, sticky tongues. These shared characteristics, however, are analogous, not homologous. Deeper analysis of other characteristics, such as skeletal structures and nucleotide sequences in several genes, indicate that all three anteating taxa—pangolins, anteaters, and aardvarks—occupy quite different branches on the mammalian phylogenetic tree. Therefore, systematists place pangolins closer to the group that includes dogs and bears than to either aardvarks or anteaters.

FIGURE 17-7

This phylogenetic diagram is based on analyses of the DNA of many kinds of mammals. These analyses do not support a systematic grouping of pangolins with either African aardvarks or South American anteaters. Instead, pangolins seem to be most closely related to carnivores, such as bears and dogs. Biologists sometimes revise their classifications in light of such new evidence.

SECTION 2 REVIEW

1. Identify two types of information that modern taxonomists use to classify organisms.
2. What is systematics, and what kinds of data are used by a systematist?
3. Define the term *phylogenetic diagram*.
4. What is a shared, derived character?
5. Compare cladistic taxonomy with more traditional taxonomy.
6. How can a comparative analysis of the karyotypes of two species provide clues to the degree of relatedness between the species?

CRITICAL THINKING

7. **Analyzing Models** Examine the phylogenetic model shown in Figure 17-3, and explain why the Class Reptilia is not a clade.
8. **Making Inferences** Explain how a taxonomist might use embryological evidence in classifying an organism.
9. **Evaluating Concepts** How could a biologist use cladistics to deduce whether the flippers of a sea lion and a whale are homologous or analogous characters?

SECTION 3

OBJECTIVES

- **Describe** the evidence that prompted the invention of the three-domain system of classification.
- **List** the characteristics that distinguish between the domains Bacteria, Archaea, and Eukarya.
- **Describe** the six-kingdom system of classification.
- **Identify** problematic taxa in the six-kingdom system.
- **Explain** why taxonomic systems continue to change.

VOCABULARY

Bacteria
Archaea
Eukarya
Eubacteria
Archaeobacteria
Protista
Fungi
Plantae
Animalia

MODERN CLASSIFICATION

Biologists continue to develop taxonomies to organize life's enormous diversity. They regularly revise the many branches of the "tree of life" to reflect current hypotheses of the evolutionary relationships between groups. They have even revised the largest and most fundamental categories of the Linnean-inspired classification system—domains and kingdoms.

THE TREE OF LIFE

Looking at the cells of any organism under a microscope, biologists can easily identify one of two fundamental types of cells. On this basis, biologists may divide all life into two large groups. One of these groups consists of *eukaryotes*, or organisms whose cells possess a membrane-enclosed nucleus and many other cell organelles. Eukaryotes include organisms such as pangolins, people, and amoebas. The second group consists of *prokaryotes*. Prokaryotic cells are smaller than eukaryotic cells and have neither a nucleus nor membrane-enclosed organelles. Prokaryotes include bacteria.

Revising the Tree

Although the differences between eukaryotes and prokaryotes are obvious, systematists cannot assume that all organisms in one of these groups are equally related. A systematist can draw inferences about evolutionary relationships only by identifying homologous features. Because all members of one of these groups have few other physical traits in common, it is difficult to make comparisons between them. However, eukaryotes and prokaryotes do contain homologous macromolecules, such as proteins and RNAs used for storing and translating genetic information. When methods became available to compare such macromolecules, biologists began to use macromolecules in phylogenetic analyses.

Such analyses led molecular microbiologist Carl Woese of the University of Illinois to propose a major revision of the system that had classified Earth's diversity into six kingdoms. In 1977, Woese began comparing ribosomal RNA (rRNA) sequences from many different organisms and then grouping these organisms according to their similarities. All of today's living organisms, even prokaryotes, have ribosomes. Furthermore, the nucleotide sequences of the genes that encode rRNA seem to have changed little throughout most of evolutionary history. Thus, rRNA genes are uniquely useful for studying the most basic evolutionary relationships between Earth's diverse organisms.

THREE DOMAINS OF LIFE

Phylogenetic analyses of rRNA genes gave scientists new insights about the relationships between major groups and suggested a new “tree of life.” Three of the most important insights were as follows:

1. The data are consistent with the hypothesis that all living organisms inherited their rRNA genes from an ancient organism or form of life. Scientists refer to this unknown ancestor as the *last universal common ancestor*.
2. At the broadest level, all living things seem to be related by ancestry to one of three major lineages, or *domains*. The three domains are named *Bacteria*, *Archaea*, and *Eukarya*. Most of the organisms that we are familiar with, such as plants and animals, belong to just one of the domains—Eukarya. Figure 17-8 shows a phylogenetic diagram for the three domains.
3. The most surprising insights were those related to the domain Archaea. The species in this domain were identified and studied more recently than those in the other two domains. At first, the archaea were classified as bacteria. However, scientists have found that archaea differ greatly from bacteria in many important ways. More recently, scientists hypothesized that modern archaea descended from a unique kind of prokaryotes that existed early in Earth’s history.

Domain Bacteria

The domain **Bacteria** is made up of small, single-celled prokaryotic organisms that usually have a cell wall and reproduce by cellular fission. Each bacterium has a cell wall, a plasma membrane, a cytoplasm that lacks complex organelles, and at least one circular chromosome. Bacteria do not have a membrane-bound DNA and thus lack a true nucleus. Most bacteria are small—many are just 2 μm long. By comparison, human cells can be 6 μm long or more. The oldest known fossils of cells appear to be bacterial cells.

Domain Archaea

The second domain also consists of prokaryotes and is named **Archaea**. The archaea have distinctive cell membranes and other unique biochemical and genetic properties. Some archaeal species are autotrophic and are able to produce food by chemosynthesis. Some species produce flammable gases, such as methane, as waste products. Many archaeal species inhabit harsh environments, such as sulfurous hot springs, deep-sea thermal vents, salty lakes, wastewater from mining, and the intestines of some animals.

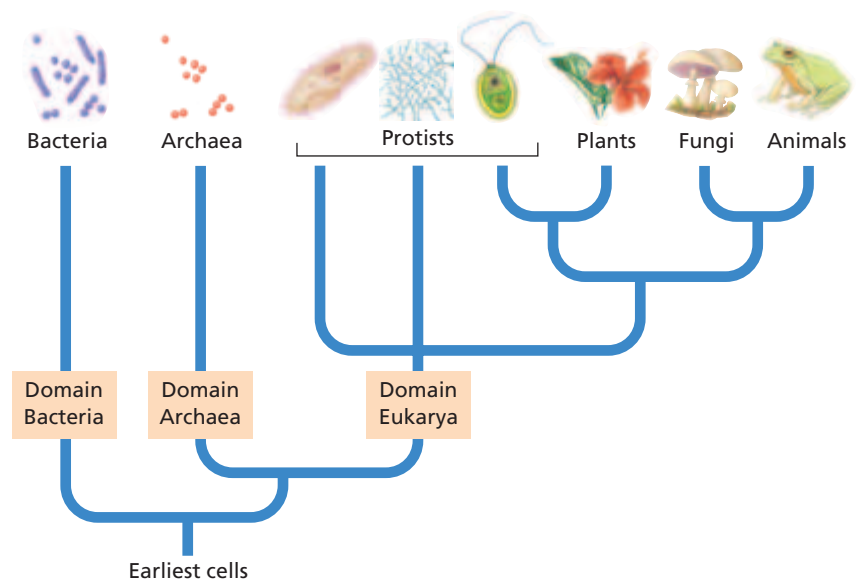
Word Roots and Origins

domain

from the Latin *dominium*, meaning “right of ownership”

FIGURE 17-8

This phylogenetic diagram represents hypotheses of the evolutionary relationships between the major recognized groups of organisms. Notice the alignment of the three domain names (*Bacteria*, *Archaea*, and *Eukarya*) with three major “branches” of the “tree” of all life. For updates on phylogenetic information, visit go.hrw.com and enter the keyword **HM6 Phylo**.



Because of the unique adaptations of the archaea, scientists think that archaea were among the earliest organisms on Earth. Furthermore, scientists think that archaea and bacteria have co-evolved during Earth's history, meaning that the cells have interacted closely within environments and that the cells could have exchanged nutrients and genetic material. These possibilities, along with biochemical evidence, have led to the *theory of endosymbiosis*, which holds that eukaryotic cells arose when ancient prokaryotic cells began to live together as one cell.

Domain Eukarya

The most familiar groups of organisms are members of the domain **Eukarya** (yoo-KAR-ee-uh), which consists of eukaryotic organisms. The cells of these organisms are large and have a true nucleus and complex cellular organelles. The domain Eukarya includes plants, animals, fungi, and a variety of single-celled organisms.

SIX KINGDOMS

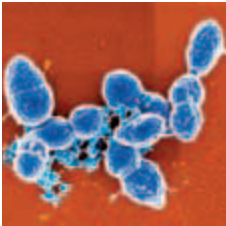
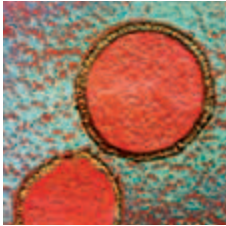
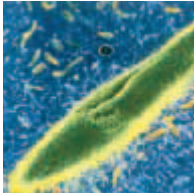
The system of three domains generally aligns with the more traditional system of six kingdoms, as represented in Figure 17-9 and summarized in Table 17-3. In the six-kingdom system, the first kingdom aligns with the domain Bacteria, the second kingdom aligns with the domain Archaea, and the remaining four kingdoms align with the domain Eukarya.

FIGURE 17-9

The six-kingdom system of classification can be aligned with the newer system of three domains. However, biologists have proposed adding, subdividing, or replacing some kingdoms. Biologists have also proposed other levels of taxa.

Kingdom Eubacteria

The kingdom **Eubacteria** (yoo-bak-TIR-ee-uh) aligns with the domain Bacteria. The name *Eubacteria* means “true bacteria,” distinguishing this group from the archaea, which are no longer considered to be bacteria.

Domain Bacteria	Domain Archaea	Domain Eukarya	
			
Kingdom Eubacteria	Kingdom Archaeobacteria	Kingdom Protista	Kingdom Plantae
			
		Kingdom Fungi	Kingdom Animalia

Kingdom Archaeobacteria

The kingdom **Archaeobacteria** (AHR-kee-bak-TIR-ee-uh) aligns with the domain Archaea. The name *Archaeobacteria* means “ancient bacteria,” but archaea are so different from bacteria that many biologists now prefer to use only the name *archaea* for this group.

Kingdom Protista

One of the four kingdoms of eukaryotes is **Protista** (proh-TIST-uh), whose members are called *protists*. This kingdom has been defined as those eukaryotes that are not plants, animals, or fungi. This definition is troublesome because it defines organisms based on what they are *not* rather than on what they *are*. Most protists are unicellular organisms, but it is difficult to make other generalizations about them. Molecular analyses indicate that many protists are less related to each other than plants are to animals. For all of these reasons, many biologists think that the taxon *Protista* is no longer useful, and several new taxa are being considered.

Common examples of unicellular protists are amoebas and paramecia. Examples of multicellular protists include some kinds of seaweeds and molds. Each of these protists is similar to animals, plants, or fungi in some ways yet different in other ways.

TABLE 17-3 *Kingdom and Domain Characteristics*

Taxon	Cell type	Cell surfaces	Body plan	Nutrition
Domain Bacteria (aligns with Kingdom Eubacteria)	prokaryotic; lack nucleus and other organelles	cell wall: contains peptidoglycans; cell membrane: contains fatty acids	unicellular	heterotrophic and autotrophic by chemosynthesis or photosynthesis
Domain Archaea (aligns with Kingdom Archaeobacteria)	prokaryotic; lack nucleus and other organelles	cell wall: lacks peptidoglycan; cell membrane: contains hydrocarbons other than fatty acids	unicellular	heterotrophic and autotrophic by chemosynthesis
Domain Eukarya Kingdom Protista	eukaryotic; have nucleus and complex organelles	cell wall: made of cellulose or other materials; cell membrane: contains fatty acids	mostly unicellular; multicellular forms: lack tissue organization	autotrophic by photosynthesis, some heterotrophic by phagocytosis, or both
Domain Eukarya Kingdom Fungi	eukaryotic; have nucleus and complex organelles	cell wall: made of chitin; cell membrane: contains fatty acids	unicellular and multicellular	heterotrophic by secreting digestive enzymes into environment
Domain Eukarya Kingdom Plantae	eukaryotic; have nucleus and complex organelles	cell wall: made of cellulose; cell membrane: contains fatty acids	multicellular; develop from embryos	autotrophic by photosynthesis
Domain Eukarya Kingdom Animalia	eukaryotic; have nucleus and complex organelles	cell wall: none cell membrane: contains fatty acids	multicellular; develop from embryos	heterotrophic by phagocytosis

Kingdom Fungi

The second kingdom of eukaryotes is **Fungi** (FUHN-jie). The kingdom Fungi consists of eukaryotic, heterotrophic organisms that are unicellular or multicellular and that gain nutrients in a unique way. Unlike animal cells and some protists, fungi absorb rather than ingest nutrients. The about 70,000 species of fungi include mushrooms, puffballs, rusts, smuts, mildews, and molds.

Kingdom Plantae

The third kingdom of eukaryotes is **Plantae** (PLAN-tee), which consists of eukaryotic, multicellular plants. Except for a few parasitic species, most plants are autotrophic, use photosynthesis as a source of energy, and develop from embryos. Most plants live on land and include mosses, ferns, conifers, and flowering plants.

Kingdom Animalia

The fourth kingdom of eukaryotes is **Animalia** (AN-uh-MAH-lee-uh). Animals are eukaryotic, multicellular, and heterotrophic organisms that develop from embryos. Most animals have symmetrical body organization and move around their environment to find and capture food.

Future Taxonomic Systems

Because taxonomic systems are scientific models, they are subject to change. Since the time of Linnaeus, biologists have proposed, used, evaluated, and modified many classification systems. Modern biologists have proposed alternatives to the six-kingdom and three-domain systems. For example, some biologists have proposed three or more new kingdoms to replace Kingdom Protista. Some use taxonomic categories such as “subkingdom” and “super-order” to adapt the more traditional Linnaean categories to newer systematic models. Some favor pure cladistics over Linnaean categories. Surely, new systems will result from future investigations into the millions of known and unknown species on Earth.

SECTION 3 REVIEW

1. What kind of evidence supports the classification of all organisms into the three-domain system?
2. Contrast bacteria with archaea.
3. List and briefly describe the kingdoms in the six-kingdom system of classification.
4. What is problematic about the kingdom name *Archaeobacteria*?
5. What is problematic about defining the kingdom Protista?
6. Why do protists, fungi, plants, and animals share a single domain in the three-domain system?

CRITICAL THINKING

7. **Evaluating Viewpoints** Biologists once classified all species of prokaryotes in a single kingdom called *Monera*. Justify their reasoning.
8. **Applying Information** You have discovered a new organism. It is unicellular and has mitochondria, chloroplasts, and a nucleus. To what kingdom does this organism belong? Justify your answer.
9. **Recognizing Differences** Describe the characteristics that differentiate plants from fungi. In what ways are plants and fungi similar?

CHAPTER HIGHLIGHTS

SECTION 1

Biodiversity

- Naturalists have invented several systems for categorizing biodiversity, which is the variety of organisms considered at all levels from populations to ecosystems.
- Naturalists replaced Aristotle's classification system because it did not adequately cover all organisms and because his use of common names was problematic.
- Taxonomy is the science of describing, naming, and classifying organisms.
- Carolus Linnaeus devised a seven-level hierarchical system for classifying organisms according to their form and structure. From the most general to the most specific, the levels are kingdom, phylum, class, order, family, genus, and species. An adaptation of this system is still in use today.
- An important part of Linnaeus's system was assigning each species a two-part scientific name—a genus name, such as *Homo*, and a species identifier, such as *sapiens*.

Vocabulary

biodiversity (p. 337)
taxonomy (p. 338)
taxon (p. 338)
kingdom (p. 339)

domain (p. 339)
phylum (p. 339)
division (p. 339)
class (p. 339)

order (p. 339)
family (p. 339)
genus (p. 339)
species (p. 339)

binomial nomenclature
(p. 339)
subspecies (p. 339)

SECTION 2

Systematics

- A modern approach to taxonomy is systematics, which analyzes the diversity of organisms in the context of their natural relationships.
- When classifying organisms, scientists consider fossils, homologous features, embryos, chromosomes, and the sequences of proteins and DNA.
- A phylogenetic diagram displays how closely related a subset of taxa are thought to be.
- Homologous features as well as similarities in patterns of embryological development provide information about common ancestry.
- Cladistics uses shared, derived characters as the only criterion for grouping taxa.
- Molecular similarities, such as similar amino acid or nucleotide sequences, as well as chromosome comparisons can help determine common ancestry.

Vocabulary

systematics (p. 341)
phylogenetics (p. 341)

phylogenetic diagram (p. 341)
cladistics (p. 342)

shared character (p. 342)
derived character (p. 342)

clade (p. 342)
cladogram (p. 342)

SECTION 3

Modern Classification

- The phylogenetic analysis of rRNA nucleotide sequences led to a new "tree of life" consisting of three domains aligned with six kingdoms.
- The three domains are Bacteria, Archaea, and Eukarya.
- Domain Bacteria aligns with Kingdom Eubacteria, which consists of single-celled prokaryotes that are true bacteria.
- Domain Archaea aligns with Kingdom Archaeobacteria, which consists of single-celled prokaryotes that have distinctive cell membranes and cell walls.
- Domain Eukarya includes the kingdoms Protista, Fungi, Plantae, and Animalia. All members of this domain have eukaryotic cells.

Vocabulary

Bacteria (p. 347)
Archaea (p. 347)
Eukarya (p. 348)

Eubacteria (p. 348)
Archaeobacteria (p. 349)

Protista (p. 349)
Fungi (p. 350)

Plantae (p. 350)
Animalia (p. 350)

CHAPTER REVIEW

USING VOCABULARY

1. Use each of the following terms in a separate sentence: *biodiversity*, *binomial nomenclature*, *species identifier*, and *subspecies*.
2. Explain the relationship between a phylum and a division.
3. Choose the term that does not belong in the following group, and explain why the term does not belong: *homologous*, *derived character*, *shared character*, and *analogous*.
4. **Word Roots and Origins** The word *taxonomy* is derived from the Greek words *taxis*, which means “to put in order,” and *nomia*, which means “a set of rules or laws.” Using this information, explain why the term *taxonomy* is a good name for the process that the term describes.

UNDERSTANDING KEY CONCEPTS

5. **Explain** the primary difficulty of organizing Earth’s biodiversity.
6. **Describe** how Aristotle’s and Linnaeus’s classification systems for organisms were similar.
7. **Explain** why a consistent naming system for species is important in scientific work.
8. **List** the seven levels of Linnaeus’s classification hierarchy from most general to most specific.
9. **Name** the kinds of evidence that systematists use in constructing a phylogenetic tree.
10. **Identify** the system of classification that is based on an analysis of shared, derived characters.
11. **Describe** how amino acid and nucleotide sequences function as a molecular clock.
12. **Summarize** one way that cladistic taxonomy differs from traditional taxonomy.
13. **Name** the three domains and six kingdoms, and indicate the relationships of the two sets of taxa.
14. **Describe** the molecular evidence that led to the three-domain system of classification.
15. **Differentiate** between bacteria and archaea.
16. **Compare** plants and fungi, and describe what they have in common with animals.
17.  **CONCEPT MAPPING** Use the following terms to create a concept map that describes how biologists classify a new species: *genus*, *species*, *binomial nomenclature*, *kingdom*, *homologous structures*, *taxonomy*, *derived characters*, *cladogram*, and *systematics*.

CRITICAL THINKING

18. **Evaluating Viewpoints** Scientists sometimes disagree about the phylogenetic histories of organisms. Cladistic taxonomists see reptiles in a different light than traditional taxonomists do. Why might scientists disagree with each other about the history of evolution?
19. **Evaluating Models** The evolutionary history of reptiles can be studied by comparing their macromolecules. The degree of difference can be related to the time that has passed since any two species descended from a common ancestor. Would the phylogenetic tree derived from macromolecular comparisons more closely resemble the results of cladistic analysis or the traditional classification?
20. **Evaluating Information** Biologists think that there may be millions of undescribed and unclassified species on Earth. Why might so many species still be undescribed or unclassified today?
21. **Analyzing Concepts** Legs are an example of a shared, derived character in vertebrates. Arthropods, such as lobsters and crickets, also have legs, but their legs are not homologous to the legs of vertebrates. Explain this difference.
22. **Justifying Conclusions** Several years ago, scientists found a living coelacanth, a fish that was thought to have become extinct 65 million years ago. The earliest fossils of coelacanths are about 350 million years old. The appearance of this species has changed little in 350 million years. If you could obtain macromolecules, such as proteins, from a 350 million-year-old fossil coelacanth and compare them with those of a freshly caught coelacanth, what would you expect to find and why?
23. **Interpreting Graphics** A taxonomist has recorded the data below.

Derived Characters in Plants

Plants	Seeds	Vascular system
Horsetails	no	yes
Liverworts	no	no
Pine trees	yes	yes

Identify the least common derived character. List the order that the plants in the table would be placed on a cladogram.

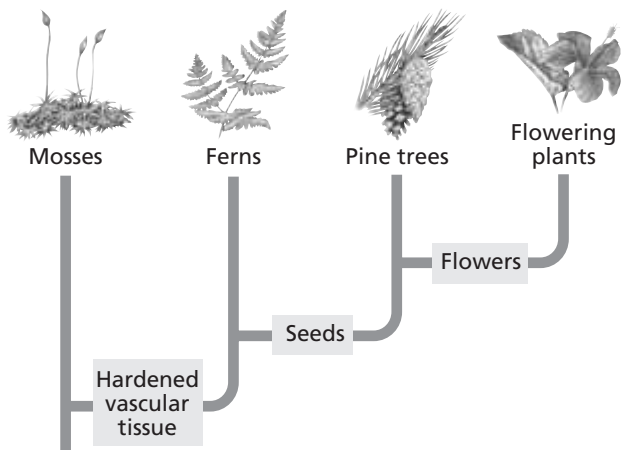


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which information is given in a species name?
 - genus and order
 - division and genus
 - genus and species identifier
 - species identifier and phylum
- To which level of classification does a group of closely related species of organisms belong?
 - class
 - order
 - genus
 - kingdom
- Eukaryotic organisms that have a nucleus and organelles, have a cell wall made of chitin, and secrete digestive enzymes belong to which kingdom?
 - Fungi
 - Plantae
 - Protista
 - Animalia

INTERPRETING GRAPHICS: The cladogram below shows the phylogenetic relationships among four kinds of plants. Use the cladogram to answer the question that follows.

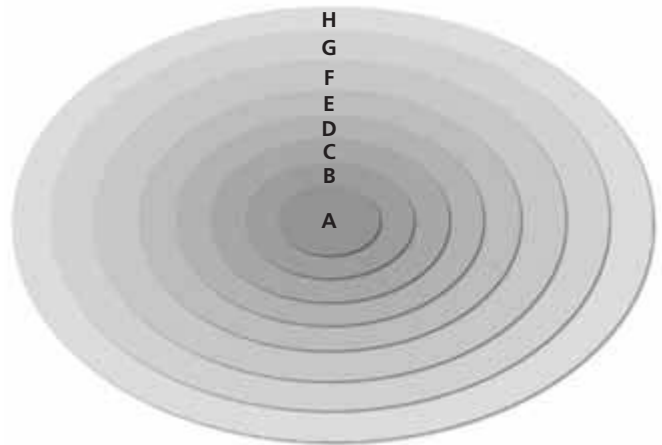


- On the basis of this cladogram, which plants share the most recent common ancestor?
 - mosses and ferns
 - mosses and pine trees
 - ferns and flowering plants
 - pine trees and flowering plants

DIRECTIONS: Complete the following analogy.

- class : order :: kingdom :
 - genus
 - domain
 - species
 - phylum

INTERPRETING GRAPHICS: The diagram below represents the eight levels of classification. Use the diagram to answer the question that follows.



- Which level of classification represents a species?
 - A
 - C
 - D
 - G

SHORT RESPONSE

Consider the characteristics of members of Kingdom Protista.

Explain why Kingdom Protista includes so many diverse organisms.

EXTENDED RESPONSE

To study the biodiversity of a rain forest, researchers sometimes collect species in vast numbers.

Part A How would traditional taxonomy aid a researcher who found 955 beetle species in one kind of tropical tree?

Part B How could molecular phylogenetics assist that same researcher?

Test TIP

If time permits, take short mental breaks to improve your concentration during a test.

Using and Formulating Dichotomous Keys

OBJECTIVES

- Use a dichotomous key to identify leaves.
- Construct a dichotomous identification key.

PROCESS SKILLS

- identifying
- classifying
- designing
- interpreting
- organizing data
- comparing and contrasting

MATERIALS

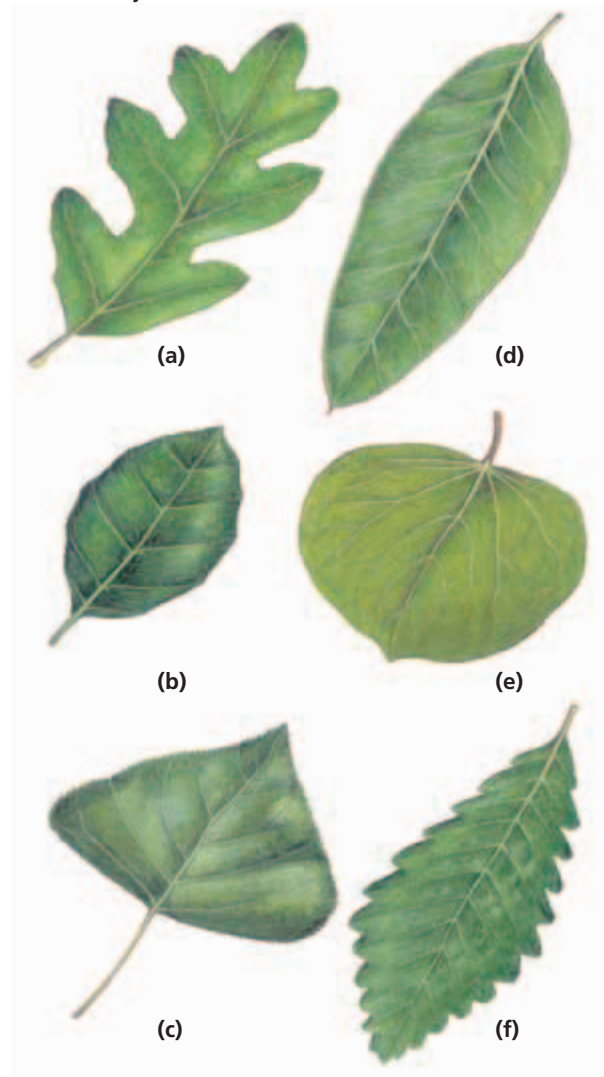
- pencil
- paper
- shoes
- masking tape
- marker

Background

1. Taxonomy is the science of naming, describing, and classifying organisms.
2. Why is classification essential to biology?
3. A dichotomous key uses pairs of contrasting, descriptive statements to lead to the identification of an organism (or other object).
4. The principle behind dichotomous keys—the forced choice—is used in many different situations to narrow the path toward an answer. If you have ever had your eyes examined for corrective lenses, you are familiar with the series of forced choices that end with the choice of the correct lenses for your eyes.
5. Do not confuse a dichotomous key with a cladogram. A cladogram represents evolutionary relationships, whereas a dichotomous key does not.

PART A Using a Dichotomous Key

1. Field guides often use dichotomous keys to help you identify organisms. Use the dichotomous key shown here to identify the tree leaves below. Begin with the paired descriptions 1a and 1b, and follow the directions. Proceed through the list of paired descriptions until you identify the leaf in question. In your lab report, write the names of the leaves as you identify them.



Dichotomous Key for Identifying Common Leaves

- 1a. If the edge of the leaf has no teeth and has no lobes, go to 2 in the key.
- 1b. If the edge of the leaf has teeth or lobes, go to 3 in the key.
- 2a. If the leaf has slightly wavy edges, the plant is a shingle oak.
- 2b. If the leaf has smooth edges, go to 4 in the key.
- 3a. If the leaf edge is toothed, the plant is a Lombardy poplar.
- 3b. If the leaf edge has lobes, go to 5 in the key.
- 4a. If the leaf is heart-shaped with veins branching from the base, the plant is a redbud.
- 4b. If the leaf is not heart-shaped, the plant is a live oak.
- 5a. If the leaf edge has a few large lobes, the plant is an English oak.
- 5b. If the leaf edge has many small lobes, the plant is a chestnut oak.

PART B Making a Dichotomous Key

- 2. Gather 10 different single shoes, and use masking tape and a marker to label the soles of the shoes with the owner's name. The labeled shoes should then be placed on a single table in the classroom.
- 3. Form small groups. Discuss the appearance of the shoes. In your lab report, make a table like the one below that lists some of the shoes' general characteristics, such as the type and size. Also list the names of the students who own the shoes. Complete the chart by describing the characteristics of each person's shoe.
- 4. Use the information in your table to make a dichotomous key that can be used to identify the owner of

- each shoe. Remember that a dichotomous key includes pairs of opposing descriptions. At the end of each description, the key should either identify an object or give directions to go to another specific pair of descriptions. Write your dichotomous key in your lab report.
- 5. After all groups have completed their key, exchange keys with a member of another group. Use the key to identify the owner of each shoe, and then verify the accuracy of your identification by reading the label on the shoe. If the key has led you to an inaccurate identification, return the key so that corrections can be made.
 - 6.  Clean up your materials before leaving the lab.

Analysis and Conclusions

- 1. What other characteristics might be used to identify leaves with a dichotomous key?
- 2. Were you able to identify the shoes using another group's key? If not, describe the problems you encountered.
- 3. How was it helpful to list the characteristics of the shoes before making the key?
- 4. Does a dichotomous key begin with general descriptions and then proceed to more specific descriptions, or vice versa? Explain your answer, giving an example from the key you made.
- 5. Are dichotomous keys based on a phylogenetic or morphological approach to classification? Explain your answer.

Further Inquiry

List characteristics that might be used to identify birds or other animals using a dichotomous key. Compare your list of characteristics with those used in a dichotomous key in a field guide for identifying birds or other animals.

DISTINGUISHING FEATURES OF A SAMPLE OF SHOES

	Left/right	Men's/women's	Laced/slip-on	Color	Size	Owner
1						
2						
3						
4						
5						

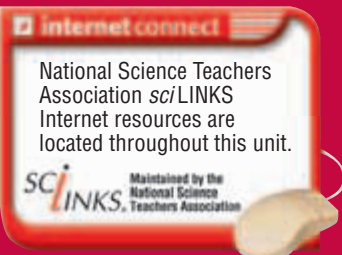
UNIT 5

CHAPTERS

- 18 **Introduction to Ecology**
- 19 **Populations**
- 20 **Community Ecology**
- 21 **Ecosystems**
- 22 **Humans and the Environment**



References to *Scientific American* project ideas are located throughout this unit.



ECOLOGY

“We and our fellow vertebrates are largely along for the ride on this planet. If we want to perpetuate the dream that we are in charge of our destiny and that of our planet, it can only be by maintaining biological diversity—not by destroying it. In the end, we impoverish ourselves if we impoverish the biota.”

From “Diverse Considerations,” by Thomas E. Lovejoy, from *Biodiversity*, edited by E. O. Wilson. Copyright © 1988 by the National Academy of Sciences. Reprinted by permission of *National Academy Press*.



Coral reef communities are second only to rain forests in diversity.



Bears are among the largest terrestrial predators.



The biosphere

Mimicry helps this mantid hide from both predators and potential prey.



Tropical rain forests are richer in species than other areas of Earth are.



INTRODUCTION TO ECOLOGY



Ecology is the study of the interactions of organisms with each other and with the nonliving parts of Earth. This image is a view of Earth as seen from space.

SECTION 1 *Introduction to Ecology*

SECTION 2 *Ecology of Organisms*

SECTION 3 *Energy Transfer*

SECTION 4 *Ecosystem Recycling*



Unit 7—Ecosystem Dynamics
Topics 1, 4–6

INTRODUCTION TO ECOLOGY

***Ecology** is the study of the interactions between organisms and the living and nonliving components of their environment. Each of the variety of organisms on Earth depends in some way on other living and nonliving things in its environment. Ecology is a broad science that involves collecting information about organisms and their environments, observing and measuring interactions, looking for patterns, and seeking to explain these patterns.*

INTERDEPENDENCE: A KEY THEME IN ECOLOGY

Although the field of ecology was not named until 1866, ecological information and understanding have always been crucial to humans. Before the development of agriculture, about 10,000–12,000 years ago, our ancestors obtained all of their food by hunting animals and gathering plants, seeds, berries, and nuts. Their survival depended on practical knowledge about the environment. Although most humans today don't survive as hunter-gatherers, they interact with the environment and other organisms every day.

Organisms and Their Environments

All organisms interact with other organisms in their surroundings and with the nonliving portion of their environment. Their survival depends on these interactions. Ecologists refer to this quality as *interconnectedness* or **interdependence**.

Interdependence is a key theme found throughout ecology. For example, you could not survive without the plants and other photosynthetic organisms that produce oxygen. Your cells need oxygen to release the energy in food, and cells will die if deprived of oxygen for even a few minutes. Conversely, photosynthetic organisms depend on the release of carbon dioxide gas by the cellular respiration of other organisms, such as humans, and geochemical processes, such as volcanic eruptions. Carbon dioxide gas is an essential raw material for making carbohydrates by photosynthesizers.

OBJECTIVES

- **Identify** a key theme in ecology.
- **Describe** an example showing the effects of interdependence upon organisms in their environment.
- **Identify** the importance of models to ecology.
- **State** the five different levels of organization at which ecology can be studied.

VOCABULARY

ecology
interdependence
ecological model
biosphere
ecosystem
community
population

Word Roots and Origins

ecology

from the Greek *oikos*, meaning "house," and *logos*, meaning "study of"

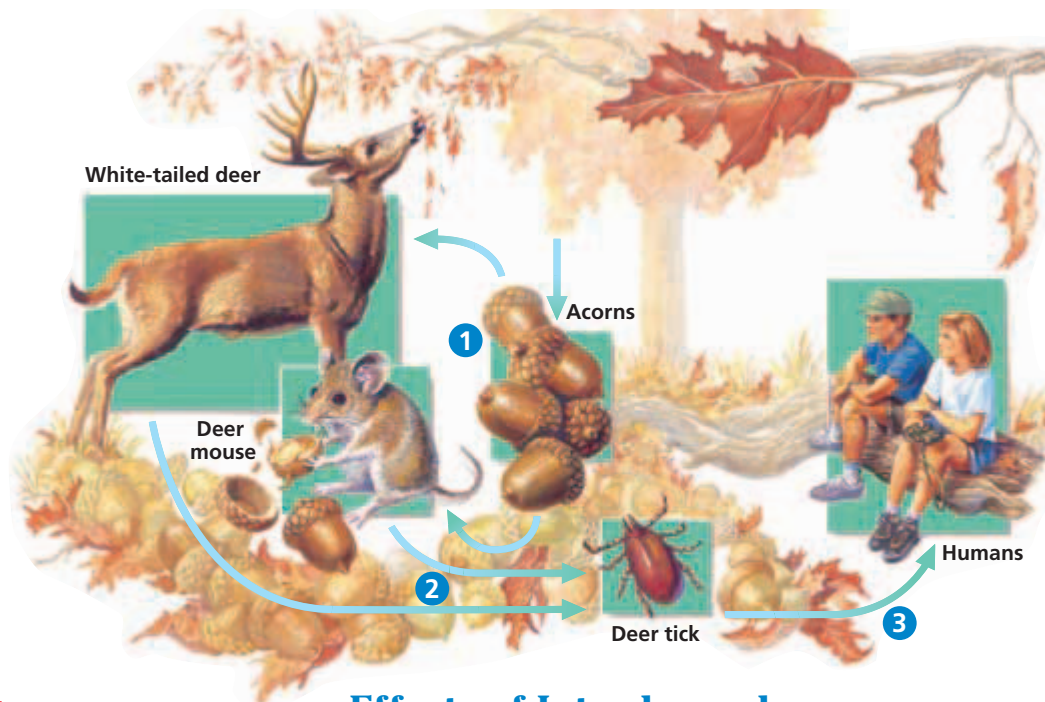


FIGURE 18-1

All of the different species shown are interconnected in the forest. ① An unusually plentiful crop of acorns helps support a large population of deer and mice. ② The deer and mice help support a large population of ticks. ③ Ticks carry the bacterium that causes Lyme disease. They pass on the disease to humans who visit the forest.

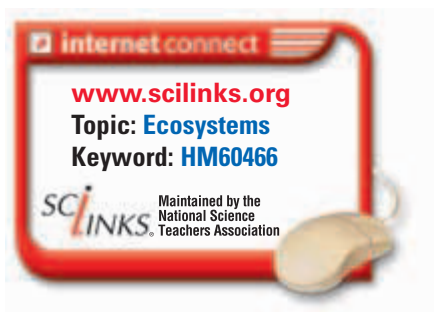
Effects of Interdependence

A consequence of interdependence is that any change in the environment can spread through the network of interactions and affect organisms that appear far removed from the change. One example is the interrelationships among species in forests in the eastern United States. Through these relationships, as shown in Figure 18-1, acorn production is connected to the spread of Lyme disease, an infection that can damage the human nervous system.

In most years, oak trees produce few or no acorns. Every few years, however, they produce a huge crop of acorns. The large number of acorns supports larger populations of deer and mice, which feed on acorns. Ticks feed on the blood of animals, so the tick population also increases. The increased number of ticks increases the chance that ticks will bite any humans in the forest. The bite of the deer tick can transmit the bacterium that causes Lyme disease to humans. So, in general, after a season of high acorn production, the cases of Lyme disease increase.

ECOLOGICAL MODELS

Ecology is extremely complex and difficult to study. One way that ecologists deal with this complexity is to use **ecological models** to represent or describe the components of an ecological system. A model may be physical, conceptual, or mathematical. Ecologists construct models to help them understand environmental interactions and to make predictions about possible changes. These predictions can be tested by comparing them with observations from the natural world. Models are widely used to help plan and evaluate solutions to environmental problems. However, an ecological model may be limited in its application, because it cannot always account for the influence of every variable in a real environment.



LEVELS OF ORGANIZATION

Scientists recognize a hierarchy of different levels of organization within organisms. Each organism is composed of one or more organs. Each organ is composed of tissues, which, in turn, are composed of cells, and so on. Likewise, ecologists recognize a hierarchy of organization in the environment, as illustrated in Figure 18-2.

Each level has unique properties that result from interactions between its components, so a complete study of ecology would look at all levels. But for practical reasons, ecologists often focus their research on one level of organization while recognizing that each level is influenced by processes at other levels.

The Biosphere

The broadest, most inclusive level of organization is the **biosphere** (BIE-oh-SFER), the thin volume of Earth and its atmosphere that supports life. All organisms are found within the biosphere. It is about 20 km (13 mi) thick and extends from about 8 to 10 km (5 to 6 mi) above the Earth's surface to the deepest parts of the oceans. In comparison, the Earth's diameter is about 12,700 km (7,900 mi), or more than 600 times the thickness of the biosphere. If Earth were the size of an apple, the biosphere would only be as thick as the apple's skin. Ecologists often describe the biosphere as a thin film of life covering an otherwise lifeless planet. Living things are not distributed evenly throughout the biosphere. Many organisms are found within a few meters of the surface of the land or oceans.

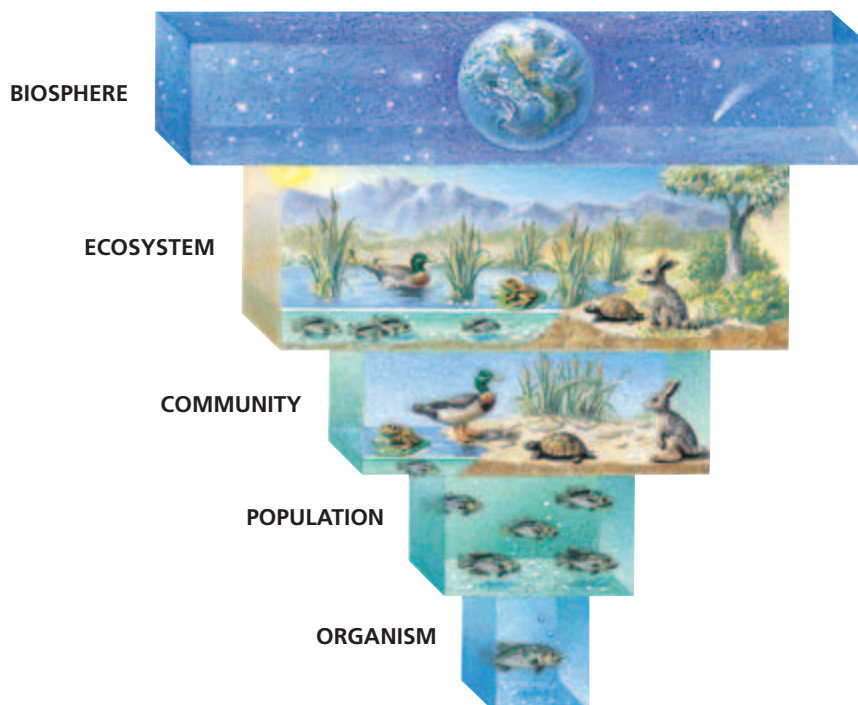


FIGURE 18-2

Ecology has been organized into five levels because of the complexity of the science. This diagram is a model illustrating the hierarchical organization of ecology.

Ecosystems

The biosphere is composed of smaller units called ecosystems. An **ecosystem** (EK-oh-SIS-tuhm) includes all of the organisms and the non-living environment found in a particular place. Consider a pond ecosystem. It contains a variety of living things, such as fish, turtles, aquatic plants, algae, insects, and bacteria. These organisms interact in ways that affect their survival. For instance, insects and fish eat aquatic plants, and turtles eat fish. The pond ecosystem also includes all the nonliving (physical and chemical) aspects of the pond that influence its inhabitants. The chemical composition of the pond—its pH, its levels of dissolved oxygen and carbon dioxide, and its supply of nitrogen—helps to determine what kinds of organisms live in the pond and how abundant they are. A very important physical factor is the amount of sunlight the pond receives, because sunlight is the ultimate source of energy for the pond's inhabitants.

FIGURE 18-3

These flowers represent a population of California poppies, *Eschscholzia californica*, living in California.



Communities, Populations, and Organisms

Whereas an ecosystem contains both living and nonliving components, a community includes only species of organisms. A **community** is all the interacting organisms living in an area. For instance, all the fish, turtles, plants, algae, and bacteria in the pond described above make up a community. Although it is less inclusive than an ecosystem, a community is still very complex, and it may contain thousands of species. Ecologists studying a community often focus on how species interact and how these interactions influence the nature of the community. Remember that the word *community* has a specific meaning in biology that differs from its everyday meaning.

Below the community level of organization is the population level, where the focus is on the members of a single species. A **population** includes all the members of a species that live in one place at one time. An example of a population of flowers is shown in Figure 18-3. The simplest level of organization in ecology is that of the organism. Research at this level concentrates on the adaptations that allow organisms to overcome the challenges of their environment.

SECTION 1 REVIEW

1. Explain why interdependence is an important theme in ecology.
2. Describe one example of the effects of interdependence upon organisms in their environment.
3. Why are models used so often in the science of ecology?
4. How does a population differ from a community?
5. Define the term *biosphere*.
6. List the five main levels of organization in ecology.

CRITICAL THINKING

7. **Predicting Results** Assuming wolves eat deer, how could a disease that kills a large portion of the wolf population affect the mice population in a forest ecosystem?
8. **Analyzing Concepts** Why is the amount of sunlight important to the animals in an ecosystem?
9. **Applying Information** Would bacteria that inhabit a cave deep inside Earth be considered part of the biosphere? Explain.

ECOLOGY OF ORGANISMS

*The place where an organism lives is its **habitat**. But why does it live there and not elsewhere? What parts of its habitat does it use? The answers to these questions depend on an organism's evolutionary history, its abilities, and its needs.*

ECOSYSTEM COMPONENTS

Ecologists separate the environmental factors that influence an organism into two types. The living components of the environment are called **biotic** (bie-AHT-ik) **factors**. Biotic factors include all of the living things that affect the organism. The nonliving factors, called **abiotic** (AY-bie-AHT-ik) **factors**, are the physical and chemical characteristics of the environment.

Biotic and Abiotic Factors

Abiotic factors include temperature, humidity, pH, salinity, oxygen concentration, amount of sunlight, availability of nitrogen, and precipitation. The importance of each factor varies from environment to environment. Abiotic and biotic factors are not independent; organisms change their environment and are influenced by those changes. For example, the availability of nitrogen in the soil affects how fast plants can grow, and plants affect nitrogen availability by absorbing nitrogen from the soil.

Abiotic factors are not constant. They vary from place to place and over time, as shown in Figure 18-4. Consider temperature, which is a very important abiotic factor. Temperature varies from hour to hour, from day to day, from season to season, and from place to place. Also important are the small differences in temperature within a habitat, such as the difference between an area in the shade of a tree and an area exposed to direct sunlight.

ORGANISMS IN A CHANGING ENVIRONMENT

Each organism is able to survive within a limited range of environmental conditions. For example, an organism may be able to function only within a specific range of temperatures. It is possible to determine this range for an organism by measuring how efficiently it performs at different temperatures. A graph of performance versus values of an environmental variable, such as temperature, is called a **tolerance curve**.

OBJECTIVES

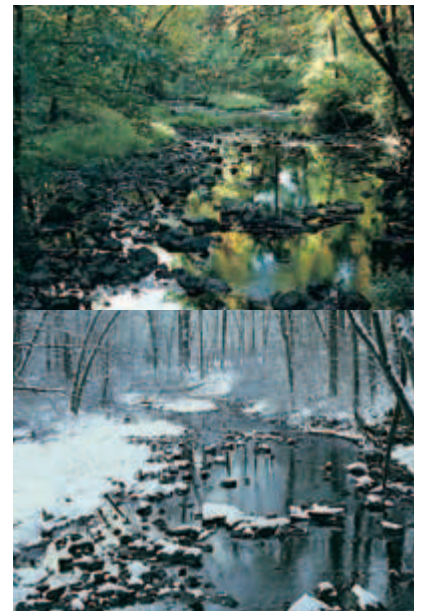
- **Compare** abiotic factors with biotic factors, and list two examples of each.
- **Describe** two mechanisms that allow organisms to survive in a changing environment.
- **Explain** the concept of the niche.

VOCABULARY

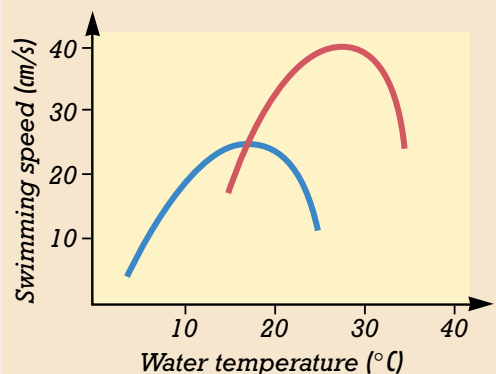
habitat
biotic factor
abiotic factor
tolerance curve
acclimation
conformer
regulator
dormancy
migration
niche
generalist
specialist

FIGURE 18-4

These pictures show the same area of forest at different times of the year. On the top, the forest displays spring foliage. On the bottom, the same area is covered with snow in the winter.



Acclimation to Temperature



■ Fish raised at 5°C
■ Fish raised at 25°C

FIGURE 18-5

Goldfish raised at 25°C are acclimated to higher temperatures, so they have a different tolerance curve than the fish raised at 5°C do.

An organism can survive and function in conditions outside its optimal range, but its performance is greatly reduced. It cannot survive under conditions that fall outside its tolerance limits. An organism's range may be determined by the levels of one or more factors, such as pH, temperature, or salinity.

Acclimation

Some organisms can adjust their tolerance to abiotic factors through the process of **acclimation** (AK-luh-MAY-shuhn). For example, goldfish raised at different temperatures have somewhat different tolerance curves, as shown in Figure 18-5. Be sure not to confuse *acclimation* with *adaptation*. Acclimation occurs within the lifetime of an individual organism. Adaptation is genetic change in a species or population that occurs from generation to generation over time.

Control of Internal Conditions

Environments fluctuate in temperature, light, moisture, salinity, and other chemical factors. There are two ways for organisms to deal with some of these changes in their environment. **Conformers** are organisms that do not regulate their internal conditions; they change as their external environment changes. The internal conditions of a conformer remain within the optimal range only as long as environmental conditions remain within that range. In contrast, **regulators** are organisms that use energy to control some of their internal conditions. Regulators can keep an internal condition within the optimal range over a wide variety of environmental conditions.

Careers in BIOLOGY

Canopy Scientist

Job Description Canopy scientists are biologists who study the forest canopy—the uppermost layer of trees. Today's researchers use a variety of equipment to reach the canopy. Canopy scientists can work in universities, botanical gardens, museums, and government and conservation agencies.

Focus On a Canopy Scientist

"I'm on the edge of discovering what is new. Plus, I get to climb trees!" Nalini Nadkarni teaches at a university and conducts research in both tropical and tem-

perate forests. She has also invited artists and musicians to visit the canopy. "I try to understand the science of the canopy, but artists and musicians help capture its aesthetic value." Nadkarni knows that her job is important. The forests she studies are important factors in world climate and are home to many unique species.

Education and Skills

- **High School**—three years of science courses and four years of math courses.
- **College**—bachelor of science in biology, including course work in botany, zoology, ecology, geography, and data



analysis; a master's (M.S.) or doctoral degree (Ph.D.) for research.

- **Skills**—self-motivation, curiosity, patience, good observation skills, research skills, computer literacy, quick thinking, and field survival skills.



For more about careers, visit go.hrw.com and type in the keyword **HM6 Careers**.

Escape from Unsuitable Conditions

Some species can survive unfavorable environmental conditions by escaping from them temporarily. For example, desert animals usually hide underground or in the shade during the hottest part of the day. Many desert species are active at night, when temperatures are much lower. A longer-term strategy is to enter a state of reduced activity, called **dormancy**, during periods of unfavorable conditions, such as winter or drought. Another strategy is to move to a more favorable habitat, called **migration**. An example of migration is the seasonal movements of birds, which spend spring and summer in cooler climates and migrate to warmer climates in the fall.

THE NICHE

Species do not use or occupy all parts of their habitat at once. The specific role, or way of life, of a species within its environment is its **niche** (NICH). The niche includes the range of conditions that the species can tolerate, the resources it uses, the methods by which it obtains resources, the number of offspring it has, its time of reproduction, and all other interactions with its environment. Parts of a lion's niche are shown in Figure 18-6.

Generalists are species with broad niches; they can tolerate a range of conditions and use a variety of resources. An example of a generalist is the Virginia opossum, found across much of the United States. The opossum feeds on almost anything, from eggs and dead animals to fruits and plants. In contrast, species that have narrow niches are called **specialists**. An example is the koala of Australia, which feeds only on the leaves of a few species of eucalyptus trees.

Some species have more than one niche within a lifetime. For example, caterpillars eat the leaves of plants, but as adult butterflies, they feed on nectar.



Word Roots and Origins

niche

from the Old French *nichier*,
meaning "to nest"

FIGURE 18-6

Plants and animals are able to share the same habitats because they each have different niches.



SECTION 2 REVIEW

1. Distinguish between biotic and abiotic factors.
2. Explain how migration allows organisms to cope with a changing environment.
3. What does a tolerance curve indicate about an organism?
4. How does an organism's niche differ from its habitat?
5. Give examples of a generalist and a specialist not mentioned in the text above.

CRITICAL THINKING

6. **Analyzing Concepts** Why do different species never occupy exactly the same niche?
7. **Applying Information** If some of the resources in a habitat are destroyed, which would be more likely to survive, a generalist species or a specialist species? Explain.
8. **Drawing Conclusions** A small rodent species and a bird species are adapted to cold temperatures. How might each species survive a major temperature increase?

SECTION 3

OBJECTIVES

- **Summarize** the role of producers in an ecosystem.
- **Identify** several kinds of consumers in an ecosystem.
- **Explain** the important role of decomposers in an ecosystem.
- **Compare** the concept of a food chain with that of a food web.
- **Explain** why ecosystems usually contain only a few trophic levels.

VOCABULARY

producer
chemosynthesis
gross primary productivity
biomass
net primary productivity
consumer
herbivore
carnivore
omnivore
detritivore
decomposer
trophic level
food chain
food web

ENERGY TRANSFER

All organisms need energy to carry out essential functions, such as growth, movement, maintenance and repair, and reproduction. In an ecosystem, energy flows from the sun to autotrophs, then to organisms that eat the autotrophs, and then to organisms that feed on other organisms. The amount of energy an ecosystem receives and the amount that is transferred from organism to organism affect the ecosystem's structure.

PRODUCERS

Autotrophs, which include plants and some kinds of protists and bacteria, manufacture their own food. Because autotrophs capture energy and use it to make organic molecules, they are called **producers**. Recall that *organic* molecules are molecules that contain carbon.

Most producers are photosynthetic, so they use solar energy to power the production of food. However, some autotrophic bacteria do not use sunlight as an energy source. These bacteria carry out **chemosynthesis** (KEE-moh-SIN-thuh-sis), in which they use energy stored in inorganic molecules to produce carbohydrates. In terrestrial ecosystems, plants are usually the major producers. In aquatic ecosystems, photosynthetic protists and bacteria are usually the major producers.

Measuring Productivity

Gross primary productivity is the rate at which producers in an ecosystem capture the energy of sunlight by producing organic compounds. Photosynthetic producers use energy and carbon dioxide to make sugar, an energy-rich organic molecule. Some of the sugar is used for cellular respiration, some is used for maintenance and repair, and some is used for making new organic material through either growth or reproduction. Ecologists refer to the organic material that has been produced in an ecosystem as **biomass**. Producers add biomass to an ecosystem by making organic molecules.

Only energy stored as biomass is available to other organisms in the ecosystem. Ecologists often measure the rate at which biomass accumulates, called the **net primary productivity**. Net primary productivity is typically expressed in units of energy per unit area per year (kcal/m²/y) or in units of dry organic mass per unit area per year (g/m²/y). Net primary productivity equals gross primary productivity minus the rate of respiration in producers.



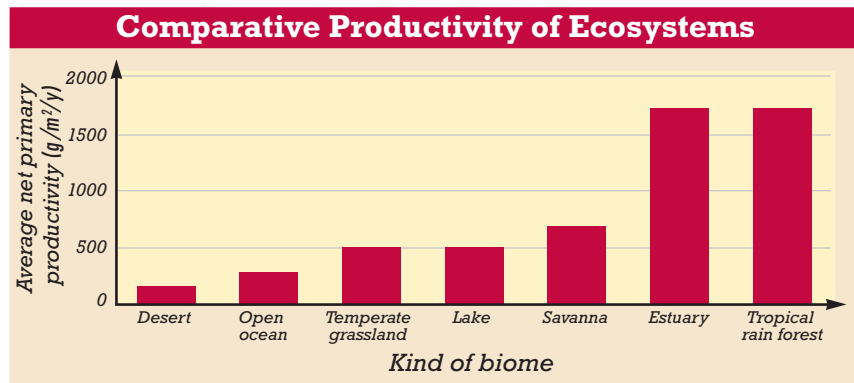


FIGURE 18-7

As the histogram shows, the net primary productivity in a tropical rain forest is very similar to the net primary productivity in an estuary. Temperate grasslands and freshwater lakes are also very similar in productivity.

Figure 18-7 shows that net primary productivity can vary greatly between ecosystems. For example, the average net primary productivity in a tropical rain forest is 25 times greater than the rate in a desert of the same size. Although rain forests occupy only about 5 percent of Earth's surface, they account for almost 30 percent of the world's net primary productivity. Variations in three factors—light, temperature, and precipitation—account for most of the variation in productivity among terrestrial ecosystems. An increase in any of these variables usually leads to a productivity increase. In aquatic ecosystems, productivity is usually determined by only two factors: light and the availability of nutrients.

Word Roots and Origins

omnivore

from the Latin *omnis*, meaning "all," and *-vore*, meaning "one who eats"

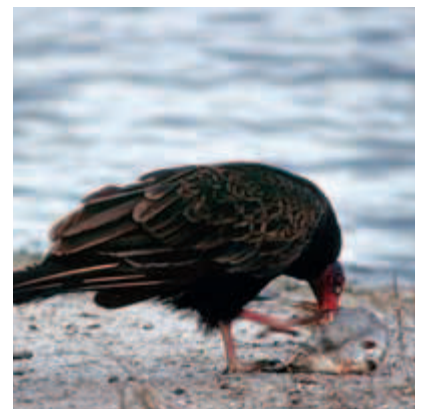
CONSUMERS

All animals, most protists, all fungi, and many bacteria are heterotrophs. Unlike autotrophs, heterotrophs cannot manufacture their own food. Instead, they get energy by eating other organisms or organic wastes. Ecologically speaking, heterotrophs are **consumers**. They obtain energy by consuming organic molecules made by other organisms. Consumers can be grouped according to the type of food they eat. **Herbivores** eat producers. An antelope that eats grass is a herbivore. **Carnivores** eat other consumers. Lions, cobras, and praying mantises are examples of carnivores. **Omnivores** eat both producers and consumers. The grizzly bear, whose diet ranges from berries to salmon, is an omnivore.

Detritivores (dee-TRIET-uh-VAWRZ) are consumers that feed on the "garbage" of an ecosystem. This waste, or *detritus*, includes organisms that have recently died, fallen leaves, and animal wastes. The vulture shown in Figure 18-8 is a detritivore. Many bacteria and fungi are detritivores that cause decay by breaking down complex molecules into simpler molecules. So, they are specifically called **decomposers**. Some of the molecules released during decay are absorbed by the decomposers, and some are returned to the soil or water. Decomposers make the nutrients that were contained in detritus available again to the autotrophs in the ecosystem. Thus, the process of decomposition recycles chemical nutrients.

FIGURE 18-8

This turkey vulture, *Cathartes aura*, is a detritivore that consumes dead animals. Detritivores play the important role of cleaning up dead organisms and aiding decomposition.



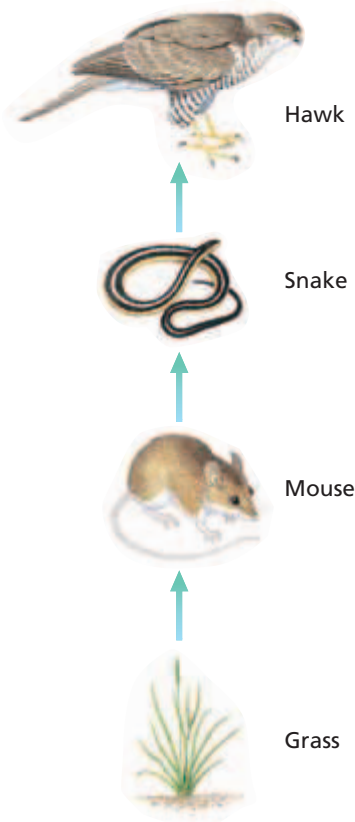


FIGURE 18-9

Energy is transferred from one organism to another in a food chain. The food chain shown above begins with a producer, grass, and ends with a carnivore, a hawk.

ENERGY FLOW

When one organism eats another, molecules are metabolized and energy is transferred. As a result, energy flows through an ecosystem, moving from producers to consumers. One way to follow the pattern of energy flow is to group organisms in an ecosystem based on how they obtain energy. An organism's **trophic** (TRAHF-ik) **level** indicates the organism's position in a sequence of energy transfers. For example, all producers belong to the first trophic level. Herbivores belong to the second trophic level, and the predators belong to the third level. Most terrestrial ecosystems have only three or four trophic levels, whereas marine ecosystems often have more.

Food Chains and Food Webs

A **food chain** is a single pathway of feeding relationships among organisms in an ecosystem that results in energy transfer. A food chain may begin with grass, which is a primary producer. The chain may continue with a consumer of grass seeds—a meadow mouse. Next, a carnivorous snake may kill and eat the mouse. A hawk then may eat the snake, as shown in Figure 18-9.

The feeding relationships in an ecosystem are usually too complex to be represented by a single food chain. Many consumers eat more than one type of food. In addition, more than one species of consumer may feed on the same organism. Many food chains interlink, and a diagram of the feeding relationships among all the organisms in an ecosystem would resemble a web, as shown in Figure 18-10. For this reason, the interrelated food chains in an ecosystem are called a **food web**.

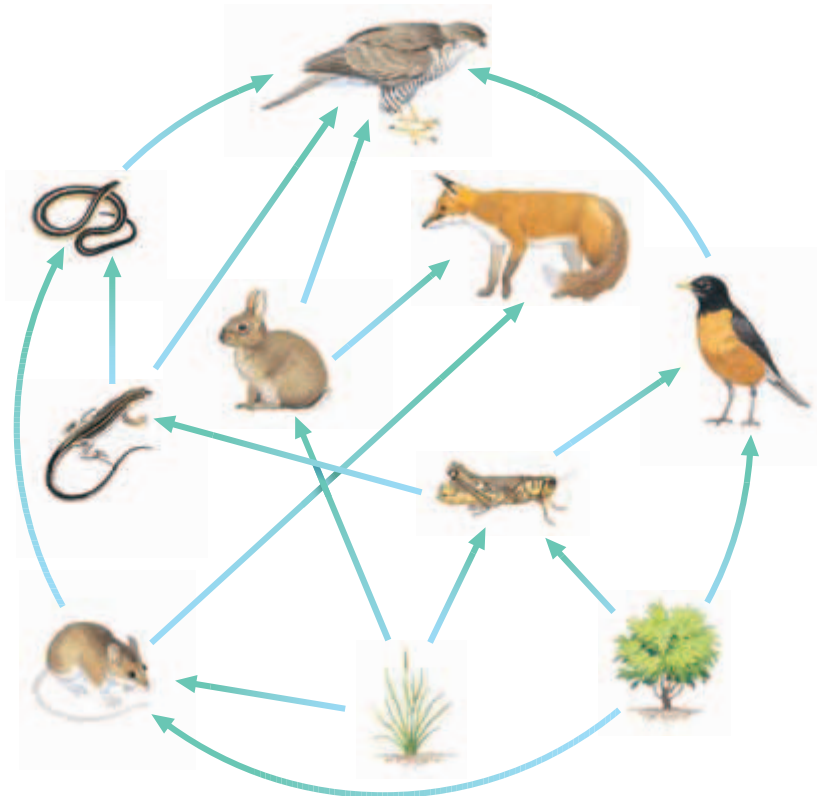


FIGURE 18-10

Because a large carnivore may be at the top of several food chains, it is helpful to show as many feeding relationships as possible in a food-web diagram. Not all organisms are listed in the food web. For example, no decomposers are shown.

Energy Transfer

Figure 18-11 represents the amount of energy stored as organic material in each trophic level in an ecosystem. The pyramid shape of the diagram indicates the low percentage of energy transfer from one level to the next. On average, 10 percent of the total energy consumed in one trophic level is incorporated into the organisms in the next.

Why is the percentage of energy transfer so low? One reason is that some of the organisms in a trophic level escape being eaten. They eventually die and become food for decomposers, but the energy contained in their bodies does not pass to a higher trophic level. Even when an organism is eaten, some of the molecules in its body will be in a form that the consumer cannot break down and use. For example, a cougar cannot extract energy from the antlers, hooves, and hair of a deer. Also, the energy used by prey for cellular respiration cannot be used by predators to synthesize new biomass. Finally, no transformation or transfer of energy is 100 percent efficient. Every time energy is transformed, such as during the reactions of metabolism, some energy is lost as heat.

TROPHIC LEVELS

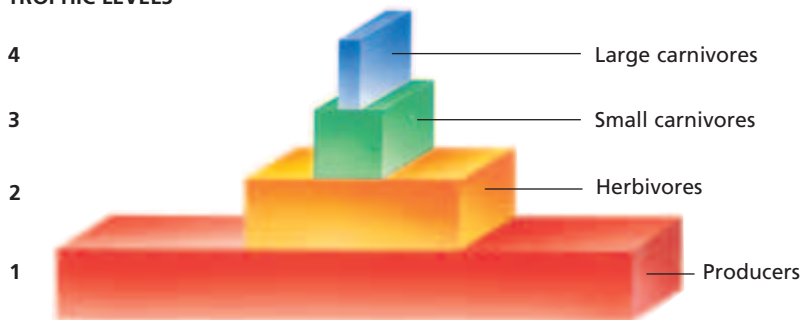


FIGURE 18-11

This diagram represents energy transfer through four trophic levels. The amount of energy transferred from one level to another can vary, so the structure shown can vary. What is always true, however, is that the top level is much smaller than the lowest level. Hence, energy-transfer diagrams are always roughly pyramid shaped.

Limitations of Trophic Levels

The low rate of energy transfer between trophic levels explains why ecosystems rarely contain more than a few trophic levels. Because only about 10 percent of the energy available at one trophic level is transferred to the next trophic level, there is not enough energy in the top trophic level to support more levels.

Organisms at the lowest trophic level are usually much more abundant than organisms at the highest level. In Africa, for example, you will see about 1,000 zebras, gazelles, and other herbivores for every lion or leopard you see, and there are far more grasses and shrubs than there are herbivores. Higher trophic levels contain less energy, so, they can support fewer individuals.

SECTION 3 REVIEW

1. How do producers and consumers obtain energy?
2. Name five types of consumers.
3. What important role do decomposers play in an ecosystem?
4. How does a food chain differ from a food web?
5. Give two reasons for the low rate of energy transfer within ecosystems.
6. Explain why food chains usually do not exceed three to four levels.

CRITICAL THINKING

7. **Predicting Results** Describe the probable effects on an ecosystem if all the plants were to die. What if all the decomposers were to die?
8. **Evaluating Models** A student has modeled a terrestrial ecosystem with seven trophic levels. Is this number reasonable? Explain.
9. **Analyzing Concepts** Explain why the same area can support a greater number of herbivores than carnivores.

Science in Action

Testing a Theory of Biogeography

In the 1960s, mathematical ecologist Robert H. MacArthur of Princeton University and taxonomist Edward O. Wilson of Harvard University developed a theory and mathematical model of island biogeography based on their study of species on islands. This model, and others inspired by it, is used to explain patterns of species distribution around the world.

HYPOTHESIS: The Number of Species on Any Island Is Constant

Ant biologist Edward O. Wilson (1929–) and mathematical ecologist Robert H. MacArthur (1930–1972) were both interested in community patterns within nature. Shortly after they met, they decided to work together on a study of species on islands.

Wilson noticed that the number of ant species on an island correlate with the size of the island. He also noticed that when a new ant species arrives on an island, one of the species already on the island becomes extinct. However, the total number of ant species remains constant. Wilson and MacArthur hypothesized that islands have a constant number of species. They proposed that the number of species on an island reflects an equilibrium—a balance between the rate at which new species colonize the island and the rate at which established species become extinct.

METHOD: Construct and Test a Model

MacArthur and Wilson developed a mathematical model to explain their observations. The mathematics of the theory is complex, but the broad outlines center on two observable patterns: (1) large islands have more species than small islands have and (2) remote islands—those located far from the mainland or from a larger island—have fewer species than less remote ones.

MacArthur and Wilson decided to test their model on Krakatau, an island in Indonesia on which a volcano had erupted in 1883, killing most forms of life on the island. The return of plant and animal life to the island had been carefully recorded since Krakatau was first revisited in 1886.

RESULTS: Species Reached Equilibrium

After examining the records of bird life at Krakatau, they learned that the number of species had climbed to 27 before leveling off.

CONCLUSION: Prediction Was Close

Using their model, MacArthur and Wilson predicted that at the point of equilibrium, the number of bird species would be about 30. Their prediction had come close.

Recent Tests of Island Biogeography

More recent studies of islands, such as those in the Sea of Cortez, suggest that many models of island biogeography may be required to explain patterns of species distributions. Researchers have generated alternative models that account for factors such as island history, climate, and species interactions.



When the volcano that forms the island of Krakatau erupted in the 1880s, it destroyed most life on the island. As life returned, scientists had a unique opportunity to study ecology in action.

REVIEW

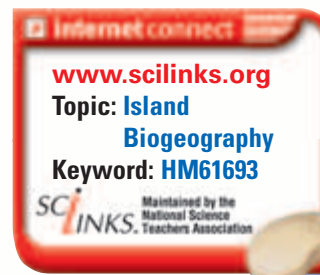
1. Describe MacArthur and Wilson's model of island biogeography.
2. Summarize the results from testing the model on Krakatau.
3. **Critical Thinking** Suggest how island factors might prevent equilibrium from being reached in MacArthur and Wilson's model.



Robert H. MacArthur



Edward O. Wilson



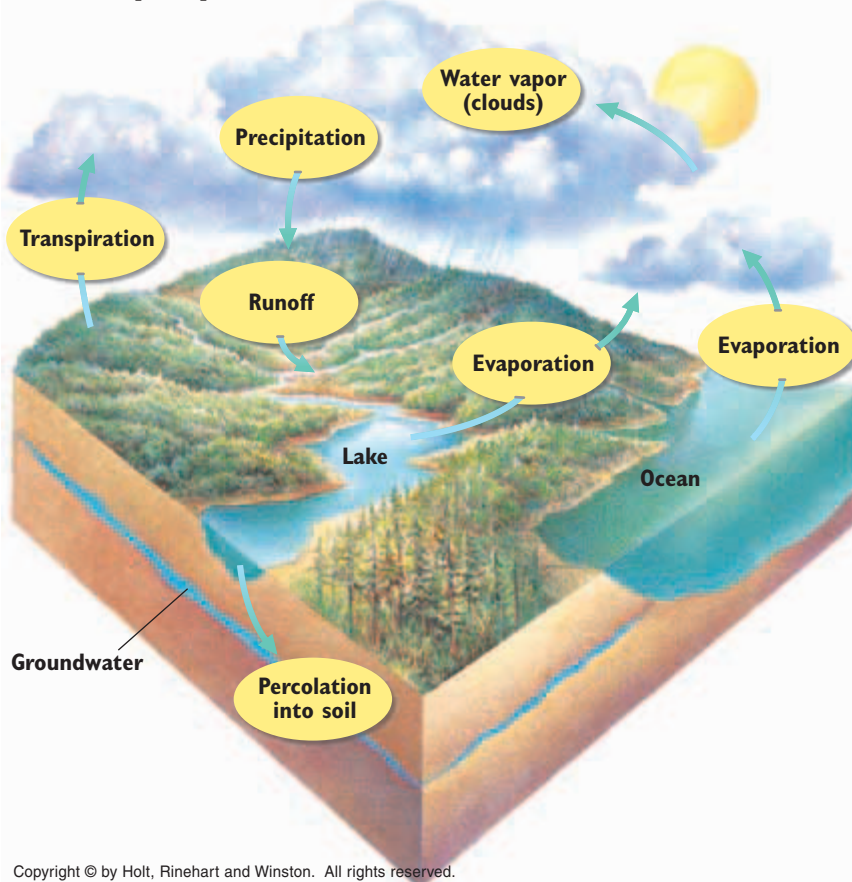
ECOSYSTEM RECYCLING

*As energy and matter flow through an ecosystem, matter must be recycled and reused. Substances such as water, carbon, nitrogen, calcium, and phosphorus each pass between the living and nonliving worlds through **biogeochemical cycles**.*

THE WATER CYCLE

Water is crucial to life. Cells contain 70 to 90 percent water, and water provides the environment in which most of life's chemical reactions occur. The availability of water is one of the key factors that regulate the productivity of terrestrial ecosystems. However, very little of the available water on Earth is trapped within living things at any given time. Bodies of water, such as lakes, rivers, streams, and oceans, contain a substantial percentage of Earth's water. The atmosphere also contains water—in the form of water vapor. In addition, some water is found below ground. Water in the soil or in underground formations of porous rock is known as **groundwater**.

The movement of water between these various reservoirs, known as the **water cycle**, is illustrated in Figure 18-12. Three important processes in the water cycle are evaporation, transpiration, and precipitation.



OBJECTIVES

- **List** four major biogeochemical cycles.
- **Summarize** three important processes in the water cycle.
- **Outline** the major steps in the carbon cycle.
- **Describe** the role of decomposers in the nitrogen cycle.
- **Summarize** the major steps of the phosphorus cycle.

VOCABULARY

biogeochemical cycle
groundwater
water cycle
transpiration
carbon cycle
nitrogen cycle
nitrogen fixation
nitrogen-fixing bacteria
ammonification
nitrification
denitrification
phosphorus cycle

FIGURE 18-12

In the water cycle, water falls to Earth's surface as precipitation. Some water reenters the atmosphere by evaporation and transpiration. Some water runs into streams, lakes, rivers, and oceans. Other water seeps through the soil and becomes groundwater. Follow the pathways of the water cycle in the figure.



Quick Lab

Modeling Groundwater

Materials disposable gloves, lab apron, 3 L plastic bottle (cut in half), small stones (250 mL), dry sod with grass, water, graduated cylinder, 500 mL beaker

Procedure



1. Put on your lab apron, goggles, and disposable gloves.
2. Invert the top half of the plastic bottle, and place it inside the bottom half of the bottle to form a column.
3. Place the stones in the bottom of the inverted top half of the bottle. Place a chunk of dry sod with grass on top of the stones.
4. Pour 250 mL of water over the sod, and observe how the water penetrates the soil and moves through the column.
5. When the water is no longer draining, remove the top half of the column, and pour the water from the bottom of the column into a beaker. Measure the volume of liquid in the beaker.

Analysis What is the volume of the water that drained through the sod? How much of the water remained in the soil? Where does the water go when applied to a real lawn or crop? What might the fate of fertilizer or pesticides be that are applied to a lawn or crop?

Evaporation adds water as vapor to the atmosphere. Heat causes water to evaporate from bodies of water, from the soil, and from the bodies of living things. The process by which water evaporates from the leaves of plants in terrestrial ecosystems is called **transpiration**. Transpiration causes plants to take in water through their roots to replace the water that is being lost through their leaves. Animals also participate in the water cycle. Animals drink water or obtain it from their food. They release this water when they breathe, sweat, or excrete.

Water leaves the atmosphere through precipitation. The amount of water the atmosphere can hold depends on abiotic factors, such as temperature and air pressure. Once the atmosphere becomes saturated with water vapor, precipitation occurs in the form of rain, snow, sleet, hail, or fog.

THE CARBON CYCLE

Photosynthesis and cellular respiration form the basis of the short-term **carbon cycle**, illustrated in Figure 18-13. In photosynthesis, plants and other autotrophs use carbon dioxide (CO_2), along with water and solar energy, to make carbohydrates. Both autotrophs and heterotrophs use oxygen to break down carbohydrates during cellular respiration. The byproducts of cellular respiration are carbon dioxide and water. Decomposers release carbon dioxide into the atmosphere when they break down organic compounds.

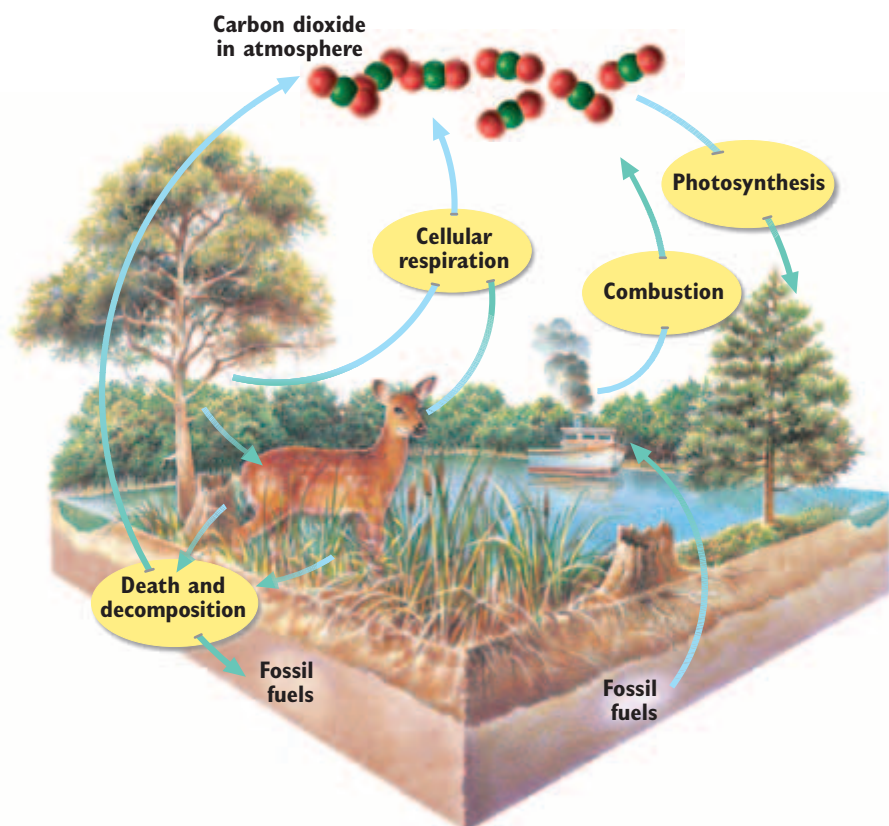
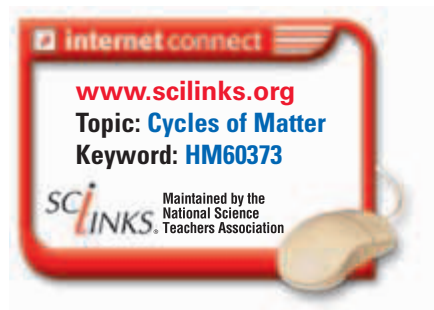


FIGURE 18-13

Carbon exists in the atmosphere as carbon dioxide. Cellular respiration, combustion, and decomposition of organic matter are the three major sources of carbon dioxide in the short-term carbon cycle. By burning large amounts of fossil fuels, humans are releasing carbon dioxide from a long-term reservoir and increasing the amount of carbon dioxide in the atmosphere.

Human Influences on the Carbon Cycle

In the last 150 years, the concentration of atmospheric carbon dioxide has risen more than 30 percent. Humans contribute to this increase by burning fossil fuels and other organic matter. Our industrial society depends on the energy released by the burning of fossil fuels—coal, oil, and natural gas. Fossil fuels are the remains of organisms that have been transformed by decay, heat, and pressure into energy-rich molecules. Burning releases the energy in these molecules, but it also releases carbon dioxide. When large areas of forest are burned each year to clear land for agriculture, less vegetation remains to absorb carbon dioxide from the atmosphere through photosynthesis.



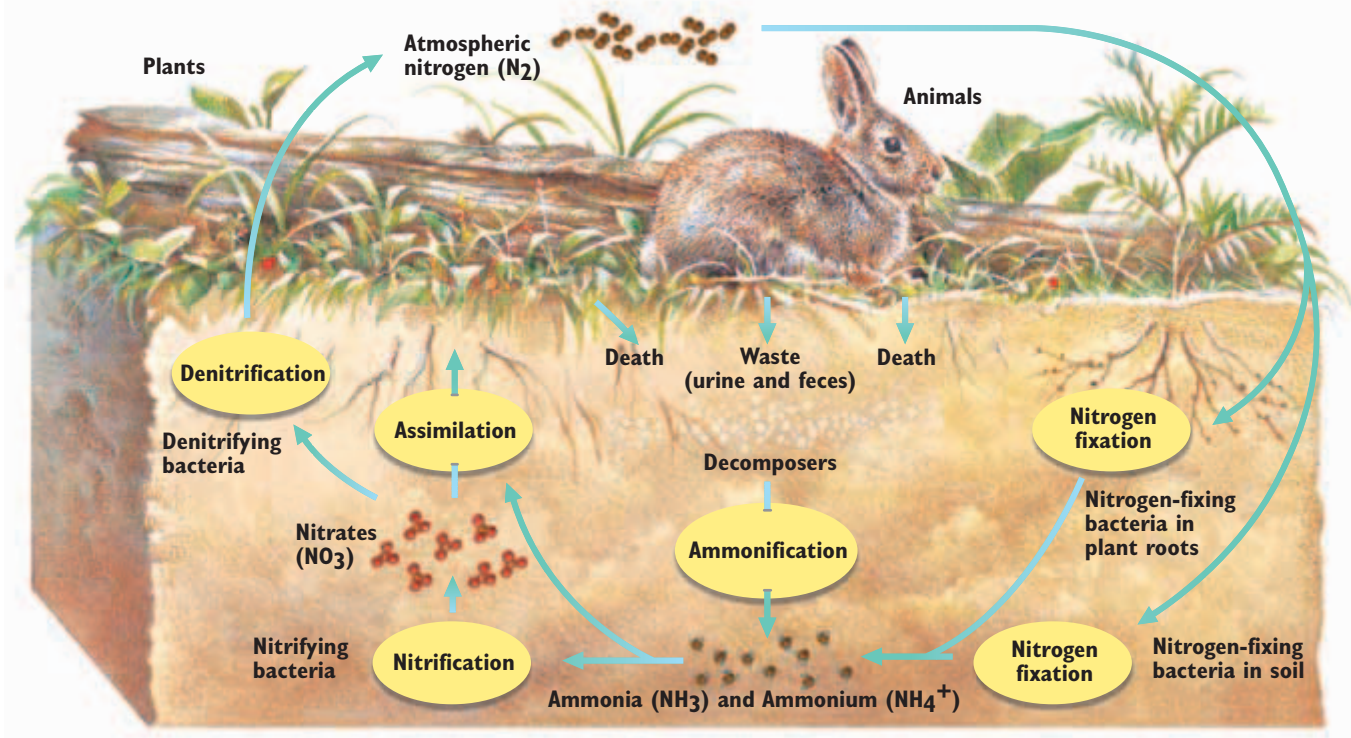
NITROGEN CYCLE

All organisms need nitrogen to make proteins and nucleic acids. The complex pathway that nitrogen follows in an ecosystem is called the **nitrogen cycle**, as shown in Figure 18-14. Nitrogen gas, N_2 , makes up about 78 percent of the atmosphere, so it might seem that it would be readily available for living things. However, most plants can use nitrogen only in the form of nitrate. The process of converting N_2 gas to nitrate is called **nitrogen fixation**.

Most organisms rely on **nitrogen-fixing bacteria** to transform nitrogen gas into a usable form. These bacteria live in the soil and inside swellings on the roots of some kinds of plants, such as beans, peas, clover, and alfalfa. These plants supply carbohydrates for the bacteria, and the bacteria produce usable nitrogen for the plant. Additional nitrogen is released into the soil.

FIGURE 18-14

This figure shows the cycling of nitrogen within a terrestrial ecosystem. Bacteria are responsible for many of the steps in the nitrogen cycle, including the conversion of atmospheric nitrogen into ammonium. Nitrogen-fixing bacteria live in the soil or in the roots of plants. These bacteria convert nitrogen gas into ammonium. Other bacteria convert the ammonium into nitrates. Plants take up the nitrates produced by the bacteria. Animals get nitrogen by eating plants or other animals.



Recycling Nitrogen

The bodies of dead organisms contain nitrogen, mainly in proteins and nucleic acids. Urine and dung also contain nitrogen. Decomposers break down these materials and release the nitrogen they contain as ammonia, NH_3 , which in soil becomes ammonium, NH_4^+ . This process is known as **ammonification**. Through this process, nitrogen is again made available to other organisms.

Soil bacteria take up ammonium and oxidize it into nitrites, NO_2^- , and nitrates, NO_3^- , in a process called **nitrification**. The erosion of nitrate-rich rocks also releases nitrates into an ecosystem. Plants use nitrates to form amino acids. Nitrogen is returned to the atmosphere through **denitrification**. Denitrification occurs when anaerobic bacteria break down nitrates and release nitrogen gas into the atmosphere. Plants can absorb nitrates from the soil, but animals cannot. Animals obtain nitrogen in the same way they obtain energy—by eating plants and other organisms and then digesting the proteins and nucleic acids.

PHOSPHORUS CYCLE

Phosphorus is an element that is an essential material needed by animals to form bones, teeth, and parts of molecules, such as DNA and RNA. Plants get the phosphorus they need from soil and water, whereas animals get their phosphorus by eating plants or other animals. The **phosphorus cycle** is the movement of phosphorus from the environment to organisms and then back to the environment. This cycle is slow and does not normally occur in the atmosphere, because phosphorus rarely occurs as a gas.

When rocks erode, small amounts of phosphorus dissolve as phosphate, PO_4^{3-} , in soil and water. Plants absorb phosphorus in the soil through their roots. Phosphorus is also added to soil and water when excess phosphorus is excreted in wastes from organisms and when organisms die and decompose. Some phosphorus applied to fields as fertilizer washes off the land into streams and groundwater.

SECTION 4 REVIEW

1. Identify four major biogeochemical cycles.
2. Through what process does most water vapor enter the atmosphere?
3. Outline the steps of the carbon cycle.
4. Describe the role of decomposers in the nitrogen cycle.
5. Identify the sources of phosphorus in the phosphorus cycle.

CRITICAL THINKING

6. **Inferring Relationships** How might the removal of vegetation affect oxygen levels in the atmosphere?
7. **Making Comparisons** Identify the role of bacteria in the carbon, nitrogen, and phosphorus cycles.
8. **Analyzing Concepts** Explain the statement that nutrients cycle, but energy flows.

CHAPTER HIGHLIGHTS

SECTION 1

Introduction to Ecology

- Species interact with both other species and their nonliving environment.
- Interdependence is a theme in ecology, and states that one change can affect all species in an ecosystem.
- Ecological models help to explain the environment.
- Ecology is usually organized into five levels: organism, population, community, ecosystem, and biosphere.

Vocabulary

ecology (p. 359)
interdependence (p. 359)

ecological model (p. 360)
biosphere (p. 361)

ecosystem (p. 362)
community (p. 362)

population (p. 362)

SECTION 2

Ecology of Organisms

- Both biotic, or living, factors and abiotic, or nonliving, factors influence organisms. Examples of nonliving things are climate, sunlight, and pH.
- A niche is a way of life, or a role in an ecosystem.
- Some species survive unfavorable environmental conditions by becoming dormant or by migrating.

Vocabulary

habitat (p. 363)
biotic factor (p. 363)
abiotic factor (p. 363)

tolerance curve (p. 363)
acclimation (p. 364)
conformer (p. 364)

regulator (p. 364)
dormancy (p. 365)
migration (p. 365)

niche (p. 365)
generalist (p. 365)
specialist (p. 365)

SECTION 3

Energy Transfer

- Most producers are photosynthetic and make carbohydrates by using energy from the sun.
- Consumers obtain energy by eating other organisms and include herbivores, omnivores, carnivores, detritivores, and decomposers.
- Decomposers feed on dead organisms and wastes, which releases the nutrients back into the environment.
- A single pathway of energy transfer is a food chain. A network showing all paths of energy transfer is a food web.
- Ecosystems contain only a few trophic levels because there is a low rate of energy transfer between each level.

Vocabulary

producer (p. 366)
chemosynthesis (p. 366)
gross primary productivity (p. 366)

biomass (p. 366)
net primary productivity (p. 366)
consumer (p. 367)

herbivore (p. 367)
carnivore (p. 367)
omnivore (p. 367)
detritivore (p. 367)

decomposer (p. 367)
trophic level (p. 368)
food chain (p. 368)
food web (p. 368)

SECTION 4

Ecosystem Recycling

- Key processes in the water cycle are evaporation, transpiration, and precipitation.
- Photosynthesis and cellular respiration are the two main steps in the carbon cycle.
- Nitrogen-fixing bacteria are important in the nitrogen cycle because they change nitrogen gas into a usable form of nitrogen for plants.
- Phosphorus moves from phosphate deposited in rock, to the soil, to living organisms, and finally to the ocean.

Vocabulary

biogeochemical cycle (p. 371)
groundwater (p. 371)
water cycle (p. 371)
transpiration (p. 372)

carbon cycle (p. 372)
nitrogen cycle (p. 373)
nitrogen fixation (p. 373)

nitrogen-fixing bacteria (p. 373)
ammonification (p. 374)

nitrification (p. 374)
denitrification (p. 374)
phosphorus cycle (p. 374)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - conformer* and *regulator*
 - community* and *ecosystem*
 - migration* and *dormancy*
 - nitrogen fixation* and *ammonification*
- Use the following terms in the same sentence: *producer*, *consumer*, *herbivore*, *omnivore*, *carnivore*, *detritivore*, and *decomposer*.
- Word Roots and Origins** The word *transpiration* is derived from the Latin *trans*, which means “through,” and *spirare*, which means “breathe.” Using this information, explain why the term *transpiration* is a good name for the process it describes.

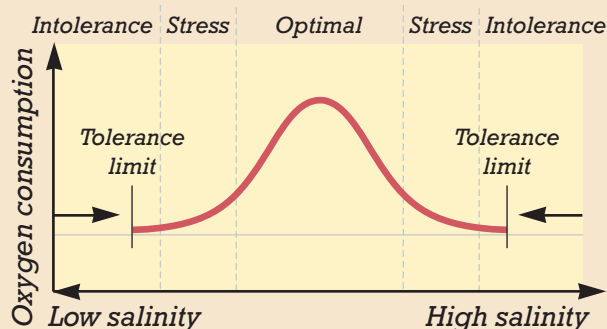
UNDERSTANDING KEY CONCEPTS

- Explain** how an understanding of interdependence in ecosystems might be important to public health officials.
- Evaluate** how models are valuable to ecologists.
- Describe** some limitations of ecological models.
- Identify** five levels of organization in ecology.
- Propose** two examples of biotic factors and abiotic factors.
- Explain** the ecological concept of a niche.
- Distinguish** between conformers and regulators in how they deal with environmental change.
- Compare** photosynthetic and nonphotosynthetic producers.
- Distinguish** between a herbivore, a carnivore, and an omnivore.
- State** an example of each of the following: a herbivore, a carnivore, and an omnivore.
- Explain** the importance of decomposers in an ecosystem.
- Describe** why a food web is a more complete picture of the feeding relationships in an ecosystem than a food chain is.
- Identify** the reasons why most ecosystems normally contain only a few trophic levels.
- Compare** the transfer of energy with the transfer of nutrients in an ecosystem.
- Explain** how plants return water to the atmosphere as part of the water cycle.
- Describe** two processes in the carbon cycle.
- List** the mutual benefits in the association between nitrogen-fixing bacteria and the plants that they inhabit.
- Summarize** the phosphorus cycle.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows some of the processes involved in the water cycle: *evaporation*, *precipitation*, *transpiration*, and *condensation*.

CRITICAL THINKING

- Interpreting Graphics** Examine the diagram below of a tolerance curve. Briefly describe the conditions in each zone of tolerance and the reactions a species may have to them.

Salinity Tolerance Curve



- Relating Concepts** Nitrogen, water, phosphorus, and carbon are recycled and reused within an ecosystem, but energy is not. Explain why energy cannot be recycled.
- Drawing Conclusions** In the fall, many kinds of songbirds migrate from the United States to Central America or South America. Explain the benefits of migration for songbirds. What are some possible costs of this behavior?
- Analyzing Concepts** Ecologists have identified several characteristics that increase the likelihood that a species will become extinct. Specialization is one such characteristic. Explain why a very specialized species is likely to be more vulnerable to extinction.
- Making Models** Farmers often grow alfalfa, clover, or bean plants in fields after they have grown a grain crop. Explain this practice in terms of biochemistry.

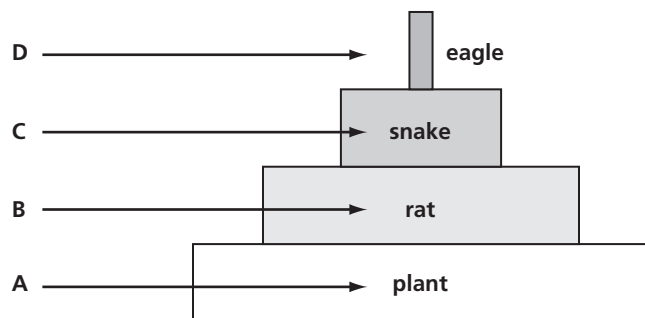


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What are the levels of organization in ecology?
 - cell, tissue, organ, organ system, body
 - organ, organism, population, community
 - organism, population, community, ecosystem, biosphere
 - population, habitat, ecosystem, biogeochemical system, planet
- What makes up an ecosystem?
 - all the habitat types on Earth
 - all parts of Earth where life exists
 - all members of a species in the same area
 - all the living and nonliving factors in an environment
- Which of the following are abiotic factors?
 - plants
 - animals
 - sunlight
 - microorganisms
- How do decomposers benefit an ecosystem?
 - by returning nutrients to the soil
 - by manufacturing energy from sunlight
 - by removing excess nutrients from the soil
 - by removing predators from the ecosystem
- Which organisms are most critical in the nitrogen cycle?
 - plants
 - nitrates
 - animals
 - bacteria

INTERPRETING GRAPHICS: The illustration below represents a trophic pyramid. Use the illustration to answer the question that follows.

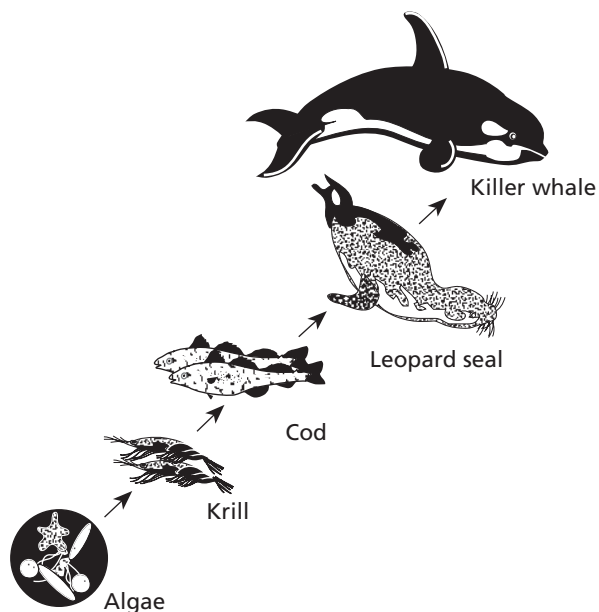


- What is the term for the kinds of organisms that make up the trophic level labeled C?
 - producers
 - consumers
 - detritivores
 - decomposers

DIRECTIONS: Complete the following analogy.

- bear : omnivore :: vulture :
 - producer
 - herbivore
 - detritivore
 - decomposer

INTERPRETING GRAPHICS: The illustration below represents a food chain. Use the illustration to answer the questions that follow.



- What role do the krill have in this food chain?
 - They are producers.
 - They are consumers.
 - They are detritivores.
 - They are decomposers.

SHORT RESPONSE

Give two reasons why the destruction of tropical rain forests can contribute to an increase in carbon dioxide levels in the atmosphere.

EXTENDED RESPONSE

Some species are generalized with regard to their niche, and other species are specialized.

Part A Compare the niche of a generalist species with one of a specialist species.

Part B Predict how two different herbivores can share the same plant resource.

Test TIP

If you are not sure about the spelling of certain words when answering the short or extended response questions, look at the question itself to see if the same word appears in the question.

Observing Habitat Selection

OBJECTIVES

- Assess the effect of light on habitat selection by brine shrimp.

PROCESS SKILLS

- observing
- measuring
- collecting data
- organizing data
- analyzing data

MATERIALS

- safety goggles
- marking pen
- clear, flexible plastic tubing (44 cm long)
- 4 test tubes with stoppers
- test-tube rack
- 2 corks to fit tubing
- graduated cylinder
- funnel
- brine shrimp culture
- aluminum foil
- 3 screw clamps
- 1 pipet
- Petri dish
- methyl cellulose
- fluorescent lamp or grow light
- 14 pieces of screen or thin cloth
- calculator

Background

1. Recall that a species' habitat is a specific area where it lives.
2. A species habitat selection depends on how well the location fits within the species' tolerance range. The more optimal all limiting factors are within a portion of an organism's range, the more likely the organism is to select that area for its habitat.
3. What limiting factors might be involved in habitat selection?
4. What is a niche?

PART A Setting Up

1. Mark the plastic tubing at 12 cm, 22 cm, and 32 cm from one end so that you will have the tube divided into four sections. Starting at one end, label the sections 1, 2, 3, and 4. Label four test tubes 1, 2, 3, and 4.
2. Place a cork in one end of the tubing. Use a graduated cylinder and a funnel to transfer about 50 mL of brine shrimp culture into the tubing. Cork the open end, and lay the tubing on the desktop.
3.  You and your partner will complete either Part B or Part C and then share your results with the other students on your team. **CAUTION** You will be working with live animals. Be sure to treat them gently and to follow directions carefully.

PART B Control Group

1. Cover the tubing with aluminum foil, and let it remain undisturbed for 30 minutes. While you are waiting, create a data table like Table A, below, in your lab report to record the numbers of shrimp in each section of the tubing.

TABLE A CONTROL GROUP

Test tube number	Count 1	Count 2	Count 3	Count 4	Count 5	Average number of shrimp in test tube
1						
2						
3						
4						

2. After 30 minutes have passed, attach screw clamps to each spot that you marked on the tubing. While your partner holds the corks firmly in place, tighten the middle clamp first, and then tighten the outer clamps.
3. Immediately pour the contents of each section of tubing into the test tube labeled with the corresponding number.

4.   **CAUTION** Put on safety goggles before handling methyl cellulose. If you get methyl cellulose in your eyes, immediately flush it out at the eyewash station while calling to your teacher. Stopper test tube 1, and invert it gently to distribute the shrimp. Use a pipet to draw a 1 mL sample of shrimp culture and transfer the culture to a Petri dish. Add a few drops of methyl cellulose to the Petri dish to slow down the shrimp. Count the live shrimp, and record the count in your lab report.
5.  Dispose of the shrimp as your teacher directs. Repeat step 4 four more times for a total of five counts from test tube 1.
6. Calculate the average number of shrimp in test tube 1, and record the result in the data table you made in your lab report.
7. Repeat steps 4–6 for the contents of each of the remaining test tubes.
8.   Clean up your materials, and wash your hands before leaving the lab.
9. In your lab report, make a histogram showing the total number of shrimp you counted in each section of tubing.

PART C Experimental Group

1. Set a fluorescent lamp 20 cm away from the tubing.
2. Cover section 1 of the tubing with eight layers of screen. Place four layers of screen on section 2 and two layers of screen on section 3. Leave section 4 uncovered. Leave this setup in place for 30 minutes. While you are waiting, create a data table in your lab report like Table B, below, to record the numbers of shrimp in each section of the tubing.

TABLE B EXPERIMENTAL GROUP

Test tube number	Count 1	Count 2	Count 3	Count 4	Count 5	Average number of shrimp in test tube
1						
2						
3						
4						

3. After 30 minutes have passed, attach screw clamps to each spot that you marked on the tubing. While your partner holds the corks firmly in place, tighten the middle clamp first, and then tighten the outer clamps.
4. Immediately pour the contents of each section of tubing into the test tube labeled with the corresponding number.
5.   **CAUTION** Put on safety goggles before handling methyl cellulose. If you get methyl cellulose in your eyes, immediately flush it out at the eyewash station while calling to your teacher. Stopper test tube 1, and invert it gently to distribute the shrimp. Use a pipet to draw a 1 mL sample of shrimp culture and transfer the culture to a Petri dish. Add a few drops of methyl cellulose to the Petri dish to slow down the shrimp. Count the live shrimp, and record the count in your lab report.
6.  Dispose of the shrimp as your teacher directs. Repeat step 5 four more times for a total of five counts from test tube 1.
7. Calculate the average number of shrimp in test tube 1, and record the result in the data table you made in your lab report.
8. Repeat steps 5–7 for the contents of each of the remaining test tubes.
9.   Clean up your materials, and wash your hands before leaving the lab.
10. In your lab report, make a histogram showing the number of shrimp in each section of tubing. Identify each section with the amount of screen.

Analysis and Conclusions

1. Describe the differences between the histogram of the control group and the histogram of the experimental group.
2. Why was a control (Part B) necessary?
3. How did the brine shrimp react to differences in light? Justify your conclusion.

Further Inquiry

Design an experiment to test the reaction of brine shrimp to a gradient of heat.

POPULATIONS



These bottlenose dolphins, *Tursiops truncatus*, are part of a population.

SECTION 1 *Understanding Populations*

SECTION 2 *Measuring Populations*

SECTION 3 *Human Population Growth*



Unit 7—Ecosystem Dynamics
Topic 2

UNDERSTANDING POPULATIONS

The human population of the world was about 6.3 billion in 2003, over three times its size in 1900. During this period of rapid human population growth, populations of many other species have decreased dramatically. Will the human population continue to grow? Will populations of other species continue to get smaller? An understanding of populations is crucial to answering these questions.

PROPERTIES OF POPULATIONS

A **population** is a group of organisms that belong to the same species and live in a particular place at the same time. All of the bass living in a pond during a certain period of time make up a population because they are isolated in the pond and do not interact with bass living in other ponds. The boundaries of a population may be imposed by a feature of the environment, such as a lake shore, or they can be arbitrarily chosen to simplify a study of the population. The humans shown in Figure 19-1 are part of the population of a city. The properties of populations differ from those of individuals. An individual may be born, it may reproduce, or it may die. A population study focuses on a population as a whole—how many individuals are born, how many die, and so on.

Population Size

A population's size is the number of individuals that the population contains. Size is a fundamental and important population property but can be difficult to measure directly. If a population is small and composed of immobile organisms, such as plants, its size can be determined simply by counting individuals. Often, though, individuals are too abundant, too widespread, or too mobile to be counted easily, and scientists must estimate the number of individuals in the population.

Suppose that a scientist wants to know how many oak trees live in a 10 km² patch of forest. Instead of searching the entire patch of forest and counting all the oak trees, the scientist could count the trees in a smaller section of the forest, such as a 1 km² area. The scientist could then use this value to estimate the population of the larger area.

OBJECTIVES

- **Describe** the main properties that scientists measure when they study populations.
- **Compare** the three general patterns of population dispersion.
- **Identify** the measurements used to describe changing populations.
- **Compare** the three general types of survivorship curves.

VOCABULARY

population
population density
dispersion
birth rate
death rate
life expectancy
age structure
survivorship curve

FIGURE 19-1

A population can be widely distributed, as Earth's human population is, or confined to a small area, as species of fish in a lake are.





FIGURE 19-2

These migrating wildebeests in East Africa are too numerous and mobile to be counted. Scientists must use sampling methods at several locations to monitor changes in the population size of the animals.

Word Roots and Origins

dispersion

from the Latin *dis-*, meaning "out," and *spargere*, meaning "to scatter"

If the small patch contains 25 oaks, an area 10 times larger would likely contain 10 times as many oak trees. A similar kind of sampling technique might be used to estimate the size of the population shown in Figure 19-2. To use this kind of estimate, the scientist must assume that the distribution of individuals in the entire population is the same as that in the sampled group. Estimates of population size are based on many such assumptions, so all estimates have the potential for error.

Population Density

Population density measures how crowded a population is. This measurement is always expressed as the number of individuals per unit of area or volume. For example, the population density of humans in the United States is about 30 people per square kilometer. Table 19-1 shows the population sizes and densities of humans in several countries in 2003. These estimates are calculated for the total land area. Some areas of a country may be sparsely populated, while other areas are very densely populated.

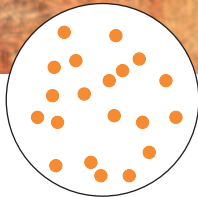
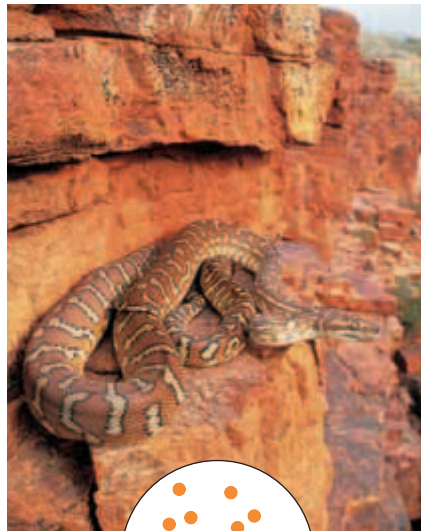
Dispersion

A third population property is dispersion (di-SPUHR-zuhn). **Dispersion** is the spatial distribution of individuals within the population. In a *clumped* distribution, individuals are clustered together. In a *uniform* distribution, individuals are separated by a fairly consistent distance. In a *random* distribution, each individual's location is independent of the locations of other individuals in the population. Figure 19-3 illustrates the three possible patterns of dispersion.

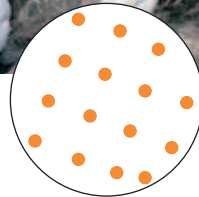
Clumped distributions often occur when resources such as food or living space are clumped. Clumped distributions may also occur because of a species' social behavior, such as when animals gather into herds or flocks. Uniform distributions may result from social behavior in which individuals within the same habitat stay as far away from each other as possible. For example, a bird may locate its nest so as to maximize the distance from the nests of other birds.

TABLE 19-1 Population Size and Density of Some Countries

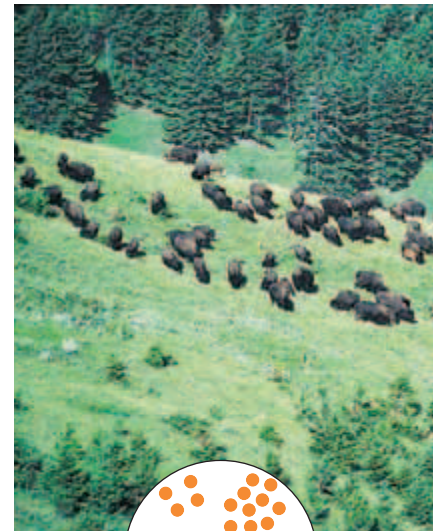
Country	Population size (in millions)	Population density (in individuals/km ²)
China	1,289	135
India	1,069	325
United States	292	30
Russia	146	8
Japan	128	337
Mexico	105	54
Kenya	32	54
Australia	20	3



(a) RANDOM



(b) UNIFORM



(c) CLUMPED

The social interactions of birds called *gannets*, which are shown in Figure 19-3b, result in a uniform distribution. Each gannet chooses a small nesting area on the coast and defends it from other gannets. In this way, each gannet tries to maximize its distance from all of its neighbors, which causes a uniform distribution of individuals.

Few populations are truly randomly dispersed. Rather, they show degrees of clumping or uniformity. The dispersion pattern of a population sometimes depends on the scale at which the population is observed. The gannets shown in Figure 19-3b are uniformly distributed on a scale of a few meters. However, if the entire island on which the gannets live is observed, the distribution appears clumped because the birds live only near the shore.

FIGURE 19-3

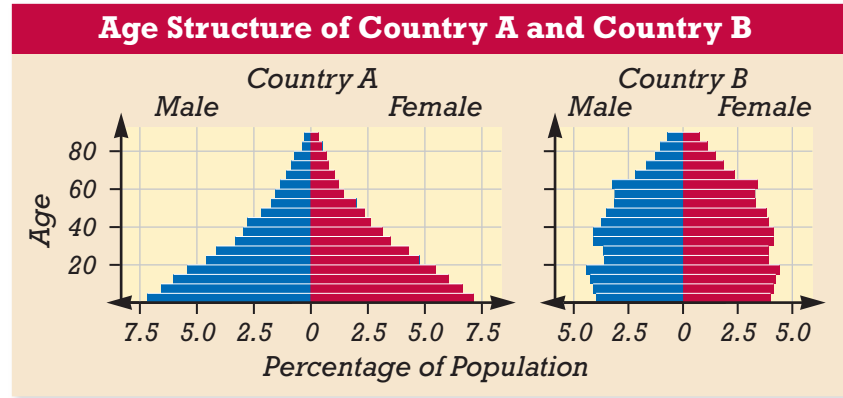
The three dispersion patterns are *random* (a), *uniform* (b), and *clumped* (c). However, the observed dispersion of a population sometimes depends on the scale at which the population is observed.

POPULATION DYNAMICS

All populations are dynamic—they change in size and composition over time. To understand these changes, scientists must know more than the population's size, density, and dispersion. One important measure is the **birth rate**, the number of births occurring in a period of time. In the United States, for example, there are about 4 million births per year. A second important measure is the **death rate**, or *mortality rate*, which is the number of deaths in a period of time. The death rate for the United States is about 2.6 million deaths per year. Another important statistic is **life expectancy**, or how long on average an individual is expected to live. In the United States in 2003, the life expectancy for a man was 74 years, and for a woman it was 80 years.

FIGURE 19-4

These two diagrams show the age structure by gender of two countries. A comparison indicates that Country A has a higher percentage of young people and a lower percentage of elderly people than Country B does.



Age Structure

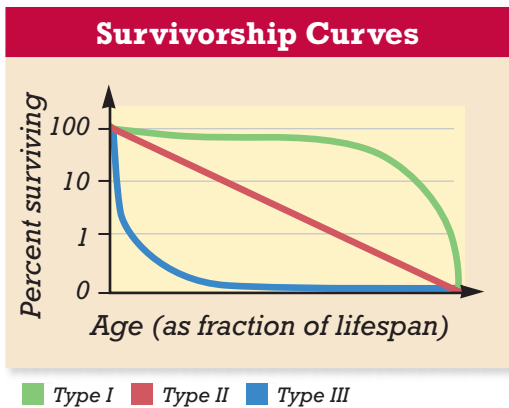
The distribution of individuals among different ages in a population is called **age structure**. Age structures are often presented in graphs, as in Figure 19-4. Many important population processes vary with age. In humans, very old individuals do not reproduce. If human populations have a high percentage of young individuals, they may have a greater potential for rapid growth.

Patterns of Mortality

The mortality data of different species tend to match one of three curves on a graph, as shown in Figure 19-5. These curves are called **survivorship curves** because they show the probability that members of a population will survive to a certain age. In humans or elephants, for instance, the likelihood of dying is small until late in life, when mortality increases rapidly. This pattern of mortality produces the Type I survivorship curve. For other organisms, such as some species of birds, the probability of dying does not change throughout life, giving a linear, or Type II, survivorship curve. Finally, many organisms are very likely to die when young. If an individual survives this early period, however, it has a good chance of surviving to old age. This type of survivorship curve, called Type III, is characteristic of animals such as oysters and salmon, and of many insects.

FIGURE 19-5

Humans have a Type I survivorship curve. Some species of birds have a Type II survivorship curve. Some species of fish are examples of a Type III survivorship.



SECTION 1 REVIEW

1. Explain how two populations can be the same size but have different densities.
2. Explain how uniform distributions could result from social interactions between individuals.
3. How can the dispersion of one population be described as both uniform and clumped?
4. Explain what *birth rate* and *death rate* mean.
5. In Figure 19-4, which country has a higher percentage of elderly people?
6. Compare the three types of survivorship curves.

CRITICAL THINKING

7. **Relating Concepts** Explain why natural selection might favor a high reproduction rate in organisms with Type III survivorship curves.
8. **Analyzing Methods** Explain two difficulties an ecologist might have in counting a population of migratory birds. Develop and explain a method for estimating the size of such a population.
9. **Predicting Patterns** Which pattern of dispersion does the global human population have?

MEASURING POPULATIONS

Charles Darwin calculated that a single pair of elephants could increase to a population of 19 million individuals within 750 years. The fact that the world is not overrun with elephants is evidence that some factor or factors restrain the population growth of elephants. In this section you will study how populations grow and what factors limit their growth.

POPULATION GROWTH RATE

Demographers, scientists who study population dynamics, define the **growth rate** of a population as the amount by which a population's size changes in a given time.

Whether a population grows, shrinks, or remains the same size depends on four processes: birth, death, emigration, and immigration. **Immigration** (IM-uh-GRAY-shuhn) is the movement of individuals into a population, and **emigration** (EM-i-GRAY-shuhn) is the movement of individuals out of the population. Two of these processes—birth and immigration—add individuals to a population, while the other two processes—death and emigration—subtract individuals from the population. For simplicity's sake, demographers usually assume that immigration and emigration are zero when calculating a population's growth rate. By making this assumption, they can describe a population's growth rate in mathematically simple terms.

Population Size

It is customary for demographers to divide large populations into groups of 1,000 and to present data *per capita*, meaning per individual. Birth rates, death rates, and growth rates for a large population are usually expressed per capita. For example, if there are 52 births and 14 deaths per 1,000 individuals in a large population in one year, the birth rate would be $\frac{52}{1,000}$, or 0.052 births per capita per year. The death rate would be $\frac{14}{1,000}$, or 0.014 deaths per capita per year.

The growth rate can be found by the following simple equation:

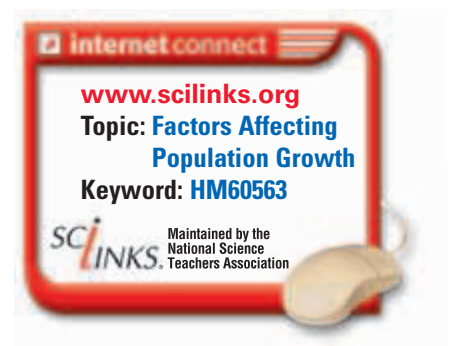
$$\text{birth rate} - \text{death rate} = \text{growth rate}$$

OBJECTIVES

- **Identify** the four processes that determine population growth.
- **Compare** the exponential model and the logistic model of population growth.
- **Differentiate** between density-dependent and density-independent regulation of populations.
- **Explain** why small populations are more vulnerable to extinction.

VOCABULARY

growth rate
immigration
emigration
exponential model
limiting factor
logistic model
carrying capacity
density-independent factor
density-dependent factor
inbreeding



The Exponential Model

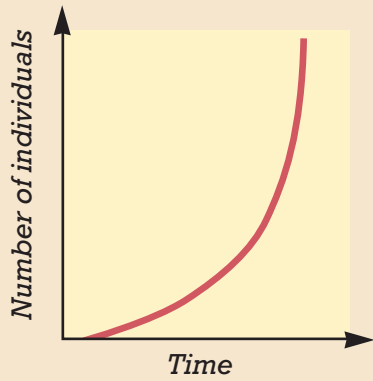


FIGURE 19-6

The graph of exponential population growth has a characteristic J shape. The exponential model indicates constantly increasing population growth.

Using the same example, we can calculate the per capita growth rate as follows:

$$0.052 \text{ (births per capita)} - 0.014 \text{ (deaths per capita)} \\ = 0.038 \text{ (growth per capita)}$$

To find the number of new individuals that will be added to the population in a year, simply multiply the per capita growth rate by the number of individuals in the population. If the population in our example numbers 50,000, the population will increase by 1,900 individuals in one year.

$$0.038 \times 50,000 = 1,900$$

If the growth rate is a positive number, the population is increasing. If it is a negative number, the population is shrinking.

THE EXPONENTIAL MODEL

As long as the birth rate of a population exceeds the death rate, the population size will continue to increase. At a steady, positive per capita growth rate, the population will add a larger number of individuals with each generation. So, a population can increase rapidly with even a small growth rate. A pattern of increase in number due to a steady growth rate is called *exponential growth*. The observation that populations can grow in this pattern is called the **exponential** (EKS-poh-NEN-shuhl) **model** of population growth.

One way to understand the exponential model is to study a graph of population size over time. A graph of exponential growth makes the characteristic J-shaped curve shown in Figure 19-6. With exponential growth, population size grows slowly when it is small, but growth speeds up as individuals join the population. The exponential model leads us to predict that the population size will increase indefinitely and by a greater number with each time period.

Applying the Exponential Model

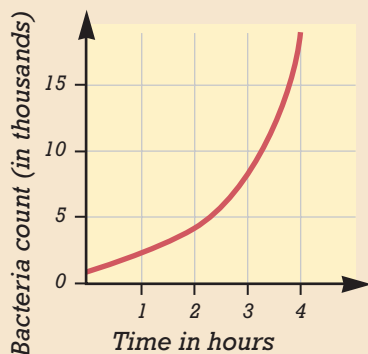
A scientific model is useful if it helps to predict or explain patterns that can be observed in reality. Indeed, the exponential model matches observed patterns of growth of real populations, but only under certain conditions and for limited periods of time. For example, a population of microorganisms can grow exponentially if provided with an abundance of food and space and if waste is removed. Figure 19-7 shows the growth of bacteria in a laboratory.

However, the exponential model does not apply to most populations. In natural environments, populations cannot grow indefinitely because the resources they depend on become scarce and harmful wastes accumulate. Any factor, such as space, that restrains the growth of a population is called a **limiting factor**. All populations are ultimately limited by their environment.

FIGURE 19-7

The population increase of bacteria in the laboratory produces a characteristic graph of exponential growth. With this kind of graph, the size of the population of bacteria at any future time can be predicted if the culture is provided with unlimited resources, such as food.

Exponential Growth of Bacteria



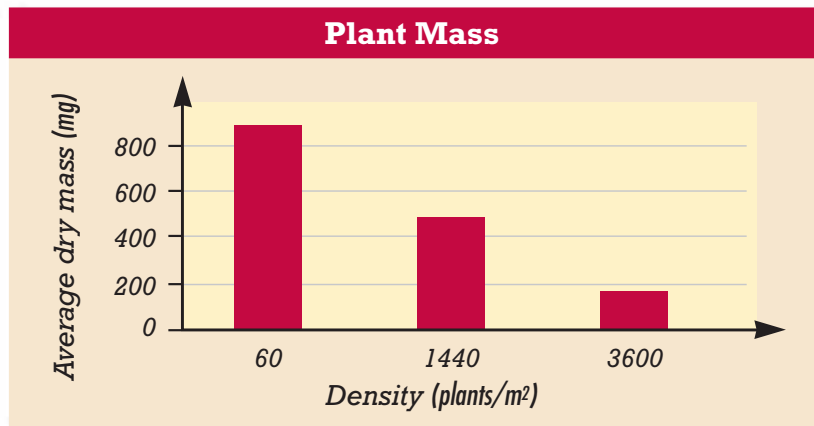


FIGURE 19-8

The graph displays the results of an experiment that tested how the growth of a group of plants was affected under three conditions of crowding. Under the condition of least crowding (fewer plants per square meter), the plants were observed to grow larger on average.

As a population grows, competition among individuals for the shrinking supply of resources intensifies, and each individual, on average, obtains a smaller share. Thus, each individual's ability to fight off disease, grow, and reproduce decreases. As a result, the population's birth rate declines and death rate increases. Figure 19-8 shows how increasing population size affected the growth of a plant species in a limited area.

THE LOGISTIC MODEL

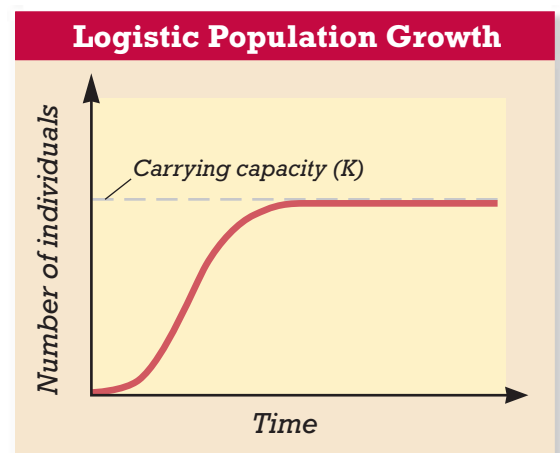
Birth rates and death rates are not constant but vary with population size: birth rates decline and death rates rise as the population grows. The **logistic** (loh-JIS-tik) **model** of population growth builds on the exponential model but accounts for the influence of limiting factors. The logistic model includes a new term, **carrying capacity** (symbolized by K), the number of individuals the environment can support over a long period of time.

A graph of logistic growth looks like a stretched-out letter S. Examine Figure 19-9. When the population size is small, birth rates are high and death rates are low, and the population grows at very near the exponential rate. But as the population size approaches the carrying capacity, the population growth rate slows because of the falling birth rate and the increasing death rate. When a population size is at its carrying capacity, the birth rate equals the death rate and growth stops. This pattern of growth is known as *logistic growth*.

The logistic model, like the exponential model, contains some assumptions. One such assumption is that the carrying capacity is constant and does not fluctuate with environmental changes. In reality, carrying capacity does fluctuate. It is greater when prey is abundant, for instance, and smaller when prey is scarce. The logistic and exponential models are not universal representations of real populations, but they are an important tool that scientists use to explain population growth and regulation.

FIGURE 19-9

This graph of logistic population growth is typical of populations in new environments. In the first phase, the population shows rapid, nearly exponential growth. In the second phase, the growth rate slows until the carrying capacity, K , is reached. In the third phase, the population has become stable, neither increasing nor decreasing in size. Real populations may fit this pattern for some period of time but rarely remain stable.



POPULATION REGULATION

Two kinds of limiting factors, which control population size, have been identified. **Density-independent factors**, such as weather, floods, and fires, reduce the population by the same proportion, regardless of the population's size. For example, if a forest fire destroys a population of chipmunks, it does not matter if the population of chipmunks is 1 or 100. An unseasonable cold snap is a density-independent factor because its severity and duration are completely independent of population size. **Density-dependent factors** include resource limitations, such as shortages of food or nesting sites, and are triggered by increasing population density. With density-dependent factors, an individual's chance of surviving or reproducing depends on the number of individuals in the same area.

Population Fluctuations

All populations fluctuate in size. Some population fluctuations are clearly linked to environmental changes. For example, a drought may reduce a population of deer living in a forest. Some population fluctuations are not obviously connected to environmental fluctuations, and explaining their occurrence is much more difficult. For example, consider the population changes shown in Figure 19-10. These cycles of change were first described by Charles S. Elton (1900–1991), one of the pioneers of ecology. Elton obtained more than 70 years of records showing the number of snowshoe hare pelts the Hudson's Bay Company of Canada purchased from trappers. He assumed that the number of pelts purchased in a year indicated the size of the snowshoe hare population. The records showed that the hare population underwent a very regular cycle, with about 10 years between peaks in population size. When Elton examined the records for the number of lynx pelts purchased, he found that the lynx, a medium-sized species of cat that preys on snowshoe hares, also followed a population cycle. The peaks in the lynx population usually occurred near the peaks in the hare population.

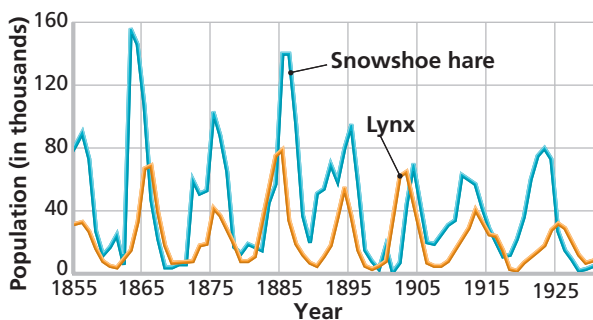
Elton thought that each species was the cause of the other's cycle. Thus, when the population of snowshoe hares increased, providing more food for the lynxes, the lynx population also increased. The increased lynx population then ate more hares, so the hare population decreased. With less food, more lynxes starved and the lynx population declined, allowing the hare population to increase and start the cycle over again. However, the observation that the same cycles occur in snowshoe hare populations living on islands without lynxes indicates that this explanation is insufficient. Another possible explanation is that the lynx cycle is dependent on the hare population but the hare cycle is dependent on some other factor.

FIGURE 19-10

The hare and the lynx (a) were observed by Elton to have parallel changes in their population cycles. The graph below (b) shows the data recorded by Elton supporting his idea that each animal controlled the other animal's cycle. You can see that the cycles fluctuate together. Because hares show the same population cycles when there are no lynxes present, it is now known that lynxes are not controlling factors in the hares' cycles.



(a)



(b)



(a)

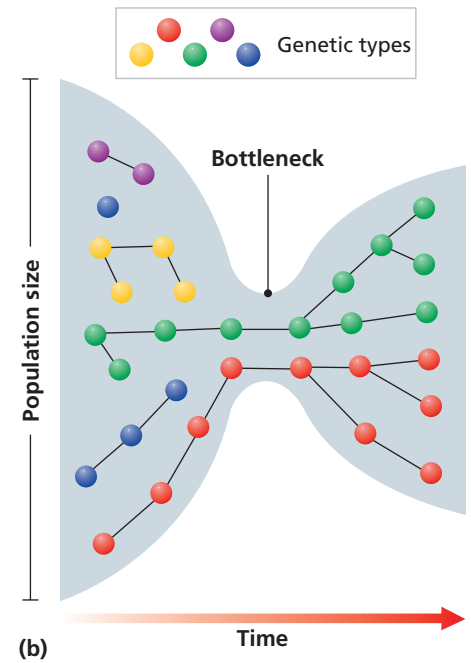


FIGURE 19-11

The genetic diversity of cheetahs (a) is so low that biologists think the cheetah population may have been reduced to a very small size in the past. Such a reduction is called a *genetic bottleneck*, illustrated in (b). The diagram shows how the genetic variation in a population can be reduced when the size of that population is reduced.

Perils of Small Populations

The rapidly growing human population has caused extreme reductions in the populations of some other species and subspecies. For example, only about 200 Siberian tigers remain in the wild because of hunting and habitat destruction. Even greater reductions have been experienced by the California condor, which was once found throughout the southwestern United States. In the 1980s, the condor's wild population had been reduced to nine individuals, although recovery efforts have since increased the numbers.

Small populations, such as the cheetahs shown in Figure 19-11a, are particularly vulnerable to extinction. Environmental disturbances, such as storms, fires, floods, or disease outbreaks, can kill off the entire population or leave too few individuals to maintain the population. Also, the members of a small population may be descended from only a few individuals, increasing the likelihood of **inbreeding**, or mating with relatives. Inbreeding in small populations often leads to decreased genetic variability, as shown in Figure 19-11b. Over evolutionary time, populations with low variability are less likely to adapt to changing environmental conditions.

SECTION 2 REVIEW

1. Explain the relationship between birth rate, death rate, and growth rate.
2. Explain the pattern described by the exponential model of population growth.
3. According to the logistic model, how do birth and death rates change with population size?
4. List two density-independent factors that could limit population growth.

CRITICAL THINKING

5. **Relating Concepts** Explain how inbreeding can threaten the survival of a small population.
6. **Predicting Results** What type of limiting factor is a disease that is transmitted by parasites?
7. **Evaluating Methods** What unknown factors might make it hard to predict the future size of the human population of a country?

SECTION 3

OBJECTIVES

- **Explain** how the development of agriculture changed the pattern of human population growth.
- **Describe** changes in human population size in the past 10,000 years.
- **Compare** observed patterns of population growth in developed and developing countries.

VOCABULARY

hunter-gatherer lifestyle
agricultural revolution
developed country
developing country
demographic transition

HUMAN POPULATION GROWTH

In the time it takes you to read this chapter, the human population will grow by about 10,000 people. The rapid growth of the human population over the last several centuries is unprecedented in history. What caused this rapid growth? How long can it continue? This section examines these questions.

HISTORY OF HUMAN POPULATION GROWTH

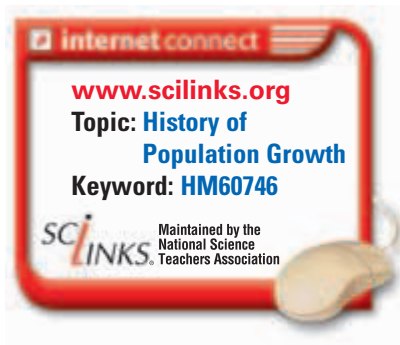
From the origin of *Homo sapiens*, more than 500,000 years ago, until about 10,000–12,000 years ago, the human population grew very slowly. During this time, humans lived in small nomadic groups and obtained food by hunting animals and gathering roots, berries, nuts, shellfish, and fruits. This way of life is called the **hunter-gatherer lifestyle**. By studying the few hunter-gatherer societies that exist today, scientists have learned that a low rate of population growth results from small populations and high mortality rates. Population growth is slowed especially when mortality is high among infants and young children, because fewer individuals reach reproductive maturity.

The Development of Agriculture

The hunter-gatherer lifestyle began to change about 10,000 to 12,000 years ago, when humans began to domesticate animals and cultivate certain plants for food. This dramatic change in lifestyle is called the **agricultural revolution**, and it led to profound changes in every aspect of life. Most important, the practice of agriculture greatly stabilized and increased the available food supply. As a result, the human population began to grow faster. About 10,000 years ago, there were between 2 million and 20 million people on Earth. By about 2,000 years ago, the population had increased to between 170 million and 330 million.

The Population Explosion

As you can see in Figure 19-12, human population growth continued through the Middle Ages despite some short-term reversals. The outbreak of bubonic plague in 1347–1352 is thought to have killed about 25 percent of the population of Europe.



Human population growth began to accelerate after 1650, primarily because of a sharp decline in death rates. Reasons for this decline in death rates included better sanitation and hygiene, control of disease, increased availability of food, and improved economic conditions. While death rates fell, birth rates remained high, resulting in rapid population growth. The human population was about 500 million in 1650 and had risen to about 1 billion by 1800 and 2 billion by 1930.

Mortality rates fell sharply again in the decades immediately following World War II because of improvements in health and hygiene in the world's poorer countries. Birth rates in these countries remained high, pushing the per capita growth rate to its highest values. It took most of human history for the human population to reach 1 billion, but the population grew from 3 billion to 5 billion in just the 27 years between 1960 and 1987.

Population Growth Today

The global growth rate peaked in the late 1960s at about 0.021 per capita. Because birth rates have decreased in many countries, the growth rate has gradually declined slowly to its 2004 level of about 0.012 per capita. This decline has led some people to mistakenly conclude that the population is not increasing. In fact, the number of people that will be added to the world population this year is larger than it was when the growth rate was at its peak. This is simply a function of today's greater population size. For example, in 1970 there were about 3.7 billion people, and the growth rate was about 0.0196. In 1970, therefore, about $3,700,000,000 \times 0.0196$, or about 73 million people, were added to the world's population. In 1999 there were about 6 billion people and the growth rate was 0.014 per capita, so the number of people added to the population was $6,000,000,000 \times 0.014$, or 84 million.

Today about 20 percent of the world's population live in **developed countries**. This category includes all of the world's modern, industrialized countries, such as the United States, Japan, Germany, France, the United Kingdom, Australia, Canada, and Russia. On average, people in developed countries are better educated, healthier, and live longer than the rest of the world's population. Population growth rates in developed countries are very low—about 0.003 per capita. The populations of some of these countries, such as Russia, Germany, and Italy, are shrinking because death rates exceed birth rates.

Most people (about 80 percent of the world's population) live in **developing countries**, a category that includes most countries in Asia, Central America, South America, and Africa. In general, these countries are poorer, and their populations are growing faster—at a rate of about 0.015 per capita.

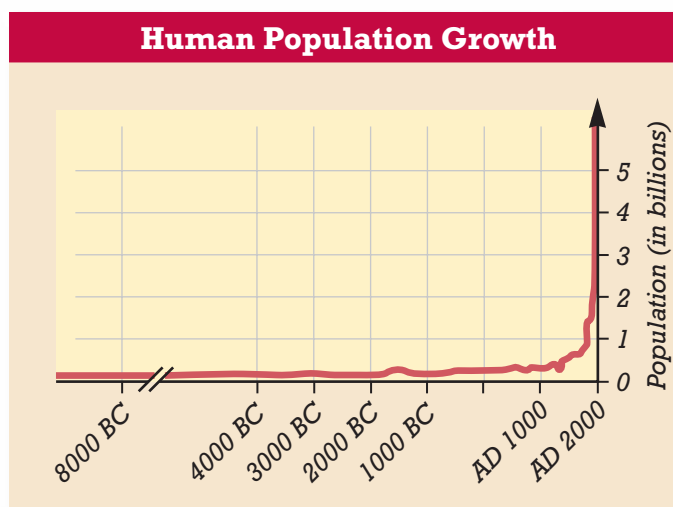


FIGURE 19-12

The J shape of the graph is characteristic of exponential growth. Many ecologists agree that the current human population growth rate is not sustainable.



Quick Lab

Demonstrating Population Doubling

Materials pencil, paper, sheet of newspaper

Procedure

1. Make a data table. Label the columns "Fold number," "Number of layers," and "Power of 2." Write the numbers 1–10 in the first column.
2. Fold a sheet of newspaper repeatedly in half, as your teacher demonstrates. Fill in your data table after each fold.

Analysis If each layer in the paper represented 100 million people and each fold in the paper represented one human generation (about 35 years), how quickly could a starting population of 100 million grow to exceed 6 billion?

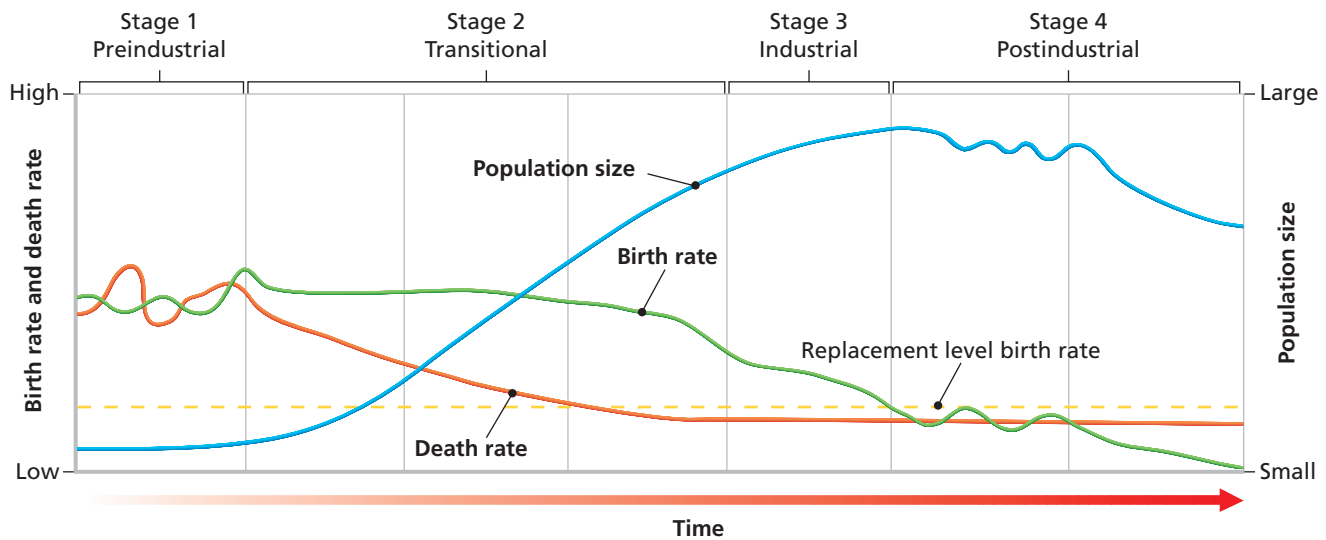


FIGURE 19-13

The demographic transition model consists of four stages. Note the relative changes in birth rates, death rates, and population size.

DEMOGRAPHIC TRANSITION

Human populations have undergone rapid growth, yet in some developed countries, populations have stopped growing. The **demographic transition** model shows how these population changes happen. The theory behind the model is that industrial development causes economic and social progress that then affects population growth rates. Figure 19-13 compares general trends in birth rates, death rates, and population sizes during four stages.

In the first stage of the model, the birth rate and the death rate are both at high levels, and the population size is stable. In the second stage, a population explosion occurs. Death rates decline as hygiene, nutrition, and education improve. But birth rates remain high, so the population grows very fast. In the third stage, population growth slows because the birth rate decreases. As the birth rate becomes close to the death rate, the population size stabilizes. In the fourth stage, the birth rate drops below replacement level, so the size of the population begins to decrease. It has taken from one to three generations for the demographic transition to occur in most developed countries.

Word Roots and Origins

demographic

from the Greek *demos*, meaning "the people," and *graphein*, meaning "to write"

SECTION 3 REVIEW

1. What effect did the agricultural revolution have on the growth of the human population?
2. Explain why mortality rates began to decline rapidly around 1650.
3. Why did population growth rates increase rapidly after World War II?
4. Compare living standards in developing countries with those in developed countries.
5. Summarize the demographic transition model.

CRITICAL THINKING

6. **Evaluating Models** Which general model of population growth most closely resembles that seen in Figure 19-13? Explain your answer.
7. **Predicting Results** How might vaccines against diseases affect population growth rates?
8. **Analyzing Data** How is it possible for some countries with low birth rates to have high rates of population growth?

CHAPTER HIGHLIGHTS

SECTION 1

Understanding Populations

- Populations can be measured in terms of size, density, dispersion, growth rate, age structure, and survivorship.
- A population's size is the number of individuals that the population contains. Density is a measure of how crowded the population is.
- Dispersion describes the distribution of individuals within the population and may be random, uniform, or clumped.
- A population's age structure indicates the percentage of individuals at each age.
- Populations show three patterns of mortality: Type I (low mortality until late in life), Type II (constant mortality throughout life), and Type III (high mortality early in life followed by low mortality for the remaining life span).

Vocabulary

population (p. 381)
population density (p. 382)

dispersion (p. 382)
birth rate (p. 383)

death rate (p. 383)
life expectancy (p. 383)

age structure (p. 384)
survivorship curve (p. 384)

SECTION 2

Measuring Populations

- The exponential model describes perpetual growth at a steady rate in a population. The model assumes constant birth and death rates and no immigration or emigration.
- In the logistic model, birth rates fall and death rates climb as the population grows. When the carrying capacity is reached, the population becomes stable.
- Population-limiting factors are density-dependent if the effect on each individual depends on the number of other individuals present in the same area.
- Small populations have low genetic diversity and are subject to inbreeding, so they are less likely to adapt to environmental changes.

Vocabulary

growth rate (p. 385)
immigration (p. 385)
emigration (p. 385)

exponential model (p. 386)
limiting factor (p. 386)
logistic model (p. 387)

carrying capacity (p. 387)
density-independent
factor (p. 388)

density-dependent
factor (p. 388)
inbreeding (p. 389)

SECTION 3

Human Population Growth

- About 10,000 to 12,000 years ago, the development of agriculture increased the growth rate of the human population.
- Around 1650, improvements in hygiene, diet, and economic conditions further accelerated population growth.
- After World War II, the human population grew at the fastest rate in history, largely because of better sanitation and medical care in poorer countries.
- Today, developing countries have faster human population growth and lower standards of living than developed countries do.

Vocabulary

hunter-gatherer
lifestyle (p. 390)

agricultural
revolution (p. 390)

developed country (p. 391)
developing country (p. 391)

demographic
transition (p. 392)

CHAPTER REVIEW

USING VOCABULARY

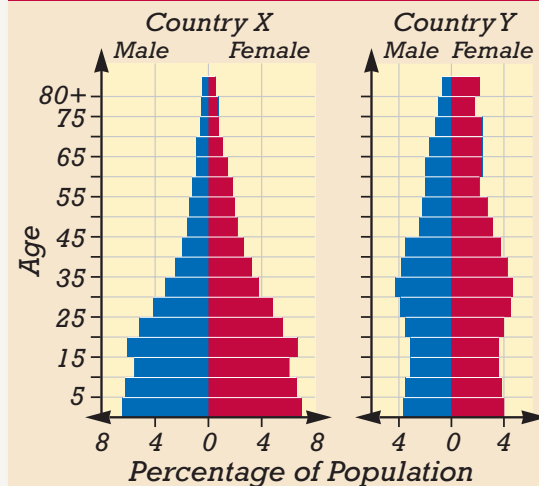
- For each pair of terms, explain how the meanings of the terms differ.
 - density* and *dispersion*
 - exponential model* and *logistic model*
 - density-dependent factor* and *density-independent factor*
 - developing country* and *developed country*
- Use the following terms in the same sentence: *growth rate*, *birth rate*, *death rate*, *immigration*, and *emigration*.
- Describe what is meant by the term *age structure*.
- Word Roots and Origins** The word *population* is derived from the Latin word *populus*, which means “people.” Using this information, explain how the term *population* has been adapted to fit its ecological definition.

UNDERSTANDING KEY CONCEPTS

- Describe** how a uniform distribution differs from a random distribution.
- Explain** two reasons that a population of turtles in a pond might have a clumped distribution.
- Identify** four kinds of measurements used to describe changing populations.
- Differentiate** between the three general types of survivorship curves.
- Name** the four processes that determine population growth.
- Compare** the main assumptions and predictions of the exponential model of population growth with those of the logistic model.
- Predict** the possible outcome of placing on a ranch a population of livestock that exceeds the carrying capacity of the ranch.
- Determine** if a volcanic eruption would result in density-dependent or density-independent regulation of populations that lived near the volcano.
- Explain** three reasons that small populations are particularly vulnerable to extinction.
- Summarize** the agricultural revolution’s effects on how people obtained food and on human population growth.
- Describe** three factors that caused the human population to begin to grow rapidly about 1650.
- Identify** the main causes of the decline in death rates following World War II.
- Compare** population growth and standards of living in developed and developing countries.
- CONCEPT MAPPING** Use the following terms to create a concept map that compares models of population growth: *logistic model*, *exponential model*, *density-dependent factors*, *density-independent factors*, *limited growth*, *unlimited growth*, *disease*, *drought*, *food availability*, and *forest fires*.
- Analyzing Concepts** Because we humans alter our environment more than other animals do, we can affect the carrying capacity of our environment. How do we increase or decrease the carrying capacity of our local area?
- Applying Information** The cause of the population cycle of the snowshoe hare is still a subject of debate. Suggest a hypothesis to explain this cycle, and suggest a way to test it.
- Drawing Conclusions** How could disease be a density-dependent factor in a population?
- Interpreting Graphics** The population of country X is projected to grow rapidly in the next few decades, while slow growth is projected for country Y. Explain these projections based on the age structure graphs shown below.

CRITICAL THINKING

Age Structure of Country X and Country Y



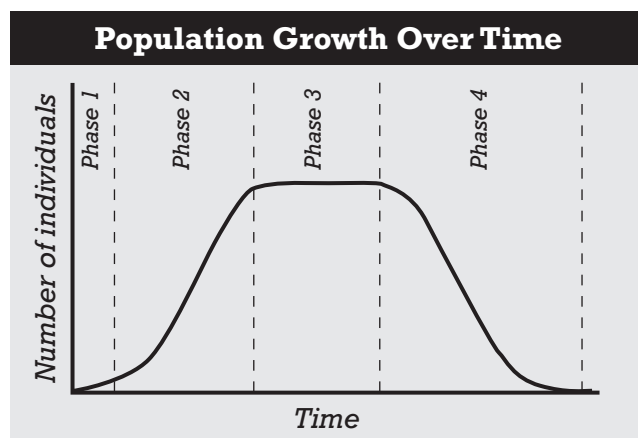


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following is a population?
 - all the fish in a pond
 - all the birds in New York City
 - all the members of a family of humans
 - all the fish of the same species in a lake
- Which of the following is true in the exponential model of population growth?
 - Population growth continues indefinitely.
 - Population growth stops at the carrying capacity.
 - Population growth increases and then decreases.
 - The immigration rate falls with increasing population size.
- Which of the following refers to the population size that can be sustained by an environment over time?
 - bell curve
 - allele frequency
 - carrying capacity
 - exponential growth
- Which of the following is a density-dependent factor for a population of deer in a forest?
 - a drought
 - a landslide
 - a period of freezing weather
 - the number of cougars in the forest

INTERPRETING GRAPHICS: The graph below shows the size of a particular population over time. Use the graph to answer the question that follows.

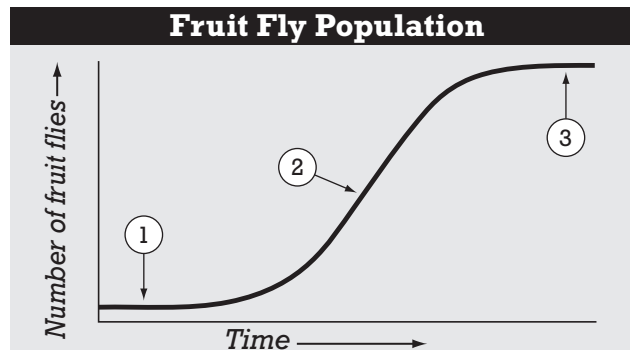


- In the graph, which time period shows negative growth of the population?
 - phase 1
 - phase 2
 - phase 3
 - phase 4

DIRECTIONS: Complete the following analogy.

- birth rate : death rate :: immigration :
 - mortality
 - migration
 - emigration
 - growth rate

INTERPRETING GRAPHICS: The graph below shows the growth of a population of fruit flies over time. Use the graph to answer the question that follows.



- At which point would a density-dependent limiting factor have a greater impact on the population?
 - 1
 - 2
 - 3
 - Both 1 and 3

SHORT RESPONSE

Study the graph of fruit fly population growth above. Explain why the population stops increasing after it reaches the point labeled 3 on the curve.

EXTENDED RESPONSE

Study the graph of fruit fly population growth above. Use the graph to support your answers to the following questions.

Part A Name one limiting factor that could affect this population of fruit flies.

Part B At which point on the curve would this limiting factor have the greatest effect on the population? Explain your reasoning.

Test TIP

For a question involving graphs, try to understand the graph by reading the graph's title and the labels on the graph's axes. For graphs that show change in some variable over time, keep in mind that the steepness and direction of a curve indicate the relative rate of change at a given point in time.

Studying Population Growth

OBJECTIVES

- Observe the growth and decline of a population of yeast cells.
- Determine the carrying capacity of a yeast culture.

PROCESS SKILLS

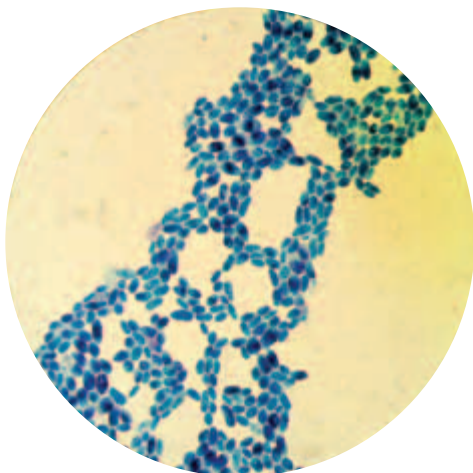
- using a microscope
- collecting data
- graphing data
- analyzing data
- calculating

MATERIALS

- safety goggles
- lab apron
- yeast cell culture
- 2 1 mL pipets
- 2 test tubes
- 1% methylene blue solution
- ruled microscope slide (2 x 2 mm)
- coverslip
- compound microscope

Background

1. What is a limiting factor and how does it affect population size?
2. How are the terms population growth, birth rate and death rate all interrelated?



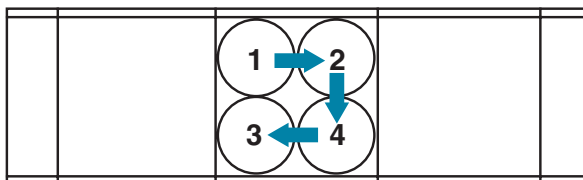
These yeast cells have been stained with methylene blue and magnified with a high-power microscope. Methylene blue gives a deep blue color to dead yeast cells, but live yeast cells will actively remove the stain.

Procedure

Counting Yeast Cells

1.   **CAUTION** Always wear safety goggles and lab apron to protect your eyes and clothing. Put on safety goggles and lab apron.
2.   **CAUTION** Do not touch or taste any chemicals. Know the location of the emergency shower and eyewash station and know how to use them. Methylene blue will stain your skin and clothing. If you get a chemical on your skin or clothing wash it off at the sink while calling to the teacher. Notify the teacher immediately of any spills. Spills should be cleaned up promptly, according to your teacher's directions. Glassware is fragile. Notify the teacher of broken glass or cuts. Do not clean up broken glass or spills with broken glass unless the teacher tells you to do so.
3. Transfer 1 mL of yeast culture to a test tube. Add 2 drops of methylene blue to the tube. The methylene blue will remain blue in dead cells but will turn colorless in living cells.
4. Make a wet mount by placing 0.1 mL, or about 1 drop, of the yeast culture and methylene blue mixture on a ruled microscope slide. Cover the slide with a coverslip.

- Observe the wet mount under the low power of a compound microscope. Notice the squares on the slide. Then switch to the high power. *Note: Adjust the light so that you can clearly see both stained and unstained cells.* Move the slide so that the top left-hand corner of one square is in the center of your field of view. This will be area 1, as shown in the diagram below.



- Count the live (unstained) cells and the dead (stained) cells in the four corners of a square using the pattern shown in the diagram above. In a data table similar to the one below, record the number of live cells and dead cells that you counted in the entire square.
- Repeat step 5 until you have counted all 6 squares on the slide.
-  Dispose of solutions and broken glass in the designated waste containers. Do not pour chemicals down the drain or put lab materials in the trash unless your teacher tells you to do so.
-  Clean up your work area and all lab equipment. Return lab equipment to its proper place. Wash

your hands thoroughly with soap and water before you leave the lab and after you finish all work.

- Refer back to your data table. Find the total number of live cells in the 6 squares. Divide this total by 6 to find the average number of live cells per square. Record this number in your data table. Repeat this procedure for the dead cells.
- Estimate the population of live cells in 1 mL (the amount in the test tube) by multiplying the average number of cells per square by 2,500. Record this number in your data table. Repeat this procedure for dead cells.
- Repeat steps 1 through 8 each day for 4 more days.

Analysis and Conclusions

- Why were several areas and squares counted and then averaged each day?
- Graph the changes in the numbers of live yeast cells and dead yeast cells over time. Plot the number of cells in 1 mL of yeast culture on the y -axis and the time (in hours) on the x -axis.
- What limiting factors probably caused the yeast population to decline?

Further Inquiry

- Write a question about population growth that could be explored in another investigation.

DATA TABLE

Time (hours)	Number of cells per square		Population size (cells/mL)
	Squares 1–6	Average	
0			
24			
48			
72			
96			

COMMUNITY ECOLOGY

A large variety of organisms interact within this coral reef community.

SECTION 1 *Species Interactions*

SECTION 2 *Patterns in Communities*



Unit 7—Ecosystem Dynamics
Topic 3



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SPECIES INTERACTIONS

Just as populations contain interacting members of a single species, communities contain interacting populations of many species. Many species have specific types of interactions with other species. This chapter introduces the five major types of interactions among species: predation, competition, parasitism, mutualism, and commensalism. These categories are based on whether each species causes any benefit or harm to the other species in a given relationship.

PREDATION

In **predation** (pree-DAY-shuhn), an individual of one species, called the *predator*, eats all or part of an individual of another species, called the *prey*. Predation is a powerful force in a community. The relationship between predator and prey influences the size of each population and affects where and how each species lives. Examples of predators include *carnivores*—predators that eat animals—and *herbivores*—predators that eat plants. Many types of organisms can act as predators or prey. All heterotrophs are either predators or parasites or both.

Predator Adaptations

Natural selection favors the evolution of predator adaptations for finding, capturing, and consuming prey. For example, rattlesnakes have an acute sense of smell and have heat-sensitive pits located below each nostril. These pits enable a rattlesnake to detect warm-bodied prey, even in the dark. Many snakes use venom to disable or kill their prey. A venomous rattlesnake is shown in Figure 20-1.

Other predator adaptations include the sticky webs of spiders, the flesh-cutting teeth of wolves and coyotes, the speed of cheetahs, and the striped pattern of a tiger's coat, which provides camouflage in a grassland habitat. Many herbivores have mouthparts suited to cutting and chewing tough vegetation.

A predator's survival depends on its ability to capture food, but a prey's survival depends on its ability to avoid being captured. Therefore, natural selection also favors adaptations in prey that allow the prey to escape, avoid, or otherwise ward off predators.

OBJECTIVES

- **Identify** two types of predator adaptations and two types of prey adaptations.
- **Identify** possible causes and results of interspecific competition.
- **Compare** parasitism, mutualism, and commensalism, and give one example of each.

VOCABULARY

predation
interspecific competition
symbiosis
parasitism
mutualism
commensalism

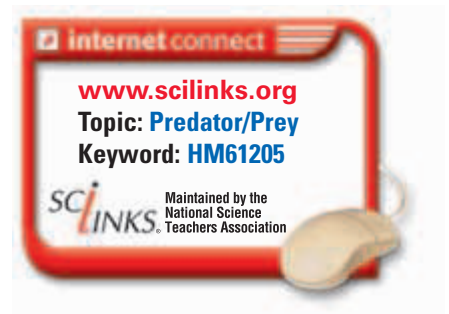


FIGURE 20-1

A rattlesnake has several adaptations that make it an effective predator.





(a) Green leaf mantid, *Choeradodis rhombicollis*



(b) Amazonian poison frog, *Dendrobates ventrimaculatus*

FIGURE 20-2

Coloration is an adaptation in prey as well as predators. In (a), the mantid cannot readily be detected among the leaves. In (b), the frog's bright colors warn other organisms that the frog is toxic if eaten.



Quick Lab

Modeling Predation

Materials 4.1 m white string, 4 stakes, 40 colored toothpicks, stopwatch or timer, meterstick

Procedure

1. Use the stakes and string to mark off a 1 m square in a grassy area.
2. One partner should scatter the toothpicks randomly throughout the square. The other partner will have 1 minute to pick up as many toothpicks as possible, one at a time. This procedure should be repeated until each team has performed five trials.
3. Record your team's results in a data table.

Analysis Toothpicks of which colors were picked up most often? Which toothpicks were picked up least often? How do you account for this difference?

Adaptations in Animal Prey

Animals may avoid being eaten by carnivores in different ways. Some organisms flee when a predator approaches. Others escape detection by hiding or by resembling an inedible object, as shown in Figure 20-2a. Some animals use deceptive markings, such as fake eyes or false heads, to startle a predator. Some animals have chemical defenses. Animals such as the frog shown in Figure 20-2b produce toxins and use bright colors to warn would-be predators of their toxicity.

In *mimicry* (MIM-ik-ree), one species closely resembles another species. For example, the harmless king snake is a mimic of the venomous coral snake, as shown in Figure 20-3. This form of mimicry is called *Batesian mimicry*. Another form of mimicry called *Müllerian mimicry*, exists when two or more dangerous or distasteful species look similar. For example, many kinds of bees and wasps have similar patterns of alternating yellow and black stripes. This kind of mimicry benefits each species involved because predators learn to avoid similar-looking individuals.

Adaptations in Plant Prey

Plants cannot run away from a predator, but many plants have evolved adaptations that protect them from being eaten. Physical defenses, such as sharp thorns, spines, sticky hairs, and tough leaves, can make plants more difficult to eat. Plants have also evolved a range of chemical defenses that are poisonous, irritating, or bad-tasting. These chemicals are often byproducts of the plants' metabolism and are called *secondary compounds*. Some examples of secondary compounds that provide a defensive function are strychnine (STRIK-nin), which is produced in plants of the genus *Strychnos*, and nicotine, which is produced by the tobacco plant. Poison ivy and poison oak produce an irritating chemical that causes an allergic reaction on some animals' skin.



(a) Scarlet king snake, *Lampropeltis triangulum*



(b) Eastern coral snake, *Micrurus fulvius*

COMPETITION

Interspecific competition is a type of interaction in which two or more species use the same limited resource. For example, both lions and hyenas compete for prey such as zebras. Likewise, many plant species compete for soil or sunlight. Some species of plants prevent other species from growing nearby by releasing toxins into the soil. If two populations compete for a resource, the result may be a reduction in the number of either species or the elimination of one of the two competitors. More often, one species will be able to use a resource more efficiently than the other. As a result, less of the resource will be available to the other species.

Competitive Exclusion

Ecologist George F. Gause was one of the first scientists to study competition in the laboratory. In test tubes stocked with a food supply of bacteria, Gause raised several species of paramecia in various combinations. When grown in separate test tubes, *Paramecium caudatum* and *Paramecium aurelia* each thrived. But when the two species were combined, *P. caudatum* always died out because *P. aurelia* was a more efficient predator of bacteria. Ecologists use the principle of *competitive exclusion* to describe situations in which one species is eliminated from a community because of competition for the same limited resource. Competitive exclusion may result when one species uses the limited resource more efficiently than the other species does.

Joseph Connell's study of barnacles along the Scottish coast in the 1960s documented competition in the wild. Connell studied two species, *Semibalanus balanoides* and *Chthamalus stellatus*. These barnacles live in the intertidal zone, the portion of the seashore that is exposed during low tide.

FIGURE 20-3

The king snake in (a) may avoid predators because of its mimicry of the color patterns of the coral snake in (b). A closer look reveals the differences: the king snake has a red snout and a black ring separating its red and yellow rings, and the coral snake has a black snout and adjacent red and yellow rings. The ring patterns of other species of coral snakes may differ from the patterns shown in the photo above.

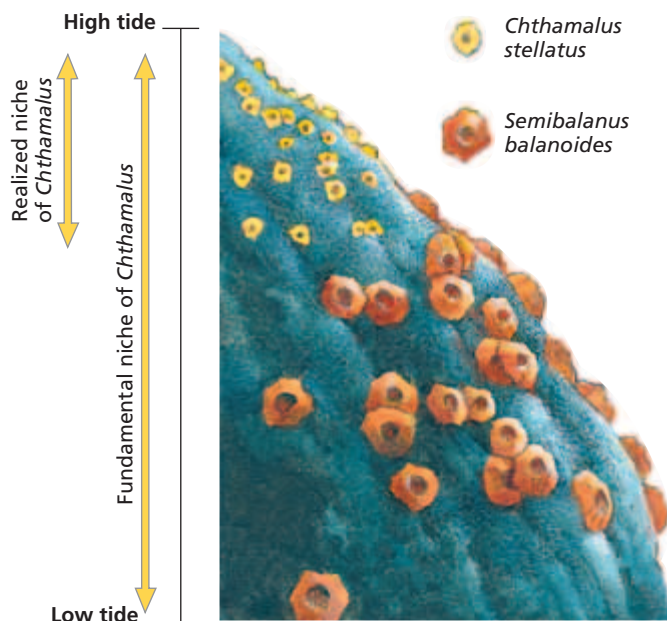


FIGURE 20-4

The realized niche of the *Chthamalus* barnacle species is smaller than its fundamental niche because of competition from the *Semibalanus* species. Although *Chthamalus* can survive at all levels of the intertidal zone, it is usually crowded out at the lower level by the faster-growing *Semibalanus*. But *Semibalanus* cannot survive in the upper level, which is left dry for longer periods.

As Figure 20-4 shows, each species of barnacle occupied a distinct band within the intertidal zone. *Chthamalus* lived higher on the rocks than *Semibalanus* did. Connell demonstrated that this difference was partly due to competition. When a rock covered with *Chthamalus* was transplanted to the lower zone, *Chthamalus* was clearly able to tolerate the conditions in the lower zone. However, *Semibalanus* would soon settle on the rock and eventually crowd out *Chthamalus*. Connell concluded that competition restricted the range of *Chthamalus*. Although *Chthamalus* could survive lower on the rocks, competition from *Semibalanus* prevented it from doing so. Higher on the rocks, *Chthamalus* is free from competition because it can tolerate drying out more than *Semibalanus* can.

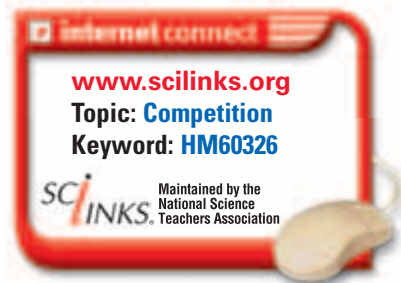
Reduced Niche Size

Recall that a species' *niche* (NICH) is the role that the species plays in its environment. But this role may vary due to interactions with other species. So, ecologists differentiate between the fundamental niche and the realized niche of a species. The *fundamental niche* of a species is the range of conditions that it can potentially tolerate and the range of resources that it can potentially use. Often, predation and competition limit the species' use of these ranges. Thus, the *realized niche* is the part of the niche that the species actually uses, as shown in Figure 20-4.

Character Displacement

Competition has the potential to be an important influence on the nature of a community. The composition of a community may change as competitors win, lose, or evolve differences that lessen the intensity of competition. Natural selection favors differences between competitors, especially where the niches of the competitors overlap. This evolution of differences in a characteristic due to competition is called *character displacement*. Character displacement is a way of reducing niche overlap.

The beaks of Galápagos Island finches provide an example of character displacement. Many of these closely related finch species eat seeds that come from the small number of plant species on the islands. Furthermore, the size of each bird's beak determines the size of the seeds that the bird can eat. When the average beak size of each species is compared with the average beak size of each of the other species, the greatest difference in beak size is observed between species that share an island. Differences in beak size reduce competition by enabling species that share an island to favor different food sources.



Resource Partitioning

As Charles Darwin noted, competition is likely to be most intense between similar species that require the same resources. When similar species coexist, each species may avoid competition with others by using a specific part of an available resource. This pattern of resource use is called *resource partitioning*. For example, the pioneering ecologist Robert MacArthur (1930–1972) studied several species of warblers that feed on insects in spruce and fir trees. MacArthur discovered that when more than one species of warbler is foraging within the same tree, each species hunts for insects only in a particular section of the tree. As a result, competition among the species is reduced.

SYMBIOSIS

A **symbiosis** (SIM-bie-OH-sis) is a close, long-term relationship between two organisms. Three examples of symbiotic relationships include: parasitism, mutualism, and commensalism. **Parasitism** (PAR-uh-SIET-IZ-UHM) is a relationship in which one individual is harmed while the other individual benefits. **Mutualism** (MYOO-choo-uhl-IZ-uhm) is a relationship in which both organisms derive some benefit. In **commensalism** (kuh-MEN-suhl-IZ-uhm), one organism benefits, but the other organism is neither helped nor harmed.

Parasitism

Parasitism is similar to predation in that one organism, called the *host*, is harmed and the other organism, called the *parasite*, benefits. However, unlike many forms of predation, parasitism usually does not result in the immediate death of the host. Generally, the parasite feeds on the host for a long time rather than kills it. Parasites such as aphids, lice, leeches, fleas, ticks, and mosquitoes that remain on the outside of their host are called *ectoparasites*. Parasites that live inside the host's body are called *endoparasites*. Familiar endoparasites are heartworms, disease-causing protists, and tapeworms, such as the one shown in Figure 20-5. Natural selection favors adaptations that allow a parasite to exploit its host efficiently. Parasites are usually specialized anatomically and physiologically for a parasitic lifestyle.

Parasites can have a strong negative impact on the health and reproduction of the host. Consequently, hosts have evolved a variety of defenses against parasites. Skin is an important defense that prevents most parasites from entering the body. Tears, saliva, and mucus defend openings through which parasites could pass, such as the eyes, mouth, and nose. Finally, the cells of the immune system may attack parasites that get past these defenses.

Word Roots and Origins

parasite

from the Latin word *parasitus*, meaning "one who eats at the table of another"

FIGURE 20-5

Tapeworms are endoparasites that can grow to 20 m or greater in length. Tapeworms are so specialized for a parasitic lifestyle that they do not have a digestive system. They live in the host's small intestine and absorb nutrients directly through their skin. Tapeworms reproduce by producing egg-filled chambers, which are released in their host's feces to be unknowingly picked up by a future host.





FIGURE 20-6

Acacia trees in Central America have a mutualistic relationship with certain types of ants. The trees provide food and shelter to the ants, and the ants defend the tree from insect herbivores.

Mutualism

Mutualism is a relationship in which two species derive some benefit from each other. Some mutualistic relationships are so close that neither species can survive without the other. An example of mutualism, shown in Figure 20-6, involves ants and some species of *Acacia* plants. The ants nest inside the acacia's large thorns and receive food from the acacia. In turn, the ants protect the acacia from herbivores and cut back competing vegetation.

Pollination is one of the most important mutualistic relationships on Earth. Animals such as bees, butterflies, flies, beetles, bats, and birds that carry pollen between flowering plants are called *pollinators*. A flower is a lure for pollinators, which are attracted by the flower's color, pattern, shape, or scent. The plant usually provides food—in the form of nectar or pollen—for its pollinators. As a pollinator feeds in a flower, it picks up a load of pollen, which it may then carry to other flowers of the same species.

Commensalism

Commensalism is an interaction in which one species benefits and the other species is not affected. Species that scavenge for leftover food items are often considered commensal species. However, a relationship that appears to be commensalism may simply be mutualism in which the mutual benefits are not apparent.

An example of a commensal relationship is the relationship between cattle egrets and Cape buffaloes in Tanzania. The birds feed on small animals such as insects and lizards that are forced out of their hiding places by the movement of the buffaloes through the grass. Occasionally, the cattle egrets also feed on ectoparasites from the hide of the buffaloes, but the buffaloes generally do not benefit from the presence of the egrets.

SECTION 1 REVIEW

1. List the five major kinds of species interactions.
2. Describe one adaptation of a herbivore and one adaptation of a carnivore for obtaining food.
3. Identify two possible results of interspecific competition.
4. What is the competitive exclusion principle?
5. Identify the ways that parasites are similar to predators.
6. In the relationship between cattle egrets and Cape buffaloes, if the egrets regularly removed ticks from the buffaloes, would this relationship still be considered commensal? Explain your answer.

CRITICAL THINKING

7. **Analyzing Concepts** Explain how two similar species of birds could nest in the same tree and yet occupy different niches.
8. **Making Comparisons** How does predation on plants differ from predation on animals, in terms of the usual effect on the prey?
9. **Analyzing Concepts** Species A and B have a very similar niche. Species A recently arrived in a location where species B previously lived and carried a disease that killed all members of species B. Is this situation an example of competitive exclusion? Explain your answer.

PATTERNS IN COMMUNITIES

The investigation of community properties and interactions is an active area of ecology. Which properties are most significant in structuring a community? What determines the number of species in a community? How do communities recover from disturbance? These questions are central to a study of communities.

SPECIES RICHNESS

One characteristic of a community is **species richness**, the number of species in the community. A related measure is **species evenness**, which is the relative abundance of each species. These two measures provide slightly different information. Species richness is a simple count of the species in the community. Each species contributes one count to the total regardless of whether the species' population is one or 1 million.

In contrast, species evenness takes into account how common each species is in the community. To calculate the species evenness of a community, ecologists must measure or estimate the population size of all species in the community. In general, ecologists study both species richness and species evenness when they investigate communities.

Latitude and Species Richness

Species richness varies with latitude (distance from the equator). As a general rule, the closer a community is to the equator, the more species it will contain. Species richness is greatest in the tropical rain forests. For example, entomologists Edward O. Wilson and Terry Erwin identified nearly as many species of ants in a single tree in Peru as can be found in the entire British Isles.

Why do the Tropics contain more species than the temperate zones do? One hypothesis is that temperate habitats, having formed since the last Ice Age, are younger. Therefore, tropical habitats were not disturbed by the ice ages, but habitats closer to the poles were disturbed. Also, the climate is more stable in the Tropics. This stability allows species to specialize to a greater degree than they could in temperate regions, where the climate is more variable.

OBJECTIVES

- **Describe** the factors that affect species richness in a community.
- **Explain** how disturbances affect community stability.
- **Distinguish** between types of succession, and explain why succession may not be predictable.

VOCABULARY

species richness
species evenness
species-area effect
disturbance
stability
ecological succession
primary succession
secondary succession
pioneer species
climax community



FIGURE 20-7

This species-richness map of North American and Central American birds shows that fewer than 100 species of birds inhabit arctic regions, whereas more than 600 species occupy some tropical regions. This evidence suggests that species richness increases closer to the equator. Equatorial rain forests are biologically the richest habitats on Earth.



Another hypothesis suggests that because plants can photosynthesize year-round in the Tropics, more energy is available to support more organisms. The high richness of species in the Tropics, as shown in Figure 20-7, is likely the result of several factors.

Habitat Size and Species Richness

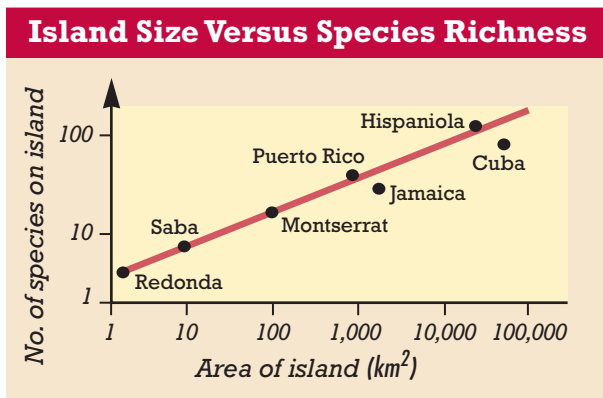
Another pattern of species richness is that larger areas usually contain more species than smaller areas do. This relationship is called the **species-area effect**. The species-area effect is most often applied to islands, where area is clearly limited by geography. In the Caribbean, for example, more species of reptiles and amphibians live on large islands, such as Cuba, than on small islands, such as Redonda, as shown in Figure 20-8. Because all of these islands are close together, differences in species richness cannot be due to

differences in latitude. Why does species richness increase as area increases? Larger areas usually contain a greater diversity of habitats and thus can support more species.

The species-area effect has one very important practical consequence: reducing the size of a habitat reduces the number of species that the habitat can support. Today, natural habitats are shrinking rapidly under pressure from the ever-growing human population. About 2 percent of the world's tropical rain forests are destroyed each year, for example. The inevitable result of the destruction of habitats is the extinction of species.

FIGURE 20-8

In a 1971 survey, the large islands of Cuba and Hispaniola each had about 100 species of reptiles and amphibians, whereas the small island of Redonda only had about 5 species. In general, species richness increases as available habitat increases. This principle is true for inland forests as well as islands.



Species Interactions and Species Richness

Interactions among species sometimes affect species richness. Several studies have demonstrated that predators can prevent competitive exclusion among their prey. In the 1960s, zoologist Robert Paine showed the importance of the sea star *Pisaster ochraceus*, shown in Figure 20-9, in maintaining the species richness of communities on the Washington coast. Paine removed all *Pisaster* individuals from one site and for several years prevented any new *Pisaster* individuals from settling there. This change caused a dramatic shift in the community. The mussel *Mytilus californianus*, which had previously coexisted with several other species, became much more abundant, spread over the habitat, and crowded out other species. The number of other species fell from almost 20 to fewer than 5 within a decade. Evidently, *Mytilus* was the superior competitor for space on the rocks, but its population was normally held in check by predation from *Pisaster*.

Community Stability and Species Richness

One of the most important characteristics of a community is how it responds to disturbance. **Disturbances** are events that change communities, remove or destroy organisms from communities, or alter resource availability. Examples of abiotic disturbances are droughts, fires, floods, volcanic eruptions, earthquakes, and storms. Examples of animal disturbances include elephants tearing up trees while feeding and prairie dogs moving soil around while digging their burrows. Human disturbances include bulldozing, clear-cutting, paving, plowing, and mowing land.

Disturbances affect practically all communities at some point. A number of organisms may even depend on a certain type of disturbance in order to survive. For example, the lodgepole pine, shown in Figure 20-10, depends on periodic fires to disperse its seeds. The cones of this tree contain a tough resin seal that is cracked open by intense heat. Disturbances may also create opportunities for species that have not previously occupied a habitat to become established.

Stability is the tendency of a community to maintain relatively constant conditions. Stability therefore relates to the community's resistance to disturbances. Ecologists suspect that a community's stability is also related to its species evenness and species richness, because communities that have more species contain more links between species. These links may spread out the effects of the disturbance and lessen disruption of the community. One line of evidence cited in support of this view is the vulnerability of agricultural fields to outbreaks of insect pests, given that these fields usually consist of one species of plants. Further evidence came from a study of grassland plots in the 1980s. Ecologists observed that during a drought, species-rich plots lost a smaller percentage of plant mass than did species-poor plots. The species-rich plots also took less time to recover from the drought.

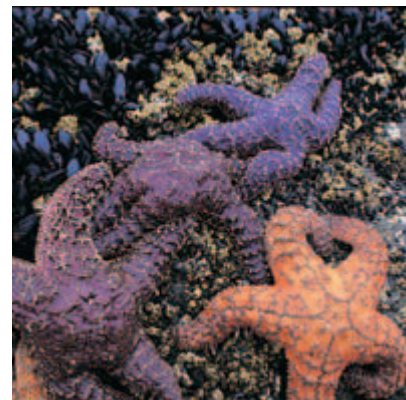


FIGURE 20-9

Predation by the sea star *Pisaster ochraceus* on the mussel *Mytilus californianus* promoted diversity of mussel species by controlling the *Mytilus* population.

FIGURE 20-10

These young lodgepole pines have started growing after a devastating forest fire. The heat of the fire helped release the pine seeds from their cones, which allowed the seeds to germinate.



Word Roots and Origins

succession

from Latin *succedere*,
meaning “to go beneath” or
“to follow after”

SUCCESSIONAL CHANGES IN COMMUNITIES

During the summer and early fall of 1988, fires burned large areas of Yellowstone National Park. Approximately 320,900 hectares (793,000 acres) were affected. If you visit Yellowstone today, you will find that regrowth is well underway in the burned areas. In time, if no further major disturbances take place, the burned areas of Yellowstone National Park will undergo a series of regrowth stages. The gradual, sequential regrowth of a community of species in an area is called **ecological succession** (EK-uh-LAHJ-i-kuhl suhk-SESH-uhn). You can see early stages of succession in vacant lots, along roads, and even in sidewalks or parking lots where weeds are pushing up through cracks in the concrete. Figure 20-11 shows the successional changes that took place after the eruption of Mount St. Helens in 1980.

Ecologists recognize two types of succession. **Primary succession** is the development of a community in an area that has not supported life previously, such as bare rock, a sand dune, or an island formed by a volcanic eruption. In primary succession, soil is not initially present. **Secondary succession** is the sequential replacement of species that follows disruption of an existing community. The disruption may stem from a natural disturbance, such as a forest fire or a strong storm, or from human activity, such as farming, logging, or mining. Secondary succession occurs where soil is already present.

Any new habitat, whether it is a pond left by heavy rain, a freshly plowed field, or newly exposed bedrock, is an invitation to many species that are adapted to the new conditions. The species of organisms that predominate early in succession are called **pioneer species**. Pioneer species tend to be small, to grow quickly, and to reproduce quickly, so they are well suited for invading and occupying a disturbed habitat. They also may be very good at dispersing their seeds, which enables them to quickly reach disrupted areas.

FIGURE 20-11

Successional changes in communities are most apparent after a major disturbance, such as a volcanic eruption. The aftermath of the 1980 Mount St. Helens eruption is shown in (a). The photo in (b) was taken 12 years after the eruption. Many herbaceous plants and young trees had grown up.

(a)

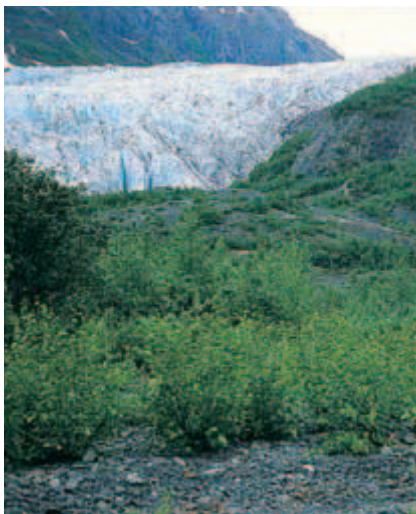


(b)

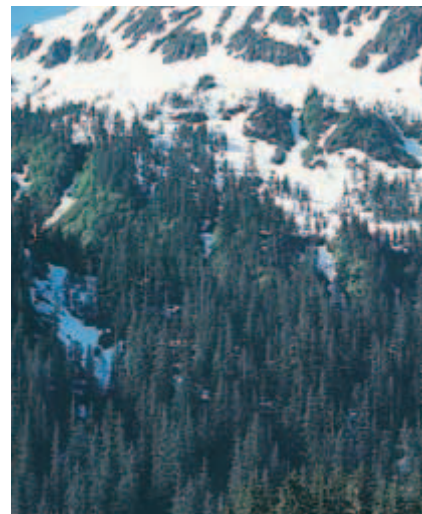




(a)



(b)



(c)

Primary Succession

Primary succession often proceeds very slowly because the soil is too thin or lacks enough minerals to support plant growth. For example, when glaciers last retreated from eastern Canada, about 12,000 years ago, they left a huge stretch of barren bedrock from which all the soil had been scraped. This geologic formation, called the Canadian Shield, was a place where plants and most animals could not live. Repeated freezing and thawing broke this rock into smaller pieces. In time, lichens—mutualistic associations between fungi and either algae or cyanobacteria—colonized the barren rock. Acids in the lichens and mildly acidic rain washed nutrient minerals from the rock. Eventually, the dead organic matter from decayed lichens, along with minerals from the rock, began to form a thin layer of soil in which a few grasslike plants could grow. These plants then died, and their decomposition added more organic material to the soil. Soon, mosses and then larger plants began to grow. Today, much of the Canadian Shield is densely populated with pine, balsam, and spruce trees, whose roots cling to soil that in some areas is still only a few centimeters deep. A similar series of changes has been documented at Glacier Bay, Alaska, shown in Figure 20-12.

Secondary Succession

Secondary succession occurs where an existing community has been cleared by a disturbance, such as agriculture, but the soil has remained intact. In secondary succession, the original ecosystem returns through a series of well-defined stages. In eastern temperate regions, secondary succession typically begins with weeds, such as annual grasses, mustards, and dandelions, whose seeds may be carried to the site by wind or by animals. If no major disturbance occurs, succession in these regions proceeds with perennial grasses and shrubs, continues with trees such as dogwoods, and eventually results in a deciduous forest community. The complete process takes about 100 years.

FIGURE 20-12

Ecologists study the process of primary succession by examining a variety of areas at different successional stages. These photos were taken at different locations at Glacier Bay, Alaska; the changes that occur between the stage shown in (a) and the stage shown in (c) take about 200 years. Shown in (a) is lifeless glacial “till” (pulverized bare rocks) left in the wake of the retreating glacier. Shown in (b) is an early stage of succession in which small plants and shrubs are growing on the site. A mature forest is shown in (c), an end stage of succession.

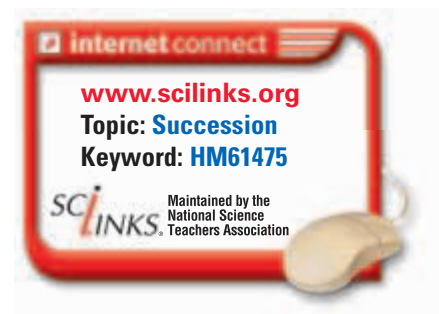




FIGURE 20-13

A recently abandoned agricultural field is being pioneered by weeds. Eventually, taller plants and shrubs will compete with the pioneers. If no further disturbances occur, a forest of pine or cottonwood may follow, succeeded by a hardwood forest. The whole process will take about 100 years.

Human disturbances such as mining, logging, farming, and urban development often start the process of succession. A recently abandoned farm field is shown in Figure 20-13. After areas such as woodlands and prairies are cleared by humans, grasses and weeds often begin to grow, thus beginning the process of secondary succession.

THE COMPLEXITY OF SUCCESSION

The traditional description of succession is that the community proceeds through a predictable series of stages until it reaches a stable end point, called the **climax community**. The organisms in each stage alter the physical environment in ways that make it less favorable for their own survival but more favorable for the organisms that eventually succeed them. In a sense, each stage paves the way for the next, leading ultimately to the climax community, which remains constant for a long period of time.

When ecologists began to study and document many instances of succession, they found a more complex picture. Some so-called climax communities, for example, are not stable and continue to change. Instead of proceeding inevitably toward the climax community, succession may be regularly “reset” by disturbances. For example, many grasslands give way to forests, but periodic fires prevent the forests from developing. There may be many possible successional pathways in a particular area. The actual path followed may depend on the identities of the species present, the order in which the species arrive, the climate, and many other factors. Ecologists agree that the idea of a single successional pathway ending in a stable climax community is too simple to describe what actually occurs in nature.

SECTION 2 REVIEW

1. What is the difference between species richness and species evenness?
2. Explain the relationship between species richness and latitude.
3. Identify examples of how animal and human disturbances can affect a community.
4. Explain how species richness can affect community stability.
5. Identify differences between primary and secondary succession.

CRITICAL THINKING

6. **Applying Information** Why is the species-area effect important in efforts to conserve species?
7. **Distinguishing Relevant Information** Explain how the example of agricultural fields relates species richness to stability.
8. **Evaluating Viewpoints** Do you agree with the following statement: “Communities are usually in a state of recovery from disturbance”? Consider both animals and plants in your answer.

CHAPTER HIGHLIGHTS

SECTION 1

Species Interactions

- Ecologists recognize five major kinds of species interactions in communities: predation, parasitism, competition, mutualism, and commensalism.
- Predation is an interaction in which one organism (the predator) captures and eats another organism (the prey).
- Predators have adaptations to efficiently capture prey, whereas prey species have adaptations to avoid capture. Mimicry is an adaptation in which a species gains an advantage by resembling another species or object.
- Competition may cause competitive exclusion, the elimination of one species in a community. Competition may also drive the evolution of niche differences among competitors.
- In parasitism, one species (the parasite) feeds on, but does not always kill, another species (the host).
- In mutualism, both interacting species benefit.
- In commensalism, one species benefits, and the other is not affected.

Vocabulary

predation (p. 399)

interspecific competition
(p. 401)

symbiosis (p. 403)
parasitism (p. 403)

mutualism (p. 403)
commensalism (p. 403)

SECTION 2

Patterns in Communities

- Species richness is the number of species in a community. Species evenness is the relative abundance of each species.
- In general, species richness is greatest near the equator, and larger areas support more species. Species interactions such as predation can promote species richness.
- Disturbances can alter a community by eliminating or removing organisms or altering resource availability.
- Species richness may improve a community's stability. Areas of low species richness may be less stable in the event of an ecological disturbance.
- Ecological succession is a change in the species composition of a community over time. Primary succession is the assembly of a community on newly created habitat. Secondary succession is the change in an existing community following a disturbance.
- Primary succession occurs in areas that have been recently exposed to the elements and lack soil. Primary succession typically proceeds from lichens and mosses to a climax community.
- Secondary succession occurs in areas where the original ecosystem has been cleared by a disturbance. Secondary succession typically proceeds from weeds to a climax community.

Vocabulary

species richness (p. 405)
species evenness (p. 405)
species-area effect (p. 406)

disturbance (p. 407)
stability (p. 407)
ecological succession (p. 408)

primary succession (p. 408)
secondary succession (p. 408)
pioneer species (p. 408)

climax community (p. 410)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - mutualism* and *commensalism*
 - parasitism* and *predation*
 - species richness* and *species evenness*
 - primary succession* and *secondary succession*
 - pioneer species* and *climax community*
- Word Roots and Origins** The word *niche* is derived from the Old French word *nichier*, which means “to nest.” Using this information, explain why *niche* is a good term for the role of an organism in its environment.

UNDERSTANDING KEY CONCEPTS

- Compare** the five major types of relationships between species by creating a table.
- Describe** two evolutionary adaptations that enable organisms to be efficient predators.
- Identify** two evolutionary adaptations that enable prey species to avoid being eaten.
- Explain** some of the adaptations that may enable a host species to defend itself against parasites.
- Describe** the experiments that Gause conducted on competition in paramecia, and explain what the results demonstrated.
- State** why, in the study of competition between two species of barnacles, *Semibalanus balanoides* was the superior competitor, yet *Chthamalus stellatus* was not excluded from the community.
- Identify** the benefits that certain ants derive from their relationship with *Acacia* plants and the benefits that the plants receive from the ants.
- Explain** the possible consequences of habitat loss in terms of the species-area effect.
- Describe** how a disturbance could benefit some species in a community.
- Summarize** one view on how species richness affects community stability.
- State** some reasons why the process of succession may be more complex than once thought.
-  **BIOLOGY** Write a short report summarizing how artificial ecosystems used in the management and treatment of wastewater and pollutants can demonstrate succession.

-  **CONCEPT MAPPING** Use the following terms to create a concept map that compares the two types of succession: *primary succession*, *secondary succession*, *pioneer species*, *climax community*, *bare rock*, *lichens*, *organic matter*, *soil*, *weeds*, *shrubs*, and *trees*.

CRITICAL THINKING



- Interpreting Graphics** Examine the figure above. These three warbler species each feed on insects in spruce trees at the same time. However, each species tends to forage in a different area of the tree.
 - What ecological process is demonstrated by the feeding patterns of these species?
 - In some areas, there are five species of warblers that feed in spruce trees. Form a hypothesis about the feeding patterns of the other two species.
- Analyzing Concepts** Can two species that never come in contact with each other compete for the same resource? Explain your answer.
- Analyzing Relationships** Some plants are pollinated by only one pollinator. Explain why this situation might benefit the plant. How could this relationship be a danger to the plant species?
- Making Inferences** Explain why measuring the species evenness of a community is usually harder than measuring species richness.

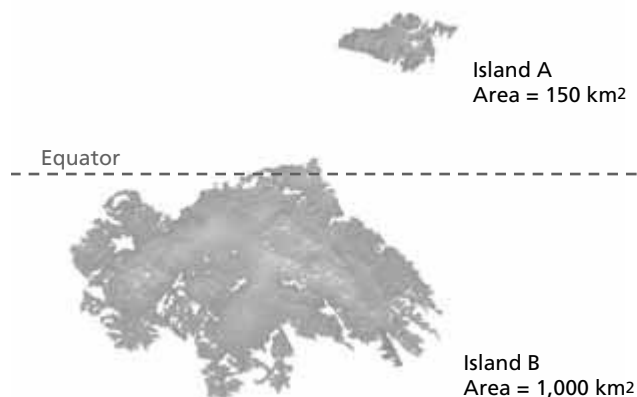


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question or completes the sentence.

1. A certain tropical tree has a fruit that is eaten by only one species of bats. As the bat digests the fruit, the seeds are made ready to sprout. When the bat excretes the wastes of the fruit, it drops seeds in new locations. Which of the following is the correct term for the relationship between the bat and the tree?
 - A. predation
 - B. mutualism
 - C. competition
 - D. commensalism
2. Which of the following is a parasite?
 - F. a lion hunting a zebra
 - G. a deer grazing on grass
 - H. a tick sucking blood from a dog
 - J. a snake swallowing a bird's egg
3. Three species of birds forage for insects in the same tree. However, each species tends to forage in different parts of the tree. This pattern of foraging is best explained as an adaptation to which of the following relationships?
 - A. predation
 - B. mutualism
 - C. competition
 - D. commensalism

INTERPRETING GRAPHICS: The map below shows two islands. Use the map to answer the question that follows.

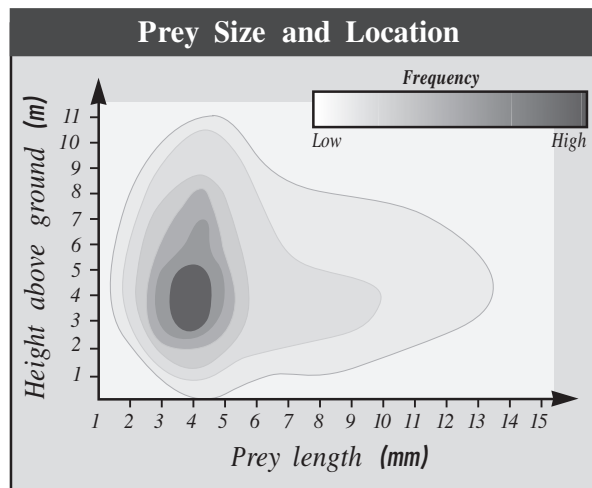


4. What can you infer about the number of species on each of these islands?
 - F. Island A has more species.
 - G. Island B has more species.
 - H. Island A and Island B will have the same number of species.
 - J. Both islands will have fewer species than islands that are located farther north.

DIRECTIONS: Complete the following analogy.

5. predator : prey :: herbivore :
 - A. carnivore
 - B. plant
 - C. parasite
 - D. predation

INTERPRETING GRAPHICS: The shading in the graph below indicates the frequency with which a certain bird species obtains prey, by prey size and location. Use the graph to answer the question that follows.



6. Which of the following statements is best supported by this graph?
 - F. Most often, the bird eats insects.
 - G. Most often, the bird nests above ground.
 - H. Most often, the bird finds prey at ground level.
 - J. Most often, the bird eats prey that is between 3 and 5 mm long.

SHORT RESPONSE

Some plants produce chemicals that are irritating or poisonous to some animals.

Explain the role of these adaptations in an ecological community.

EXTENDED RESPONSE

The gradual, sequential change in species in an area is called *ecological succession*.

Part A Describe the stages of primary succession.

Part B Compare primary succession and secondary succession.

Test TIP Carefully read all instructions, questions, and answer options completely before attempting to choose an answer.

Observing Symbiosis: Root Nodules

OBJECTIVES

- Examine root nodules in legumes.
- Investigate the differences between a legume (bean) and a nonlegume (radish).
- View active cultures of symbiotic *Rhizobium*.

PROCESS SKILLS

- recognizing relationships
- hypothesizing
- comparing and contrasting

MATERIALS

- protective gloves
- 2 three-inch flowerpots
- 2 cups of soil
- 2 mixing sticks
- 1.2 mL (1/4 tsp) of *Rhizobium* bacteria per pot
- 2 bean seeds
- 2 radish seeds
- 2 microscope slides
- 2 coverslips
- 1 prepared slide of a soybean root nodule with symbiotic *Rhizobium* bacteria
- compound light microscope
- stereoscope or magnifying glass
- scalpel

Background

1. Define *symbiosis*, and give examples.
2. Nitrogen-fixing bacteria and leguminous plants have a symbiotic relationship.
3. *Rhizobium* is a genus of nitrogen-fixing bacteria. *Rhizobium* species exist in soil and in the root nodules of leguminous plants such as soybeans.
4. Green root nodules indicate actively reproducing bacteria that are not fixing nitrogen. Pink nodules indicate bacteria that are actively fixing nitrogen but are not reproducing.
5. The process of nitrogen fixation is the basis of the nitrogen cycle. Only a few kinds of organisms are capable of converting nitrogen gas into a form that is usable by other organisms.

PART A Growing the Test Plants

1. Fill two flowerpots with soil. Using a mixing stick, stir approximately 1/4 tsp of the *Rhizobium* mixture into each pot.
2. Plant two bean seeds in one pot, and label the pot "Bean." Plant two radish seeds in the other pot, and label this pot "Radish." Water each pot so that the soil is moist but not saturated.
3. Place the plants where they will receive direct sunlight. Water the soil when necessary to keep the soil moist but not saturated. Do not fertilize the plants.
4. After approximately one week, check to see if both seeds germinated in each pot. If they did, remove the smaller seedling. The plants will be ready to be examined after six to eight weeks. Monitor the plants each day. (Note: The radish seeds will germinate faster than the bean seeds.)
5.   Clean up your materials, and wash your hands before leaving the lab.

PART B Observing the Roots of Beans

6. Prepare a data table similar to the one below. As you work, record your observations in your data table.
7.  **CAUTION** Wear disposable gloves while handling plants. Do not rub any plant part or plant juice on your eyes or skin. Remove the bean plant from the pot by grasping the bottom of the stem and gently pulling the plant out. Be careful not to injure the plant. Carefully remove all dirt from the roots.
8. View the roots of the bean plant under a stereoscope. Compare the appearance of the bean root system with the photograph above. Note the formation of any

OBSERVATIONS OF ROOT NODULES

Color of nodule	
Shape of nodule	
Number of nodules	
Number of pink nodules	
Number of green nodules	



nodules on your bean plant's roots. Draw a root with a nodule in your lab report, and label each structure.

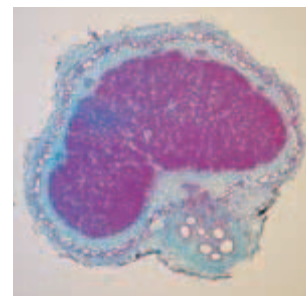
9.  **CAUTION** Use the scalpel with care. A scalpel is a very sharp instrument. When cutting, make sure that the blade faces away from your body. If you cut yourself, quickly apply direct pressure to the wound, and call for your teacher. Remove a large nodule from the bean root, and carefully cut the nodule in half with a scalpel. The pink nodules contain active nitrogen-fixing bacteria. The green nodules contain bacteria but cannot fix nitrogen because they are actively reproducing. *Rhizobium* bacteria will begin fixing nitrogen only after they stop reproducing. View the cross section under a stereoscope. Note the arrangement of symbiotic cells within the nodule. Draw and label this arrangement.

PART C Observing the Roots of Radishes

10.  **CAUTION** Wear disposable gloves while handling plants. Do not rub any plant part or plant juice on your eyes or skin. Remove the radish plant from the pot by grasping the bottom of the stem and gently pulling the plant out. Be careful not to injure the plant. Carefully remove all dirt from the roots.
11. Examine the roots of the radish plant under a stereoscope. Compare the radish roots with the roots of the bean plant that you have already examined. Do the roots of the radish plant contain any nodules? Draw the root of the radish plant in your lab report. Label the drawing "Radish."

PART D Preparing a Wet Mount of *Rhizobium*

12.  **CAUTION** Handle the slide and coverslip carefully. Glass slides break easily, and the sharp edges can cut you. Prepare a wet mount by placing part of a pink nodule on a microscope slide, adding a drop of water, and covering the slide with a coverslip.
13. Place the slide on a flat surface. Gently press down on the slide with your thumb. Use enough pressure to squash the nodule. Make sure that the coverslip does not slide.
14. Examine the slide under a microscope. Draw and label your observations in your lab report. Note the power of magnification used.
15. Compare your wet mount preparation with the prepared slide of *Rhizobium* bacteria and the photograph on the right. Cells with active *Rhizobium* bacteria should look like the reddish cells in the photograph at right.
16.   Clean up your materials, and wash your hands before leaving the lab.



Analysis and Conclusions

1. Which plant had the most nodules?
2. How many nodules were found on the radish plants?
3. How do legumes become inoculated with bacteria in nature?
4. What kind of relationship exists between the legume plant and *Rhizobium* bacteria? How does this relationship benefit the legume plant? How does this relationship benefit the bacteria?
5. If you were to try to grow legumes without root nodules to use as experimental controls, why should you plant the seeds in sterile soil?

Further Inquiry

Perform the experiment by using beans with and without *Rhizobium* bacteria. Count the number of leaves on each plant, and measure the mass of each whole plant as well as the masses of roots, stems, and leaves separately. Predict which part of the plant will differ the most.

ECOSYSTEMS



Tropical rain forests are the most biologically diverse of all of the biomes on Earth. This scarlet macaw, *Ara macao*, lives in the rain forests of the Americas.

SECTION 1 *Terrestrial Biomes*SECTION 2 *Aquatic Ecosystems*

Unit 7—Ecosystem
Dynamics,
Topic 1



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TERRESTRIAL BIOMES

Biomes (BIE-OHMZ) are very large climatic regions that contain a number of smaller but related ecosystems within them. A certain biome may exist in more than one location on Earth, but each of those locations has a similar climate and inhabitants with similar adaptations.

THE MAJOR BIOMES

Biomes are distinguished by the presence of characteristic plants and animals, but they are commonly identified by their dominant plant life. For example, hardwood trees, such as oaks and maples, are the dominant form of plant life in the deciduous forest biome. Most ecologists recognize eight major biomes, shown on the map in Figure 21-1, and several minor biomes. In this section, you will learn about the characteristics of these major biomes: tundra, tropical forest, temperate forest, taiga, temperate grassland, savanna, chaparral, and desert.

Because abiotic factors change over a landscape, biomes seldom have distinct boundaries. As climate varies over the Earth's surface, for example, deserts tend to gradually change into grasslands, tundra into taiga, and so on. Figure 21-1 shows how the major biomes are distributed over the Earth. Because climate varies with elevation, mountains contain a variety of communities and do not belong to any one biome. Table 21-1 describes the major biomes and lists their average annual temperature and rainfall.

OBJECTIVES

- Identify the eight major biomes.
- Compare tundra with taiga.
- Compare the different kinds of forests.
- Compare the different kinds of grasslands.
- Describe the adaptations of desert organisms.

VOCABULARY

biome
tundra
permafrost
tropical forest
canopy
epiphyte
coniferous tree
deciduous tree
temperate deciduous forest
taiga
temperate grassland
savanna
chaparral
desert

FIGURE 21-1

The eight biomes cover most of Earth's land surface. Antarctica is not shown because it has no major biomes.

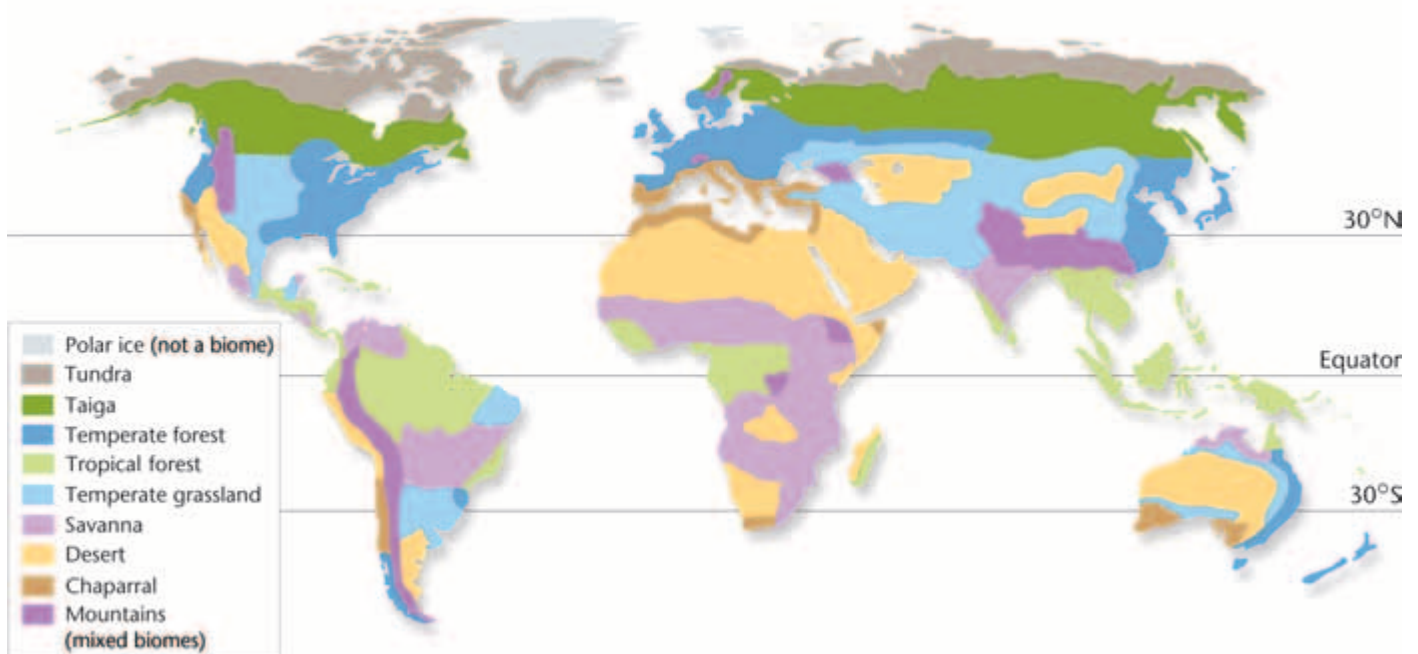


TABLE 21-1 *Characteristics of the Major Biomes*

Biome	Average yearly temperature range	Average yearly precipitation	Soil	Vegetation
Tundra	−26°C to 12°C	<25 cm	moist, thin topsoil over permafrost; low in nutrients; slightly acidic	mosses, lichens, grasses, and dwarf woody plants
Taiga	−10°C to 14°C	35–75 cm	low in nutrients; highly acidic	coniferous evergreen trees
Temperate forest	6°C to 28°C	75–125 cm	moist, moderately thick topsoils; moderate nutrient levels	broad-leaved deciduous trees and shrubs or evergreen coniferous trees
Tropical forest	20°C to 34°C	200–400 cm	moist, thin topsoil; low in nutrients	broad-leaved evergreen trees and shrubs
Temperate grassland	0°C to 25°C	25–75 cm	deep layer of topsoil; very rich in nutrients	dense, tall grasses in moist areas; short grasses in drier areas
Savanna	16°C to 34°C	75–150 cm	dry, thin topsoil; porous; low in nutrients	tall grasses and scattered trees
Chaparral	10°C to 18°C	<25 cm	rocky, thin topsoil; low in nutrients	evergreen shrubs and small trees
Desert	7°C to 38°C	<25 cm	dry, often sandy; low in nutrients	succulent plants and scattered grasses

TUNDRA

FIGURE 21-2

This herd of caribou lives in the treeless tundra in Canada. The photo shows the tundra in the summer, after the top layer of soil has thawed enough for plants to grow.



The **tundra** (TUHN-druh) is a cold and largely treeless biome that forms a continuous belt across northern North America, Europe, and Asia. It is the largest and northernmost biome, covering about one-fifth of the world's land surface. **Permafrost**, a permanently frozen layer of soil under the surface, characterizes the tundra. Even the surface soil above the permafrost remains frozen for all but about eight weeks of the year.

Trees do not usually grow in the tundra because the winters are long and extremely cold and permafrost prevents their roots from penetrating far into the soil. The tundra receives little precipitation and has a very short growing season of about two months. Cold temperatures slow down decay, so few nutrients are available in the soil. For these reasons, most tundra plants are small and grow slowly. Grasses, sedges, and mosses are common. Animals that inhabit the tundra include caribou, musk oxen, snowy owls, arctic foxes, lemmings, and snowshoe hares. Figure 21-2 shows some caribou in the Canadian tundra. In the brief summer, the upper layers of ice and soil thaw, creating ponds and bogs. Swarms of insects appear, and ducks, geese, cranes, other waterfowl, and predatory birds arrive in great numbers to feed.

FORESTS

Scientists have classified up to 26 types of forests worldwide. Forest biomes are divided into three main types: tropical, temperate, and boreal forests, or taiga.

Tropical Forests

Tropical forests occur near the equator, in the region between 23.5°N and 23.5°S, known as the *tropics*. This region includes parts of Asia, Africa, South America, and Central America. Stable temperature and abundant rainfall make tropical forests the most productive biome type. Tropical forests have only two seasons—one wet and one dry. *Tropical rain forests* are characterized by long wet seasons and tall trees and plants that grow year-round. *Tropical dry forests* have long dry seasons during which trees lose their leaves.

Competition for light is intense in a tropical rain forest. Most of the plants are trees, and some have evolved to grow as tall as 50 to 60 m (164 to 197 ft). The treetops form a continuous layer, called the **canopy**, which shades the forest floor. Though you may think of the tropical rain forest as an impenetrable jungle, much of the forest floor is relatively free of vegetation because so little sunlight reaches the ground. The very dense growth known as *jungle* is found along riverbanks and in disrupted areas where sunlight can reach the forest floor. Because of the intense competition for sunlight, many small plants live on the branches of tall trees. These plants are called **epiphytes** (EP-uh-FIETS) and include mosses, orchids and bromeliads, such as the one shown in Figure 21-3. Epiphytes use other organisms as support, but they are not parasitic because they make their own food.

Tropical forests have the highest species richness of all the biomes. One hectare of tropical rain forest (about the size of two football fields) may contain as many as 300 species of trees. An area of temperate deciduous forest of the same size, by contrast, would probably contain fewer than 12 species of trees. Animal life is also very diverse in the tropical rain forest. Rain-forest vertebrates include many kinds of monkeys, snakes, lizards, and colorful birds, such as parrots. Insect species are particularly diverse in tropical rain forests. There may be more than 1 million species of tree-dwelling beetles in the tropical rain forest biome alone. Overall, tropical rain forests probably contain about one-half of the world's species.

Temperate Forests

Temperate forests occur in eastern North America, western and central Europe, and northeastern Asia. This biome is characterized by distinct seasons and a moderate climate. Temperate forests can be characterized by the type of tree that is most common, such as **coniferous trees**, which bear seeds in cones and tend to be evergreen, or **deciduous trees**, which shed their leaves each year.



FIGURE 21-3

The tall trees in a rain forest are home to many epiphytic species, such as this bromeliad. The red structure in the middle of the bromeliad shown here is a fruiting structure similar to a pineapple.

Word Roots and Origins

epiphyte

from the Greek word *epi*, meaning “on” or “upon” and *phyton*, meaning “a plant”





FIGURE 21-4

A temperate deciduous forest is characterized by deciduous trees. Being deciduous is an adaptation that conserves the tree's energy and avoids freezing during winter.

Word Roots and Origins

taiga

from the Russian word *taiga*, meaning "forest"

FIGURE 21-5

Organisms of the taiga are adapted for dry, cold conditions and a reduced availability of food in the winter season. The conifers have needle-shaped leaves, which are adapted to conserve water.

Temperate Deciduous Forests

Trees that lose all of their leaves in the fall and regrow them each spring characterize **temperate deciduous forests**. Deciduous forests stretch across eastern North America, much of Europe, and parts of Asia and the Southern Hemisphere. These regions have pronounced seasons, with precipitation unevenly distributed throughout the year. Compared with taiga, discussed below, temperate deciduous forests have warmer winters and longer summers and receive more precipitation. Deciduous trees have broad, thin leaves with a large surface area that permits maximum light absorption. Familiar deciduous trees include the birch, beech, maple, oak, hickory, sycamore, elm, ash, willow, and cottonwood. Bears, wolves, white-tailed deer, foxes, raccoons, and squirrels are typical mammals of the temperate deciduous forests. Large areas of temperate deciduous forest in the United States, Europe, and Asia have been cut for timber or cleared to make way for farms, towns, and cities. Figure 21-4 shows a stand of trees in a temperate deciduous forest.

Taiga

South of the tundra and north of the temperate regions is the **taiga** (TIE-guh), a forested biome dominated by coniferous trees, such as pines, firs, and spruces. Taiga, also called *boreal forest*, stretches across large areas of northern Europe, Asia, and North America between 50°N and 60°N. During the long winter, snow covers and insulates the ground, protecting tree roots against freezing.

Plants living in the taiga are adapted for long, cold winters; short summers; and nutrient-poor soil. On a coniferous tree, the waxy, needle-shaped leaves remain on the tree all winter long. The shape of the needle is a leaf adaptation that reduces water loss, because the small holes through which the leaves exchange air are partially enclosed in the needle. Typical mammals of this biome include moose, bears, wolves, lynxes, and hares. Animals that are adapted to survive the winter may stay in the taiga year-round, but others migrate to warmer climates in the fall and return in the spring. Many species hibernate six to eight months of the year. Figure 21-5 shows a representative area of taiga.



GRASSLANDS

Grasslands are, as the name suggests, dominated by a variety of grasses. Grasslands are known by different names in different parts of the world: *prairies* in North America, *steppes* in Asia, *pampas* in South America, and *veldts* in southern Africa.

Temperate Grasslands

Temperate grasslands usually form in the interior of continents, at about the same latitude as temperate deciduous forests. However, rainfall patterns make these areas too dry to support trees. This biome once covered large areas of North America, Asia, Europe, Australia, and South America.

Temperate grasslands have rich, fertile soil. In areas that have remained relatively undisturbed by humans, grasslands support large herds of grazing mammals, such as the North American bison shown in Figure 21-6a. Grass can survive repeated grazing by animals and occasional fires that sweep across the area because the actively growing part of the plant is at or below the ground rather than at the tip of the stem. Because grasslands have such rich soil, much of the world's temperate grassland has been transformed into farmland for growing crops such as wheat and corn. Only fragments of undisturbed prairie remain in the midwestern United States.

Savanna

Savannas (suh-VAN-uhz) are tropical or temperate grasslands that have scattered deciduous trees and shrubs. The savannas of Africa are the best known, but this biome also occurs in South America and Australia. Savannas receive more rainfall than deserts do but less rainfall than tropical or temperate forests do. Alternating wet and dry seasons characterize savannas. Like temperate grasslands, savannas support large numbers of herbivores, such as zebras, wildebeests, giraffes, and gazelles, as Figure 21-6b shows. Large carnivores, such as lions, leopards, and cheetahs, feed on these herbivores.

Because most of the rain falls during the wet season, the plants and animals of the savanna must be able to deal with prolonged periods of drought. Some trees of the savanna shed their leaves during the dry season to conserve water, and the above-ground parts of grasses often die during the dry season and regenerate after a period of rain.

Chaparral

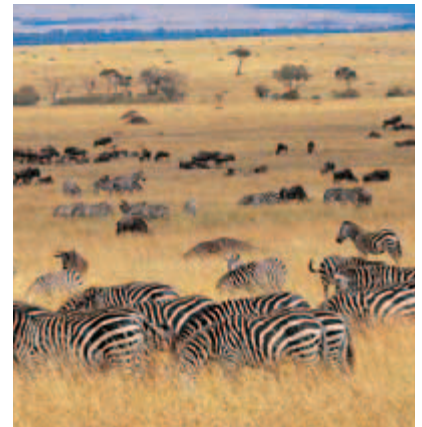
Chaparral (SHAP-uh-RAL) is a biome that is dominated by dense, spiny shrubs and has scattered clumps of coniferous trees, as Figure 21-6c shows. Chaparral is characterized by mild, rainy winters and hot, dry summers with periodic fires. This biome is located in the middle latitudes, about 30° north and 30° south of the equator. Chaparral is found primarily in coastal regions—for example, around the Mediterranean Sea and in southern California.

FIGURE 21-6

Temperate grasslands once covered a large portion of the United States and supported huge herds of herbivores, such as bison (a). An area rich in wildlife, the savanna biome supports great herds of large herbivores (b). The chaparral biome supports grasses, shrubs, and clumps of small trees (c).



(a) Temperate grassland



(b) Savanna



(c) Chaparral

DESERTS

FIGURE 21-7

All of the organisms of the desert biome are adapted to endure dry, often hot conditions by conserving energy and water. This saguaro cactus is full of water, which it has stored from the infrequent rains in the desert.



Deserts are areas that receive an average of less than 25 cm (9.9 in.) of rainfall per year. Large parts of North Africa, central Australia, southwestern North America, and eastern Asia are desert. Contrary to popular belief, deserts are not hot all the time. So-called cold deserts, such as the Great Basin in the western United States and the Gobi in eastern Asia, are hot in summer but cold in winter. Even in hot deserts, temperatures may fall by as much as 30°C (54°F) at night because the dry air is a poor insulator, allowing the heat that builds up during the day to escape.

Desert vegetation is often sparse and consists mainly of plants that have adapted to the dry climate. The leaves of the creosote bush of the United States and Mexico, for example, have a waxy coating that reduces evaporation. Plants in moist environments normally allow a great deal of water to escape through evaporation and transpiration from *stomata*, tiny openings in their leaves. In contrast, desert plants may have few stomata or may open the stomata only at night. A common type of desert plant is a cactus, such as the saguaro cactus shown in Figure 21-7. It has an expandable body that can store water and leaves that have evolved into sharp spines that protect the plant from herbivores. A single saguaro can hold about 1,000 kg (2,200 lb) of water. The saguaro cactus is found in Arizona, California, and Mexico.

Like desert plants, desert animals must conserve water. Many animals avoid the heat of the day by hiding in small spots of shade or by burrowing into the ground. Others, such as kit foxes and some kinds of lizards and snakes, are active only at night, when loss of water to evaporation is low.

SECTION 1 REVIEW

1. List at least one animal that lives in each of the eight major biomes.
2. Compare the level of biodiversity in a tropical rain forest with that in a temperate deciduous forest.
3. Why do so few trees grow in the tundra?
4. What other name is used for grasslands in North America?
5. Where in the world could you travel to see savanna? chaparral?
6. Describe three adaptations of desert organisms for water conservation.

CRITICAL THINKING

7. **Evaluating Information** For deciduous trees, compare the benefits and possible disadvantages of shedding leaves in the fall.
8. **Comparing Concepts** Discuss the similarities between temperate grassland and savanna.
9. **Comparing Concepts** Discuss the differences between tundra and taiga.
10. **Calculating Information** Summarize the conditions that determine which type of biome is present in a particular location. Provide at least two abiotic factors and one biotic factor.

AQUATIC ECOSYSTEMS

As terrestrial organisms, we tend to focus on the other land-dwelling organisms we see around us. But on the geologic time scale, life seems to have originated and existed in the oceans long before it came onto land. Water covers about three-fourths of Earth and is home to a variety of organisms.

OCEAN ZONES

The ocean covers about 70 percent of Earth's surface and has an average depth of 3.7 km (2.3 mi). The deepest parts of the ocean are about 11 km (6.8 mi) deep. The water contains about 3 percent salt, mostly sodium chloride, a factor that strongly affects the biology of the organisms that live there.

Another important factor affecting marine organisms is the availability of light. Most of the ocean is cold and dark. This zone, where sunlight cannot penetrate and photosynthesis cannot occur, is called the **aphotic** (ay-FOHT-ik) **zone**. Because water absorbs light, sunlight penetrates only the upper few hundred meters of the ocean. The part of the ocean that receives sunlight is the **photic** (FOHT-ik) **zone**. The depth of the photic zone varies.

Ecologists recognize three zones relative to the ocean's edges, as illustrated in Figure 21-8. The **intertidal zone** is the area of shoreline that is twice daily covered by water during high tide and exposed to air during low tide. Farther out is the **neritic** (nee-RIT-ik) **zone**, which extends from the intertidal zone over the continental shelf and to relatively shallow water depths of about 180 m.

OBJECTIVES

- **Identify** the major ocean zones.
- **Compare** the aphotic zone with the photic zone.
- **Compare** the neritic zone with the oceanic zone.
- **Describe** estuaries.
- **Compare** eutrophic lakes with oligotrophic lakes.
- **Explain** the significance of gradient in rivers and streams.
- **Describe** freshwater wetlands.

VOCABULARY

aphotic zone
 photic zone
 intertidal zone
 neritic zone
 oceanic zone
 pelagic zone
 benthic zone
 plankton
 estuary
 eutrophic lake
 oligotrophic lake
 freshwater wetland

FIGURE 21-8

This diagram shows the major ecological zones of the ocean.

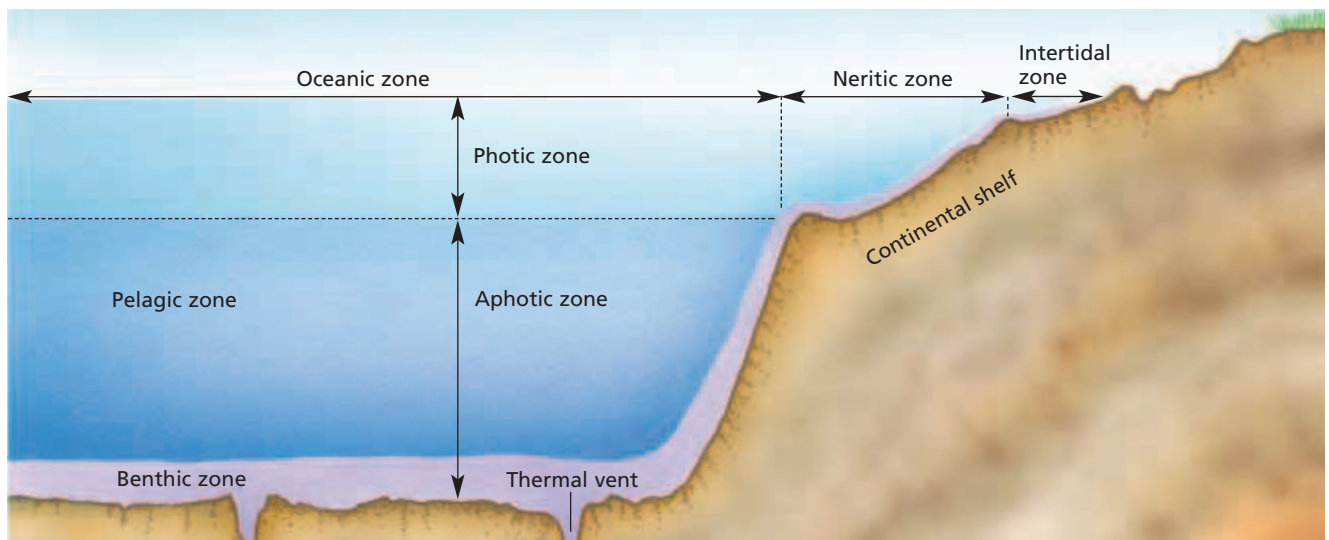




FIGURE 21-9

The intertidal zone shown here is similar to intertidal zones all over the planet. A sandy beach is part of the intertidal zone, but it is usually distinguished from the rocky intertidal zones and not as rich in species as the rocky intertidal zones are.

FIGURE 21-10

Coral reefs, such as the one shown here, are among the most diverse ecosystems on Earth. Coral reefs are extremely sensitive to pollution. Many reefs around the world are threatened, and many have been damaged or destroyed as a result of human activities.

Beyond the continental shelf is the **oceanic zone**, which is the deep water of the open sea. The neritic and oceanic zones are further divided. The open ocean is known as the **pelagic** (pi-LAJ-ik) **zone**, and the ocean bottom is known as the **benthic zone**.

The Intertidal Zone

Organisms in the intertidal zone are adapted to periodic exposure to air during low tide. Crabs avoid dehydration by burrowing into the sand or mud. Clams, mussels, and oysters retreat into their shells at low tide. Organisms living in the intertidal zone must be able to withstand the force of crashing waves. Sea anemones cling to rocks with a muscular disk, and sea stars use tube feet to adhere to surfaces. Figure 21-9 shows some of the organisms living in the rocky intertidal zone of the California coast.

The Neritic Zone

The neritic zone is the most productive zone in the ocean, supporting more species and numbers of organisms than any other zone. The water throughout most of the neritic zone is shallow enough for photosynthesis to occur. Strong currents called *upwellings* carry nutrients from the ocean bottom and mix them with nutrients contained in runoff from land. These waters are rich in **plankton**, communities of small organisms that drift with the ocean currents. Plankton is consumed by many larger animals. Numerous fishes, sea turtles, and other animals also live in these waters.

Coral reefs form in the neritic zone of tropical areas. Like tropical rain forests, coral reefs are very productive and rich in species. A coral reef is built by coral animals over a long time period. They construct external skeletons of calcium carbonate. As the animals grow and die, the skeletons accumulate to form the base of a reef, as shown in Figure 21-10. A reef is home to many species of fishes, crustaceans, mollusks, and other animals. Most species of reef-forming coral have a mutualistic relationship with photosynthetic protists, from which they receive food.



The Oceanic Zone

The oceanic zone encompasses the deepest parts of the ocean and contains fewer species than the neritic zone does. Even in photic areas, nutrient levels are too low to support as much life. Although the productivity per square meter of open ocean is very low, the total productivity of the oceanic zone is high because the ocean covers such a vast area. About half of the photosynthesis that occurs on Earth takes place in the oceanic zone. The producers of the upper parts of the oceanic zone are microscopic protists, bacteria, plants, and invertebrates in the plankton. Animals include fishes, mammals such as whales, and many large invertebrates.

In the aphotic zone, animals feed primarily on sinking plankton and dead organisms. Organisms living deep in the ocean must cope with near-freezing temperatures and crushing pressure. Deep-sea organisms, such as the squid shown in Figure 21-11, have slow metabolic rates and reduced skeletal systems. Fishes in these depths have large jaws and teeth and expandable stomachs that can accommodate the rare prey that they can catch.

In the 1970s, scientists found diverse communities living near volcanic vents 2,500 m (8,200 ft) below the surface. These vents release water that is rich in minerals and often exceeds 750°C. The producers for this ecosystem are chemosynthetic bacteria that use energy contained in hydrogen sulfide, H_2S , from the vents. Unique clams, crabs, and worms feed on these bacteria. Tube worms living near the vents have lost their digestive system over evolutionary time and receive all of their food directly from chemosynthetic bacteria living in their bodies.

Estuaries

An **estuary** (ES-tyoo-er-ee) occurs where freshwater rivers and streams flow into the sea. Examples of estuary communities include bays, mud flats, mangrove swamp forests, and salt marshes. The shallow water receives plenty of light, and rivers deposit large amounts of mineral nutrients. However, the interaction between fresh water and salt water causes great variation in temperature and salinity. In addition, like the surface of the intertidal zone, the surface of an estuary is partly exposed during low tide. Inhabitants of estuaries are adapted for frequent change. For example, some kinds of mangrove trees have special glands on their leaves that eliminate excess salt water taken up by the roots. Softshell clams lie buried in mud with only their long siphons protruding above the surface. The siphon filters plankton from the salt water at high tide and detects predators at low tide, contracting whenever it senses danger. Figure 21-12 shows an example of an estuary.



FIGURE 21-11

Organisms in the deep ocean have many adaptations to their environment. The squid in the photograph, *Vampyroteuthis* sp., is adapted to eating a large quantity of food at once because prey is hard to find.

FIGURE 21-12

Estuaries are almost as rich in species as tropical rain forests are. Estuaries serve as stopovers and nurseries for both terrestrial and marine organisms. Organisms that spend at least part of their lives in estuaries include shrimp, oysters, mullet, redfish, anchovies, striped bass, and other animals that people consume as food.



RESTORING AN ECOSYSTEM: Molasses Reef

For years, Molasses Reef looked like a colorful carpet of living coral with thousands of tropical fish species darting in and out of the reef, as the photo at the bottom of the page shows. This barrier reef stretches for miles south-east of Key Largo, Florida, and is part of the third-largest reef system in the world.

Disaster Strikes the Reef

The *Wellwood*, a cargo ship measuring 366 feet long, wrecked on Molasses Reef on August 4, 1984. It took 12 days for tugboats to remove the ship with steel cables, as the photo below shows. This motion crushed large areas of coral. Furthermore, the ship created shade over the coral, starving its symbiotic photosynthesizers for days. As a result, the shaded coral lost its living layer and lost its color, a condition known as

bleaching. Animals such as fishes and crabs left the area. By the end of summer, one-half of a square mile of Molasses Reef was pulverized, and an additional 6 square miles were damaged.

Restoration and Stabilization of Molasses Reef

The owners of the *Wellwood* agreed to fund the restoration of Molasses Reef. Soon after the accident, divers put some of the fallen coral back in place. They later transplanted several species of living coral from nearby reef sites. The most extreme repair involved filling large holes and cracks in the reef with concrete and building 22 structures to restore the basic shape of the reef. Each structure weighed more than 3,500 pounds. Ultimately, biologists hoped to restore the appearance of the reef and

stabilize the biological community. More than 20 years later, marine biologists are seeing the regrowth of many, but not all, of the original coral species.

Future Protection of the Florida Keys Reef Ecosystem

People can damage reef systems in other ways as well. More than 3 million people visit the Keys each year, and about 80,000 people live there. The U.S. Coast Guard and the National Oceanic and Atmospheric Administration have ongoing goals for protecting Molasses Reef, which is now part of the Florida Keys National Marine Sanctuary. One goal is to control sewage runoff. Another goal is to educate residents and visitors about damage from boats, divers, fishers, and beachgoers. Programs are underway to protect this unique ecosystem.



In the 12 days it took to remove the wrecked Wellwood from Molasses Reef, much of the reef community was lost. But efforts to restore the community are beginning to have positive results.

REVIEW

1. What events led to coral reef bleaching at Molasses Reef?
2. Why is education important in protecting Molasses Reef?
3. **Critical Thinking** Do you think boats should be allowed in a marine sanctuary? Explain your answer.



FRESHWATER ZONES

Low levels of dissolved salts characterize freshwater ecosystems. The salt content of fresh water is about 0.005 percent. Examples of freshwater ecosystems include lakes, ponds, freshwater wetlands, clear mountain streams, and slow, sediment-rich rivers.

Lakes and Ponds

Ecologists divide lakes and ponds into two categories. **Eutrophic** (yoo-TRAHF-ik) **lakes** are rich in organic matter and vegetation, so the waters are relatively murky. As the number of plants and algae in a lake grows, the number of bacteria feeding on decaying organisms also grows. These bacteria use the oxygen dissolved in the lake's waters. Eventually, the reduced amount of oxygen kills organisms that need oxygen to survive. Lakes naturally become eutrophic over a long period of time.

In contrast, **oligotrophic** (AHL-i-goh-TRAHF-ik) **lakes** contain little organic matter. The water is much clearer, and the bottom is usually sandy or rocky. Fishes inhabit both eutrophic and oligotrophic lakes. Freshwater lakes and ponds support a variety of other animals, including mammals such as the otter and muskrat, birds such as ducks and loons, reptiles such as turtles and snakes, and amphibians such as salamanders and frogs.

Rivers and Streams

A river or stream is a body of freshwater that flows down a gradient, or slope, toward its mouth, as shown in Figure 21-13. Water flows swiftly down steep gradients, and organisms are adapted to withstand powerful currents. For example, the larvae of caddis flies and the nymphs of mayflies cling to the rocky bottom, and brook trout and other fishes have evolved the strength to swim upstream. Slow-moving rivers and their backwaters are richer in nutrients and therefore support a greater diversity of life. Rooted plants and the fishes that feed on them are adapted to the weaker currents of slow-moving rivers.



(a)



(b)



Quick Lab

Comparing Organisms in Terrestrial and Aquatic Biomes

Materials paper, scissors, magazines, glue

Procedure Cut out and collect images of plants, animals, and other organisms from magazines. Sort out the images based on which major biome each organism would live in. Develop a collage of each major biome.

Analysis Of the images you collected, which organisms live in terrestrial biomes (specify the type of terrestrial biome)? in aquatic biomes (specify the type)? What types of adaptations do each of the organisms have that allow them to live in the biome?

FIGURE 21-13

Streams in mountain areas flow rapidly down steep slopes. These streams usually have rocky bottoms and clear water (a). Rivers that flow slower carry more fine sediment and commonly have muddy bottoms (b).



FIGURE 21-14

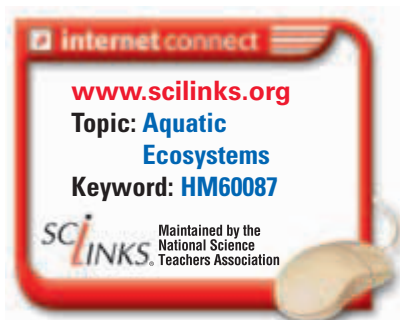
The American alligator, *Alligator mississippiensis*, is the largest reptile in North America. It lives in marshes and swamps in Florida, Louisiana, South Carolina, Georgia, Alabama, and Texas. The alligator is protected by law because of its similarity to the very rare American crocodile.

Freshwater Wetlands

Freshwater wetlands are areas of land that are covered with fresh water for at least part of each year. The two main types of freshwater wetlands are *marshes* and *swamps*. Marshes contain non-woody plants, such as cattails, and swamps are dominated by woody plants, such as trees and shrubs. Another type of freshwater wetland is a *bog*, which is dominated by sphagnum mosses.

Freshwater wetlands are the most productive freshwater ecosystem. They contain a wide variety of birds, fishes, mammals, amphibians, invertebrates, and reptiles. Wetlands in the United States are home to rare, large predators such as the whooping crane, the Florida panther, the American crocodile, and the American alligator, shown in Figure 21-14. Many wetlands are important as stopovers for migratory birds. Wetlands, like estuaries, provide protection for spawning organisms, such as fishes. Wetlands are also important to people economically and environmentally. Wetlands act as filters to clean pollutants out of the water flowing through them and also act as flood control when they absorb large quantities of water that could otherwise flood homes, farms, and businesses.

For many decades, wetlands have been diverted, drained, and filled in. But now, the vital importance of wetlands to the economic and environmental health of many areas is being recognized. In some cases, government agencies and private organizations are working together to protect wetlands and to restore areas that have been damaged. One of the most important wetlands in the world is the Everglades National Park in southern Florida. There is currently a major effort to restore the flow of water through the Everglades, upon which the entire ecosystem depends. The Everglades has been designated a World Heritage Site, an International Biosphere Reserve, and a Wetlands of International Importance.



SECTION 2 REVIEW

1. Distinguish between the photic and aphotic zones.
2. Describe the intertidal zone, and give examples of organisms living there.
3. Describe the differences between the neritic and oceanic zones in the ocean.
4. What are two sources of nutrients in the neritic zone?
5. What role do chemosynthetic bacteria play in deep-sea volcanic-vent ecosystems?
6. What is the main difference between oligotrophic and eutrophic lakes?

CRITICAL THINKING

7. **Applying Information** What could happen to the organisms that live in a fast-moving river if a dam were built on the river?
8. **Evaluating Information** Estuaries serve as breeding sites and nurseries for thousands of species of marine animals. What are the possible advantages and disadvantages to the organisms using estuaries for these purposes?
9. **Drawing Conclusions** Why are freshwater wetlands considered to be economically and environmentally important?

CHAPTER HIGHLIGHTS

SECTION 1

Terrestrial Biomes

- The major types of terrestrial ecosystems, known as biomes, are tundra, tropical forest, temperate forest, taiga, temperate grassland, savanna, chaparral, and desert.
- Tundra is a cold biome characterized by permafrost under the surface of the ground.
- Tropical forests receive abundant rainfall and have stable temperatures. They have a greater species richness than any other biome.
- Temperate biomes have four annual seasons. The trees in temperate deciduous forests shed all of their leaves in the fall.
- Taiga is cold but is warmer than tundra and receives more precipitation. Taiga is dominated by coniferous forests.
- Temperate grasslands occur in areas with cold winters and hot summers. They are dominated by grasses and herds of grazing animals.
- Savannas are tropical grasslands with alternating wet and dry seasons. They are dominated by herds of grazing animals.
- Chaparral is found in coastal regions with warm, dry summers and mild winters. It is dominated by dense, spiny shrubs.
- Deserts receive less than 25 cm (9.9 in.) of precipitation per year. Desert inhabitants have adaptations for conserving water.

Vocabulary

biome (p. 417)
tundra (p. 418)
permafrost (p. 418)
tropical forest (p. 419)

canopy (p. 419)
epiphyte (p. 419)
coniferous tree (p. 419)
deciduous tree (p. 419)

temperate deciduous
forest (p. 420)
taiga (p. 420)
temperate grassland (p. 421)

savanna (p. 421)
chaparral (p. 421)
desert (p. 422)

SECTION 2

Aquatic Ecosystems

- The photic zone in the ocean receives light, but the aphotic zone does not. Other zones of the ocean are defined based on their relative locations.
- In the intertidal zone, organisms must be able to tolerate drying and pounding by waves.
- The neritic zone receives nutrients from the bottom of the ocean and from land. It is the ocean's richest zone in terms of the number of species and individuals.
- Production in the oceanic zone is limited by a shortage of nutrients.
- Estuaries are very productive areas where rivers and streams flow into the sea.
- Oligotrophic lakes are clear and lacking in nutrients. Eutrophic lakes are rich in nutrients and are often murky.
- Rivers and streams are bodies of water that flow down an elevation gradient within a watershed.
- Freshwater wetlands are areas of land, such as marshes and swamps, that are covered with fresh water for at least part of each year.

Vocabulary

aphotic zone (p. 423)
photic zone (p. 423)
intertidal zone (p. 423)

neritic zone (p. 423)
oceanic zone (p. 424)
pelagic zone (p. 424)

benthic zone (p. 424)
plankton (p. 424)
estuary (p. 425)

eutrophic lake (p. 427)
oligotrophic lake (p. 427)
freshwater wetland (p. 428)

CHAPTER REVIEW

USING VOCABULARY

- For each pairs of terms, explain how the meanings of the terms differ.
 - savanna* and *desert*
 - tundra* and *taiga*
 - photic zone* and *aphotic zone*
 - estuary* and *freshwater wetland*
- For each pair of terms, explain the relationship between the terms.
 - tropical rain forest* and *canopy*
 - tundra* and *permafrost*
 - neritic zone* and *plankton*
- Word Roots and Origins** The word *epiphyte* is derived from the Greek *phytos*, which means “plant,” and the prefix *epi-*, which means “on top of.” Using this information, explain why the term *epiphyte* is a good name for the type of organism than the term describes.

UNDERSTANDING KEY CONCEPTS

- Explain** why tundra soil is poor in nutrients.
- Describe** a tropical rain forest.
- Explain** how the destruction of tropical rain forests can contribute to an increase in carbon dioxide levels in the atmosphere.
- Compare** the dominant vegetation of temperate deciduous forests with that of taiga.
- Identify** the adaptive function of the needle-like leaves on a coniferous tree.
- Contrast** a temperate grassland with a savanna.
- Describe** the typical plants and soil of a desert.
- List** the approximate latitudes where the following biomes are mostly located: tropical rain forests, tundra, and chaparral.
- Identify** which terrestrial biomes are the most productive.
- Identify** the major zones in the oceans.
- Identify** which ocean zones are most productive.
- Explain** why the oxygen content of eutrophic lakes decreases over time.
- Explain** why estuaries are more productive than most other biomes.
- Explain** how gradient affects rivers and streams.
-  **Write** a report summarizing how artificial aquatic ecosystems are used to eliminate pollutants from wastewater. Find out which kinds of modern urban facilities use artificial ecosystems.

-  **CONCEPT MAPPING** Use the following terms to create a concept map that shows how biomes can be classified based on climate: *desert*, *savanna*, *tropical rain forest*, *temperate grassland*, *temperate deciduous forest*, *tundra*, *taiga*, *climate*, *arid*, *cold*, *temperate*, *tropical*, *hot*, *dry*, *trees*, and *grass*.

CRITICAL THINKING

- Interpreting Graphics** The rare species of squid in the photo below has several adaptations for living in very deep water. Explain some of the selective pressures that exist at great depths.



- Analyzing Information** Which biome occurs in all of the following locations: Spain, Australia, Canada, and the northeastern United States?
- Evaluating Information** Which soil is better suited for agricultural use: the soil of a tropical rain forest or the soil of a temperate deciduous forest? Explain your answer.
- Inferring Relationships** The organisms found near deep-sea volcanic vents do not use sunlight to get energy. What are possible energy sources for these organisms?
- Predicting Results** Would a herd of large herbivores be able to survive for many generations in a desert biome? Explain your answer.
- Evaluating Viewpoints** Imagine that a local government is debating whether a large freshwater wetland should be drained and filled in so that a new road can be built or whether the area should be turned into a wildlife refuge. Both of these options will have costs and benefits to the community. Draft some arguments that could be made for each side of the debate.

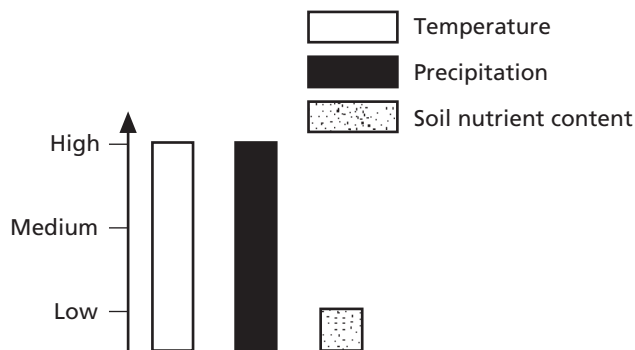


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Why are estuaries more productive than most other biomes?
 - Estuaries contain vast coniferous forests.
 - Estuaries have shallow, nutrient-laden water.
 - Estuaries get more sunlight than other biomes.
 - The majority of land on Earth is covered by estuaries.
- Which of the following characterizes the neritic zone of the ocean?
 - It receives little sunlight.
 - It supports very few species.
 - It is exposed to the air by low tide.
 - It receives nutrients washed from land.
- Which of the following is true of temperate deciduous forests?
 - They are found near the equator.
 - They have the lowest rainfall of any biome.
 - They undergo seasonal changes in temperature.
 - They have the highest species richness of any biome.
- Which of the following best describes the water of all eutrophic lakes?
 - cold
 - salty
 - murky
 - lifeless

INTERPRETING GRAPHICS: The graph below shows the relative temperature, precipitation, and soil nutrient content in a specific biome. Use the graph to answer the question that follows.



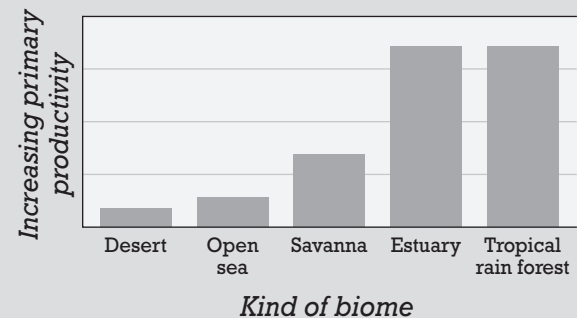
- Which of the following biomes is best represented by this graph?
 - tundra
 - desert
 - tropical rain forest
 - temperate grassland

DIRECTIONS: Complete the following analogy.

- grasses : savanna :: coniferous trees :
 - taiga
 - tundra
 - desert
 - temperate deciduous forest

INTERPRETING GRAPHICS: The graph below ranks several types of biomes in terms of their relative productivity. Use the graph to answer the question that follows.

Comparative Productivity of Ecosystems



- If a scientist needed to add a bar representing the temperate grassland biome to this graph, where should the bar be placed?
 - to the left of desert
 - between savanna and estuary
 - between estuary and tropical rain forest
 - to the right of tropical rain forest

SHORT RESPONSE

Biomes are very large climatic regions that contain a number of smaller but related ecosystems.

Describe how the major biomes are characterized.

EXTENDED RESPONSE

Ecologists recognize the following ecological zones in the ocean: *photic*, *aphotic*, *intertidal*, *neritic*, *oceanic*, *pelagic*, and *benthic*.

Part A Describe the factors used as a basis for distinguishing the zones.

Part B Relate these factors to the types of organisms that inhabit each zone.

Test TIP

Allow a few minutes at the end of the test-taking period to check for mistakes in marking answers.

Constructing and Comparing Ecosystems

OBJECTIVES

- Observe the interaction of organisms in a closed ecosystem.
- Compare this ecosystem with others observed in nature.

PROCESS SKILLS

- observing
- recognizing relationships
- hypothesizing

MATERIALS

- large glass jar with lid for each ecosystem
- pond water or dechlorinated tap water
- gravel, rocks, and soil
- graph paper
- $8\frac{1}{2} \times 11$ in. acetate sheets
- several colored pens for overhead transparencies

Ecosystem 1

- pinch of grass seeds
- pinch of clover seeds
- 10 mung-bean seeds
- 3 earthworms
- 4 to 6 isopods (pill bugs or sow bugs)
- 6 mealworms (darkling beetle larvae)
- 6 crickets

Ecosystem 2

- 5 to 15 strands of aquatic plants
- 1 ounce duckweed
- culture or mixture of phytoplankton
- 1 or 2 ramshorn snails
- 2 to 4 guppies, platys, or comet goldfish

Ecosystem 3

- small, clear glass container for inner water
- 3 to 7 strands of aquatic plants
- culture or mixture of phytoplankton
- 10 *Daphnia* sp. (water fleas)

- 2 to 4 pond snails (Lymnaeidae family)
- 2 to 4 guppies, platys, or comet goldfish
- pinch of grass seeds
- pinch of clover seeds

Background

1. How are living things affected by nonliving things in the environment?
2. Relate the theme of interdependence among organisms to ecosystems.
3. How do different types of organisms interact?

Experiment Setup

1. In this investigation, you will observe organisms in one of three different ecosystems. Look at the organisms listed in the materials list, and choose the ecosystem you would like to observe. Which organisms might be the most numerous? Which organisms might decrease in number? Hypothesize how the organisms will interact.
2. Form a group with classmates who have all chosen the same ecosystem. As a group, prepare the environment in a jar with a chosen substrate. Go to Part A, Part B, or Part C for set-up instructions for your ecosystem.
3. Put lids on the containers, and let the ecosystem remain undisturbed in indirect sunlight for a week. Go to Part D after one week.

PART A Setting Up Ecosystem 1

1. Place the soil in the bottom of the jar about 4 or 5 in. deep. Clean soil off the inside of the jar.
2. Place rocks in and on top of the soil in a natural-looking arrangement.
3. Moisten the soil carefully. Do not saturate the soil.
4. Plant the seeds according to the package instructions.
5.   Clean up your materials, and wash your hands before leaving the lab.



PART B Setting Up Ecosystem 2

6. Place about 2 in. of clean gravel in the bottom of the jar.
7. Gently fill the jar with pond water or dechlorinated tap water until the jar is about three-fourths full.
8. Add the plants and algae to your ecosystem.
9.   Clean up your materials, and wash your hands before leaving the lab.

PART C Setting Up Ecosystem 3

10. Place a small, clear glass container, such as a mayonnaise jar, inside the larger jar and against one side.
11. Carefully place soil in the larger jar about 4 or 5 in. deep, filling around the smaller jar so that you can see through both jars on one side. Do not place any soil on the side where the two jars touch. Clean any soil from the sides of the jar that may obstruct your view.
12. Place some rocks on the soil, and moisten the soil carefully.
13. Place 1 in. of gravel inside the smaller jar, and gently fill with pond water or dechlorinated tap water to the level of the soil.
14. Place the algae and the aquatic plants in the water. Plant the grass seeds and clover seeds in the soil in separate areas of the jar.
15.   Clean up your materials, and wash your hands before leaving the lab.

PART D Observing Your Ecosystem

16. Place the chosen animals in the jar, and loosely replace the lid.
17. Observe the jar for a few minutes daily.
18. Make a chart in your lab report to record the original number of each species in your ecosystem. Record daily any changes you observe.
19. Make a graph for each species in your chart, and plot the number of organisms as a function of time. Place a clear acetate sheet over each graph. Using a pen used for overhead transparencies, trace each graph onto the acetate sheet. Use a different colored pen and a different acetate sheet for each organism.
20. Compare the acetate sheets of two organisms that you hypothesized would interact—a predator and its prey, for example. Hold one sheet on top of the other, and analyze both graphs. Record the results in your lab report.
21.   Clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

1. What happened to the organisms in your ecosystem? How did their population sizes change?
2. What are some possible causes of the change in the populations you observed?
3. Construct a food chain or food web for the ecosystem you observed. Which organisms were producers? Which organisms were consumers?
4. What could you learn if you set up more than one jar in an identical manner?
5. How does your ecosystem resemble a natural ecosystem? How does it differ?
6. How did your observation compare with your hypotheses? If the results differed from what you expected, explain what might have caused the difference.
7. Look at your graphs, and determine what kind of relationship exists between predator and prey populations.
8. How would you modify the ecosystem if you were to repeat this investigation?

Further Inquiry

Develop an experiment to study the effects of abiotic factors—such as temperature, light, and moisture—on the organisms in the ecosystem you constructed.

HUMANS AND THE ENVIRONMENT

This aircraft is leading a flock of whooping cranes in order to teach the birds to migrate between Wisconsin and Florida. The whooping crane, *Grus americana*, is a species that has almost disappeared because of human actions. But scientists and others have made efforts to save this species. Their efforts have included protecting the cranes with laws, preserving the cranes' habitat, breeding the cranes in captivity, and then teaching the cranes to migrate.



SECTION 1 *An Interconnected Planet*

SECTION 2 *Environmental Issues*

SECTION 3 *Environmental Solutions*



Unit 7—Ecosystem
Dynamics
Topics 1–6

AN INTERCONNECTED PLANET

*The study of the interaction between humans and their own environment is **environmental science**. Like all other organisms, humans depend on their environment for food, water, air, shelter, and other resources. Earth and its inhabitants are interconnected in complex ways. However, the ability of humans to understand, manipulate, and possibly damage ecosystems exceeds that of other organisms. Thus, environmental science may be critical to our own survival.*

EARTH'S LAYERS

One way to understand our unique planet is to think of Earth as a giant sphere made up of layers, or spheres within spheres. Scientists describe three major layers, as Figure 22-1 shows. The **geosphere** is Earth's rock interior that extends from the molten center of the planet's core to the solid surface of its crust. Above the geosphere lies the **hydrosphere**, the portion of Earth that is water, including all oceans, lakes, and rivers. Above the hydrosphere and geosphere floats the **atmosphere**, the mixture of gases that surrounds Earth. Living organisms are known to exist in parts of each of Earth's three main spheres. Thus, scientists also refer to the **biosphere**, meaning the part of Earth where life exists.

OBJECTIVES

- **Explain** the importance of the study of environmental science.
- **Describe** Earth's major layers.
- **Explain** the natural functions of the ozone layer and greenhouse effect.
- **Summarize** the ways in which biologists measure biodiversity.
- **Discuss** the value of biodiversity.

VOCABULARY

environmental science

geosphere

hydrosphere

atmosphere

biosphere

ozone layer

greenhouse effect

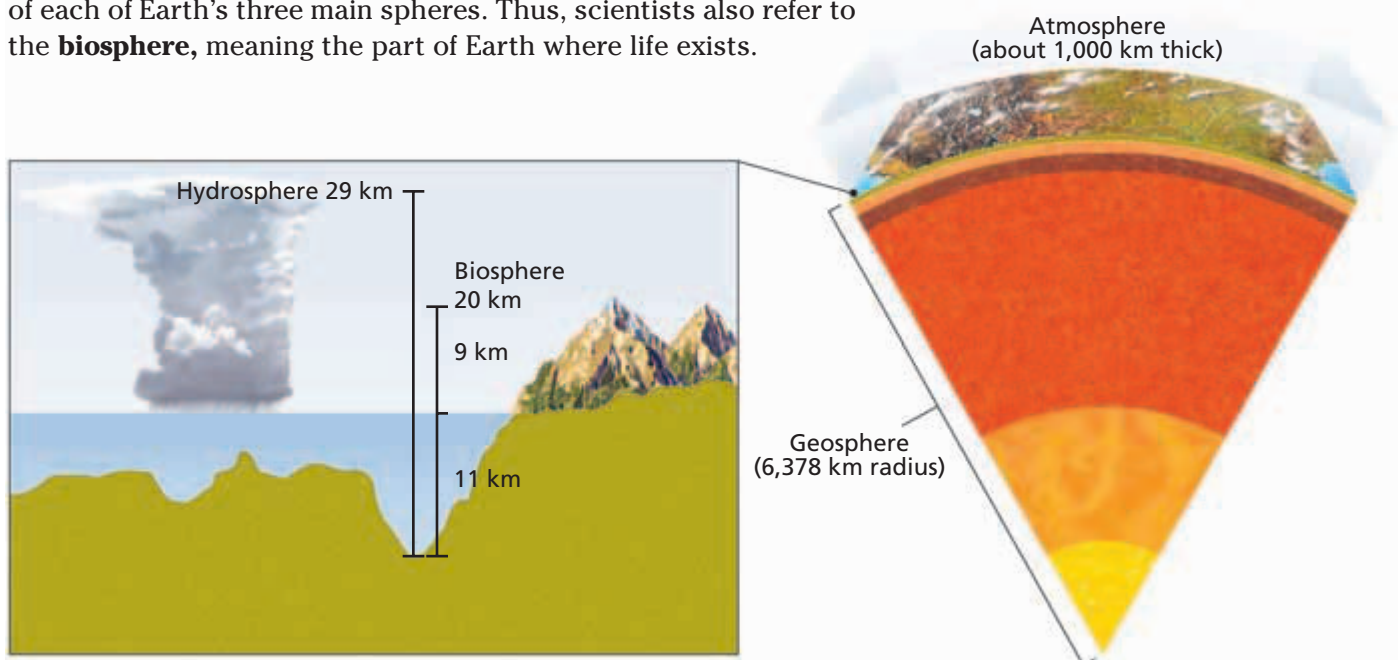
biodiversity

species diversity

genetic diversity

FIGURE 22-1

Layers of air (atmosphere), water (hydrosphere), and rock and soil (geosphere) interact with each other. Living things make up the biosphere, which includes parts of each of these three layers.



Word Roots and Origins

ozone

from the Greek *ozein*, meaning
"to smell"

Atmosphere

The atmosphere consists of about 78 percent nitrogen, N_2 , about 21 percent oxygen, O_2 , and about 1 percent other gases, including water vapor, H_2O , and carbon dioxide, CO_2 . Atmospheric gases become less dense at greater altitudes. For example, the density (number of molecules per liter) of oxygen at 18,000 ft is half of its density at sea level. While the density of each gas decreases with altitude, the relative proportion of each gas remains about the same. An exception is ozone (OH-zohn), O_3 , a naturally occurring gas that is vital to life on Earth. The concentration of ozone is greatest in the **ozone layer**, a region that is about 20 km above Earth's surface.

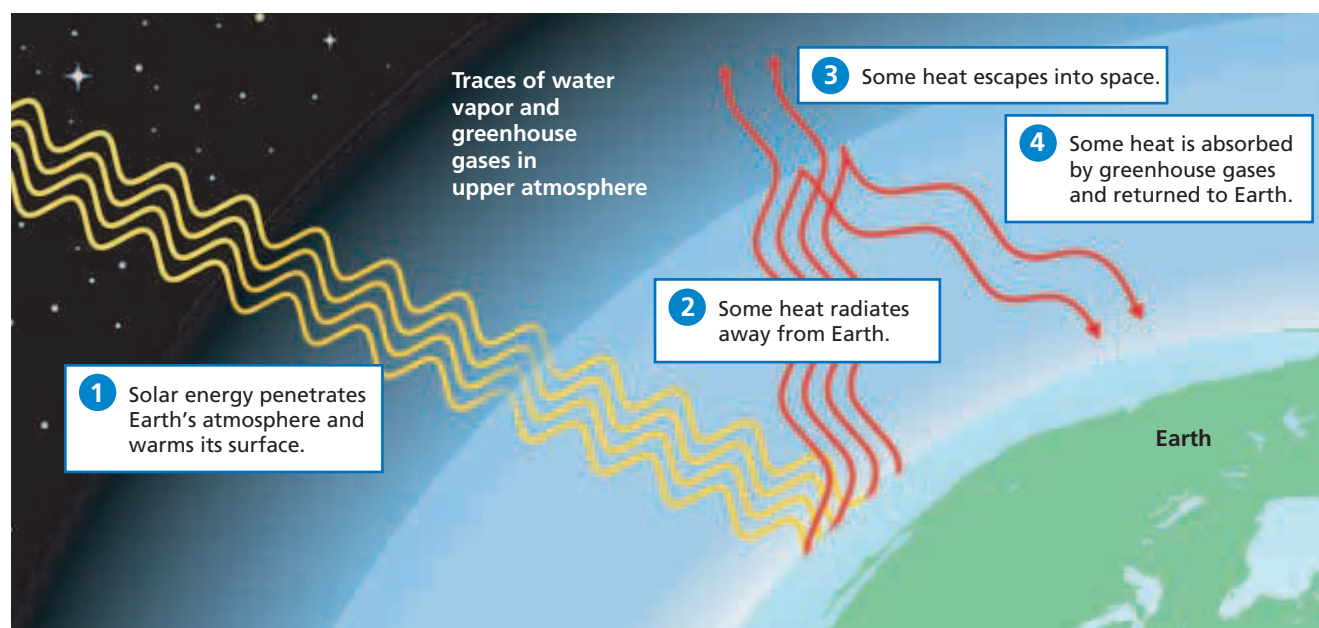
Climate and Atmosphere

The ozone layer is important because it absorbs most of the sun's ultraviolet radiation before it reaches Earth's surface. Most organisms on Earth depend on this protection because ultraviolet radiation can damage DNA and cause mutations. In contrast, some gases in the atmosphere direct energy toward Earth's surface. Energy from the sun reaches Earth's surface as light but may leave the surface as heat. Some gases in the atmosphere function to radiate this heat back toward Earth's surface. The atmosphere's ability to trap heat in this way is called the **greenhouse effect**. The atmospheric gases that contribute to this effect are sometimes called *greenhouse gases*.

As Figure 22-2 illustrates, Earth can be heated by sunlight in the same way that air in a greenhouse can be. **1** First, solar energy passes through the atmosphere and strikes Earth's surface. Some of this light is absorbed and converted to heat. **2** A portion of the heat radiates back through the atmosphere. **3** Some of the radiated heat escapes into space. **4** However, some heat is absorbed by gases in the atmosphere and then radiated back to Earth's surface. As a result, Earth's surface is kept warm.

FIGURE 22-2

Gases in Earth's atmosphere trap heat near the planet's surface just as the panes of glass on a greenhouse trap heat. Solar energy is trapped near the Earth's surface by greenhouse gases such as carbon dioxide and methane.



The concentrations of greenhouse gases, such as CO₂ and methane, affect the amount of the sun's heat that is trapped by the atmosphere. CO₂ cycles between the atmosphere and living things through the processes of photosynthesis and cellular respiration. Additional CO₂ enters the atmosphere when organic matter is burned. Increasingly, humans burn organic matter that is in the form of *fossil fuels*, such as natural gas, coal, and petroleum.

Hydrosphere

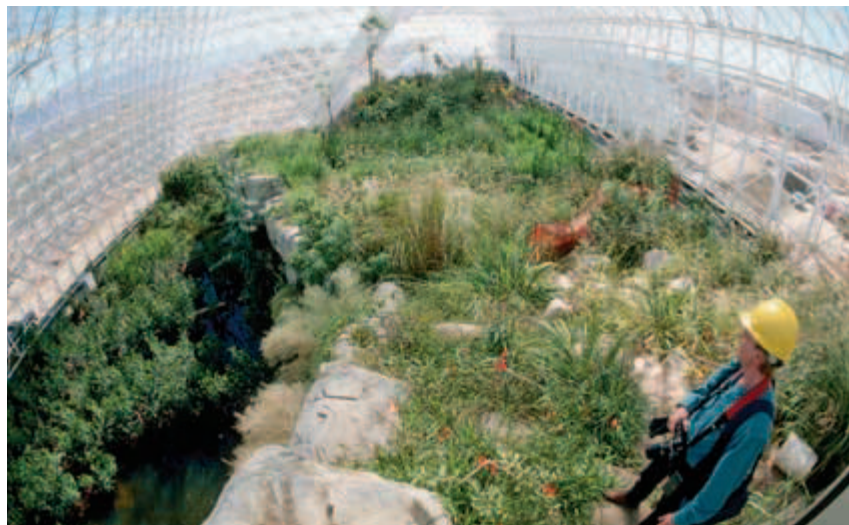
When astronauts in space look at Earth, they see a mostly blue planet because oceans cover about 70 percent of Earth's surface. Water is very important to life on Earth. Especially important is *freshwater*, or water that is not salty and so is suitable for uses such as drinking, bathing, and watering crops. Only about 3 percent of surface water is freshwater, and most of this freshwater is tied up in glaciers at the poles.

Geosphere

The geosphere exchanges materials with the atmosphere, hydrosphere, and biosphere. Recall that photosynthetic organisms take carbon from the atmosphere and incorporate it into living tissue. After the organisms die, some of the carbon may become coal or oil and thus enter the geosphere. Sulfur, phosphorus, and other elements also cycle between the biosphere and geosphere.

Biosphere

The biosphere includes all parts of Earth where life exists. The biosphere includes the part of the atmosphere between Earth's surface and approximately 9 km above the surface, most of the hydrosphere, and at least 11 km of the top crust of the geosphere. Scientists occasionally discover life in new places and thus expand the known boundaries of the biosphere. Attempts to replicate an Earth-like system, such as the project shown in Figure 22-3, have shown that Earth's spheres form a complex and delicately balanced system.



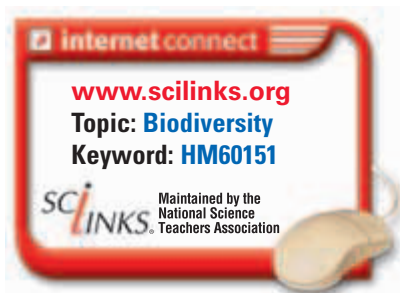
Word Roots and Origins

biosphere

From the Greek *bios*, meaning "life," and the Latin *sphaera*, meaning "globe."

FIGURE 22-3

With an ambitious project in the Arizona desert called *Biosphere 2*, researchers attempted to build a small, self-contained, Earth-like system that included a biosphere, geosphere, hydrosphere, and atmosphere capable of sustaining human life for long periods. The project encountered many difficulties, showing that Earth's biosphere is not easy to replicate.



BIODIVERSITY

The term **biodiversity** refers to the variety of forms of life in an area. Biologists study and describe biodiversity in several ways. One measure of biodiversity is **species diversity**. Species diversity, in turn, is reflected by both species richness and species evenness. *Species richness* refers to the number of unique species within an area. This kind of diversity is easiest to notice and is most often what is meant when referring to biodiversity. *Species evenness* refers to the relative number of individuals of each species in an area.

Species evenness is harder to study than species richness but may be more important to understanding ecosystems. Two locations may have the same species richness yet differ in species evenness. For example, suppose that five tree species grow in each of two forests. In one forest, the numbers of each kind of tree are about the same, but in the other forest, one kind of tree far outnumbers the other kinds. A disruption such as a drought would affect the two forests quite differently.

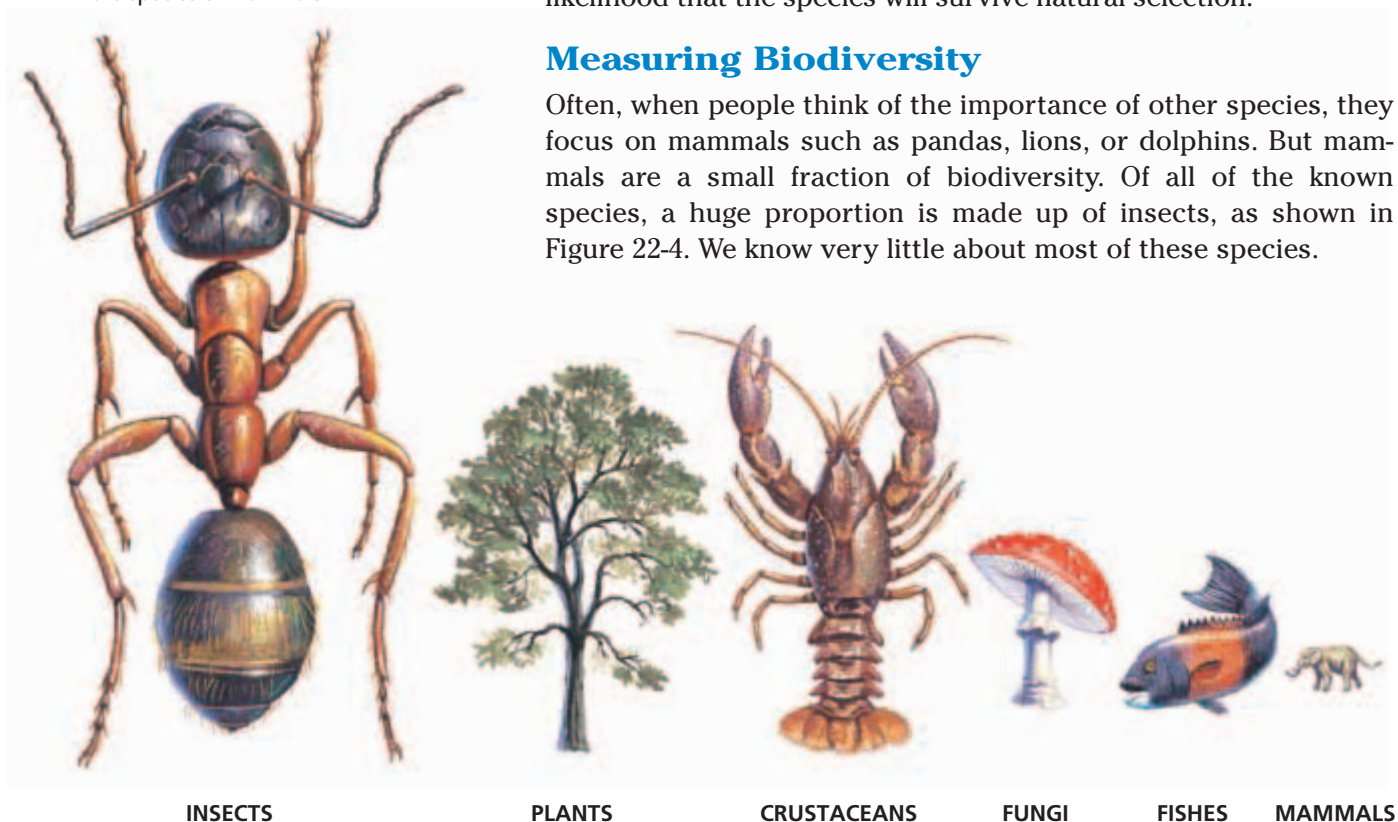
Another measure of biodiversity is **genetic diversity**, or the amount of variation in the genetic material within all members of a population. Genetic diversity is important because it affects a population's ability to adapt in the face of environmental change. Recall that evolution by natural selection acts upon genetic variation within populations. When populations are reduced to small numbers, the amount of genetic variation is reduced for many generations to come. A reduction in a species' genetic variation reduces the likelihood that the species will survive natural selection.

FIGURE 22-4

This diagram represents the relative numbers of species of different categories on Earth. The sizes of the organisms drawn here are proportional to the estimated number of species in each category. There are many more species of insects and plants than there are species of mammals.

Measuring Biodiversity

Often, when people think of the importance of other species, they focus on mammals such as pandas, lions, or dolphins. But mammals are a small fraction of biodiversity. Of all of the known species, a huge proportion is made up of insects, as shown in Figure 22-4. We know very little about most of these species.



In terms of species richness only, how much biodiversity exists? Various biologists have made widely different estimates for the number of species on Earth, ranging from 2 million to 100 million species. Yet in about 200 years of cataloging, scientists have named and described about 2 million species.

Because of the frequency and ease with which new species are discovered, scientists acknowledge the likelihood that there are many undescribed species. For example, in recent years, several new species of primates were discovered in Madagascar, and an entirely new order of insects was recognized. In 1982, biologist Terry Erwin announced that his team had studied all of the species of insects in one kind of tree in a tropical rain forest. They counted 1,200 beetle species, 163 of which seemed to inhabit only that one kind of tree.

Our knowledge of smaller organisms and organisms in remote or extreme locations is especially poor. Scientists must use innovative approaches to find and describe these lesser-known organisms. For example, they may build robotic vehicles to explore remote and extreme environments, such as caves, volcanoes, hot springs, and deep sea trenches. It seems that everywhere they look, scientists find new varieties of organisms.

Valuing Biodiversity

Biodiversity provides important benefits to people. For example, thousands of plant and animal species can serve as food. Trees provide wood for homes and fuel. Many species are sources of medicines and useful chemicals. Undiscovered species may someday supply other benefits. And ecosystems recycle human wastes, including CO₂. Writing about such benefits, biologist E. O. Wilson stated, "Biological diversity is the key to the maintenance of the world as we know it."

Some people think that organisms and ecosystems are important for reasons other than their use to humans. These opinions may be based on moral, aesthetic, or religious beliefs that are beyond the scope of biology. People may value biodiversity for multiple reasons.



Quick Lab

Estimating Microscopic Diversity

Materials microscope, dissecting scope, 50 mL sample of pond water

Procedure Gently stir the pond water. Place one drop of the pond water on a microscope slide. Examine the slide under the dissecting scope and under the microscope at low magnification. Try to find as many different types of organisms as possible. Quickly draw or describe each one. Make a tally of the total number of unique organisms you find.

Analysis Compare your total and your descriptions with those of others in your class. How many unique kinds of organisms might there be in the total sample of pond water?

SECTION 1 REVIEW

1. Explain the importance of the study of environmental science.
2. Name the three major layers between Earth's center and 20 km above Earth's surface.
3. Define the biosphere.
4. How does the greenhouse effect occur?
5. Compare species richness and species evenness.
6. Compare species diversity and genetic diversity.
7. List several reasons to value biodiversity.

CRITICAL THINKING

8. **Calculating Information** Compare the density of oxygen in the atmosphere at 36,000 ft above sea level to its density at 18,000 ft above sea level.
9. **Analyzing Models** The amount of greenhouse gases released into the atmosphere by human activities has increased over recent years. Predict a possible result of this trend.
10. **Applying Information** Explain why measuring biodiversity is difficult.

SECTION 2

OBJECTIVES

- **Describe** major consequences of air pollution.
- **Relate** air pollution to effects on global climate.
- **Describe** how chemical pollutants may undergo the process of biological magnification.
- **Identify** the primary causes of modern extinctions.
- **Explain** why extinctions and ecosystem disruption are of concern to humans.
- **Relate** human resource use to its impacts on ecosystems.

VOCABULARY

pollution
smog
chlorofluorocarbon (CFC)
biological magnification
acid precipitation
extinction
keystone species
sustainability

ENVIRONMENTAL ISSUES

*As the human population increases, so does the human impact on the environment. Humans often cause **pollution** by putting substances that cause unintended harm into air, water, or soil. Many human activities disrupt ecosystems.*

POLLUTION

In many urban areas, the air is visibly polluted with **smog**, water vapor mixed with chemicals that result from human activities. These activities include burning fuels and using chemicals in vehicles, homes, and industries. Smog may contain nitrogen oxides, NO_x , sulfur oxides, SO_x , organic chemicals, small particles, and ozone, O_3 . Ozone is produced when nitrogen oxides react with oxygen in air under sunlight. Animals and plants are harmed by respiring the pollutants in smog. Other results of air pollution include ozone thinning, global warming, and acid precipitation.

Ozone Thinning

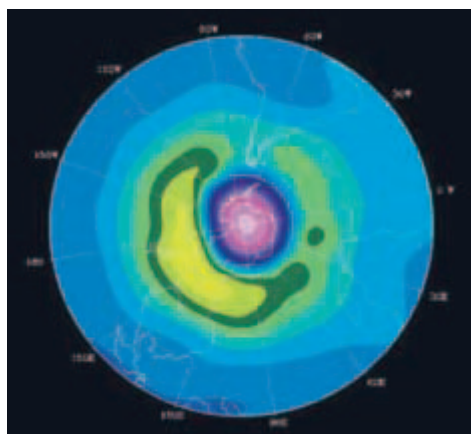
Ozone in the lower atmosphere is harmful, but ozone floating 20 km above Earth in a zone called the *stratosphere* shields the planet's surface from harmful ultraviolet radiation, or *UV light*. Unfortunately, several kinds of human-made chemicals contribute to the destruction of the ozone layer. The most important of these are **chlorofluorocarbons** (KLAWR-oh-FLOOR-uh-KAHR-buhnzh), or **CFCs**. Gaseous CFCs were once widely used as coolants in refrigerators and air conditioners, as propellants in aerosol spray cans, and in the manufacture of plastic-foam products. After floating up to the stratosphere, CFCs act as catalysts for reactions that break down ozone. Scientists have estimated that a single CFC molecule can help destroy up to 100,000 ozone molecules.

Beginning in the 1980s, scientists discovered thinning areas in the ozone layer over Earth's polar regions, as shown in Figure 22-5, and declining ozone levels elsewhere in the layer. An international study estimated that a 10 percent drop in ozone levels would cause 300,000 new cases of human skin cancer worldwide. UV light also harms plants and photosynthetic algae, so ozone depletion could modify entire ecosystems over time.

Damage to the ozone layer led most countries to stop producing CFCs by 1995. Environmental scientists estimate that it will take the ozone layer 50 to 100 years to recover completely. The effort to protect the ozone layer shows that scientists, the public, and policy makers can work together to solve environmental problems.

FIGURE 22-5

An ozone "hole," a region of abnormally low ozone concentration, has formed over Antarctica (pink region in image). This lets more ultraviolet light strike the Earth's surface, where it damages DNA and other molecules of life.



Global Warming

Scientists have noticed a relationship between average levels of CO_2 in the atmosphere and average global temperatures. As CO_2 levels have increased, so have temperatures, as shown in Figure 22-6. This pattern, in which two variables change in similar ways, is called a *correlation*. However, a correlation can be difficult to interpret. A correlation could result from a *cause-and-effect relationship*, in which a change in one variable causes a change in another variable. However, a correlation does not necessarily indicate which variable, if any, is the cause. That is, rising CO_2 might cause temperature to rise, rising temperature might cause CO_2 to rise, or other variables might cause both CO_2 and temperature to rise.

Scientists cannot conduct a global experiment to test these hypotheses. However, scientists have strong evidence that CO_2 and certain other gases do cause Earth's atmosphere to trap more solar energy. For example, scientists have developed computer models to simulate Earth's climate and predict possible climate changes. Climate models are very complex because they include many factors, such as wind patterns and the effects of oceans on the atmosphere. By varying CO_2 levels in climate models, scientists can simulate the possible effects of CO_2 on global temperatures. The results of such simulations consistently show that doubling atmospheric CO_2 leads to an increase in average global temperatures of 1.0°C to 4.5°C (2°F to 8°F), although the exact temperature change depends on the particular model. Most scientists studying this problem have concluded that increased CO_2 levels in the atmosphere cause increases in global temperatures.

The year 2003 was the second-warmest year on record at that time. Average global temperatures were 0.56°C (1.01°F) above the past century's average. The warmest year before 2003 was 1998. An international panel of scientists predicts that temperatures will rise an additional 2°C (4°F) in the next 100 years. This small change could alter rainfall patterns, soil moisture, and sea level around the world. Also, this change could shift agricultural regions and disrupt both terrestrial and aquatic ecosystems. Rising temperatures are already changing biological communities in measurable ways.

Acid Precipitation

Recall that in the global water cycle, water enters the atmosphere as vapor and returns to Earth's surface in the form of precipitation, such as rain or snow. Some air pollutants combine with water in the atmosphere and form acids. The result is precipitation that is acidic, or **acid precipitation**. Because organisms are adapted to the normal pH range of their environment, increased acidity of soil and water causes disease or death in trees, fish, and other organisms.

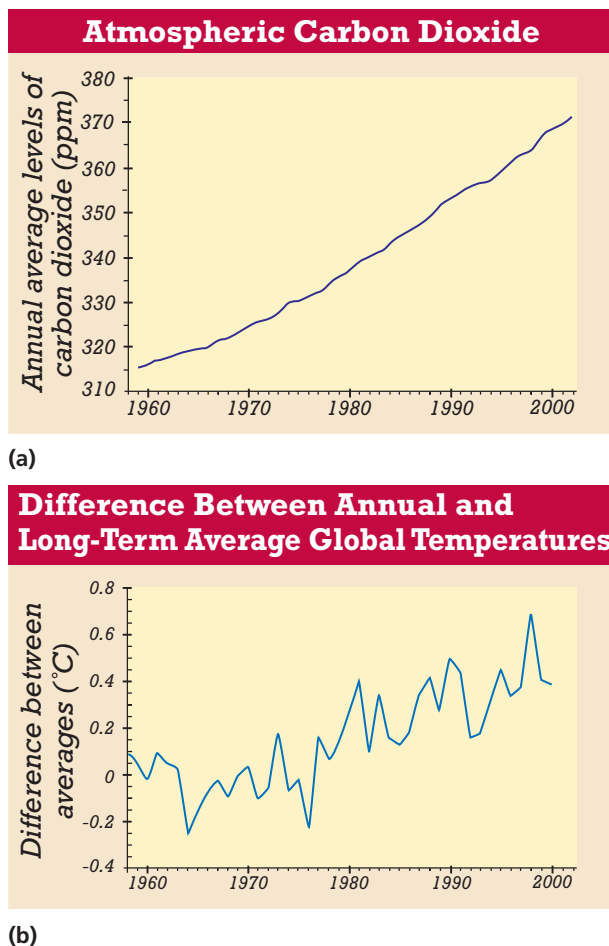
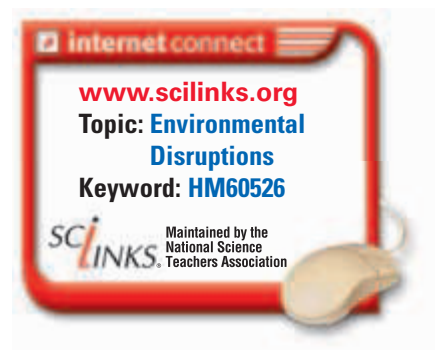


FIGURE 22-6

(a) Atmospheric CO_2 concentration has been increasing in recent decades, as measured from the top of Mauna Loa, a large volcano in Hawaii. (b) Average global temperatures have been rising in a similar pattern. The graph shows the difference between recent short-term and past long-term temperature averages.



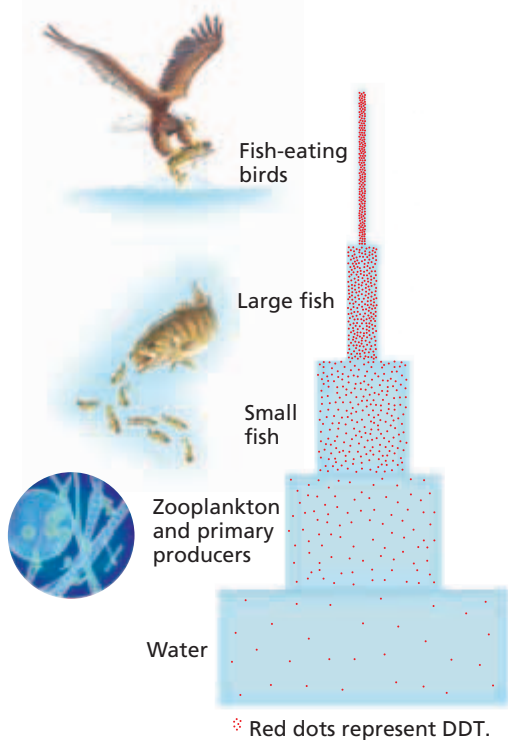


FIGURE 22-7

In this example of biological magnification, the insecticide DDT runs off farms and into water sources, where it is taken up by algae. Then, the DDT passes up the food chain from the algae to zooplankton, to small fish, to bigger fish, and finally to predators such as eagles. DDT accumulates in fatty tissue, and its concentration increases as it moves up the food chain. In predatory birds, DDT can reach levels that cause the production of thin eggshells and leads to the death of offspring. The banning of DDT in the United States may have saved the bald eagle from extinction.

At the top of the food chain, DDT concentration has increased almost 10 million times.

DDT is stored in primary producers, such as algae.

DDT enters the water through ground water and runoff from land.

Land and Water Pollution

Human activities can pollute land and water in several ways. Humans produce and must dispose of waste in the form of sewage and unused materials. Humans produce and use a variety of chemicals in industries, farms, ranches, stores, homes, schools, workshops, and cleaning services. Many of these chemicals are toxic, and some also cause cancer. Some chemicals are not directly harmful but cause damage to ecosystems if they are disposed of improperly or spilled by accident. Some chemicals enter ecosystems and undergo **biological magnification**, a process in which chemicals become more concentrated in an organism the higher on a food chain that organism is, as Figure 22-7 shows. As environmental problems are identified, governments increasingly regulate the manufacture, use, and disposal of chemicals.

ECOSYSTEM DISRUPTION

The human impacts on air, climate, land, and water described above are happening at the same time. These impacts contribute to *ecosystem disruption*, the destruction or substantial change in the functioning of natural ecosystems. Ecosystem disruption is evident as species—and sometimes entire communities—disappear.

Extinction

Organisms generally require specific habitats in order to thrive. For example, when whooping cranes are in their winter nesting grounds in Texas, they eat blue crabs. Along their migration routes, they eat various invertebrates in the marshes where they stop to rest. If people drain the marshes for farmland, the crane could experience **extinction**, the death of every member of the species. In fact, whooping cranes are in danger of extinction, or *endangered*. Worldwide, thousands of species have become endangered or extinct as a result of human activities.

One way that species become endangered is when their habitat is destroyed. Habitat destruction occurs as humans convert complex natural ecosystems into simplified systems that do not sustain as many species, such as farmland and urban areas. Since the development of agriculture, people have cut down more than half of the world's forests. Other examples of habitat destruction include damming rivers, draining swamps, surface mining, logging, and clearing land for buildings or roads.

In addition to habitat destruction, the main human causes of the endangerment of species include hunting or harvesting and the transfer of invasive species into new habitats. An increasing number of species are endangered, as Table 22-1 shows. Scientists estimate that perhaps 20 percent of Earth's plant and animal species may become extinct in the next 50 years, a situation called the *biodiversity crisis*. Living species have not been lost at such a high rate since the mass extinction of dinosaurs 65 million years ago. The mass extinction currently underway is unique in that humans are its primary cause.

Ecosystem Imbalances

No one can predict the results of the loss of millions of species. We do know that some species are critical to the functioning of ecosystems. For example, a species of sea otters, shown in Figure 22-8, once lived in great numbers along the Pacific coast. These otters feed heavily on sea urchins, whereas sea urchins feed on giant kelp (algae) that form the basis of an underwater “forest” community. When people hunted the otters to near-extinction, the sea urchins increased in number and overfed on the kelp. Entire kelp communities began to disappear. Later, biologists reintroduced and helped secure legal protections for the sea otters in some areas, and the kelp recovered. Species such as the sea otter that affect many other species in a community are called **keystone species**. Extinction of keystone species has serious and sometimes unknown effects on ecosystems.

Overusing resources can also unbalance ecosystems. For millennia, the waters off what is now Cape Cod supported huge populations of cod, flounder, haddock and other fish. Until the 1960s, people harvested the fish at about the same rate that the fish reproduced. When newer fishing methods brought in larger harvests, the fish populations decreased rapidly and have not yet recovered. Similar overuse of terrestrial resources has caused the formation of desertlike areas in Africa and the erosion of topsoils in the midwestern United States.



FIGURE 22-8

The southern sea otter, *Enhydra lutris nereis*, has the thickest fur of any mammal. Sea otters live in coastal kelp bed communities. They eat sea urchins, other invertebrates, and fish. As people hunted otters to near-extinction in the 1800s and 1900s, the kelp communities where they lived started to disappear.

TABLE 22-1 Species Known to Be Threatened or Extinct Worldwide

Group of species	Number threatened (all categories of risk)	Number extinct (since ~1800)	Percent of known species that are threatened
Mammals	1,130	78	23
Birds	1,194	132	12
Reptiles	293	22	4
Amphibians	146	7	3
Fishes	750	91	3
Insects	553	70	0.06
Mollusks	967	303	1
Other invertebrates	439	8	1
Plants	6,774	106	2

Word Roots and Origins

sustainable

from Latin *sub*, meaning
"from below" and *tenere*,
meaning "to hold"

HUMAN RESOURCE USE

The global human population uses an estimated 10 to 55 percent of Earth's *primary production*, the total energy stored through photosynthesis by terrestrial organisms. People use this resource in the form of fuel and food for both humans and domestic animals. As the human population increases, so does its use of this resource.

Another analysis of human impact on ecosystems is known as an *ecological footprint*. This analysis accounts for people's use of food and natural resources, such as land and water, as well as people's production of wastes and pollution. Ecologists are concerned that the biosphere has a limited capacity to renew or repair itself in the face of these impacts. One group of ecologists and economists calculated that since the 1980s, human demand has exceeded the biosphere's renewal capacity.

Although making such analyses is difficult, it is clear that Earth's capacity to support humans has an upper limit. This problem is a question of sustainability. **Sustainability** means the ability to meet human needs in such a way that a human population can survive indefinitely. To live sustainably, humans must close the gaps between Earth's renewable resources and capacities, our own needs, and the needs of Earth's other living species.

One approach is to develop new technologies for uses such as energy production, transportation, agriculture, housing, and waste disposal. Another approach is to slow or reverse human population growth. A third approach is to reduce our consumption of resources. For example, the U.S. Department of Energy estimates that increasing energy efficiency could cut national energy use by up to 20 percent in a few decades. This kind of change would require government policies as well as private efforts that reduce the use and improve the efficiency of cars, appliances, buildings, factories, and the systems that produce and distribute energy.

SECTION 2 REVIEW

1. Identify some of the components of smog.
2. Explain how the release of chlorofluorocarbons affects the ozone layer.
3. Explain atmospheric scientists' predictions for global temperatures over the next 100 years.
4. Describe how pesticides, such as DDT, undergo the process of biological magnification.
5. Identify the primary causes of modern species extinctions.
6. Describe the factors considered in an analysis of a person's or population's ecological footprint.

CRITICAL THINKING

7. **Applying Information** Choose a species that might be affected by global warming, and predict how that species might be affected.
8. **Calculating Information** The human population doubles every 36 years. If people currently use 25 percent of Earth's primary production, in what year might humans use 100 percent?
9. **Analyzing Models** Sketch a model of your own ecological footprint. Include boxes to represent the resources you use, their original sources, and any activities that affect your environment. Draw arrows to indicate relationships between factors.

MILESTONES

IN

Environmental Protection

Timeline

1835 Ralph Waldo Emerson writes essay "Nature."



1865 Frederick Law Olmsted advocates preservation of Yosemite.

1872 The world's first national park, Yellowstone, is established.

1892 John Muir founds the Sierra Club.



1905 U.S. Forest Service is established under Theodore Roosevelt.

1916 National Park Service is established under Woodrow Wilson.

1962 Rachel Carson publishes *Silent Spring*.



1967 Congress passes Clean Air Act.

1968 Paul Ehrlich publishes *The Population Bomb*.

1970 The Environmental Protection Agency is established, and the first Earth Day is celebrated.



2002 The World Summit on Sustainable Development occurs.

Throughout U.S. history, people have demonstrated and written about their concern for the environment. Thousands of citizens have tried to stop pollution, promote public health, and preserve America's natural assets. Prompted largely by a few individuals and events, environmental awareness has increased over the decades.

In 1835, Ralph Waldo Emerson wrote the essay "Nature" and thus began a tradition of writing about nature. Henry D. Thoreau, Walt Whitman, and others continued this literary tradition. These and other public figures drew attention to the value of ecosystems.

In 1865, landscape architect Frederick Law Olmsted (best known for designing New York's Central Park) submitted a report on preserving Yosemite Valley in California. This was the first justification for public preservation and management of natural areas that was based on the areas' value to humans. Soon after, the U.S. government established Yellowstone National Park and thus began the national park system.

Scottish-born naturalist and writer John Muir advocated for preserving western lands as wilderness. He founded the first major environmental organization, the Sierra Club, in San Francisco in 1892. This group was and continues to be dedicated to the preservation of wilderness and natural areas.

In 1903, President Theodore Roosevelt created the first National Bird Preserve on Pelican Island, Florida. This was the start of the National Wildlife Refuge system. He also established the U.S. Forest Service in 1905. By 1909, the Roosevelt administration had created 42 million acres of national forests and 53 national wildlife refuges. In 1916, the National Park Service was established with 40 national parks and monuments.

In 1962, biologist Rachel Carson published a landmark book, *Silent Spring*. The

book warned people about the dangers of the increasing use of toxic pesticides, such as DDT. In 1968, Stanford ecologist Paul Ehrlich warned of the hazards of rapid population growth with his book, *The Population Bomb*. These books contributed to a surge of environmental awareness in the United States.

In 1970, the U.S. Environmental Protection Agency (EPA) was created to enforce new environmental policies, such as the Clean Water Act and Clean Air Act. The first Earth Day was celebrated internationally. The Endangered Species Act became law in 1973.

In 2002, the World Summit on Sustainable Development met in South Africa. Leaders there agreed that the future of human survival will depend on an international commitment to work together to protect Earth's ecosystems.

Review

1. Describe the role of literature in the environmental movement.
2. **Applying Concepts** What environmental issues affect your life right now?
3. **Conducting Research** Use the Web site below to create a timetable of the history of U.S. environmental laws.



SECTION 3

OBJECTIVES

- **State** the goals of conservation and restoration biology.
- **Describe** examples of efforts to protect species and their habitats.
- **Summarize** international strategies for protecting entire ecosystems.
- **Discuss** the roles of governments and laws in addressing environmental problems.
- **List** several things that individuals can do to help solve environmental problems.

VOCABULARY

conservation biology
restoration biology
bioindicator
biodiversity hotspot
ecotourism
urban ecology

ENVIRONMENTAL SOLUTIONS

Our planet's air, water, land, and organisms are interconnected. Environmental problems connect across national boundaries, so solutions may require worldwide efforts. Scientists, governmental bodies, and individual citizens must work together to live responsibly within the biosphere.

CONSERVATION AND RESTORATION BIOLOGY

As human populations have increased rapidly, so has the human impact on ecosystems. Scientific understanding of these impacts is weak, but is improving. Meanwhile, individuals, such as the students in Figure 22-9, can contribute to scientific understanding and also take responsibility for minimizing human impacts on ecosystems.

In a discipline called **conservation biology**, scientists seek to identify, protect, and manage natural areas that still retain much biodiversity. Where humans have had the largest impacts—agricultural areas, former strip mines, and drained wetlands, for example—biologists must often devise plans to reverse changes and replace missing ecosystem components. In **restoration biology**, scientists deal with extreme cases of ecosystem damage. For example, restoring a grassland community to an area that was strip-mined may involve recontouring the land surface, reintroducing bacteria to the soil, planting grass and shrub seedlings, and using periodic controlled fires to manage the growth of vegetation.

FIGURE 22-9

Dwarf wedge mussels are an endangered species in the Aschuelot River of New Hampshire. These high school students from a nearby town are making a census of mussel populations to help save the species from extinction and to monitor water quality in the river.



Even the best scientific efforts may not be able to completely restore an ecological community. But biologists can encourage restoration by applying their understanding of processes such as energy flow, species interactions, and biogeochemical cycling.

Species and Habitats

One principle that biologists have learned to apply is that species and their habitats are interconnected. Figure 22-9 shows students studying dwarf wedge mussels (*Alasmidonta heterodon*) in the Ashuelot River in New Hampshire. In 1990, government biologists classified this mollusk as an endangered species because its population was declining rapidly. The students looked for clues to explain the decline. The dwarf wedge mussel is a **bioindicator**, a species that is especially sensitive to ecological change. Because the mollusk is harmed by tiny amounts of water pollution, it acts as an early warning signal for environmental problems. Its decline suggested a pollution problem in the Ashuelot River, and the students' research helped to find the source of the pollution.

Case Study: Saving the Whooping Crane

The whooping crane, or “whooper,” is an example of a species that is on the road to recovery, thanks to protective laws, international cooperation, and the efforts of volunteers. Before 1870, about 20,000 whoopers migrated annually between Canada and the Gulf Coast. Then, American settlers began to hunt the birds for their feathers and drain the marshes where birds fed during migrations.

By 1937, only about 15 whoopers remained. Meanwhile, the federal government had begun to regulate the hunting of migratory birds and then established a winter refuge for the cranes in Texas. In 1967, the U.S. Fish and Wildlife Service established a captive breeding program for the cranes, and people learned how to encourage whooper reproduction. For example, Figure 22-10 shows a man imitating a courtship dance with a female crane. This ritual dance prompts female cranes to produce eggs, which can then be fertilized artificially with a male crane's sperm. The resulting offspring may survive in the wild, but may not know how to migrate.

FIGURE 22-10

(a) To help save whooping cranes from extinction, conservationists have tried to recreate many aspects of the crane's natural lives, including the courtship dance shown here. (b) Several generations of cranes have been bred and raised in captivity and released into the wild, although with mixed success.



(a)



(b)

At one point, biologists tried placing whooping crane eggs in the nests of wild sandhill cranes in the hopes that the sandhill cranes would raise the whooper chicks and teach them to migrate. The whooper chicks did learn to migrate but did not learn how to court other whoopers for successful mating.

Captive breeding programs now produce about 30 whooping crane chicks each year. To be sure that the chicks learn to live with other whoopers, the people who raise the chicks wear crane costumes or use crane puppets. The young cranes are released into the wild, but they will not migrate on their own. Since 2001, pilots have used small aircraft to lead cranes along their historical 2000-mile migration route. Like many migratory birds, the cranes learn the route after one trip.

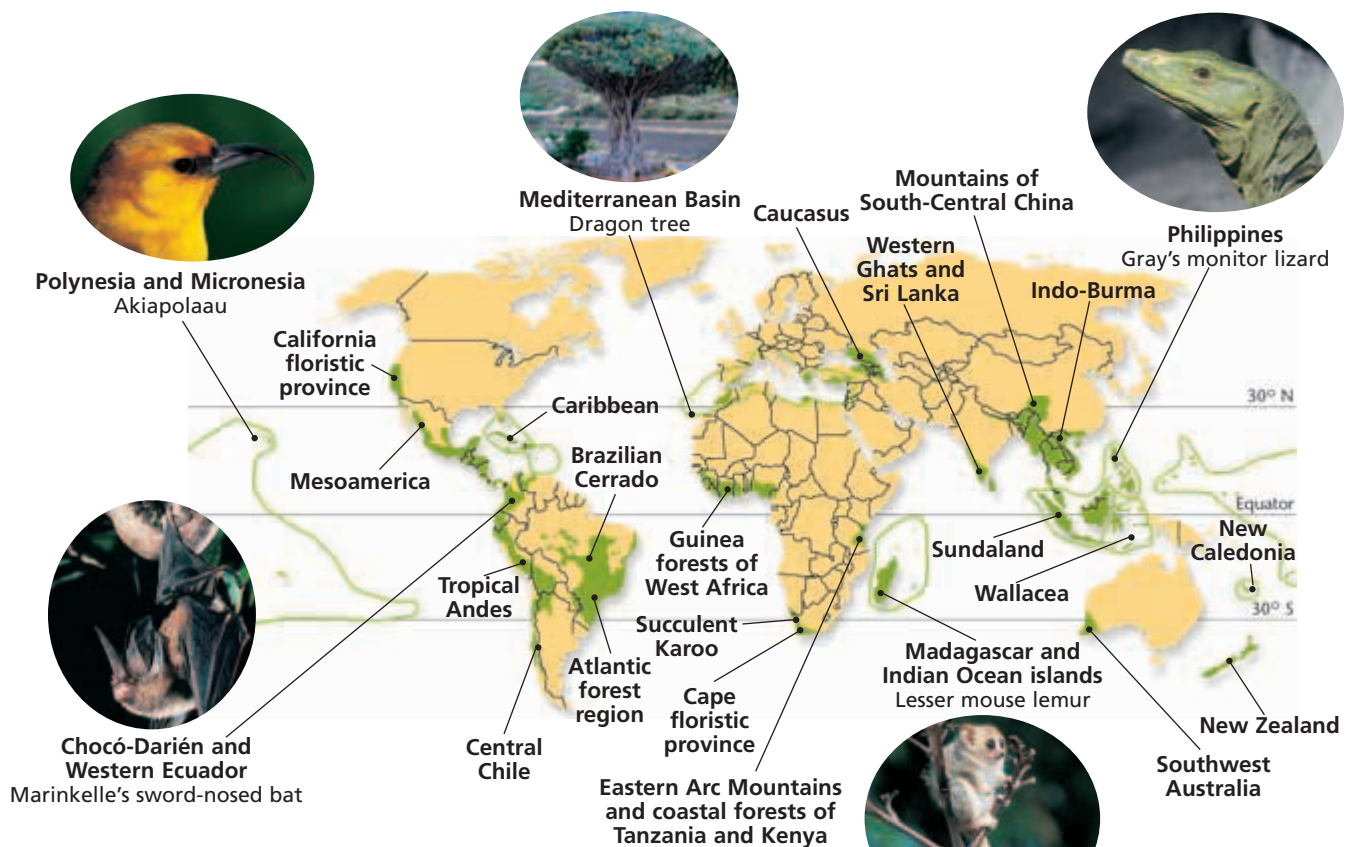
The whooping crane case shows that protecting species from extinction is not easy. But it can be done with creative ideas, cooperation, an understanding of each organism's biology, and the efforts of ordinary citizens.

PROTECTING ECOSYSTEMS

FIGURE 22-11

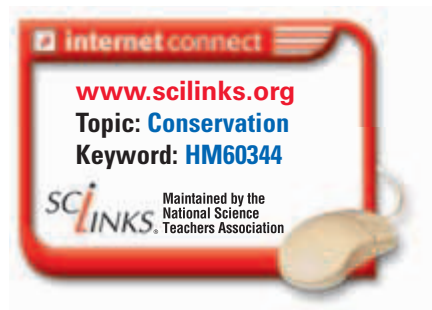
An international group of conservation biologists identified these 25 regions (shaded in green) as *biodiversity hotspots*. These regions contain especially high densities of unique but threatened species.

Ecosystem protection is clearly very important in the long term. But in the short term, economic costs and human freedoms should also be considered. How can society protect ecosystems while protecting individuals' property rights and economic opportunities? One strategy is to focus protection efforts on the regions that currently harbor most of Earth's biodiversity.



Biodiversity Hotspots

In the face of crisis, conservationists look for efficient strategies to protect biodiversity. Many have decided to focus on ecological regions that are richest in unique species and that are most in need of protection. In 2000, an international group identified 25 of these **biodiversity hotspots**, shown in Figure 22-11. These regions contain about 44 percent of all species of vascular plants and 35 percent of all species in four groups of vertebrates. Yet, these regions cover less than 2 percent of the globe. Three of the hotspots include parts of the United States. Internationally, governments and conservation organizations are looking for ways to preserve large enough portions of these hotspots to ensure the survival of a majority of species.



GOVERNMENTS AND LAWS

Governments and laws can play a critical role in solving environmental problems. In the United States, governments at the local, state, and federal levels have used laws, policies, and public funds to address environmental issues. Government actions include legally protecting endangered species and habitats, setting aside land for public use, cleaning up pollution, regulating destructive activities, conducting scientific studies, and encouraging responsible resource use through education and economic incentives.

However, many environmental problems cannot be solved by a single government or law. Ecosystems cross political boundaries. Whooping cranes migrate between the United States and Canada. Pollutants emitted by factory smokestacks in Ohio can fall as acid rain in New York. Storms in Africa blow dust and microorganisms into Europe and out to sea. Thus, to solve many environmental problems, governments must cooperate with each other.

Internationally, countries regularly meet and make agreements to decrease pollution, protect endangered species, or plan for future human activities. Sometimes, many countries agree and solve problems successfully. For example, an agreement to reduce CFC use worldwide seems to have slowed the thinning of the ozone layer. However, countries have had less success agreeing on ways to decrease the threat of global warming. Most countries agree that they need to decrease the amount of greenhouse gases being added to the atmosphere, but they do not agree on how much responsibility each country should have. With many environmental issues, the interests of countries that are more developed differ from the interests of countries that are less developed.

Many developing countries have little money but large amounts of unique habitat and natural resources. This situation presents the opportunity of a *debt-for-nature swap*. In this exchange, richer countries or private organizations pay some of the debts of a developing country and the developing country takes steps such as preserving undeveloped land or educating its citizens.



Quick Lab

Evaluating Environmental Issues in the News

Materials newspapers, magazines, television news broadcasts, or other news sources

Procedure Find a news article or watch a news broadcast about a current environmental issue. First, record its title or topic and its date of broadcast or publication. Then, as you read or watch, take notes, and write down your immediate thoughts and questions. Finally, reread your notes, and write any additional thoughts that you have.

Analysis Did the report relate to any of the information in this chapter? Did the report present the issue accurately and fairly? Was more than one view of the issue represented? Was scientific information included in the report? Describe how you might have presented the report differently.

Urban Ecologist

Job Description Urban ecologists are scientists who study the relationships between humans, animals, and the cities in which they live. Some urban ecologists manage parks, reserves, or water quality. Urban ecologists can work in universities, government agencies, parks, botanical gardens, and private industry.

Focus On an Urban Ecologist

"I like big cities and I like the outdoors!" says urban ecologist Charles Nilon. As a boy, Nilon loved to catch frogs and turtles. "Ecology deals with the things that I like to do." Today, Nilon studies urban wildlife ecology and conservation. "The 'wildlife' portion of my job focuses on learning how different kinds of animals

live in urban areas. The 'conservation' side studies how to manage the contact—both good and bad—between people and animals," says Nilon. Nilon helps local governments manage park areas to protect key wildlife sites. "I also work with private landowners and developers to help them design subdivisions in ways that will enable wildlife to thrive even when there are lots of houses." Nilon's job is all about finding the right balance.

Education and Skills

- **High school**—three years of science courses and four years of math courses
- **College**—bachelor of science in biology, including course work in



botany, zoology, ecology, and wildlife biology; a master's (M.S.) or doctoral degree (Ph.D.) to conduct research

- **Skills**—strong writing skills, strong social skills, excellent observation skills, patience, and research skills



For more about careers, visit go.hrw.com and type in the keyword **HM6 Careers**.

FIGURE 22-12

Rachel Carson awakened millions of readers to important environmental issues, such as the consequences of using the insecticide DDT.



PRIVATE EFFORTS AND COOPERATION

Conservationists have encouraged a variety of strategies to help developing countries derive economic benefit from their land's biodiversity. One approach is **ecotourism**, a form of tourism that supports the conservation of ecologically unique areas while bringing economic benefit to local people. In ecotourism, tourists pay for nature guides, food, and lodging in exchange for the opportunity to experience the ecosystem and its unique organisms.

Cooperation between conservation groups, individuals, and governments is crucial in identifying and addressing environmental issues. In the United States, conservation groups such as the Nature Conservancy, Sierra Club, and Ducks Unlimited make efforts to educate people, protect land, and influence the making of laws.

Individuals and the media also play an important role in raising awareness of environmental issues. In 1962, biologist Rachel Carson, shown in Figure 22-12, made millions of people aware of the dangers of pesticides in her book *Silent Spring*. French oceanographer Jacques Cousteau was internationally famous for his books and television programs about the undersea world.

CONSERVATION CASE STUDY: THE EVERGLADES

The Everglades ecosystem is essentially a wide, shallow, slow-moving river that flows south across much of southern Florida, as shown in Figure 22-13. The area is a swampy patchwork of islands among water that is knee deep in grasses and reeds. The area is home to panthers, alligators, storks, and a wealth of other unique and rare species. The Everglades National Park is one of the largest protected areas in the United States. However, many human interests continue to compete for use of the ecosystem's resources.

Early in the 20th century, the beautiful beaches and pleasant climate of southern Florida attracted land developers. However, most of the land was too wet to build on, and mosquitoes discouraged people from living there. Over time, the developers dug drainage canals to divert water flows toward the ocean and dry out the land. They also planted non-native melaleuca trees because these trees take up large amounts of water from the soil.

At that time, few people could foresee the ecological consequences of these actions. However, journalist Marjory Stoneman Douglas (1890–1998) saw them clearly. She worked to help establish Everglades National Park. In 1947, she wrote a book that described the Everglades ecosystem and explained its importance. Douglas spent a lifetime trying to protect the Everglades.

Still, by the end of the century, half of the area's wetlands had been drained. Ninety percent of the wading birds had disappeared. Because salt concentrations had doubled in Florida Bay, sea grass and shrimp nurseries had died. Fertilizers from agricultural fields flowed along with the water, which poisoned many species and made fish dangerous to eat. Development had diverted so much water that many farmers and homeowners experienced shortages of groundwater. And the melaleuca trees had become an invasive weed.

FIGURE 22-13

The Everglades ecosystem (a) is a large region with enough natural habitat for large predators, such as the highly endangered Florida panther (b). Little islands called *hummocks* rise above the river and are home to a diversity of organisms, including trees, birds, alligators, snakes, and insects. The Everglades National Park (c) contains only about 20 percent of the ecosystem.



(c)



(a)



(b)

TABLE 22-2
Things You Can Do
to Be Environmentally
Responsible

Use cars less by riding a bike, walking, carpooling, or using public transportation instead.

Recycle or reuse packaging and materials such as cans, bottles, boxes, and paper; buy items that have less packaging and that have recyclable packaging.

Use less water by turning off faucets when they are not in use; use water-saving devices and appliances.

Use less energy in your home; look for ways to use less air conditioning or heat; turn off lights and appliances when they are not in use.

Find out the environmental impacts of the products that you buy; ask for environmentally responsible products.

Plant native trees and plants, which provide shade and wildlife habitat and produce O₂ while absorbing CO₂ from the atmosphere.

Conservationists have since established a 20-year plan for the Everglades ecosystem. The plan includes eliminating some of the drainage canals, restoring the Kissimmee River to its original channel, cutting back stands of melaleuca trees, and purchasing more than 40,000 hectares (100,000 acres) of land surrounding Everglades National Park. This plan is the most ambitious ecosystem restoration project so far attempted in the United States.

YOUR ROLE IN THE ENVIRONMENT

A new environmental field, called **urban ecology**, is the study of biodiversity in areas that are densely populated by humans. You can be an amateur urban ecologist by learning about your local environment. For example, try to identify five plants that are native to your area, determine their growing seasons, and learn which plants can be used for landscaping homes or businesses. Find out the names of local birds, how the birds survive, and whether any special laws protect them. Learn which crops and livestock are raised locally, and find out how farmers or ranchers obtain water. Identify endangered species in your area, and find out the reasons that they are endangered. Investigate where your garbage goes after it is collected, and learn whether your sanitation department supports recycling. If you want to study further and do more, find out if there are parks, organizations, or projects in your local area that need volunteers to help with conservation efforts.

No one can predict our environment's future, but it is clear that individuals can make a difference in its fate. Thus, it is important for you, as an individual, to understand your role in the environment and to take responsibility for that role. Taking responsibility can involve simple efforts, such as those listed in Table 22-2. In addition, a planet full of undiscovered species, unknown ecological interactions, and unanswered questions awaits your exploration.

SECTION 3 REVIEW

1. Contrast conservation biology with restoration biology.
2. Give an example of a bioindicator species, and explain its importance.
3. Outline the whooping crane recovery effort.
4. Explain why different governmental bodies must work together to address environmental problems.
5. What is a debt-for-nature swap?
6. Explain the idea of ecotourism.

CRITICAL THINKING

7. **Applying Information** What benefits might people living near Everglades National Park gain from efforts to preserve the ecosystem?
8. **Critiquing Arguments** A student argues that we should not worry about the loss of species because extinction is natural. Evaluate this argument.
9. **Conducting Research** Research an area of high biodiversity near you, and summarize your findings.

CHAPTER HIGHLIGHTS

SECTION 1

An Interconnected Planet

- Earth's geosphere, hydrosphere, and atmosphere are interconnected in many ways. Life exists in parts of each sphere. Together, these parts make up the biosphere.
- Important parts of the atmosphere are greenhouse gases, which trap heat on Earth, and the ozone layer, which shields Earth from UV radiation.
- A very small portion of the hydrosphere is fresh water, and much of this fresh water is not easily usable.
- Biodiversity refers to the variety of life found in an area and can be measured in different ways, including by species richness, species evenness, and genetic diversity.
- Biodiversity is valued for various reasons.

Vocabulary

environmental science (p. 435)
geosphere (p. 435)
hydrosphere (p. 435)

atmosphere (p. 435)
ozone layer (p. 436)
greenhouse effect (p. 436)

biosphere (p. 437)
biodiversity (p. 438)
species diversity (p. 438)

genetic diversity (p. 438)

SECTION 2

Environmental Issues

- Over a short time period, human activities have affected global ecosystems in ways that harm humans and other species. Human impacts range from local pollution to global change in ecosystems.
- Industrial chemicals called chlorofluorocarbons (CFCs) act as catalysts in chemical reactions that break down O_3 molecules in the ozone layer. Most countries have banned CFCs, and the ozone layer seems to be recovering.
- The correlation of increasing atmospheric CO_2 and rising global temperature suggests a cause-and-effect relationship. Considering several types of evidence, many scientists have concluded that increased CO_2 levels have caused warmer surface temperatures on Earth.
- Certain air pollutants cause acid precipitation, which harms or kills many organisms.
- The release of toxic chemicals, such as DDT, into the biosphere can impact ecosystems in many ways, especially when chemicals undergo biological magnification.
- Human impacts on the environment are causing an increasing number of extinctions. Important causes of extinctions are habitat destruction, the transfer of invasive species to new habitats, and overharvesting or hunting. This loss of species has both known and unknown effects on ecosystems.
- Current levels of human resource use are probably not sustainable.

Vocabulary

smog (p. 440)
chlorofluorocarbon (CFC)
(p. 440)

acid precipitation (p. 441)
biological magnification
(p. 442)

extinction (p. 442)
keystone species (p. 443)

sustainability (p. 444)

SECTION 3

Environmental Solutions

- Conservation biologists are concerned with identifying and maintaining ecosystems, while restoration biologists are usually involved with repairing badly damaged ecosystems.
- Populations of many migratory birds, such as the whooping crane, are in decline because of human activities. However, some populations are recovering as a result of legal protection, breeding programs, habitat restoration, and international partnerships.
- International and cooperative efforts to preserve habitat and prevent extinctions include identifying biodiversity hotspots, making debt-for-nature swaps, and promoting ecotourism.
- Environmental problems can be addressed through the combined efforts of governments, scientists, businesses, and individuals.
- The Everglades restoration project is the most ambitious ecosystem-wide restoration project attempted in the United States.

Vocabulary

conservation biology (p. 446)
restoration biology (p. 446)

bioindicator (p. 447)
biodiversity hotspot (p. 449)

ecotourism (p. 450)
urban ecology (p. 452)

CHAPTER REVIEW

USING VOCABULARY

1. In your own words, write a definition for the term *environmental science*.
2. Use the following terms in the same sentence:
 - a. *geosphere*, *hydrosphere*, *atmosphere*, and *biosphere*
 - b. *biodiversity*, *species diversity*, and *genetic diversity*
3. For each pair of terms, explain how the meanings of the terms differ.
 - a. *keystone species* and *bioindicator*
 - b. *conservation biology* and *restoration biology*
4. **Word Roots and Origins** The word *conservation* is derived from the Latin *conservare*, which means “to keep” or “to preserve”. Using this information, explain why the term *conservation* is a good name for the act it represents.

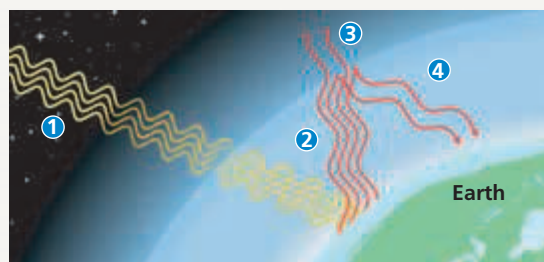
UNDERSTANDING KEY CONCEPTS

5. **Define** the biosphere in relation to Earth’s three major layers.
6. **Explain** Earth’s greenhouse effect.
7. **Identify** three major consequences of air pollution.
8. **Explain** how the chemical ozone can be both beneficial and harmful to living things.
9. **Explain** why scientists must distinguish between a correlation and a cause-and-effect relationship.
10. **Describe** the biological magnification of DDT.
11. **Identify** the three primary causes of modern extinctions.
12. **Give** examples of several ways that people could value a single species.
13. **Define** an ecological footprint.
14. **Summarize** the efforts to save the whooping crane from extinction.
15. **Identify** three regions of the United States that are parts of biodiversity hotspots.
16. **Justify** the need for a conservation plan for the Everglades ecosystem.
17. **List** four things that individuals could do to reduce their ecological footprint.
18. **BIOLOGY INTERACTIVE TUTOR** Write a short report that explains why efficient use of groundwater is important and that summarizes several ways that people can work to reduce the depletion and pollution of groundwater.

19. **CONCEPT MAPPING** Use the following terms to create a concept map that describes two ways in which human activities have an impact on the entire planet: *pollution*, *atmosphere*, *carbon dioxide*, *chlorofluorocarbons*, *greenhouse effect*, *ozone layer*, and *climate*.

CRITICAL THINKING

20. **Analyzing Concepts** What do environmental scientists mean by *interdependence*? Give an example of interdependence from this chapter.
21. **Applying Information** Suggest some species interactions that are likely to be influenced by global warming.
22. **Inferring Relationships** The formation of the ozone layer depended on the presence of oxygen in the atmosphere. Drawing on what you have learned about the history of life, explain how organisms effected and were affected by the ozone layer as it was first forming.
23. **Predicting Patterns** Predict how future human population growth could affect the biosphere.
24. **Evaluating Methods** Humans need clean, fresh water. Environmental scientists think that fresh water may become a limiting factor for human population growth. Explain how you could estimate Earth’s carrying capacity for humans based on the availability of fresh water. What information would you need to make this estimate? How might technological advances change your estimate?
25. **Recognizing Relationships** As part of the global treaty to eliminate CFCs, developed countries are contributing to a fund to help developing countries buy CFC substitutes. What benefits do developed countries receive from this investment?
26. **Forming Reasoned Opinions** After considering Table 22-2, what other ideas could you add to the list? Explain your reasoning.
27. **Interpreting Graphics** Using the illustration below, explain the greenhouse effect. In your explanation, refer to the numbered parts.



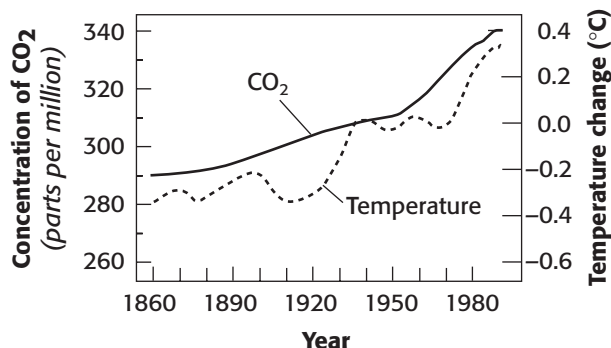


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following is the term for the parts of Earth where water is located?
 - biosphere
 - geosphere
 - atmosphere
 - hydrosphere
- What is the term for the natural ability of Earth's atmosphere to trap energy from the sun?
 - global warming
 - ozone depletion
 - greenhouse effect
 - biological magnification
- What does Earth's ozone layer shield its inhabitants from?
 - solar heating
 - meteor impacts
 - ozone depletion
 - ultraviolet radiation
- Which of the following groups of organisms contains the largest estimated number of species?
 - birds
 - plants
 - insects
 - mammals
- What term describes a measure of the number of species in an area?
 - species richness
 - species evenness
 - bioindicator species
 - biological magnification

INTERPRETING GRAPHICS: The graph below shows trends in two global measurements over several decades. Use the graph to answer the question that follows.



- What is the term commonly used to describe the trend shown in this graph?
 - water pollution
 - global warming
 - ozone depletion
 - biodiversity crisis

DIRECTIONS: Complete the following analogy.

7. thinning of ozone layer : chlorofluorocarbons :: biological magnification :
- pesticides
 - wastewater
 - carbon dioxide
 - greenhouse gases

INTERPRETING GRAPHICS: The table below compares the measurements of two environmental factors for the populations of three countries. Use the table to answer the question that follows.

	United States	Japan	Indonesia
People per square mile	78	829	319
Garbage produced per person per year	720 kg	400 kg	43 kg

8. Which country produces the greatest amount of garbage per square mile?
- Japan
 - Indonesia
 - United States
 - They all produce the same amounts.

SHORT RESPONSE

The term *biodiversity* can be used in several ways. Explain two measures of biodiversity.

EXTENDED RESPONSE

Imagine that your local government has asked for citizen input to plan for the future use of a currently roadless area that is near a body of water. Some citizens would like a new mall to be built there, and some citizens would like the area to be made into a public park.

Part A Make a table of the possible problems or benefits that could result from each option.

Part B Write a letter to the local government that expresses your opinion about this issue and explains your reasons for that opinion.

Test TIP

Before choosing an answer to a question, try to answer the question without looking at the answer choices on the test.

Testing the Effects of Thermal Pollution

OBJECTIVES

- Model the effects of thermal pollution on living organisms.
- Apply the underlying scientific principles to environmental issues.

PROCESS SKILLS

- hypothesizing
- experimenting
- observing
- organizing data
- analyzing data

MATERIALS

- 400 mL beakers, 2
- ice
- hot water
- thermometer
- U-shaped glass tubing, 30 cm long
- 2 corks to fit both ends of the tubing
- 125 mL beaker
- distilled or non-chlorinated water
- culture of paramecia
- hand lens or dissecting scope
- stopwatch or clock
- glass-marking pen or wax pencil

Background

1. Some power plants use water from rivers as a coolant. After the water is used, it is much hotter than the water in the river.
2. What is a pollutant?
3. How can heat be considered a pollutant?
4. How can power plants release nonharmful water?

Procedure

1. Discuss the objectives of this investigation with your partners. Develop a hypothesis concerning the effect of temperature on paramecia.

2. Design an experiment using the given materials to test your hypothesis. In your experiment, the paramecia will be contained in the U-shaped tube. One large beaker will be filled with ice, and the other large beaker will be filled with hot water. Other materials that you can use in your experiment are listed in your materials list.
3. In designing your experiment, decide which factor will be an independent variable. Plan how you will vary your independent variable. In your lab report, list your independent variable and your method of varying it.
4. Decide which factor will be the dependent variable in your experiment. Plan how you will measure your dependent variable. In your lab report, list your dependent variable and your method of measuring it.
5. In most experiments, a control is necessary. Plan your control, and describe it in your lab report.
6. Discuss your planned experiment with your teacher. Proceed with your experiment only after you have received your teacher's permission to do so.
7.  **CAUTION** Water hotter than 60°C can scald. Be careful handling hot water, and alert your teacher if you burn yourself. Fill a 400 mL beaker with ice and water. Make sure that ice remains in the beaker for the entire experiment. Fill another 400 mL beaker with 60°C tap water.
8. In a 125 mL beaker, gently swirl 20 mL of water and 20 mL of a culture of paramecia. Your teacher will provide aged tap water or distilled water for you to use during this step. (Chlorinated water would kill the paramecia.)
9. While your partner holds the test tube steady, carefully pour the paramecia-and-water mixture into the U-shaped tube. Fill the tube completely, leaving just enough room for a cork at each end of the tube. Make sure there are no large air bubbles in the tube. Place a cork at each end of the tube.
10. Create a table similar to the one shown to record your data. For example, the table shown is designed

to record the number of paramecia in three parts of the U-shaped tube over time. Design your data table to fit your own experiment. Remember to allow plenty of space for recording your data.

11. Proceed with your experiment, using the tube, ice water, hot water, and hand lens to observe any response of the paramecia to the environment.
12. As you conduct your experiment, record your results, including the number of paramecia and the time involved, in your data table. Organize your data so that others reading your lab report will be able to understand the results of your experiment.
13.   Clean up your materials and wash your hands before leaving the lab.



Analysis and Conclusions

1. Did the results of your experiment support your hypothesis? Explain your answer.
2. What effect did heat and cold have on the paramecia in your experiment?
3. What evidence do you have that paramecia preferred one temperature range to another?
4. What are some possible sources of error in your experiment?

5. How might a pollutant cause an increase in the number of organisms? Explain.
6. Judging from your experiment, how do you think other organisms might react to a change in water temperature?
7. How could a power plant change the type of organisms that live in the water where it releases its cooling water?

Further Inquiry

Develop a hypothesis about the effects of acid rain on paramecia, and design an experiment to test your hypothesis.

NUMBER OF PARAMECIA			
	Cold end	Hot end	Middle
Elapsed time	0		
	15		
	30		
	45		
	60		
	75		

UNIT 6

CHAPTERS

23 Bacteria

24 Viruses

25 Protists

26 Fungi

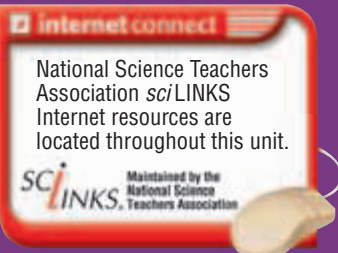
MICROBES, PROTISTS, AND FUNGI

“In the leaves of every forest, in the flowers of every garden, in the waters of every rivulet, there are worlds teeming with life...”

From *Thoughts on Animalcules*, or *A Glimpse of the Invisible World Revealed by the Microscope*, by G. A. Mantell, as quoted in Primo Levi's book *Other People's Trades*.



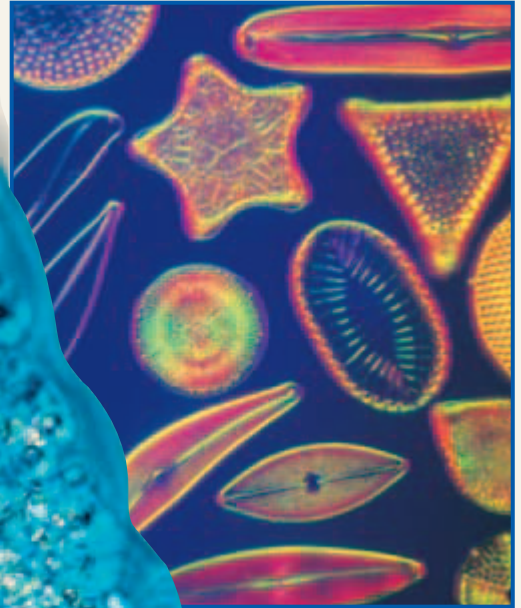
Mushrooms, members of the kingdom Fungi, are important decomposers in nature.





These rod-shaped bacteria are members of the kingdom Eubacteria and the species Escherichia coli.

Diatoms, members of the Kingdom Protista, are unicellular, photosynthetic organisms that live in oceans and lakes.



Pelomyxa carolinensis

These Petri dishes contain bacterial cultures. Scientists often grow bacteria in the laboratory to study the bacteria more easily.



A scanning electron microscope (SEM) image showing numerous pink, comma-shaped Helicobacter pylori bacteria. The bacteria are clustered together and are positioned on a green, textured surface that represents human stomach cells. The background is a dense field of these green cells, with the pink bacteria standing out prominently.

This is a scanning electron microscope (SEM) image of *Helicobacter pylori* cells on human stomach cells. *H. pylori* causes about 80 percent of peptic ulcers. *H. pylori* bacteria survive the acidic environment of the stomach by secreting an enzyme that neutralizes stomach acids. Irritation is caused by the bacteria burrowing into the lining of the stomach and duodenum.

SECTION 1 *Prokaryotes***SECTION 2** *Biology of Prokaryotes***SECTION 3** *Bacteria and Humans*

PROKARYOTES

Prokaryotes are the most numerous organisms on Earth. They are found almost everywhere, from the skin of a fingertip to the waters of a thermal geyser to the freezing landscape of the Antarctic. The earliest fossils of prokaryotes, which are about 3.5 billion years old, indicate that prokaryotes lived long before other forms of life evolved.

TWO MAJOR DOMAINS: ARCHAEA AND BACTERIA

Prokaryotes are single-celled organisms that do not have a membrane-bound nucleus. They live in nearly every environment on Earth. Many live in places where no other organisms can live, such as scalding hot water and inside solid rock. Prokaryotes are a major source of food for many organisms. They also help many organisms, including humans, cattle, and insects, digest food. Prokaryotes also play an important role as decomposers of dead organic matter in the environment.

Even though most prokaryotes are tiny organisms, they differ greatly in their genetic traits, in their sources of energy and nutrients, and in their habitats. Through DNA technology, scientist Carl Woese and his colleagues found in the late 1970s that there are two major branches, or domains, of prokaryotes. One branch, formerly called *Eubacteria* (YOO-bak-TIR-ee-uh), is called *Bacteria*. The other branch is called *Archaea* (ahr-KEE-uh), formerly called *Archaeobacteria*. The name *archaea* means “archaic” or “ancient.” Most known species of archaea live in extreme environments thought to resemble harsh environments present millions of years ago.

Figure 23-1 shows the phylogenetic relationship between living organisms based on comparisons of ribosomal RNA (rRNA) sequences. The rRNAs are well suited for such studies because they make up part of ribosomes, which are organelles that all organisms have. Also, the structure of rRNA changes very slowly over time. Surprisingly, rRNA analysis has shown that archaea are more closely related to eukaryotes than they are to bacteria. In addition, archaea have some genes that are the same as bacterial genes, but most archaeal genes more closely resemble genes found in eukaryotes. This similarity of genes suggests that archaea and eukarya likely share a more recent common ancestor than eukarya and bacteria do.

OBJECTIVES

- **Explain** the phylogenetic relationships between the domains Archaea, Bacteria, and Eukarya.
- **Identify** three habitats of archaea.
- **Describe** the common methods used to identify bacteria.
- **Identify** five groups of bacteria.
- **Explain** the importance of nitrogen-fixing bacteria for many of Earth's ecosystems.

VOCABULARY

prokaryote
peptidoglycan
methanogen
halophile
thermoacidophile
bacillus
coccus
spirillum
streptococcus
staphylococcus
Gram-negative bacterium
Gram-positive bacterium
antibiotic

FIGURE 23-1

Living organisms have been categorized into three main groups, or domains, based on ribosomal RNA analysis: Bacteria and Archaea, which contain prokaryotic cells, and Eukarya, which contains individual eukaryotic cells and includes humans.

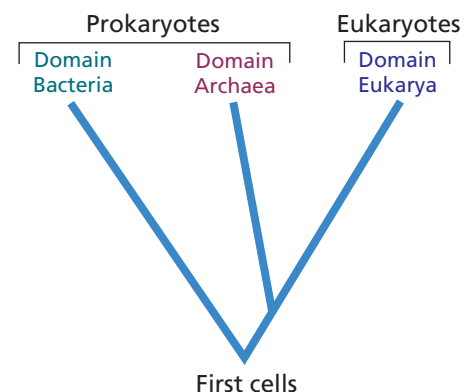




FIGURE 23-2

The fluid gushing from a hydrothermal vent, as this one seen through the window of the miniature submarine *Alvin*, can reach temperatures up to 300°C (572°F). The fluid often contains dissolved sulfides from which thermoacidophilic archaea extract energy. Vents such as this one are called *black smokers* because the vent fluids are dark in color and contain high levels of iron, manganese, and copper.

DOMAIN ARCHAEA

Archaea are not like bacteria in many ways. Archaea differ in the makeup of their cell walls. They also differ in their membrane lipids and in their genetics and metabolism. For example, archaeal cell walls do not have **peptidoglycan** (PEP-ti-doh-GLIE-KAN), a protein-carbohydrate compound found in bacterial cell walls. Archaeal cell walls have different amino acids than bacterial cell walls do. Archaea also have different types of lipids in their cell membranes. Archaea are like eukaryotic cells in that archaeal genes have introns, portions of DNA that do not code for amino acids and that are transcribed into RNA but are removed before being translated into proteins.

Archaea were first discovered in extreme environments, such as swamps, salt lakes, and hot springs. Until recently, scientists believed that archaea lived only in these extreme environments. However, after finding archaeal genetic material in samples of surface waters from the North Pacific and Antarctic Oceans, scientists now think that archaea may be more common than they once thought.

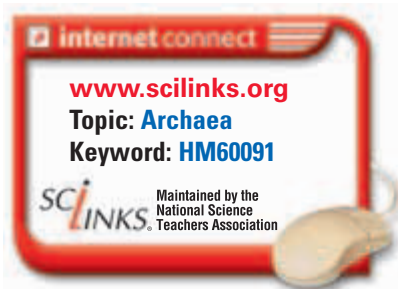
Archaeal Groups

Genetic analysis of archaea has revealed at least three groups of archaea: methanogens, halophiles, and thermoacidophiles.

Methanogens (muh-THAN-uh-JENZ) are named for their unique way of getting energy: they convert hydrogen gas and carbon dioxide into methane gas. Oxygen is poisonous to them, so methanogens can live only in anaerobic environments (environments that lack oxygen), such as in deep fresh water, marine mud, swamp mud, and sewage. The methane that bubbles out at sites such as swamps is called *marsh gas*. Methanogens also thrive in the intestinal tracts of organisms such as cows and termites. A cow can belch between 200 and 400 liters of methane per day.

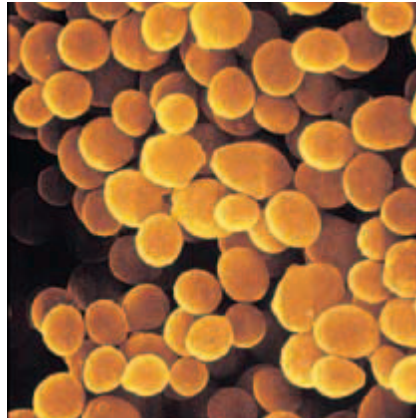
Halophiles (HAL-oh-FIELZ) are “salt-loving” archaea that live in environments that have very high salt concentrations, such as the Great Salt Lake and the Dead Sea. High salt concentrations would kill most bacteria but favor the growth of halophiles because halophiles have adapted to live in very salty water.

Thermoacidophiles (THUHR-moh-uh-SID-uh-FIELZ) live in very acidic environments that have very high temperatures, such as the hot springs of Yellowstone National Park. Some thermoacidophiles live at temperatures up to 110°C (230°F) and at a pH of less than 2. Thermoacidophiles also live near volcanic vents on land or near hydrothermal vents called *black smokers*. Black smokers are cracks in the ocean floor that leak very hot, dark-colored, acidic water, as shown in Figure 23-2. Scientists have found large communities of worms, clams, crabs, and mussels living near these thermal vents. These communities live at great depths and in total darkness, where photosynthesis cannot take place, so they depend on thermoacidophilic archaea as a primary source of food.





(a) Bacillus (rod-shaped)
Escherichia coli



(b) Coccus (round-shaped)
Micrococcus luteus



(c) Spirillum (spiral-shaped)
Spirillum volutans

DOMAIN BACTERIA

Most known prokaryotes are bacteria. They occur in many shapes and sizes and have distinct biochemical and genetic characteristics. Most bacteria have one of three basic shapes, as shown in Figure 23-3. Rod-shaped bacteria are called **bacilli** (buh-SIL-IE). Sphere-shaped bacteria are called **cocci** (KAHK-SIE), and spiral-shaped bacteria are called **spirilla** (spie-RIL-uh). When cocci occur in chains, they are called **streptococci** (STREP-tuh-KAHK-SIE). Grapelike clusters of cocci are called **staphylococci** (STAF-uh-loh-KAHK-SIE).

Gram Stain

Biologists group most species of bacteria into two categories based on the structure of their cell walls as shown by a laboratory technique called the *Gram stain*. Figure 23-4a shows that Gram-negative bacteria have cell walls that are complex and have relatively small amounts of peptidoglycan. **Gram-negative bacteria** take up the second, red dye of the Gram stain process which makes the cells appear reddish pink under a microscope. Figure 23-4b shows that the walls of Gram-positive bacteria are simpler and have more peptidoglycan. **Gram-positive bacteria** retain the purple dye in their cell walls and appear purple.

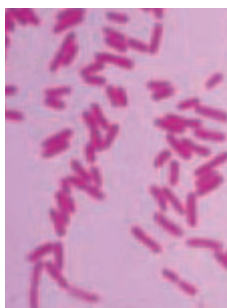
FIGURE 23-3

The most common bacterial shapes are (a) bacillus, (b) coccus, and (c) spirillum.

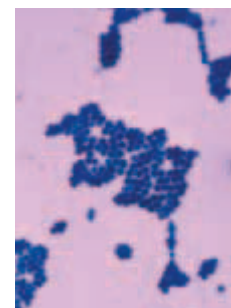
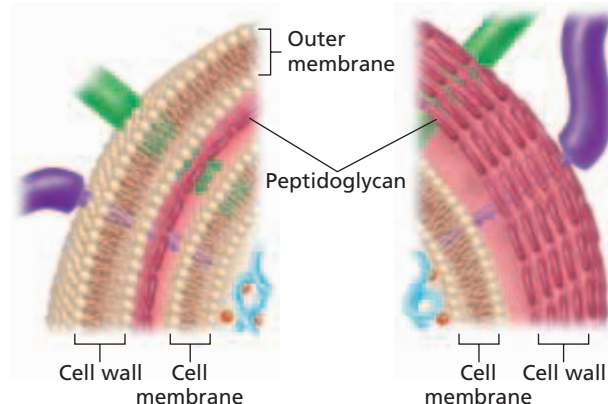
FIGURE 23-4

After the Gram-staining procedure, a Gram-negative bacterium (a) appears pink because the cells become counterstained by the safranin red stain after being decolorized by an alcohol/acetone wash. Gram-positive cells (b) retain the crystal violet stain and appear purple.

(a) GRAM-NEGATIVE



(b) GRAM-POSITIVE



Microbiologist

Job Description Microbiologists are scientists who study microorganisms—organisms that are too small to be seen by the naked eye. Microbiologists interact with health workers and agencies to reduce the frequency and spread of diseases caused by microorganisms. They also work in agriculture, education, research, wastewater management, pharmaceuticals, biotechnology, and other industries.

Focus on a Microbiologist

James Rogers, Ph.D., majored in microbiology. “Learning to look closely is key to any microbiologist’s success. Keen observation may lead you to a new discovery,” says Rogers. Rogers and his colleagues at the U.S. Army’s Chemical

and Biological Command are helping the United States prepare against bioterrorism. They develop methods to test environmental samples for the presence of deadly viruses and bacteria such as the anthrax bacterium (*Bacillus anthracis*). After new methods are devised, Rogers conducts additional research to make the tests run faster and more efficiently. “We are helping the United States with what we are developing. We might be saving lives one day in the future.”

Education and Skills

- **High school**—three years of science courses and four years of math courses
- **College**—bachelor of science in biology, including course work in



chemistry, physics, microbiology, bacteriology, and virology; a master’s (M.S.) or doctoral degree (Ph.D.) is needed for research

- **Skills**—good communication skills, good observation skills, curiosity, ability to use fine tools, and patience



For more about careers, visit go.hrw.com and type in the keyword **HM6 Careers**.

FIGURE 23-5

Bacteria provide essential nutrients to many organisms. Members of the genus *Rhizobium* live symbiotically within these soybean root nodules. They fix nitrogen from the air into ammonia, a form of nitrogen that both bacteria and plants can use to make molecules such as amino acids, proteins, and nucleic acids.



IMPORTANT BACTERIAL GROUPS

In addition to classifying bacteria based on their shape and reaction to Gram staining, biologists also classify bacteria by their biochemical properties and evolutionary relationships. Studies of DNA and rRNA similarities have improved biologists’ understanding of how bacteria are related to each other. But questions about the flow of genes from one species to another and other issues remain. Prokaryotes can pick up genes from DNA in the environment by transformation. This passing of DNA from one species to another can sometimes confuse true evolutionary relationships. However, most rRNA comparison data support the bacterial lineages that are discussed below.

Proteobacteria

Proteobacteria make up one of the largest and most diverse groups of bacteria. This group contains several subgroups that have several nutritional needs and includes both aerobic and anaerobic bacteria.

Many proteobacteria live symbiotically with other organisms. For example, the nitrogen-fixing bacteria (genus *Rhizobium*) seen in Figure 23-5 live in nodules inside the roots of legumes. Legumes are plants of the pea and bean family. Legumes also include alfalfa and clover.

Nitrogen-fixing bacteria are important to the success of many ecosystems. Even though almost 80 percent of Earth's atmosphere is made up of nitrogen gas, N_2 , plants and animals cannot use it directly to make proteins, nucleic acids, and other nitrogen-containing molecules. In a process called *nitrogen fixation*, species of the genus *Rhizobium* convert nitrogen in the atmosphere to ammonia, which plants can use. These species are a major source of soil nitrogen for plants. The huge populations in root nodules can make about 250 kg (114 lb) of fixed nitrogen per hectare (2.5 acres) of alfalfa in a single year. Soil bacteria of the genus *Nitrosomonas*, another genus of proteobacteria, play an important part in the nitrogen cycle. Nitrosomonads form nitrite from ammonia in the soil. Nitrites are then converted into nitrates by other bacteria. Plants also use nitrates as a source of nitrogen.

Some proteobacteria cause diseases in plants and animals. For example, species of the genus *Agrobacterium* cause tumors in plants. Rickettsiae cause diseases such as Rocky Mountain spotted fever in humans. *Helicobacter pylori*, shown in the chapter opener photo, is a bacterium that commonly causes stomach ulcers.

Some species of proteobacteria are enteric bacteria—they live in human and animal intestines. Enteric bacteria such as *Escherichia coli* make vitamin K and help digestive enzymes in the breakdown of foods in the intestines. Some strains of *E. coli* and species of the genus *Salmonella* cause foodborne illnesses either by invading the cells that line the intestines or by making toxins.

Gram-Positive Bacteria

Most members of this group are Gram-positive, but not all are. Biologists place a few species of Gram-negative bacteria in this group because these species are genetically similar to Gram-positive bacteria. This group is very large and has many members. Members include the streptococcal species that causes strep throat and *Clostridium botulinum*, the bacterium that makes the toxin that causes botulism. Botulinum toxin is used medicinally to treat painful muscle spasms and, more recently, to erase “frown lines” from the face. Lactic acid bacterial species, such as species of the genus *Lactobacilli*, which turn milk sour and make yogurt, are also members of this phylum. Anthrax is caused by the Gram-positive rod *Bacillus anthracis*, shown in Figure 23-6. *B. anthracis* is often linked to its use as a biological weapon.

Actinomycetes (AK-tuh-noh-MIE-SEETS) are Gram-positive bacteria that form branching filaments of colonies. Many species of actinomycetes grow in soil and make **antibiotics**, chemicals that inhibit the growth of or kill other microorganisms. Antibiotics kill neighboring bacteria and fungi that compete for resources. Streptomycin (which is made by species of the genus *Streptomyces*) and tetracycline are examples of antibiotics that are used medicinally. Members of the genus *Mycobacteria*, another genus of actinomycetes, cause tuberculosis and Hansen's disease (leprosy).

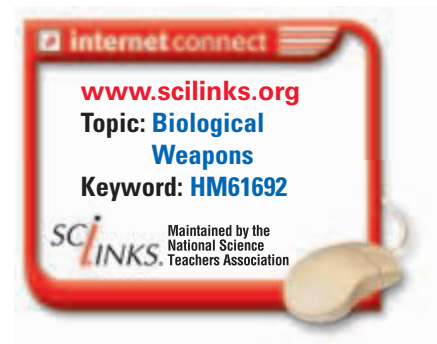


FIGURE 23-6

Human infection by Gram-positive *Bacillus anthracis* (a) is usually through a cut in the skin (cutaneous anthrax). However, inhaling endospores may result in pulmonary anthrax. If endospores reach the intestine, gastrointestinal anthrax may result. Photo (b) shows the ulcerated lesion, called an *eschar* (ES-KAHR), of cutaneous anthrax on a shoulder.



(a)



(b)



FIGURE 23-7

Cyanobacteria, such as this filament of cells of the genus *Anabaena*, play an important role in the carbon cycle because they take up large amounts of carbon from the atmosphere during photosynthesis. The orange cells are the heterocysts in which nitrogen fixation occurs.

Cyanobacteria

Cyanobacteria use photosynthesis to get energy from sunlight and make carbohydrates from water and the carbon dioxide in the air. They give off oxygen as a waste product. Certain cyanobacteria, such as those in the genus *Anabaena* shown in Figure 23-7, grow in filaments. Some of the organisms in the genus *Anabaena* form specialized cells called *heterocysts*. Heterocysts have enzymes for fixing atmospheric nitrogen into ammonia, which plants can use. Nitrogen-fixing cyanobacteria have nature's simplest nutritional requirements. They need only light energy, carbon dioxide, nitrogen, water, and some minerals in order to grow.

Cyanobacteria are very numerous and offer a large amount of food to marine and freshwater ecosystems. They were Earth's first oxygen-producing organisms. They are believed to have been responsible for making the oxygen-rich atmosphere in which aerobic (oxygen-using) organisms evolved. Once called *blue-green algae*, cyanobacteria are now known to be bacteria because they lack a membrane-bound nucleus and chloroplasts.

Spirochetes

Spirochetes (SPIE-roh-KEETS) are Gram-negative, spiral-shaped bacteria. Some are aerobic, and some are anaerobic. They move by means of a corkscrew-like rotation. Spirochetes live freely or as pathogens. The spirochete *Treponema pallidum* causes syphilis, a sexually transmitted infection (STI), and *Borrelia burgdorferi* causes Lyme disease.

Chlamydia

Gram-negative, coccoid pathogens of the group Chlamydia live only inside animal cells. They depend on the animal cells for protection and nutrients. Unlike the cell walls of other bacteria, the cell walls of chlamydia do not have peptidoglycan. *Chlamydia trachomatis* causes nongonococcal urethritis, or chlamydia, an STI.

SECTION 1 REVIEW

1. Describe the major ways in which archaea differ from bacteria and eukarya.
2. Name three habitats where archaea live.
3. Explain how a Gram stain helps identify bacteria.
4. Name one organism from each of the five groups of bacteria discussed.
5. Discuss the importance of nitrogen-fixing bacteria.
6. Describe the significance of cyanobacteria in the formation of Earth's atmosphere.

CRITICAL THINKING

7. **Applying Information** If Earth suddenly lost its light source but kept the same temperature, what forms of life would most likely remain?
8. **Relating Information** Would you expect to find thermoacidophilic bacteria and enteric bacteria living in the same environment? Explain.
9. **Inferring Relationships** Certain antibiotics kill only Gram-positive bacteria. Why is this information important to consider when treating a patient who has a bacterial infection of unknown cause?

BIOLOGY OF PROKARYOTES

A look at prokaryotic cells under a light microscope shows that they are very small. Such a view, however, leaves unexplained the complexity of the cells' internal workings. Here, we will study the structure of prokaryotic cells and the many ways that they gather nutrients and energy and reproduce.

STRUCTURE AND FUNCTION

In general, prokaryotes are made up of a cell wall, a cell membrane, and cytoplasm that has ribosomes, DNA, and small molecules and ions. Recall that prokaryotes do not have membrane-bound organelles. Additional distinctive structures help some bacteria adapt to their own niches.

Cell Wall

Most members of the domains Bacteria and Archaea have a cell wall. The cell wall gives a cell its shape and protects the cell from toxic substances. The cell walls of many disease-causing bacteria have structures that add to the disease-causing abilities of the organisms. Bacterial cell walls are made of peptidoglycan, not cellulose, which makes up plant cell walls. Archaeal cell walls do not have peptidoglycan; instead, some of them contain *pseudomurein*, a compound made of unusual lipids and amino acids.

Cell Membrane and Cytoplasm

Just inside the cell wall, prokaryotes have a cell membrane, or *plasma membrane*. Both bacterial and archaeal cell membranes are lipid bilayers that have proteins. However, the lipids and proteins of archaeal cell walls differ from those of bacterial cell walls.

The cell membrane acts as a selective barrier by controlling the molecules that move into and out of the cell. In this way, the cell membrane helps keep the internal environment of the organism stable. The cell membrane is also the place where many important metabolic functions occur. For example, enzymes that catalyze the reactions of cellular respiration are found in the bilayer.

Photosynthetic bacteria have cell membranes whose internal foldings are called *thylakoids* (THIE-luh-KOYDZ). These foldings are much like the stacked membranes of plant chloroplasts. Like thylakoids in plant chloroplasts, bacterial thylakoids have photosynthetic pigments that trap light energy.

OBJECTIVES

- **Describe** the internal and external structure of prokaryotic cells.
- **Identify** the need for endospores.
- **Compare** four ways in which prokaryotes get energy and carbon.
- **Identify** the different types of environments in which prokaryotes can live.
- **List** three types of genetic recombination that prokaryotes use.

VOCABULARY

plasmid
capsule
glycocalyx
pilus
endospore
heterotroph
autotroph
phototroph
chemotroph
obligate anaerobe
facultative anaerobe
obligate aerobe
transformation
conjugation
transduction

Word Roots and Origins

pseudomurein

from the Greek *pseudes*, meaning "false," and the Latin *murus*, meaning "wall"

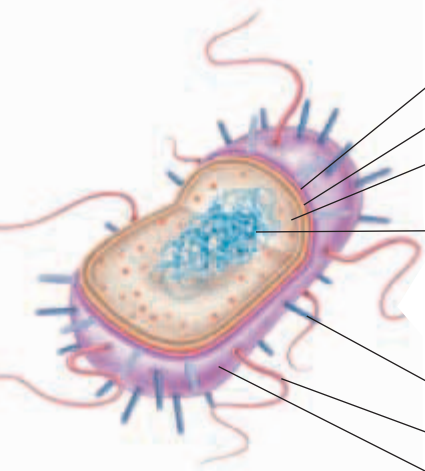


TABLE 23-1 *Structural Characteristics of a Bacterial Cell*

Structure	Function
Capsule	protects the cell and helps the cell attach to other cells and surfaces
Cell wall	protects the cell and gives the cell its shape
Cell membrane	regulates the types of molecules that move into and out of the cell
Cytoplasm	contains DNA, ribosomes, and organic compounds that are needed for life
Chromosome	carries genetic information from one generation to the next
Plasmid	carries genes that are transferred through genetic recombination; is a small, circular DNA loop
Endospore	contains DNA; is a thick-coated, resistant structure
Pilus	helps the cell attach to surfaces and other cells during conjugation
Flagellum	propels the cell by rotating in a whiplike motion
Outer membrane	protects the cell against some antibiotics (present only in Gram-negative bacteria)

Bacterial cells do not have membrane-bound organelles in their cytoplasm. Instead, the cytoplasm is a semifluid solution that contains ribosomes, DNA, small organic and inorganic molecules, and ions, as shown in Table 23-1.

DNA

Prokaryotic DNA is a single closed loop of double-stranded DNA attached at one point to the cell membrane. Unlike eukaryotic DNA, prokaryotic DNA is not enclosed in a nucleus. Along with this single main chromosome, some prokaryotes have plasmids. **Plasmids** are small, circular, self-replicating loops of double-stranded DNA. Plasmids are usually not necessary for the cell's growth and reproduction, but some plasmids carry genes that enable the bacterium to cause disease. Other plasmids carry genes that provide the bacterium with resistance to certain antibiotics.

Capsules and Pili

Many bacteria have an outer covering of polysaccharides called a **capsule**. These sugars bind to the cell wall and protect the cell against drying or harsh chemicals. The capsule also helps protect a pathogenic (disease-causing) bacterium from the host's white blood cells, which could otherwise engulf the bacterium. A capsule made up of a fuzzy coat of sticky sugars is called a **glycocalyx** (GLIE-koh-KAY-likes). This structure allows bacteria to connect to the surface of host cells and tissues.

Pili (PIL-ee) (singular, *pilus*) are short, hairlike protein structures on the surface of some bacteria. Pili help bacteria connect to each other and to surfaces, such as those of a host cell. Pili can also serve as a bridge to pass genetic material between bacteria.

Word Roots and Origins

pilus

from the Latin *pilus*,
meaning "hair"

Endospores

Some Gram-positive bacteria can form a thick-coated, resistant structure called an **endospore** when environmental conditions become harsh. The harsh conditions may destroy the original cell, but the endospore containing the cell's DNA can survive. Endospores can resist high temperatures, strong chemicals, radiation, drying, and other environmental extremes. When good conditions return, the endospore gives rise to a normal bacterial cell. Species of the genera *Bacillus* and *Clostridium* can form endospores. Figure 23-8 shows a *C. botulinum* cell with an endospore. The endospores of *C. botulinum* can germinate in improperly sterilized canned foods. The bacteria make a toxin that if eaten can cause the nerve disease *botulism*. While rare, botulism is often fatal if it is not treated promptly.

Prokaryotic Movement

Many prokaryotes have long flagella that allow the prokaryotes to move toward food sources or away from danger. Movement toward or away from a stimulus is called *taxis* (TAKS-is). In *chemotaxis*, prokaryotes react to chemical stimuli by moving toward food or away from a toxin. Prokaryotic flagella rotate and move the prokaryote in a “run-and-tumble” motion. Prokaryotes can have a single whiplike flagellum or many flagella.

Some bacteria do not have flagella but can move by other means. Species of the genus *Myxobacteria*, for example, form a layer of slime. Wavelike contractions of the outer membrane move the organisms through the slime. Some spiral-shaped bacteria move by a corkscrew-like rotation. Filaments inside the organism's cell walls contract and cause the bacterium to turn and move ahead.

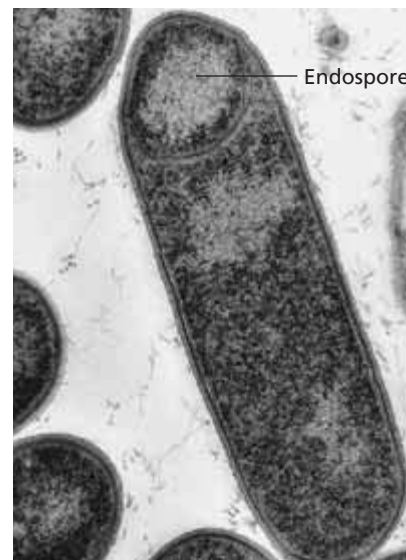


FIGURE 23-8

Unfavorable conditions can cause some bacteria, such as the Gram-positive rod *Clostridium botulinum*, to form endospores. After forming the endospore, the bacterial cell disintegrates, and the endospore containing the cell's DNA is released. Endospores can survive in harsh environments for a long time.

NUTRITION AND METABOLISM

Prokaryotes have two chief nutritional needs: a source of carbon to build the organic molecules of their cells and a source of energy. They have many ways of getting both carbon and energy from the environment. Organisms that get their carbon from other organisms are called **heterotrophs**. **Autotrophs** are organisms that get their carbon directly from the inorganic molecule carbon dioxide, CO_2 . Organisms that get energy from light are called **phototrophs**. **Chemotrophs** get energy from chemicals taken from the environment. Table 23-2 shows how these ways of getting nutrients can be used to divide prokaryotes into four nutritional groups.

TABLE 23-2 Major Bacterial Nutritional Modes

Nutritional mode	Energy and carbon source
Heterotroph	
Photoheterotroph	uses light energy but gets its carbon from other organisms
Chemoheterotroph	obtains both energy and carbon from other organisms
Autotroph	
Photoautotroph	uses light energy and gets carbon from CO_2
Chemoautotroph	extracts energy from inorganic compounds and uses CO_2 as a carbon source



FIGURE 23-9

Prokaryotes that have different pigments live at different temperatures in a Yellowstone hot spring. Some are photoautotrophs, some are chemoautotrophs, and others are chemoheterotrophs.



FIGURE 23-10

Thermus aquaticus cells form a long, stringy mass in the outflow channel of a hot spring. *T. aquaticus* was first isolated from a hot spring in Yellowstone by Thomas D. Brock. Later, the heat-tolerant enzyme *Taq* polymerase was isolated from *T. aquaticus*. Biotechnologists now routinely use *Taq* polymerase to rapidly replicate pieces of DNA in the polymerase chain reaction (PCR).

Most prokaryotes that we have studied so far are photoautotrophs (FOHT-oh-AWT-oh-TROHFS) and chemoheterotrophs (KEE-moh-HET-uhr-oh-TROHFS). Prokaryotes such as cyanobacteria that gather sunlight for photosynthesis are photoautotrophs. Photoautotrophs harvest energy with light-trapping compounds similar to those of plants. Chemoautotrophs, a group of autotrophs, oxidize inorganic compounds to get energy. For example, members of the genus *Nitrosomonas* oxidize ammonia, NH_3 , forming nitrite, NO_2^- , and use the released energy. Methanogens living in a cow's digestive tract make methane gas from inorganic molecules. Pathogenic (disease-causing) bacteria are chemoheterotrophs. Figure 23-9 shows an environment in which various prokaryotes that have different kinds of nutritional needs live close together.

PROKARYOTIC HABITATS

Prokaryotes live in certain habitats based on their biochemical abilities. Many bacteria are **obligate anaerobes** (AHB-luh-git AN-uhr-OHBS), organisms that cannot live where molecular oxygen, O_2 , is present. The obligate anaerobe *Clostridium tetani* causes the nerve disease *tetanus*. **Facultative anaerobes** (FAK-uhl-TAYT-iv AN-uhr-OHBS) can live with or without oxygen. *Escherichia coli*, which is common in the human digestive tract, is a facultative anaerobe. Prokaryotes that need oxygen to live are called **obligate aerobes**. The obligate aerobe *Mycobacterium tuberculosis* causes the disease tuberculosis (TB).

Prokaryotes have varying temperature requirements for growth. Psychrophilic (SIE-kroh-FIL-ik) (cold-loving) prokaryotes grow well at 0°C to 20°C (32°F to 68°F). Their cellular enzymes and cell membranes work well at cold temperatures. Some prokaryotes of the Antarctic survive very cold temperatures by living just under the surfaces of rocks. The weak sunlight warms the rocks just enough so that the psychrophiles have a slow-growing life cycle.

Prokaryotes that grow best at temperatures between 20°C (68°F) and 40°C (104°F) are called *mesophiles*. Humans are also classified as mesophiles. Prokaryotes that grow at very hot temperatures—between 45°C (113°F) and 110°C (230°F)—are called *thermophiles* (THUHR-moh-FIELZ). Thermophilic species have evolved proteins that resist being destroyed by heat. The development of biotechnology owes much to the purification of the heat-stable, DNA-replicating enzyme *Taq* polymerase from the thermophile *Thermus aquaticus*, shown in Figure 23-10. *T. aquaticus* was first isolated from a hot spring in Yellowstone National Park.

Most bacterial species grow best at a pH of 6.5 to 7.5 (7.0 is neutral). However, certain species, such as those that make yogurt and sour cream from milk, favor acidic environments that have a pH of 6.0 or lower. Acid-loving prokaryotes are called *acidophiles*.